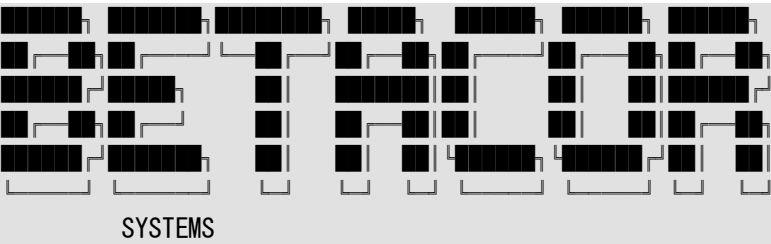


MACHINE BETA BC-500X

Comprehensive Operation, Maintenance, and Service Manual

Manufacturer: BetaCorp Systems, Inc. **Model:** BC-500X CNC Machining Center **Document Number:** BCS-MAN-BC500X-REV4.2 **Revision Date:** September 2026 **Language:** English (EN-US)



BC-500X CNC MACHINING CENTER COMPLETE TECHNICAL MANUAL

IMPORTANT SAFETY NOTICE This manual must be read in its entirety before installing, operating, or servicing the BC-500X machine. Failure to follow all instructions, warnings, and cautions contained herein may result in serious personal injury, death, or significant property damage. BetaCorp Systems, Inc. assumes no liability for injuries or damages resulting from failure to comply with the procedures described in this document.

BetaCorp Systems, Inc. 4500 Industrial Parkway Greenfield, OH 45123 United States of America

Technical Support: +1 (800) 555-BETA (2382) International: +1 (740) 555-9200 Fax: +1 (740) 555-9201 Email: techsupport@betacorpssystems.com Web: www.betacorpssystems.com

Parts & Service: parts@betacorpssystems.com **Emergency Hotline (24/7):** +1 (800) 555-9911

Copyright © 2024–2026 BetaCorp Systems, Inc. All rights reserved. No part of this manual may be reproduced, transmitted, stored in a retrieval system, or translated into any language in any form or by any means without the prior written permission of BetaCorp Systems, Inc. BetaCorp Systems reserves the right to make changes to this manual and to the products described herein without notice.

Printed in the United States of America Document Part Number: BCS-DOC-10042-EN

WARNINGS, CAUTIONS, AND NOTES — HOW TO USE THIS MANUAL

Throughout this manual, the following signal words are used to alert the reader to conditions that

may affect personal safety or equipment integrity:

DANGER — Indicates an imminently hazardous situation which, if not avoided, **WILL** result in death or serious injury.

WARNING — Indicates a potentially hazardous situation which, if not avoided, **COULD** result in death or serious injury.

CAUTION — Indicates a potentially hazardous situation which, if not avoided, **MAY** result in minor or moderate injury, or equipment damage.

NOTE — Provides additional information that may be helpful but is not safety-critical.

REVISION HISTORY

Revision	Date	Author	Changes
1.0	2024-01-15	J. Harrington	Initial release
1.1	2024-03-20	M. Kowalski	Added hydraulic section updates
2.0	2024-07-01	Engineering Team	Major revision: Tool changer service chapter added
2.1	2024-09-14	R. Okonkwo	Fault code table expanded to 150+ codes
3.0	2025-01-10	J. Harrington	Parameter reference revised for firmware v3.x
3.1	2025-04-22	M. Kowalski	Coolant system chapter rewritten
4.0	2025-10-01	Engineering Team	Encoder section updated; spare parts catalog revised
4.1	2026-02-14	R. Okonkwo	Safety chapter expanded per OSHA 2026 updates
4.2	2026-09-01	J. Harrington	Current release; hydraulic diagram revised

TABLE OF CONTENTS

PART 1 — INTRODUCTION AND SAFETY

Chapter 1 — Introduction and General Information

- 1.1 About This Manual 28
- 1.2 Machine Identification 29
- 1.3 Intended Use 30
- 1.4 Foreseeable Misuse 31
- 1.5 Residual Risks 32
- 1.6 Applicable Standards and Regulations 33
- 1.7 Environmental Conditions for Operation 34
- 1.8 Noise Emission Data 35
- 1.9 Electromagnetic Compatibility (EMC) 36
- 1.10 Warranty Information 37
- 1.11 Disposal and Recycling 38

Chapter 2 — Safety Instructions 40

- 2.1 General Safety Philosophy 40
- 2.2 Qualified Personnel Requirements 42
- 2.3 Personal Protective Equipment (PPE) 44
- 2.4 Electrical Safety 48
 - 2.4.1 Arc Flash Hazard Assessment 50
 - 2.4.2 Lockout/Tagout (LOTO) Procedures 54
 - 2.4.3 Capacitor Discharge Waiting Periods 61
 - 2.4.4 Ground Fault Protection 63
- 2.5 Mechanical Safety 65
 - 2.5.1 Rotating Component Hazards 67
 - 2.5.2 Pinch Point Identification 69
 - 2.5.3 Stored Energy — Springs and Counterweights 72
- 2.6 Hydraulic Pressure Hazards 74
 - 2.6.1 High-Pressure Injection Injury Risk 76
 - 2.6.2 Hydraulic Line De-pressurization 78

• 2.6.3 Hot Oil Burns	80
• 2.7 Pneumatic System Safety	82
• 2.8 Coolant and Chemical Hazards	84
• 2.9 Chip and Swarf Hazards	87
• 2.10 Emergency Stop Procedures	89
• 2.10.1 E-Stop Locations	90
• 2.10.2 Machine Restart After E-Stop	92
• 2.11 Fire Safety	94
• 2.12 Guards and Interlocks — Never Bypass	97
• 2.13 Safe Work Practices During Maintenance	100
• 2.14 Confined Space Considerations	104
• 2.15 Chemical Right-to-Know (GHS/SDS)	106

PART 2 — TECHNICAL SPECIFICATIONS

Chapter 3 — Technical Specifications	110
• 3.1 Machine Overview	110
• 3.2 Work Envelope and Travel Specifications	112
• 3.3 Spindle Specifications	114
• 3.4 Feed and Rapid Traverse Rates	116
• 3.5 Tool Changer Specifications	117
• 3.6 Electrical Specifications	119
• 3.6.1 Main Power Supply Requirements	119
• 3.6.2 Control Cabinet Power Distribution	121
• 3.6.3 Motor Power Ratings	122
• 3.7 Hydraulic System Specifications	124
• 3.8 Pneumatic System Specifications	126
• 3.9 Coolant System Specifications	127
• 3.10 Axis Servo System Specifications	128
• 3.11 Encoder Specifications	130
• 3.12 Physical Dimensions and Weight	132
• 3.13 Ambient Environmental Specifications	134

- 3.14 Noise and Vibration 135
- 3.15 CE/UL/CSA Compliance Data 136

PART 3 — INSTALLATION

Chapter 4 — Installation Guide	140
• 4.1 Pre-Installation Planning	140
• 4.2 Site Preparation and Foundation	142
• 4.2.1 Foundation Drawing and Requirements	143
• 4.2.2 Anchor Bolt Installation	147
• 4.2.3 Leveling Pads and Vibration Isolation	150
• 4.3 Machine Delivery and Rigging	153
• 4.3.1 Lifting Points and Rigging Diagram	155
• 4.3.2 Unpacking and Inspection	158
• 4.4 Machine Leveling	161
• 4.5 Electrical Connection	164
• 4.5.1 Power Supply Wiring to Main Disconnect	165
• 4.5.2 Grounding Requirements	168
• 4.5.3 Control Transformer Tap Selection	170
• 4.6 Hydraulic System Connection	172
• 4.6.1 Hydraulic Power Unit Installation	174
• 4.6.2 Hydraulic Line Connection and Flushing	176
• 4.7 Pneumatic System Connection	179
• 4.8 Coolant System Filling	182
• 4.9 Chip Conveyor Installation	185
• 4.10 Initial Power-On and System Checkout	187
• 4.11 Commissioning Procedure	191
• 4.12 Geometry and Accuracy Verification	196
• 4.13 Final Acceptance Testing	200

PART 4 — MACHINE COMPONENTS

Chapter 5 — Machine Components	205
• 5.1 Machine Base and Column	205

• 5.2 Spindle Assembly	208
• 5.2.1 Spindle Motor	210
• 5.2.2 Spindle Bearings	212
• 5.2.3 Spindle Encoder	214
• 5.2.4 Tool Retention (Draw Bar) System	217
• 5.3 Axis Servo Motors and Drives	220
• 5.3.1 X-Axis Assembly	222
• 5.3.2 Y-Axis Assembly	224
• 5.3.3 Z-Axis Assembly	226
• 5.3.4 Ballscrew and Nut Assemblies	228
• 5.3.5 Linear Guideways	230
• 5.4 Tool Changer Carousel (24-Position)	232
• 5.4.1 Carousel Drive Motor and Gear Assembly	235
• 5.4.2 Tool Pot and Gripper Assembly	238
• 5.4.3 ATC Arm Assembly	241
• 5.4.4 ATC Position Sensors	244
• 5.5 Hydraulic Power Unit	247
• 5.5.1 Hydraulic Pump	249
• 5.5.2 Hydraulic Valves	251
• 5.5.3 Hydraulic Cylinders	254
• 5.6 Coolant System	257
• 5.6.1 Coolant Tank and Pump	259
• 5.6.2 Coolant Filters	261
• 5.6.3 Coolant Nozzles and Distribution	263
• 5.7 Chip Conveyor	265
• 5.8 Lubrication System (Way Lube)	267
• 5.9 Enclosure and Guarding	270
• 5.10 Control Panel and Operator Interface	272
• 5.11 Pendant and MPG Hand Wheel	275
• 5.12 Electrical Cabinet	277

PART 5 — OPERATION

Chapter 6 — Operation Manual	282
• 6.1 Operator Qualification and Training	282
• 6.2 Pre-Operation Safety Inspection	284
• 6.3 Machine Power-On Sequence	287
• 6.4 Machine Homing (Reference Return) Procedure	291
• 6.5 Control Panel Layout and Functions	295
• 6.6 Manual Operation Modes	300
• 6.6.1 JOG Mode	302
• 6.6.2 HANDLE (MPG) Mode	304
• 6.6.3 Single-Block Mode	306
• 6.7 Program Entry and Editing	308
• 6.7.1 MDI (Manual Data Input) Mode	310
• 6.7.2 Program Storage and Management	312
• 6.8 Tool Management	315
• 6.8.1 Tool Length Offset Setup	317
• 6.8.2 Tool Radius Offset Setup	320
• 6.8.3 Tool Life Management	322
• 6.9 Workpiece Setup and Fixture	325
• 6.9.1 Work Coordinate System Setup	327
• 6.9.2 Part Probing (Optional)	330
• 6.10 Coolant System Operation	332
• 6.11 Automatic Cycle Operation	335
• 6.11.1 Program Selection	337
• 6.11.2 Cycle Start and Feed Hold	339
• 6.11.3 Override Functions	341
• 6.12 Machine Power-Off Sequence	344
• 6.13 Emergency Situations During Operation	346

PART 6 — PARAMETER REFERENCE

Chapter 7 — Parameter Reference	350
--	------------

- 7.1 Introduction to BC-500X Parameters 350
- 7.2 Parameter Group B1.xx — Spindle Parameters 354
- 7.3 Parameter Group B2.xx — Axis Motion Parameters 368
- 7.4 Parameter Group B3.xx — Tool Changer Parameters 382
- 7.5 Parameter Group B4.xx — Hydraulic System Parameters . 396
- 7.6 Parameter Group B5.xx — Coolant System Parameters ... 408
- 7.7 Parameter Group B6.xx — Encoder/Feedback Parameters . 420
- 7.8 Parameter Group B7.xx — Communication Parameters ... 432
- 7.9 Parameter Group B8.xx — Safety and Interlock Parameters 442
- 7.10 Parameter Group B9.xx — Diagnostic Parameters 454

PART 7 — FAULT CODES AND TROUBLESHOOTING

Chapter 8 — Fault Codes and Troubleshooting 466

- 8.1 Fault System Overview 466
- 8.2 Fault Log Access and Management 469
- 8.3 Fault Code Reference Table — Complete Listing 472
- 8.4 Critical Faults (E001–E099) 475
- 8.5 Encoder and Feedback Faults (E100–E199) 490
 - 8.5.1 E101 — Spindle Encoder Signal Lost 491
- 8.6 Coolant System Faults (E200–E299) 510
 - 8.6.1 E202 — Coolant Level Below Minimum 511
- 8.7 Tool Changer Faults (E300–E399) 526
 - 8.7.1 E303 — Tool Changer Timeout 527
- 8.8 Hydraulic System Faults (E400–E499) 544
 - 8.8.1 E404 — Hydraulic Pressure Low 545
- 8.9 Servo System Faults (E500–E599) 562
 - 8.9.1 E505 — Axis Servo Overload 563
- 8.10 Communication and I/O Faults (E600–E699) 580
- 8.11 PLC and Logic Faults (E700–E799) 594
- 8.12 Thermal and Environmental Faults (E800–E899) 606
- 8.13 System Configuration Faults (E900–E999) 618

PART 8 — MAINTENANCE

Chapter 9 — Maintenance Schedule 630

- 9.1 Maintenance Philosophy and Preventive Maintenance Overview ... 630
- 9.2 Daily Maintenance Tasks 634
- 9.3 Weekly Maintenance Tasks 642
- 9.4 Monthly Maintenance Tasks 650
- 9.5 Quarterly Maintenance Tasks 660
- 9.6 Semi-Annual Maintenance Tasks 670
- 9.7 Annual Maintenance Tasks 682
- 9.8 Hydraulic Oil Change Procedure 694
- 9.9 Coolant Replacement Procedure 702
- 9.10 Spindle Bearing Inspection 710
- 9.11 Tool Changer Lubrication 716
- 9.12 Encoder Cable Inspection and Replacement 722
- 9.13 Ballscrew and Guideway Maintenance 728
- 9.14 Way Lube System Maintenance 734
- 9.15 Filter Replacement Schedule 740

PART 9 — SPARE PARTS CATALOG

Chapter 10 — Spare Parts Catalog 748

- 10.1 How to Order Parts 748
- 10.2 Spindle Assembly Parts 751
- 10.3 Axis Drive Parts 758
- 10.4 Tool Changer Parts 764
- 10.5 Hydraulic System Parts 770
- 10.6 Coolant System Parts 776
- 10.7 Electrical and Control Parts 782
- 10.8 Enclosure and Structure Parts 788
- 10.9 Lubrication System Parts 792
- 10.10 Chip Conveyor Parts 795
- 10.11 Consumables and Wear Items 798

PART 10 — HYDRAULIC SYSTEM MANUAL

Chapter 11 — Hydraulic System Manual	804
• 11.1 Hydraulic System Overview	804
• 11.2 Hydraulic Circuit Description	807
• 11.3 Hydraulic Power Unit (HPU)	812
• 11.4 Pressure Settings and Adjustment	816
• 11.5 Hydraulic Valve Descriptions	820
• 11.6 Hydraulic Cylinder Operation	826
• 11.7 Hydraulic Oil Specifications	830
• 11.8 Contamination Control	834
• 11.9 Hydraulic System Diagnostics	838

PART 11 — COOLANT SYSTEM GUIDE

Chapter 12 — Coolant System Guide	844
• 12.1 Coolant System Overview	844
• 12.2 Coolant Selection and Mixing Ratios	847
• 12.3 pH Monitoring and Control	853
• 12.4 Bacterial Contamination Testing	858
• 12.5 Coolant Maintenance Procedures	863
• 12.6 Tramp Oil Removal	868
• 12.7 Coolant Disposal Procedures	872
• 12.8 Coolant System Troubleshooting	876

PART 12 — TOOL CHANGER SERVICE

Chapter 13 — Tool Changer Service Manual	882
• 13.1 Tool Changer Overview	882
• 13.2 Carousel Positioning and Indexing	886
• 13.3 Gripper Adjustment and Replacement	892
• 13.4 ATC Sensor Setup and Adjustment	898
• 13.5 ATC Arm Timing and Adjustment	904
• 13.6 Tool Changer Troubleshooting	910

PART 13 — APPENDICES

Appendix A — Glossary of Terms	918	Appendix B — Hydraulic Fluid Specifications	928
Appendix C — Torque Specifications Chart	934	Appendix D — Electrical Schematic Index	940
Appendix E — Pneumatic Schematic Index	944	Appendix F — Service Contact Directory	948
Appendix G — Spare Parts Quick-Reference	952	Appendix H — Parameter Quick-Reference Card	956
Appendix I — Maintenance Log Template	960	Appendix J — Warranty Registration	964

PART 1 — INTRODUCTION AND SAFETY

Chapter 1 — Introduction and General Information

1.1 About This Manual

This manual provides complete technical documentation for the BetaCorp Systems BC-500X CNC Machining Center. It is intended to serve as the primary reference for all personnel involved in the installation, operation, maintenance, and service of this machine tool. The manual is organized into thirteen parts and multiple appendices, covering every aspect of the machine from initial site preparation through advanced troubleshooting.

Who Should Read This Manual:

- **Installation Engineers:** Read Chapters 1, 2, 3, and 4 completely before beginning any installation activities.
- **Machine Operators:** Read Chapters 1, 2, 3, and 6 completely. Refer to Chapter 8 for fault code resolution.
- **Maintenance Technicians:** Read all chapters. Pay particular attention to Chapters 9, 11, 12, and 13.
- **Service Engineers:** All chapters are required reading. Chapters 5, 7, 8, 11, 12, and 13 are of primary importance.
- **Supervisors and Safety Officers:** Chapters 1 and 2 are essential. Review Chapter 9 for maintenance scheduling.

Manual Organization:

This manual follows a logical progression from general information through increasingly detailed technical content. Part 1 establishes the safety foundation that must be understood before any work is performed. Subsequent parts build on this foundation with specific technical information for each machine system.

Cross-references throughout the manual direct the reader to related information in other sections.

When a procedure references a parameter number (e.g., Parameter B1.14), that parameter is fully described in Chapter 7. When a procedure references a fault code (e.g., E404), full details are found in Chapter 8.

Electronic Version:

This manual is available in PDF format on the BetaCorp Systems customer portal at docs.betacorp-systems.com. The electronic version includes hyperlinked cross-references and bookmarks for rapid navigation. Always verify you are viewing the current revision by checking the document number and revision on the cover page against the revision posted on the customer portal.

Language:

This manual is available in English, German, French, Spanish, Italian, Portuguese, Japanese, Korean, and Mandarin Chinese. Contact BetaCorp Systems to obtain translated copies. In the event of any discrepancy between the English version and a translated version, the English version shall take precedence.

1.2 Machine Identification

The BC-500X CNC Machining Center carries multiple identification numbers that are required when contacting BetaCorp Systems for technical support, spare parts, or warranty service.

Machine Serial Number Plate Location:

The primary serial number plate is mounted on the right side of the machine base, approximately 400 mm (15.75 in) from the front of the machine, at approximately 900 mm (35.4 in) above floor level. A secondary serial number label is located inside the main electrical cabinet on the door inner panel.

Information on the Serial Number Plate:

Field	Description	Example
Model	Machine model designation	BC-500X
Serial No.	Unique machine serial number	BC500X-2025-04817
Mfg. Date	Date of manufacture (YYYY-MM)	2025-08
Voltage	Machine rated voltage	480V / 3Ph / 60Hz
Max Power	Connected load in kVA	45 kVA
Weight	Machine net weight	8,500 kg
CE Mark	European conformity (if applicable)	CE

Control System Identification:

The CNC control unit carries its own serial number, which is visible on the control pendant and also accessible through the diagnostic menu: SYSTEM → DIAGNOSTIC → CONTROL INFO.

Field	Description
Control Model	BetaCorp CNC Controller Model 5000
Firmware Version	v4.2.1 or later
PLC Version	v3.8 or later
HMI Version	v4.2.0 or later

When Contacting Technical Support:

Always have the following information ready:

1. Machine model (BC-500X)
2. Machine serial number
3. Control firmware version
4. Nature of the problem or fault code displayed
5. Recent maintenance history relevant to the issue
6. Customer name, contact name, phone, and email

1.3 Intended Use

The BC-500X CNC Machining Center is a computer numerically controlled vertical machining center designed for precision metal cutting operations. The machine is intended exclusively for:

Approved Workpiece Materials:

- Carbon steel (up to 60 HRC with appropriate tooling)
- Alloy steel (including tool steels, stainless steels)
- Cast iron (gray iron, ductile iron, malleable iron)
- Aluminum alloys (cast and wrought)
- Copper and copper alloys (brass, bronze)
- Titanium alloys (with appropriate speeds and coolant flow)
- Nickel-based superalloys (with reduced parameters and appropriate tooling)
- Engineering plastics (with appropriate chip management)

Approved Machining Operations:

- Face milling
- Peripheral (end) milling
- Pocket milling
- Slot milling

- Drilling
- Boring (line boring and single-point boring)
- Reaming
- Tapping (rigid and floating)
- Contouring and profiling

Approved Tooling:

The BC-500X accepts tooling with CAT-40 (7/24 taper) or BT-40 (MAS 403 BT-40) tool holders. Maximum tool weight is 8 kg (17.6 lb). Maximum tool diameter in the tool magazine is 80 mm (3.15 in) for adjacent pockets occupied; 125 mm (4.92 in) for adjacent pockets empty. Maximum tool length from gauge line is 300 mm (11.81 in).

Approved Coolant:

Water-soluble metalworking fluids (semi-synthetic or fully synthetic) at concentrations between 5% and 12% as specified in Chapter 12. Neat (straight) cutting oil is NOT approved for this machine as configured from the factory. Contact BetaCorp Systems for information on conversion kits for minimum quantity lubrication (MQL) or through-spindle oil application.

1.4 Foreseeable Misuse

The following uses are NOT approved and may result in machine damage, injury, or voiding of warranty:

1. **Grinding operations:** The BC-500X is not designed for abrasive wheel grinding. Grinding dust will contaminate the linear guideways, ballscrews, and spindle bearings and will cause rapid degradation.
2. **Machining of explosive, radioactive, or highly toxic materials:** These materials require specialized equipment and are beyond the scope of this machine's design.
3. **Use as a press or stamping machine:** The machine structure and servo systems are not designed for the impulsive loading associated with pressing or stamping operations.
4. **Operation beyond specified work envelope:** Attempting to machine workpieces that require axis travel beyond the machine's specified limits will cause over-travel faults and potential collision damage.
5. **Overloading the table:** The maximum allowable workpiece and fixture weight on the table is 600 kg (1,323 lb). This limit must not be exceeded.
6. **Bypassing safety guards or interlocks:** Guards are integral to the machine's safety system. Operating with guards removed or interlocks defeated is expressly prohibited.
7. **Operation by unqualified personnel:** The machine must only be operated by personnel who have completed BetaCorp Systems' approved training program or equivalent training as documented by the employer.

8. **Modification of machine structure or control software:** Unauthorized hardware modifications or modification of machine control software will void the warranty and may create hazardous conditions.
-

1.5 Residual Risks

Despite comprehensive guarding and safety systems, certain residual risks cannot be completely eliminated. Users must be aware of the following:

Risk	Location	Protective Measure Required
Chip ejection during door opening	Work zone door	Do not open door until spindle has fully stopped; wear face shield
Coolant splash	Work zone	Wear eye protection; ensure door seals are in good condition
High-pressure coolant injury	Coolant nozzles	Never direct nozzle at skin; depressurize before adjusting
Pinch point — tool changer arm	Tool changer area	Never reach into tool change area during operation
Hot chips and workpiece	Work zone	Allow adequate cooling time; use chip hooks, never hands
Noise during cutting	Machine area	Wear hearing protection during prolonged exposure
Tripping on coolant overflow	Machine perimeter	Keep floor clean; use anti-slip flooring
Ergonomic strain during loading	Table area	Use appropriate lifting aids for heavy fixtures

1.6 Applicable Standards and Regulations

The BC-500X is designed and manufactured in compliance with the following standards and regulations:

United States:

- OSHA 29 CFR Part 1910.217 — Mechanical Power Presses (general principles applied)
- OSHA 29 CFR Part 1910.147 — Control of Hazardous Energy (LOTO)
- OSHA 29 CFR Part 1910.303–308 — Electrical Standards
- ANSI B11.8 — Safety Requirements for Turning, Milling, and Machining Centers
- ANSI B11.19 — Performance Criteria for Safeguarding
- NFPA 70 (NEC) — National Electrical Code

- NFPA 79 — Electrical Standard for Industrial Machinery

European Union (CE Marked Versions):

- Machinery Directive 2006/42/EC
- Low Voltage Directive 2014/35/EU
- EMC Directive 2014/30/EU
- EN ISO 16090-1:2017 — Machine tools safety — Machining Centres
- EN 60204-1:2018 — Safety of Machinery — Electrical Equipment of Machines
- EN ISO 13849-1:2015 — Safety of Machinery — Safety-Related Parts of Control Systems

Canada:

- CAN/CSA-Z432 — Safeguarding of Machinery
- CSA C22.2 No. 14 — Industrial Control Equipment

1.7 Environmental Conditions for Operation

The BC-500X is designed for installation and operation in controlled factory environments meeting the following conditions:

Parameter	Requirement
Ambient temperature (operation)	+5°C to +40°C (41°F to 104°F)
Ambient temperature (storage/transport)	-20°C to +60°C (-4°F to +140°F)
Relative humidity	20% to 75% non-condensing
Altitude	0 to 1,000 m (0 to 3,281 ft) above sea level
Altitude (derated)	Up to 2,000 m with power derating; contact BetaCorp
Vibration (installation site)	Less than 0.2 g RMS, 10–200 Hz
Pollution degree	Pollution Degree 2 (IEC 60664-1)
Floor loading capacity	Minimum 15,000 kg/m ² (3,073 lb/ft ²)
Magnetic fields	Avoid installation near strong magnetic field sources

NOTE: Operation outside the specified temperature range will result in degraded accuracy and potential component damage. The machine requires a minimum of 4 hours of thermal stabilization at room temperature before precision work is performed.

1.8 Noise Emission Data

Sound measurements were taken in accordance with ISO 3746:2010 under defined machining

conditions (dry run at maximum rapid traverse rate, no cutting).

Measurement	Value
A-weighted sound pressure level at operator position	74 dB(A) (dry run)
A-weighted sound pressure level at operator position	Up to 84 dB(A) (machining steel)
A-weighted sound power level	89 dB(A) (machining steel)
Peak C-weighted instantaneous sound pressure	Less than 130 dB(C)

WARNING: Continuous exposure to sound levels at or above 85 dB(A) requires hearing protection. At the maximum measured levels during steel cutting operations, workers must wear appropriate hearing protection. Refer to Section 2.3 for PPE requirements.

1.9 Electromagnetic Compatibility (EMC)

The BC-500X is designed to comply with applicable EMC requirements. The following practices must be observed to maintain compliance:

1. All cable routing must follow the guidelines in Chapter 4. Signal cables must be separated from power cables by a minimum of 200 mm (8 in) or run in separate grounded conduit.
 2. The machine must be connected to an adequately grounded power supply.
 3. Modifications to the electrical system, including the addition of non-BetaCorp components, may void EMC compliance.
 4. The machine generates conducted and radiated emissions. Sensitive electronic equipment should not be located within 2 m (6.5 ft) of the machine.
 5. If the machine is to be installed in a location subject to strong external RF fields (e.g., near broadcast antennas), consult BetaCorp Systems before installation.
-

1.10 Warranty Information

BetaCorp Systems warrants the BC-500X CNC Machining Center against defects in materials and workmanship for a period of **24 months from the date of acceptance** or **30 months from the date of shipment from our factory**, whichever occurs first.

Warranty Coverage Includes:

- Defects in materials or workmanship in all machine components
- Control system hardware defects
- Hydraulic system component defects

- Spindle assembly defects (excluding normal bearing wear)

Warranty Does NOT Cover:

- Normal wear items (cutting tools, filters, seals, V-belts, coolant)
- Damage caused by failure to follow maintenance procedures in this manual
- Damage caused by use of incorrect coolant, lubricant, or hydraulic fluid
- Damage caused by operation outside specified environmental conditions
- Damage caused by unauthorized modifications
- Damage caused by crash events (tool-in-work collisions, axis over-travel events)
- Consumable items including but not limited to: way wiper seals, bellows, light bulbs, fuses

Warranty Claim Procedure:

1. Contact BetaCorp Systems Technical Support at +1 (800) 555-BETA
2. Obtain a Return Material Authorization (RMA) number
3. Document the defect with photographs and fault logs
4. Do not return parts without an RMA number

Complete warranty terms and conditions are available at www.betacorpssystem.com/warranty and in Appendix J of this manual.

1.11 Disposal and Recycling

At end of life, the BC-500X must be disposed of in accordance with all applicable local, state, federal, and international regulations.

Materials Requiring Special Handling:

Material	Location	Handling Requirement
Hydraulic oil	Hydraulic reservoir	Collect and recycle through licensed waste oil processor
Coolant fluid	Coolant tank	Collect and process through licensed waste water facility
Electrical components	Control cabinet	Recycle through certified e-waste facility (WEEE Directive in EU)
Batteries	CNC control board	Lithium batteries require separate disposal
Fluorescent lamps	Enclosure lighting	Mercury — separate disposal required
Way lube	Lubrication reservoir	Collect and recycle through licensed processor

Material	Location	Handling Requirement
Steel/cast iron	Machine structure	Metal recycling
Aluminum	Machine components	Metal recycling

Contact your local waste management authority for specific requirements in your jurisdiction. BetaCorp Systems can provide guidance on decommissioning services.

Chapter 2 — Safety Instructions

2.1 General Safety Philosophy

BetaCorp Systems designs the BC-500X with safety as the primary consideration in every aspect of the machine's design. However, no machine can be made completely safe if misused or improperly maintained. The BC-500X is a powerful industrial machine capable of generating substantial forces and operating at speeds that can cause serious injury or death if proper safety precautions are not observed.

The Hierarchy of Safety Controls:

BetaCorp Systems has applied the following hierarchy of safety controls in the design of the BC-500X, listed from highest to lowest preference:

1. **Elimination:** Design features that eliminate hazards entirely (e.g., fully enclosed work zone eliminates exposure to cutting zone)
2. **Substitution:** Use of safer materials and processes where possible
3. **Engineering Controls:** Guards, interlocks, emergency stops, safety relays
4. **Administrative Controls:** Training requirements, operating procedures, maintenance schedules
5. **Personal Protective Equipment (PPE):** Last line of defense; required in addition to all other controls

Fundamental Safety Rules:

The following rules are absolute and must never be violated:

RULE 1: Always follow the lockout/tagout procedure in Section 2.4.2 before performing any maintenance, service, or adjustment that requires access to hazardous energy sources.

RULE 2: Never operate the machine with guards removed or interlocks bypassed. If a guard is damaged or missing, tag the machine out of service and repair before resuming operation.

RULE 3: Never put any part of your body inside the machine work zone, tool changer area, or any other area identified as a hazardous zone while the machine is in an energized state (exception: de-energized and locked out per LOTO procedure).

RULE 4: If you observe an unsafe condition, stop work immediately and report the condition to

your supervisor. Do not resume work until the condition is corrected.

RULE 5: Never modify the machine's safety systems. This includes: removing or tying back safety relays, bridging interlock circuits, modifying machine parameters related to safety functions without authorization from BetaCorp Systems.

2.2 Qualified Personnel Requirements

Different tasks on the BC-500X require different levels of qualification. The following personnel classifications are defined:

2.2.1 Machine Operator

A machine operator is a person who operates the BC-500X to perform production machining tasks. The operator does not perform electrical service, mechanical disassembly, or hydraulic work.

Minimum Qualifications for Operator:

- Must be at least 18 years of age
- Must have completed BetaCorp Systems' BC-500X Operator Training course (16 hours minimum) or equivalent training documented by employer
- Must demonstrate ability to identify and respond to all fault codes in the operator's quick reference guide
- Must demonstrate ability to execute emergency stop and machine power-down procedures
- Must demonstrate knowledge of PPE requirements for machining operations
- Must NOT be under the influence of alcohol, prescription medications that impair judgment, or any other substance that may affect alertness

2.2.2 Maintenance Technician

A maintenance technician performs scheduled preventive maintenance tasks as described in Chapter 9.

Minimum Qualifications for Maintenance Technician:

- All operator qualifications, PLUS:
- Must have completed BetaCorp Systems' BC-500X Maintenance Training course (24 hours minimum) or equivalent
- Must be trained in lockout/tagout procedures per OSHA 29 CFR 1910.147
- Must have basic mechanical aptitude and ability to use hand tools correctly
- Must understand hydraulic system safety including pressure de-energization
- Must understand coolant handling and chemical safety per applicable SDS

2.2.3 Electrical Service Technician

An electrical service technician performs troubleshooting and repair of the machine's electrical and control systems.

Minimum Qualifications for Electrical Service Technician:

- All maintenance technician qualifications, PLUS:
- Must be a qualified electrician as defined by NFPA 70E
- Must have completed arc flash training and have a current arc flash PPE assessment for the specific installation
- Must hold appropriate electrical licenses as required by local jurisdiction
- Must have completed BetaCorp Systems' BC-500X Electrical Service training

2.2.4 Authorized BetaCorp Service Engineer

Certain procedures in this manual are marked "**BetaCorp Authorized Service Only.**" These procedures require factory-trained BetaCorp Systems service engineers with specialized tools and access credentials. Do not attempt these procedures without BetaCorp authorization.

2.3 Personal Protective Equipment (PPE)

2.3.1 Required PPE for All Personnel in the Machine Area

The following PPE is required at all times for anyone working in the vicinity of the BC-500X:

Safety Glasses (ANSI Z87.1+):

- Required at all times within 3 meters (10 feet) of the machine
- Must meet ANSI Z87.1 impact resistance
- Side shields must be used

Safety Footwear (ASTM F2413):

- Steel-toe or composite-toe safety shoes/boots required
- Metatarsal protection recommended for anyone who handles heavy fixturing or tools
- Anti-slip soles strongly recommended due to coolant on floors

No Loose Clothing or Jewelry:

- Loose sleeves, ties, scarves, and similar items must be removed or securely tucked before approaching the machine
- Finger rings must be removed when handling machine components
- Long hair must be tied back and secured

2.3.2 Task-Specific PPE Requirements

Task	Additional PPE Required
Opening work zone door after machining	Face shield (ANSI Z87.1, EN166) over safety glasses
Adjusting coolant nozzles	Chemical-resistant gloves (nitrile, minimum)
Coolant system maintenance	Chemical-resistant gloves + face shield
Handling coolant concentrate	Chemical-resistant gloves, face shield, chemical-resistant apron
Chip removal from work zone	Cut-resistant gloves (ANSI Level A4 minimum), face shield
Electrical panel work	Arc flash PPE per arc flash hazard analysis; minimum Category 2
Hydraulic system work	Face shield; chemical-resistant gloves (oil-resistant)
Grinding or filing metal parts	Face shield, dust mask (N95 minimum)
Noise exposure >85 dB(A)	Hearing protection (earmuffs or earplugs, NRR ≥25)
Cleaning with compressed air	Face shield, hearing protection
Hydraulic fluid cleanup	Chemical-resistant gloves, eye protection

2.3.3 PPE Inspection

All PPE must be inspected before use. Remove from service and replace any PPE that shows:

- Cracks or scratches in lenses (safety glasses, face shields)
- Tears, punctures, or degradation of glove material
- Deterioration of protective coating
- Any damage that may compromise protective function

Maintain a PPE inspection log. Replace safety glasses lenses annually at a minimum, regardless of apparent condition.

2.3.4 Face Shield Specifications

When a face shield is required (not just safety glasses), the face shield must meet:

- ANSI Z87.1 or EN166 standard
- Minimum lens rating: 3+ (impact protection)
- Full face coverage including chin
- Used OVER (not instead of) safety glasses

2.3.5 Arc Flash PPE

For electrical work, the arc flash hazard analysis for your specific installation must be reviewed and PPE selected accordingly. As a general guideline for the BC-500X main electrical cabinet at 480V:

DANGER: Arc flash hazard assessment must be conducted by a qualified electrical engineer before any electrical service is performed. The incident energy at the main electrical cabinet may be 4 to 40 cal/cm² or higher depending on the upstream power system configuration. This energy level requires Category 2 to Category 4 arc flash PPE per NFPA 70E. NEVER perform electrical work without an arc flash assessment.

Minimum Category 2 PPE (typical, must verify for your installation):

- Arc-rated face shield and balaclava, OR arc-rated arc flash suit hood
 - Arc-rated jacket and pants (minimum 8 cal/cm² arc rating)
 - Arc-rated gloves
 - Leather work boots
-

2.4 Electrical Safety

2.4.1 Overview of Electrical Hazards

The BC-500X contains multiple voltage levels that present shock and arc flash hazards:

Voltage Level	Location	Hazard Level
480 VAC, 3-phase	Main power input; main electrical cabinet	FATAL — Extreme danger
480 VAC	Spindle motor drive input	FATAL — Extreme danger
480 VAC	Axis servo drive input	FATAL — Extreme danger
480 VAC	Hydraulic pump motor	FATAL — Extreme danger
240 VAC	Control transformer secondary (some circuits)	FATAL — High danger
120 VAC	Control circuit power; lighting	SERIOUS — High danger
24 VDC	PLC I/O; sensors; solenoids	INJURY possible from short circuits
5 VDC	Encoder signals; control signals	Low voltage; normally safe
48 VDC Bus	Servo drive DC bus (after capacitor charge)	FATAL — capacitors retain charge

DANGER: The servo drive DC bus capacitors will retain dangerous voltage levels for up to **5 minutes** after the main power is disconnected. NEVER open any servo drive enclosure until the main disconnect has been locked out AND a minimum of 5 minutes has elapsed AND the voltage has been verified at zero using a properly rated voltmeter.

2.4.2 Arc Flash Hazard Assessment

What is Arc Flash?

An arc flash is an electrical explosion that occurs when electrical current travels through ionized air between conductors. Arc flash events can produce temperatures of up to 35,000°F (19,000°C) at the arc point — four times the surface temperature of the sun — and pressure waves exceeding 2,000 lb/ft². Arc flash causes severe burns, fires, explosion-related injuries, and death.

BC-500X Arc Flash Risk Factors:

- Three-phase 480 VAC power supply
- Available fault current at the machine's main disconnect depends on the facility power system; must be determined by facility electrical engineer
- Incident energy varies with available fault current and upstream overcurrent protection

Requirements Before Energized Electrical Work:

1. Obtain a current arc flash hazard analysis for the machine's specific installation location. This analysis must be performed by a Licensed Professional Engineer or Qualified Electrical Engineer using IEEE 1584 methods.
2. Review arc flash labels on all electrical enclosures. These labels must be applied as part of the installation process.
3. Select PPE based on the incident energy shown on the arc flash label, NOT based on voltage alone.
4. Establish an arc flash boundary. No unprotected personnel may be within the arc flash boundary during energized work.
5. Only qualified personnel (as defined in NFPA 70E) may perform energized electrical work.
6. When work can be performed with the equipment de-energized and locked out, that approach is ALWAYS preferred.

Arc Flash Label Information:

Each major electrical enclosure (main cabinet, servo cabinet, spindle drive cabinet) must have an arc flash label applied during installation. The label must include:

- Nominal system voltage
- Incident energy at working distance
- Working distance
- Required PPE category
- Arc flash boundary distance
- Date of analysis

Contact BetaCorp Systems or your facility's electrical engineering department if arc flash labels are missing or if the power system has been modified since the last analysis.

2.4.3 Lockout/Tagout (LOTO) Procedures

Regulatory Basis:

The lockout/tagout procedures in this section comply with OSHA 29 CFR 1910.147, "The Control of Hazardous Energy." This regulation requires that all energy sources be isolated and locked before maintenance or service activities where unexpected energization could cause injury.

DANGER: LOTO procedures MUST be followed before performing ANY maintenance task that involves: accessing the interior of any electrical cabinet, adjusting or replacing any mechanical component while the machine is energized, working on the hydraulic system, working on the pneumatic system, or any other task where unexpected machine motion could cause injury.

Energy Sources on the BC-500X:

The following energy sources must be isolated during LOTO:

1. **Electrical energy:** Main electrical supply (480 VAC, 3-phase)
2. **Hydraulic energy:** Hydraulic system pressure (up to 210 bar / 3,046 psi)
3. **Pneumatic energy:** Compressed air supply (up to 8 bar / 116 psi)
4. **Gravitational energy (stored):** Z-axis may fall if hydraulic brake is released while hydraulic power is removed
5. **Capacitive energy (stored):** Servo drive bus capacitors retain voltage after power removal
6. **Spring energy (stored):** Tool retention springs in spindle draw bar assembly; tool change arm springs

LOTO Hardware Requirements:

- Personal padlock(s) — each person working on the machine must apply their own padlock. Never allow another person to hold your lockout key.
- Hasp (multi-lock hasp for group lockout)
- Lock tags with authorized personnel name, date, and contact information
- Lockout device for main disconnect (provided with machine)
- Valve lockout devices for hydraulic and pneumatic isolation valves
- Voltage tester rated for 600 VAC minimum (CAT III or CAT IV)

LOTO Procedure — Standard Machine Lockout:

Step 1: Notify Affected Personnel Inform all personnel in the area that the machine is being taken out of service for maintenance. Post "DO NOT OPERATE — UNDER MAINTENANCE" signage at the control panel and at all machine access points.

Step 2: Identify All Energy Sources Review the energy source list above. Identify all energy isolation points for the specific maintenance task to be performed.

Step 3: Operate Machine to a Safe State

- If the machine is running an automatic cycle: Press FEED HOLD to pause the cycle, then press E-STOP.
- Navigate the spindle to a safe position using JOG mode before removing power if possible.
- Retract the Z-axis to its home position (Z=0) to prevent gravity drop.

Step 4: Isolate Electrical Energy

- Locate the main disconnect switch on the right side of the main electrical cabinet.
- Turn the main disconnect to the OFF position.
- Apply the lockout hasp to the main disconnect handle.
- Apply your personal padlock to the hasp. The lock must be your personal lock; do not share locks.
- Attach a lockout tag with your name, date, and reason for lockout.

Step 5: Isolate Pneumatic Energy

- Locate the pneumatic shutoff valve on the rear of the machine (labeled "AIR IN ISOLATION VALVE").
- Turn the pneumatic isolation valve to the OFF position.
- Apply a valve lockout device and your personal padlock.
- Exhaust residual pressure: slowly open the manual bleed valve (located adjacent to the shutoff valve, labeled "BLEED") until all pressure is released. Verify zero pressure on the pneumatic pressure gauge.

Step 6: Isolate Hydraulic Energy

- The hydraulic pump motor will stop when electrical energy is isolated (Step 4).
- Allow hydraulic system pressure to bleed down through the system. This takes approximately 2–3 minutes with no solenoid valves energized.
- Verify hydraulic pressure is zero on the hydraulic pressure gauge on the HPU.
- Apply a hydraulic line lockout device to the main hydraulic supply line shutoff valve.
- Apply your personal padlock.

Step 7: Secure Z-Axis Against Gravity The Z-axis spindle head assembly weighs approximately 280 kg (617 lb) and is retained in position by a hydraulic brake (fail-safe, spring-applied) and a servo brake. If hydraulic pressure is removed and the servo brake is not engaged:

WARNING: The Z-axis WILL fall if the servo brake is disengaged or faulty AND hydraulic brake pressure is lost. Before working under or near the Z-axis assembly, install the Z-axis safety block (Part No. BC-TOOL-4417, "Z-Axis Safety Block"). This block is stored in the machine tool cabinet. Slide the block onto the Z-axis column guide and verify it is fully engaged before working beneath the Z-axis head.

Installation procedure for Z-axis safety block:

1. Jog the Z-axis to approximately $Z = -200$ mm (mid-travel).
2. Power down and lock out the machine per Steps 1–6.
3. Open the Z-axis side access panel.
4. Insert the Z-axis safety block (Part No. BC-TOOL-4417) over the Z-axis guide rail from the front.
5. Slide the block down until it contacts the Z-axis head casting.
6. Verify the block is fully seated and cannot fall out.
7. Proceed with maintenance.
8. Remove the Z-axis safety block before restoring power.

Step 8: Wait for Capacitor Discharge After the main electrical disconnect is opened, wait a minimum of **5 minutes** before opening any servo drive enclosure or accessing servo drive terminals. This allows the servo drive DC bus capacitors to discharge to a safe voltage level.

Step 9: Verify Zero Energy Before beginning work, test for zero energy at all relevant points:

- Test for zero voltage at the main power terminals using a properly rated voltmeter (CAT III, 600V minimum rating). Test all three phases and line-to-line combinations.
- Test for zero voltage at servo drive DC bus terminals.
- Verify pneumatic pressure gauge reads zero.
- Verify hydraulic pressure gauge reads zero.
- Confirm Z-axis safety block is in place if working near Z-axis.

NOTE: "Test before touch" is an absolute rule. A voltmeter reading of zero does not guarantee safety if the meter is faulty. Test the voltmeter on a known live source before and after testing the machine to verify the meter is functioning correctly.

Step 10: Perform Maintenance Work Perform the maintenance or service task as described in this manual. Do not leave the machine partially assembled and unattended.

Step 11: Restoration of Energy — Release from Lockout

After maintenance is complete:

1. Inspect the work area. Ensure all tools, materials, and personal items are removed from inside the machine.
2. Ensure all guards, covers, and access panels are reinstalled and fastened.
3. Remove the Z-axis safety block if it was installed.
4. Inform all personnel that the machine is about to be re-energized.
5. Each person who applied a personal padlock must remove their own padlock.
6. Remove the lockout hasps from the main disconnect, pneumatic valve, and hydraulic valve.
7. Open the pneumatic isolation valve to restore air supply.

8. Verify hydraulic oil level before restoring electrical power.
9. Restore electrical power by turning the main disconnect to ON.
10. Perform a system checkout as appropriate for the work performed.

Group Lockout Procedures:

When multiple technicians are working on the machine simultaneously, each technician must apply their own personal padlock to the lockout hasp. The multi-lock hasp provided with the machine accepts up to 6 padlocks. If more than 6 technicians are involved, obtain additional hasp devices from your maintenance department.

Each technician retains control of their own lock. The machine CANNOT be re-energized until ALL technicians have removed their personal locks. This ensures no individual can re-energize the machine while another person is still working on it.

Shift Change During Lockout:

If a lockout must be maintained across a shift change:

1. The outgoing technician must NOT remove their lock until the incoming technician has applied their personal lock to the hasp.
2. Transfer of lock control requires physical handover and signing of the maintenance log.
3. Never remove another person's lockout padlock, regardless of circumstances. If a lock cannot be removed because the keyholder is unreachable, follow your facility's procedures for emergency lock removal.

2.4.4 Capacitor Discharge Waiting Periods

The BC-500X servo drives contain large electrolytic capacitors on the DC bus. These capacitors can store lethal amounts of electrical energy and will maintain dangerous voltage levels after the main power supply is disconnected.

Required Waiting Periods:

Situation	Minimum Wait Time
Main disconnect opened, servo drives were in normal operation	5 minutes
Main disconnect opened, machine was in fault/alarm condition	8 minutes
Main disconnect opened after prolonged operation at high duty cycle	8 minutes
Power failure (unexpected) — unknown capacitor state	10 minutes

DANGER: These are MINIMUM waiting periods. Always verify zero voltage with a properly rated voltmeter before accessing any servo drive internals. The capacitors in the BC-500X servo drive system are rated at 800 VDC. Contact with charged capacitor

terminals is FATAL.

Servo Drive Discharge Indicator:

The servo drive units (located in the servo cabinet on the right side of the machine) are equipped with LED indicators on the front panel:

- **Red LED lit:** DC bus voltage is present; DO NOT open the drive enclosure
- **Red LED off:** DC bus may have discharged; verify with voltmeter before proceeding

CAUTION: The LED is powered by the DC bus. When the LED extinguishes, voltage has dropped below the LED operating threshold, but may still be at a level capable of causing injury. Always verify with a voltmeter.

2.4.5 Ground Fault Protection

The BC-500X is equipped with a main ground fault circuit interrupter (GFCI) on the 120 VAC control circuit power. However, the main 480 VAC power input is protected by a ground fault relay, not a GFCI.

Grounding Requirements:

- A dedicated equipment grounding conductor (green or green/yellow striped) must be connected from the facility grounding system to the main electrical cabinet ground bus bar.
- The grounding conductor must be sized per NEC Table 250.122 based on the upstream overcurrent device.
- Ground conductor resistance from the main cabinet ground bus to the facility ground electrode system must not exceed 0.1 ohm.
- Verify ground continuity during annual maintenance (see Chapter 9).

Isolated Ground for CNC Control: The CNC control system uses a separately derived 24 VDC power supply with an isolated 0V reference. Do not connect the CNC control 0V reference to the machine chassis ground. This connection is pre-made at the factory and must not be modified.

2.5 Mechanical Safety

2.5.1 Rotating Component Hazards

The BC-500X contains several rotating components that can cause severe injury if contact is made:

Spindle:

- Maximum speed: 10,000 RPM
- At maximum speed, the tip of a 10 mm diameter end mill rotates at over 5,200 mm/second (17 feet/second)
- Contact with a rotating tool or spindle will cause immediate and severe injury

- NEVER reach into the work zone while the spindle is rotating or while the machine is energized

Tool Changer Carousel:

- The tool magazine carousel rotates at indexing speed during tool changes
- Tool pot grippers extend radially and can cause pinching injuries
- NEVER reach into the tool magazine area during operation

Chip Conveyor:

- The chip conveyor belt contains sharp metal chips from machining
- The drive sprockets and return rollers are pinch points
- NEVER reach into the chip conveyor during operation
- ALWAYS wear cut-resistant gloves when handling the chip conveyor or chip bins

Coolant Pump:

- The coolant pump motor and impeller are rotating components
- Do not attempt to clear coolant pump blockages while the system is energized

Hydraulic Pump:

- The hydraulic pump is driven by an electric motor
- Do not open the hydraulic pump or motor assembly while energized

General Rules for Rotating Components:

1. Confirm the spindle is stopped (spindle speed = 0 on control display) before opening the work zone door.
2. Even after the spindle stop command, the spindle may coast for a brief period. Wait for the spindle to completely stop before reaching into the work zone.
3. The spindle brake feature (parameter B1.22) should be enabled to minimize coasting time during door-open sequences.
4. Never manually brake a rotating spindle by contact with the tool holder or spindle nose. The forces involved will cause immediate injury.

2.5.2 Pinch Point Identification

The following pinch points exist on the BC-500X. Machine operators and maintenance personnel must be aware of these locations:

Location	Description	Guard/Protection
Z-axis head vs. workpiece/fixture	Spindle head descends during machining and tool changes	Work zone door; Z-axis over-travel limits
X-axis table vs. machine column	Table moves toward column at right side of travel	X-axis positive over-travel limit; column guard

Location	Description	Guard/Protection
Y-axis table vs. splash guard	Table moves toward rear of machine	Y-axis positive over-travel limit
ATC arm vs. spindle	ATC arm rotates rapidly during tool change	Tool change zone guard; ATC interlock
ATC arm vs. carousel	ATC arm and carousel move simultaneously	Interlock prevents simultaneous motion
Tool changer carousel vs. ATC arm standby position	Carousel indexes through area near ATC arm	Interlock; carousel guard
Chip conveyor drive sprocket	Sprocket and chain are exposed at conveyor drive end	Conveyor drive guard; interlocked access cover
Chip conveyor return roller	Return roller at conveyor discharge end	Return end guard
Hydraulic power unit coupling	Motor-to-pump coupling	HPU coupling guard
Coolant pump	Coolant pump motor and impeller	Coolant tank cover

Additional Pinch Point During Maintenance: During Z-axis maintenance, the Z-axis head (280 kg, 617 lb) can fall rapidly if the hydraulic brake is released unexpectedly. This is a crush hazard, not merely a pinch hazard. Always install the Z-axis safety block (BC-TOOL-4417) before working on or near the Z-axis assembly.

2.5.3 Stored Energy — Springs and Counterweights

Spindle Draw Bar Spring: The tool retention system in the BC-500X spindle uses a Belleville washer stack to apply clamping force to the tool holder. This spring stack is preloaded and stores significant mechanical energy. During draw bar service:

- The spring preload force is approximately 18,000 N (4,047 lb)
- Do not compress or release the spring stack without proper tooling and thorough familiarity with the procedure in Chapter 5
- The draw bar service procedure requires BetaCorp-trained service technicians

ATC Arm Return Springs: The ATC arm assembly uses torsion springs to assist the arm return motion. These springs are preloaded during assembly.

- Do not disassemble the ATC arm without following the procedure in Chapter 13
- The ATC arm torsion spring stores approximately 45 N·m (33 ft·lb) of energy
- Improper disassembly can cause the spring to release suddenly with enough force to cause bone fractures

Axis Over-Travel Hardware Stops: Each axis has adjustable mechanical over-travel stops. These stops are not intended to arrest axis motion under normal circumstances and are not designed to absorb the full kinetic energy of a crash event. Do not rely on mechanical stops as a substitute for proper programming and limit switch setup.

2.6 Hydraulic Pressure Hazards

The BC-500X hydraulic system operates at pressures up to **210 bar (3,046 psi)**. At this pressure, hydraulic fluid behaves very differently from low-pressure fluids. The following hazards are specific to high-pressure hydraulic systems:

2.6.1 High-Pressure Injection Injury

DANGER: High-pressure hydraulic fluid injection through the skin can cause catastrophic injury and death. A pinhole leak in a hydraulic line at 210 bar pressure generates a fluid jet that can penetrate human skin from distances of several centimeters. The fluid injection causes immediate tissue destruction and rapidly expanding tissue damage from both the mechanical injection force and the chemical toxicity of hydraulic fluid. **HIGH-PRESSURE INJECTION INJURIES REQUIRE IMMEDIATE EMERGENCY SURGICAL TREATMENT.** Even if the injection wound appears small and painless initially, the internal damage is severe and life-threatening.

Prevention:

- Never run your hand along a hydraulic hose or line to locate a leak
- Use a piece of cardboard or paper to detect leaks by observing wet spots
- Never approach a suspected hydraulic leak with your face
- Wear safety glasses or face shield when working around hydraulic lines
- De-pressurize the hydraulic system completely before disconnecting any hydraulic line or fitting

If High-Pressure Injection Injury Occurs:

1. Call emergency services (911) IMMEDIATELY
2. Inform medical personnel that a high-pressure hydraulic fluid injection injury has occurred
3. Tell them the type of hydraulic fluid (see Chapter 11, Section 11.7)
4. Do NOT apply a tourniquet
5. Keep the injured person calm and still
6. Treatment must be performed by a surgeon; this is NOT a "treat at first aid station" injury

2.6.2 Hydraulic Line De-pressurization Procedure

Before performing ANY work on the hydraulic system (disconnecting lines, replacing valves, replacing cylinders, changing filters while system is under pressure), the hydraulic system must be completely de-pressurized.

Hydraulic De-pressurization Procedure:

WARNING: The hydraulic system stores energy not only in the pump and lines but also in hydraulic cylinders under load. The Z-axis brake cylinder, tool clamping

cylinder, and workholding cylinders (if equipped) all contain trapped pressure that may not be fully released by simply stopping the hydraulic pump. Follow this complete procedure.

Step 1: Perform the standard LOTO procedure (Section 2.4.2) to isolate electrical energy and stop the hydraulic pump motor.

Step 2: Allow 2–3 minutes for the hydraulic system pressure to equalize through the internal system pressure relief valves.

Step 3: Verify hydraulic pump has stopped rotating (listen and observe).

Step 4: Check the hydraulic system pressure gauge on the HPU panel. The gauge should read zero. If it does not read zero within 3 minutes, there may be a check valve holding pressure in a branch circuit.

Step 5: To release trapped pressure in individual cylinders, locate the manual pressure release valve for the specific circuit (refer to hydraulic circuit diagram in Chapter 11). Slowly crack the manual release valve open (maximum 1/4 turn) and allow pressure to equalize over approximately 30 seconds. Do not fully open pressure release valves rapidly.

Step 6: Verify zero pressure on all pressure gauges before disconnecting any lines.

Step 7: When disconnecting a hydraulic fitting, use both hands — one to brace the fitting, one to turn the nut. Even at zero pressure, residual oil will drain from the line. Have absorbent material and a drain container ready.

Step 8: Cap open hydraulic fittings immediately after disconnection to prevent contamination.

2.6.3 Hot Oil Burns

During normal operation, hydraulic oil temperature in the BC-500X system reaches 40–50°C (104–122°F). After prolonged operation under heavy load, temperatures can reach 60°C (140°F). Contact with hot hydraulic oil causes thermal burns.

Prevention:

- Allow the hydraulic system to cool for a minimum of 30 minutes after machine shutdown before draining or disconnecting hydraulic lines
- Check oil temperature on the HPU temperature gauge before opening any hydraulic circuit
- Wear chemical-resistant, heat-resistant gloves when working with hydraulic oil
- Have absorbent spill materials readily available

Oil Spill and Fire Hazard: Hydraulic oil is combustible. The flash point of the specified ISO 46 hydraulic oil is approximately 200°C (392°F). Under normal circumstances, oil is not in contact with ignition sources. However:

- Keep cutting sparks away from the hydraulic power unit
- In the event of a hydraulic oil fire, use CO₂ or dry chemical extinguisher. Do NOT use water.

- Clean up oil spills immediately. Oily floors create slip hazards and fire hazards.
-

2.7 Pneumatic System Safety

The BC-500X pneumatic system operates at 6 bar (87 psi) supply pressure. Pneumatic hazards include:

Compressed Air Injection: Compressed air directed at the skin can force air under the skin, causing air embolism. Never use compressed air to clean clothing or skin. Never use compressed air to "blow off" chips unless using a properly reduced pressure (maximum 30 psi / 2 bar) and with appropriate chip guards in place.

Flying Projectiles: When a pneumatic line is disconnected under pressure, loose fittings can become projectiles. Always depressurize the pneumatic circuit (close the isolation valve and bleed the circuit) before disconnecting fittings.

De-pressurization Procedure for Pneumatic System:

1. Close the pneumatic isolation valve (LOTO).
2. Slowly open the manual bleed valve adjacent to the isolation valve.
3. Allow pressure to bleed down over 10–15 seconds.
4. Verify zero pressure on the pneumatic system pressure gauge.
5. Disconnect fittings with hands clear of the fitting face.

Stored Pneumatic Energy: Pneumatic accumulators (if equipped) and pneumatic cylinders retain pressure after the isolation valve is closed. The bleed valve releases pressure from the supply side but may not release pressure from:

- Pneumatic cylinders in the tool unclamping circuit (air cylinder holds tool release position)
- Any pneumatic circuit with a check valve on the outlet

After bleeding the supply, manually cycle each pneumatic device by pressing the manual override button on the solenoid valve (machine must be in LOTO state) to verify all cylinders have released their stored energy.

2.8 Coolant and Chemical Hazards

2.8.1 Metalworking Fluid Health Hazards

The water-soluble metalworking fluids used in the BC-500X are generally of low acute toxicity when properly maintained. However, prolonged skin contact and inhalation of mist from metalworking fluids present significant health risks:

Dermatitis: Prolonged skin contact with metalworking fluid can cause:

- Contact dermatitis (irritant type): red, dry, cracked skin; usually on hands and forearms

- Allergic contact dermatitis: immune response to fluid components; can spread beyond contact area

Prevention: Wear nitrile gloves during coolant handling. Apply a barrier cream before work. Wash hands thoroughly with soap and water after coolant contact. Do not use solvent to clean skin.

Respiratory Hazards: Metalworking fluid mist is generated during machining operations. The BC-500X enclosure contains mist during operation, but residual mist exits when the work zone door is opened. Prolonged inhalation exposure has been associated with:

- Respiratory irritation
- Hypersensitivity pneumonitis (occupational asthma)
- In historical literature (with contaminated fluids), lipoid pneumonia

Prevention: Ensure the machine mist collector (optional accessory) is operating correctly. Do not open the work zone door unnecessarily during cutting. Maintain coolant concentration and biocide levels per Chapter 12 to minimize microbial contamination and nitrosamine formation.

Bacterial Contamination: Metalworking fluids, particularly water-based semi-synthetic and synthetic fluids, are susceptible to bacterial and fungal contamination. Signs of contamination include:

- Rancid or "Monday morning" odor
- Slimy or discolored fluid
- pH below 8.5

Contaminated fluid can cause:

- Severe skin and eye irritation
- Respiratory problems from inhaling contaminated mist
- Increased corrosion of machine surfaces

Testing and treatment of metalworking fluid is described in Chapter 12.

2.8.2 Coolant Concentrate Handling

Metalworking fluid concentrate (before dilution) is significantly more hazardous than the diluted working fluid. Concentrated fluid can:

- Cause severe skin irritation and chemical burns to eyes
- Cause respiratory tract irritation if inhaled directly

When handling coolant concentrate:

- Always wear nitrile or neoprene gloves (minimum)
- Wear a face shield
- Work in a well-ventilated area
- Review the Safety Data Sheet (SDS) for the specific product

First Aid — Eye Contact: Immediately flush eyes with large amounts of clean water for 15 minutes. Seek medical attention.

First Aid — Skin Contact: Remove contaminated clothing. Wash affected area with soap and water for 15 minutes. If irritation persists, seek medical attention.

First Aid — Ingestion: Do NOT induce vomiting. Call Poison Control at 1-800-222-1222 (US). Seek medical attention with SDS in hand.

2.8.3 Hydraulic Oil Hazards

Mineral-based hydraulic oil (ISO 46) has low acute toxicity but presents:

- Skin and eye irritant
- Combustion hazard (flash point ~200°C)
- Environmental hazard (petroleum-based)

Biologically degradable hydraulic oil options are available; consult Chapter 11.

2.8.4 Chemical Storage

Store all chemicals used with the BC-500X in designated storage areas with:

- Appropriate secondary containment (spill trays)
 - Temperature within storage requirements for each product
 - Separation of incompatible chemicals
 - Proper labeling per GHS/HazCom 2012
-

2.9 Chip and Swarf Hazards

Metal chips and swarf generated during machining operations are among the most frequently overlooked hazards in the machine shop environment. The BC-500X generates chips in quantities and forms that present the following hazards:

Cutting Hazards: Metal chips, particularly from steel, stainless steel, and titanium machining, are razor-sharp. Long stringy chips can wrap around extremities. Compressed chip nests in the work zone can spring apart when disturbed, throwing sharp chip shards.

Burns: Freshly generated chips from high-speed machining operations can be extremely hot, particularly from aluminum and brass machining. Do not grab chips by hand immediately after machining.

Eye Hazards: Chip fragments can be ejected from the work zone, particularly when the door is opened immediately after machining. Always wear a face shield when opening the work zone door after machining.

Safe Chip Handling Procedures:

1. Never remove chips from the work zone by hand. Use chip hooks, magnetic sweepers (for

ferrous materials), or chip brushes.

2. Always wear cut-resistant gloves (ANSI Level A4 minimum) when handling chips.
3. Use a face shield when opening the work zone door after machining.
4. Allow chips to cool before handling.
5. Never use compressed air to blow chips. Use a coolant flush or chip hook.
6. Empty the chip bin before it overflows. Overflowing chip bins create tripping and cutting hazards on the floor around the machine.
7. Dispose of chips in designated chip bins. Never mix chip materials (aluminum chips must not be mixed with steel chips in many recycling programs).

2.10 Emergency Stop Procedures

2.10.1 Emergency Stop Button Locations

The BC-500X is equipped with four Emergency Stop (E-Stop) pushbuttons. Each E-Stop button is a red, mushroom-head pushbutton mounted on a yellow background panel. Activating ANY E-Stop button will immediately halt all machine motion and remove power from all axis drives and the spindle drive.

E-Stop Location	Description
E-Stop 1	Main operator panel, center console — primary operator position
E-Stop 2	Left side of machine enclosure, front panel — accessible from tool load position
E-Stop 3	Pendant (MPG hand wheel unit) — used during setup and maintenance
E-Stop 4	Rear of machine, service access side — for maintenance personnel

E-Stop Button Function: The E-Stop buttons are normally-closed, direct-opening safety contacts connected to a safety relay circuit (Pilz PNOZ or equivalent). When any E-Stop button is pressed:

1. The safety relay drops out immediately.
2. Power is removed from all servo drive enable circuits.
3. Power is removed from the spindle drive enable circuit.
4. All axis servos apply their internal dynamic braking.
5. The spindle applies its electromagnetic brake (if Parameter B1.22 is enabled).
6. The PLC enters fault state and logs the event.
7. All hydraulic solenoid valves return to their spring-default positions.

8. The CNC control displays: "**E-STOP ACTIVE — PRESS RESET TO RESUME**"

What E-Stop Does NOT Do:

- E-Stop does NOT remove main power from the machine
- E-Stop does NOT de-pressurize the hydraulic system
- E-Stop does NOT release the tool from the spindle
- E-Stop does NOT prevent the machine from being re-started by pressing Reset

WARNING: E-Stop is an emergency stopping device, not a lockout device. After pressing E-Stop, the machine can be re-started by pressing the Reset button. E-Stop **MUST NOT** be used as a substitute for lockout/tagout during maintenance.

2.10.2 Machine Restart After Emergency Stop

After an E-Stop has been activated, the following procedure must be followed before resuming machine operation:

Step 1: Assess the Situation Determine WHY the E-Stop was activated. If the E-Stop was activated due to a hazardous condition, do not restart the machine until the hazard is corrected.

Step 2: Ensure Safety Verify that all persons are clear of all hazardous areas (work zone, tool changer area, chip conveyor area).

Step 3: Identify and Correct Fault The CNC control will display a fault condition. Review the fault message and the fault log (SYSTEM → FAULT LOG). Correct any underlying fault condition before restarting.

Step 4: Reset E-Stop Button Rotate the E-Stop button clockwise until it releases (pops out). All four E-Stop buttons must be released before the machine will accept a reset command.

Step 5: Press RESET Press the green RESET button on the main operator panel. The safety relay will re-energize, and servo power will be restored.

Step 6: Homing (if required) In some E-Stop conditions, the CNC control may lose position reference for one or more axes. If the control displays "HOMING REQUIRED" or if any axis position indicator shows an asterisk (*), the affected axis must be homed before resuming automatic operation. Follow the homing procedure in Section 6.4.

Step 7: Inspect Work in Progress Before resuming an interrupted machining cycle, inspect the workpiece and tooling for any damage that may have occurred at the time of the E-Stop. Check for broken tool fragments in the work zone.

Step 8: Resume or Restart Program If the machining program can safely be resumed from the current position, use the CYCLE START function. If the tool is broken or the workpiece is damaged, abort the program and address the condition before restarting.

2.11 Fire Safety

2.11.1 Fire Hazards on the BC-500X

Material	Hazard	Location
Hydraulic oil (ISO 46)	Combustible liquid	HPU, hydraulic lines throughout machine
Way lube oil (ISO 68)	Combustible liquid	Way lube reservoir, distribution lines
Coolant (water-soluble)	Primarily non-flammable when properly mixed; evaporation leaves flammable residue	Coolant tank, distribution lines, work zone
Chips (magnesium alloy, titanium)	Fire/explosion risk if machined without proper coolant	Work zone chips
Electrical wiring insulation	Fire if overloaded or arc fault occurs	Throughout machine
Control cabinet	Electrical fire from component failure	Control cabinet

2.11.2 Fire Prevention

1. **Coolant flow:** Ensure coolant is flowing whenever cutting operations are in progress. Loss of coolant increases chip and work zone temperatures significantly.
2. **Chip accumulation:** Do not allow chips to accumulate excessively in the work zone or chip conveyor. Compressed chip piles can generate heat through exothermic reactions and retain heat from cutting.
3. **Electrical maintenance:** Follow all electrical maintenance procedures. Loose connections and overloaded circuits are primary electrical fire causes.
4. **Hydraulic leaks:** Repair hydraulic leaks immediately. Oil on hot chip surfaces can ignite.
5. **Magnesium and titanium:** If machining these materials, use only approved coolants and maintain aggressive flow. Consult your tooling and coolant supplier for specific recommendations.

2.11.3 Fire Response

In the event of a fire in or near the BC-500X:

1. **Activate the E-Stop** to halt machine operation.
2. **Alert personnel** in the area.
3. **Call emergency services** (911) immediately. Do not attempt to fight significant fires without professional assistance.
4. **For small, contained fires** (e.g., a small oil drip fire): Use the appropriate fire extinguisher:
 - **CO₂ extinguisher:** Preferred for electrical fires and oil fires; leaves no residue that

damages machine

- **Dry chemical extinguisher:** Effective but leaves residue that is difficult to clean and may damage electrical components
- **Water:** NOT suitable for oil fires or electrical fires

5. **Evacuate** if the fire is not immediately controlled.

6. Do not re-enter the area until cleared by fire emergency personnel.

7. Report all fire events to BetaCorp Systems Technical Support. Fire events may indicate machine malfunctions requiring inspection before return to service.

2.11.4 Fire Extinguisher Placement

A minimum of one CO₂ fire extinguisher (minimum 5 lb capacity) must be mounted within 10 meters (33 feet) of the BC-500X, in a location accessible from the operator's normal working position. The extinguisher must be inspected monthly and serviced annually per local fire code requirements.

2.12 Guards and Interlocks — Never Bypass

The BC-500X incorporates numerous machine guards and interlocks designed to prevent injury. These safeguards are integral to the machine's safety design and must NEVER be removed, modified, or bypassed.

2.12.1 Machine Guards

Guard	Description	Removal Requirement
Work zone enclosure	Full enclosure of work zone; polycarbonate window panel	LOTO required; must reinstall before resuming operation
Work zone door	Sliding or hinged door with interlock; primary operator access	Open only when authorized; interlocked
Tool changer carousel guard	Sheet metal guard enclosing the tool magazine carousel	LOTO required for removal
Chip conveyor drive guard	Guard over drive sprocket and motor coupling	LOTO required for removal
HPU coupling guard	Guard over hydraulic pump motor coupling	LOTO required for removal
Z-axis covers (bellows)	Telescoping covers protect Z-axis ballscrew and guideways	Do not remove unnecessarily; inspect for damage
X/Y-axis covers (telescoping)	Protect X and Y axis ballscrews and guideways	Do not remove unnecessarily; inspect for damage
Control cabinet door	Prevents access to high-voltage control components	Padlockable; LOTO before opening
Electrical panel covers	Covers over terminal strips and	LOTO before removal

Guard	Description	Removal Requirement
	bus bars	

2.12.2 Machine Interlocks

Interlock	Function	Safety System
Work zone door switch	Prevents axis motion and spindle start when door is open (unless in JOG mode at reduced speed)	Safety relay; dual-channel
Work zone door lock	Holds door closed when spindle is rotating above 30 RPM	Electromagnetic lock; safety relay
E-Stop chain	Removes servo and spindle power when any E-Stop is activated	Safety relay; dual-channel
Axis over-travel (+ and -)	Halts axis motion before reaching mechanical end of travel	Hardware limit switch; safety relay
Spindle encoder loss	Halts spindle and sets fault E101 if encoder signal is lost	Drive-level safety function
Hydraulic pressure low	Sets fault E404 if hydraulic pressure drops below minimum	Pressure switch; PLC logic
Tool unclamped detect	Prevents spindle start if tool is not fully clamped	Proximity sensor; PLC logic
ATC arm home detect	Prevents axis motion if ATC arm is not in home position	Proximity sensor; PLC logic
Coolant level low	Sets fault E202 if coolant level drops below minimum	Float switch; PLC logic
Z-axis brake confirmation	Confirms Z-axis brake is applied before removing hydraulic power	Pressure switch; PLC logic

WARNING: These interlocks use hard-wired safety relay circuits and cannot be bypassed through software parameter changes. Any attempt to physically bypass the wiring of safety interlock circuits is PROHIBITED and may result in fatal injury. If an interlock is malfunctioning, the machine must be taken out of service until the interlock is repaired by a qualified technician.

2.13 Safe Work Practices During Maintenance

2.13.1 Work Planning

Before beginning any maintenance task:

1. Read the procedure completely before starting. Do not rely on memory for safety-critical steps.
2. Gather all required tools, parts, and materials before beginning.
3. Identify all energy sources that must be isolated.
4. Inform your supervisor of the planned maintenance task and estimated duration.
5. Post "MACHINE UNDER MAINTENANCE — DO NOT OPERATE" notices on the control panel.
6. Prepare the maintenance log entry with planned work scope.

2.13.2 Working in the Work Zone

On occasion, maintenance requires access to the inside of the work zone (e.g., cleaning, inspection, workpiece removal after crash). The following rules apply:

1. The machine must be locked out (LOTO) before any person enters the work zone for maintenance.
2. In routine production (operator access), the work zone door may be opened to load/unload workpieces only after:
 - The spindle has come to a complete stop (speed = 0)
 - The axes are in a safe position (Z-axis at home, or at a position well clear of the workpiece)
 - The door interlock has been satisfied (door interlock clears automatically when spindle speed = 0 and no axis motion command is active)
3. If a workpiece or tool must be manually removed while the machine is still energized (not in LOTO), move to JOG mode only, confirm spindle is stopped, and use proper tools (chip hooks, end mill holder, etc.) — never use bare hands inside the work zone even when the spindle is stopped, as the machine is still energized and could be accidentally activated.

2.13.3 Tool and Toolholder Handling

Cutting tools and toolholders present cutting hazards even when not rotating:

1. Never pick up a cutting tool by the flutes. Always grasp at the shank or use a tool holder.
2. When installing or removing a toolholder from the spindle, use a spindle wrench (Part No. BC-TOOL-0224) to hold the spindle. Do not allow the spindle to rotate freely during tool installation.
3. When handling large, heavy toolholders (face mills, etc.), use two hands and/or toolholder lifting straps to prevent drops.
4. Store cutting tools in tool racks when not in use. Loose tools on work surfaces create cutting hazards.

2.13.4 Housekeeping

Good housekeeping is a fundamental element of safety:

1. Keep the floor around the machine clean and free of coolant puddles, chips, and oil.
 2. Use anti-slip mats at the operator's normal work position.
 3. Dispose of used rags, filters, and empty containers promptly.
 4. Keep chip bins from overflowing.
 5. Report any coolant or oil leaks immediately and clean up spills promptly.
 6. Store all chemicals in designated areas.
 7. Maintain clear access to all emergency stops, fire extinguishers, and exits.
-

2.14 Confined Space Considerations

The BC-500X coolant tank (capacity: 350 liters / 92 gallons) requires internal inspection and cleaning periodically. The coolant tank is large enough that a person's head and upper body could enter during cleaning.

WARNING: Evaluate whether the coolant tank cleaning operation requires a permit-required confined space entry program per OSHA 29 CFR 1910.146. Factors that may make the coolant tank a permit-required confined space include: potential for accumulation of hazardous atmospheric contaminants (H₂S gas from microbial activity in contaminated coolant), oxygen deficiency, and engulfment hazard. Have the atmosphere in the coolant tank tested before any person inserts their head or upper body into the tank.

Signs of Potential Atmospheric Hazard in Coolant Tank:

- Strong sulfur ("rotten egg") odor from the tank
- Visible sludge layer greater than 25 mm (1 in) deep at tank bottom
- Coolant that has been stagnant for more than 2 weeks

If any of these conditions are present, treat the coolant tank cleaning as a permit-required confined space entry. Contact your facility safety department.

For routine coolant tank cleaning where conditions are normal, follow the confined space pre-entry checklist in Chapter 12.

2.15 Chemical Right-to-Know (GHS/SDS)

All chemicals used with or in the BC-500X are subject to OSHA's Hazard Communication Standard (29 CFR 1910.1200) and the Globally Harmonized System of Classification and Labeling of Chemicals (GHS).

Required SDS Documents:

The facility must maintain Safety Data Sheets (SDS) for all chemicals used with the BC-500X, including:

- Metalworking cutting fluid (coolant)
- Hydraulic oil
- Way lube oil
- Spindle bearing grease
- ATC grease
- Cleaning solvents
- Rust preventive oils

SDS documents must be:

- Accessible to all employees in the work area at all times
- Available in the language of the workers using the chemicals
- Current (within 5 years for reformulated products)

Chemical Training:

All operators and maintenance personnel must receive GHS/HazCom training covering:

- How to read and interpret an SDS
- Understanding GHS label elements (hazard pictograms, signal words, hazard statements)
- Proper use of PPE for chemical handling
- Emergency response (first aid, spill response)

This training must be documented and records maintained per OSHA requirements.

End of Chapter 2 — Safety Instructions

PART 2 — TECHNICAL SPECIFICATIONS

Chapter 3 — Technical Specifications

3.1 Machine Overview

The BC-500X is a full-enclosure, vertical-spindle CNC machining center designed for high-productivity precision machining of a wide range of metallic and non-metallic materials. The machine features a rigid box-column construction with hardened and ground linear guideways on all axes, a 24-position random-access automatic tool changer, and a fully integrated hydraulic

power unit. The BC-500X is controlled by the BetaCorp CNC Controller Model 5000, a full-feature CNC system with integrated PLC.

Machine Configuration Summary:

Feature	Specification
Machine type	Vertical CNC Machining Center
Number of CNC axes	3 (X, Y, Z) — 4th and 5th axes optional
Spindle orientation	Vertical
Column type	Box column, one-piece cast iron
Guideway type	Hardened and ground linear guideways (roller type)
Drive type	AC servo motors with ball screw drives on all axes
ATC type	Random-access, dual-arm, 24-position carousel
Spindle taper	CAT-40 (ANSI/ASME B5.50) / BT-40 (MAS 403) — specify at order
Coolant type	Flood coolant; through-spindle optional
Control system	BetaCorp CNC Controller Model 5000

3.2 Work Envelope and Travel Specifications

Parameter	Value	Unit
X-axis travel	750	mm
Y-axis travel	500	mm
Z-axis travel	550	mm
X-axis travel	29.53	in
Y-axis travel	19.69	in
Z-axis travel	21.65	in
Table size (X × Y)	900 × 520	mm
Table size (X × Y)	35.43 × 20.47	in
T-slot width	18	mm
T-slot spacing	125	mm
Number of T-slots	5	—
T-slot length	860	mm
Table load capacity	600	kg
Distance: spindle nose to table (min / max)	100 / 650	mm
Distance: spindle center to	550	mm

Parameter	Value	Unit
column face		

3.2.1 Axis Coordinate System

The BC-500X uses a standard right-hand Cartesian coordinate system:

- **X-axis:** Positive direction is to the right when facing the machine from the front
- **Y-axis:** Positive direction is away from the operator (toward the rear of the machine)
- **Z-axis:** Positive direction is upward (away from the table)

Machine home position (reference point) is at X+, Y+, Z+ limits.

Program zero (workpiece coordinate system) is established by the operator using the work coordinate offset function (see Section 6.9.1).

3.3 Spindle Specifications

Parameter	Value	Unit
Spindle type	Cartridge spindle, angular contact bearings	—
Spindle taper	CAT-40 (7/24 taper) or BT-40	—
Maximum speed	10,000	RPM
Maximum power (continuous / 30-min rated)	15 / 18.5	kW
Maximum power (continuous / 30-min rated)	20.1 / 24.8	HP
Maximum torque (continuous)	95.5	N·m
Maximum torque (peak)	143.2	N·m
Spindle motor type	AC induction motor with vector drive	—
Speed range (gear 1, low)	50–4,000	RPM
Speed range (gear 2, high)	500–10,000	RPM
Gear change type	Pneumatic / hydraulic two-speed gearbox	—
Spindle acceleration time (0 to 10,000 RPM)	Less than 3.5	seconds
Spindle deceleration time (10,000 to 0 RPM)	Less than 4.0	seconds (with brake)
Tool retention force (draw bar clamping)	18,000	N
Tool retention type	Belleville washer stack,	—

Parameter	Value	Unit
	pneumatic/hydraulic unclamp	
Spindle bearing type	High-precision angular contact ball bearings (P5)	—
Spindle bearing preload	Medium preload	—
Spindle nose face run-out	Less than 3	μm
Spindle taper run-out (25 mm from gauge line)	Less than 5	μm
Spindle encoder type	Optical incremental encoder	—
Spindle encoder resolution	4,096 pulses/revolution (pre-quadrature)	PPR
Spindle encoder resolution (post-quadrature)	16,384	counts/rev
Lubrication method	Oil-air mist lubrication (continuous)	—

3.3.1 Spindle Speed/Torque Curve

The BC-500X spindle delivers torque characteristics optimized for a broad range of materials:

Speed Range	Characteristic
50–1,500 RPM	Constant torque region (95.5 N·m continuous)
1,500–10,000 RPM	Constant power region (15 kW continuous)
50–600 RPM (Gear 1 only)	Enhanced low-speed torque for heavy boring

NOTE: Operating continuously at the maximum torque rating for extended periods requires adequate coolant flow. Monitor spindle motor drive temperature during heavy-cut operations. If drive temperature alarm occurs, reduce cutting parameters.

3.4 Feed and Rapid Traverse Rates

Parameter	Value	Unit
Rapid traverse, X-axis	36	m/min
Rapid traverse, Y-axis	36	m/min
Rapid traverse, Z-axis	30	m/min
Maximum feed rate (all axes)	15	m/min
Minimum feed rate	1	mm/min
Feed override range	0–150	%
Rapid override settings	5%, 25%, 50%, 100%	—
Acceleration/deceleration type	S-curve (jerk-limited)	—

Parameter	Value	Unit
Acceleration (X, Y axes)	0.5	g
Acceleration (Z axis)	0.4	g
Positioning accuracy (ISO 230-2)	±0.005	mm
Repeatability (ISO 230-2)	±0.003	mm
Thermal compensation	Active thermal compensation (X, Y, Z)	—

3.5 Tool Changer Specifications

Parameter	Value	Unit
ATC type	Double-arm (simultaneous swap)	—
Magazine type	Rotary carousel	—
Number of tool positions	24	—
Tool selection method	Random access (T-code)	—
Maximum tool weight	8	kg
Maximum tool diameter (adjacent pockets occupied)	80	mm
Maximum tool diameter (adjacent pockets empty)	125	mm
Maximum tool length (from gauge line)	300	mm
Chip-to-chip time (standard tool)	4.2	seconds
Tool-to-tool time (carousel index only)	1.8	seconds
Carousel drive motor	AC servo motor, 0.75 kW	—
ATC arm drive	Cam-driven, hydraulically actuated	—
ATC arm rotation speed	90° in 0.9 seconds	—
Tool pot type	Spring-grip, BT-40 compatible	—
Pot retention force	85	N
Position sensors	Proximity sensors (inductive), 1 per 24 positions + home	—
ATC home sensor type	Inductive proximity, NPN, normally-open	—

3.6 Electrical Specifications

3.6.1 Main Power Supply Requirements

Parameter	Specification
Supply voltage	480 VAC $\pm$ 10%, 3-phase
Supply frequency	60 Hz $\pm$ 1 Hz
Number of phases	3-phase + ground
Neutral required	No (delta-connected machine)
Total connected load	45 kVA
Typical demand load (machining)	28 kVA
Main disconnect	100 A, 3-pole fusible disconnect (provided in machine)
Recommended upstream breaker	100 A, 3-pole, Class J fuse or HACR breaker
Minimum short circuit current rating	10,000 A at 480 VAC
Recommended wire size (service entry)	4 AWG copper (or 2 AWG aluminum)
Maximum wire size (cabinet entry)	350 kcmil
Supply conduit knockout size	63.5 mm (2.5 in)
Ground conductor requirement	8 AWG minimum (see NEC 250.122)

NOTE: The specified 100 A upstream breaker is for a dedicated circuit. If the BC-500X shares a circuit with other loads (not recommended), the upstream protection must be sized for the total connected load. Consult your electrical engineer.

3.6.2 Control Cabinet Power Distribution

The main electrical cabinet distributes power to all machine subsystems through a central power distribution bus. All distribution is from the main cabinet; no external power connections are required except for the single 480 VAC supply.

Circuit	Voltage	Protection	Load
Spindle drive input	480 VAC, 3-phase	50 A, Class J fuse	18.5 kW
X-axis servo drive	480 VAC, 3-phase	20 A, Class J fuse	3 kW
Y-axis servo drive	480 VAC, 3-phase	20 A, Class J fuse	3 kW
Z-axis servo drive	480 VAC, 3-phase	20 A, Class J fuse	4 kW
Hydraulic pump motor	480 VAC, 3-phase	15 A, Class J fuse	4 kW
Coolant pump motor	480 VAC, 3-phase	10 A, Class J fuse	2.2 kW
ATC carousel servo	480 VAC, 3-phase	10 A, Class J fuse	0.75 kW
Chip conveyor motor	480 VAC, 3-phase	5 A, Class J fuse	0.75 kW
Control transformer	480 VAC primary	15 A fuse	3 kVA
120 VAC control circuit	120 VAC	10 A circuit breaker	Control/solenoids

Circuit	Voltage	Protection	Load
24 VDC system	24 VDC	Fused at each output	PLC/sensors

3.6.3 Motor Power Ratings

Motor	Power	Speed	Torque	Frame
Spindle motor	15 kW cont. / 18.5 kW (30 min)	0–10,000 RPM	95.5 N·m cont.	IEC 112M
X-axis servo	3 kW	0–3,000 RPM	9.55 N·m	IEC 90L
Y-axis servo	3 kW	0–3,000 RPM	9.55 N·m	IEC 90L
Z-axis servo	4 kW	0–3,000 RPM	12.73 N·m	IEC 90L
ATC carousel servo	0.75 kW	0–3,000 RPM	2.39 N·m	IEC 71
Hydraulic pump motor	4 kW	1,450 RPM	26.3 N·m	IEC 100L
Coolant pump motor	2.2 kW	2,900 RPM	7.2 N·m	IEC 80
Chip conveyor motor	0.75 kW	1,450 RPM (geared)	—	IEC 71
Way lube pump motor	0.12 kW	1,450 RPM	—	IEC 56

3.7 Hydraulic System Specifications

Parameter	Value	Unit
System operating pressure	150	bar
Maximum system pressure (relief valve set)	210	bar
Hydraulic pump type	Fixed displacement gear pump	—
Hydraulic pump displacement	8.3	cc/rev
Hydraulic pump flow rate (at 1,450 RPM)	12	L/min
Hydraulic pump motor	4 kW, 1,450 RPM	—
Reservoir capacity	50	L
Hydraulic oil specification	ISO VG 46 (anti-wear type)	—
Oil filtration	Return line filter, 10 µm absolute	—
Oil temperature operating range	20–60	°C
Oil temperature maximum	65	°C

Parameter	Value	Unit
(alarm)		
Oil temperature maximum (shutdown)	70	°C
Oil temperature control	Air-cooled heat exchanger (fan-cooled)	—
Heat exchanger motor	0.18 kW	—
Hydraulic circuits served	Z-axis brake, tool unclamp, ATC arm, gear change	—
Number of solenoid valves	8	—
Solenoid valve voltage	24 VDC	—
Pressure gauge accuracy	±2.5% full scale	—
Pressure switch (low pressure alarm)	Set at 120 bar (rising), 100 bar (falling)	—

3.7.1 Hydraulic Circuit Functions

Circuit	Operating Pressure	Function
Z-axis hydraulic brake	150 bar (release) / spring apply	Holds Z-axis position; applied when power is removed
Spindle tool unclamp	150 bar	Overcomes draw bar spring to release tool
ATC arm actuation	150 bar	Drives ATC arm extension, rotation, and retraction
Gear change actuation	80 bar	Selects low/high spindle gear ratio
Workholding (optional)	0–150 bar (adjustable)	Customer-supplied workholding fixtures

3.8 Pneumatic System Specifications

Parameter	Value	Unit
Supply pressure (required)	6.0–8.0	bar
Supply pressure (required)	87–116	psi
Air consumption (average during operation)	120	NL/min
Air consumption (peak, tool change)	350	NL/min
Air filtration required	40 µm or finer	—

Parameter	Value	Unit
Air quality	Clean, dry, oil-free	—
Dew point at supply	Less than 3°C at operating pressure	—
Inlet connection size	G3/8 (ISO 228-1) or 1/2 NPT	—
Internal working pressure (regulated)	5.5 bar / 80 psi	—
Pneumatic filter-regulator-lubricator (FRL)	Provided; mounted on rear of machine	—
Pneumatic circuits served	Spindle air purge, chip blow-off, tool sensor, gear change assist	—

NOTE: The BC-500X pneumatic system uses a non-lubricated configuration for the spindle air purge and chip blow-off circuits. The FRL lubricator section is bypassed for these circuits. Do not add oil to the FRL lubricator bowl unless specifically directed by a BetaCorp service engineer.

3.9 Coolant System Specifications

Parameter	Value	Unit
Coolant tank capacity	350	L
Coolant pump type	Centrifugal	—
Coolant pump motor	2.2 kW, 2,900 RPM	—
Coolant flow rate (standard nozzles)	60	L/min
Coolant pressure at nozzles	3–5	bar
Coolant supply connections	4 adjustable nozzles (standard)	—
Coolant filtration	Magnetic chip separator + 150 µm stainless screen	—
Coolant level sensor	Float switch, 4-wire, 24 VDC	—
Low coolant level setpoint	80 mm below full level	—
Coolant temperature (normal operation)	20–35	°C
Coolant temperature (maximum)	45	°C
Recommended coolant type	Semi-synthetic or synthetic water-soluble	—
Coolant concentration (typical)	7–10	%
Through-spindle coolant (optional)	70 bar, 30 L/min	bar, L/min

Parameter	Value	Unit
Coolant pH range (normal)	8.5–9.5	—

3.10 Axis Servo System Specifications

Parameter	X-Axis	Y-Axis	Z-Axis	Unit
Servo motor power	3	3	4	kW
Servo motor speed (rated)	3,000	3,000	3,000	RPM
Servo motor torque (rated)	9.55	9.55	12.73	N·m
Drive type	BetaCorp SD-300	BetaCorp SD-300	BetaCorp SD-400	—
Ballscrew pitch	10	10	10	mm/rev
Ballscrew diameter	40	40	50	mm
Ballscrew accuracy class	ISO Class 3	ISO Class 3	ISO Class 3	—
Ballscrew support	Fixed-fixed (angular contact)	Fixed-fixed	Fixed-floating	—
Linear guideway type	Roller (IKO LWE)	Roller (IKO LWE)	Roller (IKO LWE)	—
Number of guideway carriages	4 (2 rails × 2)	4 (2 rails × 2)	4 (2 rails × 2)	—
Encoder type	Optical incremental	Optical incremental	Optical incremental	—
Encoder resolution	2,500 PPR	2,500 PPR	2,500 PPR	—
Encoder resolution (quadrature × 4)	10,000	10,000	10,000	counts/rev
Effective resolution at table	1	1	1	μm
Servo update rate	250	250	250	μs
Position loop gain (default)	See B2.01	See B2.11	See B2.21	—

3.10.1 Axis Brake Specifications

Axis	Brake Type	Brake Torque
X-axis	Electromagnetic (on servo motor)	5 N·m
Y-axis	Electromagnetic (on servo motor)	5 N·m
Z-axis	Hydraulic (fail-safe, spring-apply) + electromagnetic	Hydraulic: 120 N·m; EM: 8 N·m

The Z-axis hydraulic brake is critically important. It is spring-applied (fail-safe) and hydraulically released. When hydraulic power is removed, the brake automatically applies. This prevents the Z-axis from falling under gravity when the machine is powered down or in emergency stop.

3.11 Encoder Specifications

Encoders are used on all servo axes and on the spindle. Proper encoder operation is critical to machine accuracy and safety.

3.11.1 Axis Encoder Specifications

Parameter	Value
Encoder manufacturer	Heidenhain (standard)
Encoder model (X, Y axes)	ERN 1381
Encoder model (Z axis)	ERN 1381
Type	Optical incremental
Resolution	2,500 PPR
Output type	Differential (RS-422)
Supply voltage	5 VDC $\pm$ 5%
Maximum frequency	200 kHz
Shaft coupling	Bellows coupling (backlash-free)
Operating temperature	0–80°C
Cable length (to drive)	5 m (standard)
Cable part number	BC-CBL-ENC-5M
Connector type	12-pin circular (BetaCorp standard)

3.11.2 Spindle Encoder Specifications

Parameter	Value
Encoder manufacturer	Heidenhain
Encoder model	ROD 420

Parameter	Value
Type	Optical incremental with index
Resolution	4,096 PPR (pre-quadrature)
Resolution (post-quadrature)	16,384 counts/rev
Output type	Differential (RS-422)
Supply voltage	5 VDC $\pm$ 5%
Maximum speed	10,000 RPM
Maximum frequency (at 10,000 RPM)	682 kHz
Operating temperature	0–80°C
Cable part number	BC-CBL-SPENC-3M
Index pulse	1 per revolution

IMPORTANT: The spindle encoder is used for:

1. Spindle speed feedback for closed-loop speed control
2. Rigid tapping synchronization
3. Spindle orientation for tool changes
4. C-axis positioning (optional 4th axis applications)

Loss of the spindle encoder signal at any time results in fault **E101 — Spindle Encoder Signal Lost**. This is a safety-critical fault. The spindle drive immediately disables the output, and the spindle coasts to a stop (or applies brake if B1.22 = 1). The machine will not allow the spindle to start until the encoder fault is resolved. See Section 8.5.1 for the complete E101 troubleshooting procedure.

3.12 Physical Dimensions and Weight

Parameter	Value	Unit
Machine height (with enclosure)	2,850	mm
Machine height (with enclosure)	112.2	in
Machine length (front to rear)	3,200	mm
Machine length (front to rear)	126.0	in
Machine width (left to right)	2,400	mm
Machine width (left to right)	94.5	in
Machine net weight (without fluids)	8,200	kg

Parameter	Value	Unit
Machine net weight (without fluids)	18,078	lb
Machine weight with all fluids	8,650	kg
Machine weight with all fluids	19,070	lb
HPU weight	185	kg
Control cabinet weight	240	kg
Chip conveyor weight	95	kg
Floor loading (maximum, footpad area)	12,800	kg/m ²
Number of leveling footpads	6	—
Footpad diameter	150	mm
Footpad adjustment range	±25	mm

3.12.1 Minimum Clearance Requirements

For access, maintenance, and heat dissipation, the following minimum clearances must be maintained around the BC-500X:

Location	Minimum Clearance
Front (operator access)	1,500 mm (59.1 in)
Right side (electrical cabinet access)	1,000 mm (39.4 in)
Left side (chip conveyor access)	800 mm (31.5 in)
Rear (hydraulic / service access)	1,200 mm (47.2 in)
Above machine (for ceiling clearance with no cranes)	500 mm (19.7 in)
Above machine (if overhead crane is used for tooling)	2,000 mm (78.7 in) above machine top

3.13 Ambient Environmental Specifications

Parameter	Specification
Ambient temperature (operation)	+5°C to +40°C
Ambient temperature (storage)	-20°C to +60°C
Relative humidity (operation)	20–75%, non-condensing
Relative humidity (storage)	10–90%, non-condensing
Maximum altitude	1,000 m (derated operation to 2,000 m — contact BetaCorp)
Vibration (floor)	Less than 0.2 g RMS (10–200 Hz)

Parameter	Specification
Pollution degree	Pollution Degree 2 (IEC 60664-1)
Overvoltage category	Category III (480 VAC supply)
IP rating (control cabinet)	IP54
IP rating (machine enclosure)	IP32 (not designed for outdoor use)
Seismic zone	Designed for Seismic Zone 1 (low seismic activity) — consult BetaCorp for other zones

3.14 Noise and Vibration

Parameter	Value	Conditions
Sound pressure level (operator position, dry run)	74 dB(A)	Rapid traverse, no cutting
Sound pressure level (operator position, machining)	Up to 84 dB(A)	Steel milling, typical parameters
Sound power level	89 dB(A)	Machining steel
Uncertainty K	4 dB	Per ISO 3746
Machine vibration (table surface, idle)	Less than 0.3 μ m RMS	ISO 10816

Noise values are measured per ISO 3746:2010 in a hemianechoic environment. Actual values in a factory environment will differ due to room reflections and other machines.

3.15 CE/UL/CSA Compliance Data

The BC-500X is designed and manufactured in compliance with applicable safety standards. CE-marked versions comply with EU Machinery Directive 2006/42/EC.

Standard	Status
UL 508A (Industrial Control Panels)	UL Listed
CSA C22.2 No. 14 (Industrial Control Equipment)	CSA Certified
EN ISO 16090-1 (Machining Centres Safety)	Conformant (CE versions)
EN 60204-1 (Electrical Equipment of Machines)	Conformant (CE versions)
EN ISO 13849-1 PL d (Safety-related control, Cat. 3)	Conformant for E-Stop, door interlock
FCC Part 15 (EMI — US)	Compliant
ICES-003 (EMI — Canada)	Compliant

Compliance Documentation Location:

- CE Declaration of Conformity: Appendix K (CE versions only)
 - UL Listing certificate: Available on request
 - Test reports: Available through BetaCorp document portal
-

End of Chapter 3 — Technical Specifications

PART 3 — INSTALLATION

Chapter 4 — Installation Guide

4.1 Pre-Installation Planning

Before the BC-500X arrives at the installation site, a comprehensive pre-installation checklist must be completed. Failure to prepare the site adequately will delay installation and may result in additional costs.

Pre-Installation Checklist:

- ☐ Foundation drawing reviewed and foundation prepared per Section 4.2
- ☐ Anchor bolt locations marked and bolts cast into concrete (if epoxy anchors: holes drilled and cleaned)
- ☐ Floor loading capacity verified by structural engineer (minimum 15,000 kg/m²)
- ☐ Electrical supply brought to the machine location: 480 VAC, 3-phase, 100 A capacity
- ☐ Electrical supply terminated in a junction box within 3 m of the machine right side
- ☐ Ground conductor installed (8 AWG minimum)
- ☐ Compressed air supply brought to the machine location: 6–8 bar, 1/2 NPT or G3/8 connection
- ☐ Air dryer or desiccant filter installed upstream of machine air supply
- ☐ Coolant drainage provision made (floor drain or collection tank) near machine perimeter
- ☐ Chip storage/disposal area designated near chip conveyor discharge
- ☐ Minimum clearances verified (see Section 3.12.1)
- ☐ Overhead clearance verified for rigging during delivery (minimum 4.5 m above floor)
- ☐ Fork lift or overhead crane with minimum 10-tonne capacity available for rigging

- ☐ Certified rigger identified and briefed on lifting plan
 - ☐ BetaCorp service engineer arrival date confirmed
 - ☐ All personnel involved in installation have read Chapter 2 (Safety)
 - ☐ PPE available for all installation personnel
-

4.2 Site Preparation and Foundation

4.2.1 Foundation Drawing and Requirements

The BC-500X requires a properly prepared concrete foundation to achieve its specified accuracy. The following requirements apply:

Concrete Specification:

- Minimum compressive strength: 25 MPa (3,625 psi) at 28 days (ASTM C39 or equivalent)
- Concrete type: Normal weight concrete (density ~2,400 kg/m³)
- Reinforcement: Per structural engineer's design based on local soil conditions
- Minimum foundation slab thickness: 250 mm (10 in)
- Preferred foundation slab thickness: 400 mm (16 in) for optimum vibration damping

Foundation Dimensions (Minimum):

- Length: 3,600 mm (141.7 in)
- Width: 2,800 mm (110.2 in)
- The machine footprint is 3,200 × 2,400 mm; the foundation must extend 200 mm beyond the machine footprint on all sides

Foundation Surface:

- The foundation surface must be level within ±2 mm over the full machine footprint before installation
- Surface must be clean, free of oil, and structurally sound
- No cracks deeper than 5 mm or wider than 2 mm are acceptable in the machine footpad area

Anchor Bolt Pattern: Six (6) anchor bolt locations are specified, corresponding to the six machine leveling footpads. Anchor bolt specifications:

- Bolt size: M24 × 400 mm (length)
- Bolt grade: Grade 8.8 or equivalent
- Bolt pattern: Per BetaCorp foundation drawing BCS-FDN-BC500X (available from BetaCorp Systems)
- Cast-in method: Cast into concrete before pour using bolt templates (provided with

machine)

- Epoxy anchor method: Hilti HIT-RE 500 V3 or equivalent; follow epoxy manufacturer's instructions for cure time before loading

CAUTION: Allow epoxy anchors to cure fully before applying any load. Typical cure time is 24 hours at 20°C; longer at lower temperatures. Check epoxy manufacturer's specifications.

Vibration Isolation: For installations in facilities with significant floor vibration from nearby presses, forging equipment, or compressors:

- Consider installing the BC-500X on anti-vibration mounts (Mason Industries type N or equivalent)
- BetaCorp Systems offers a vibration isolation package (Part No. BC-OPT-VIBISO) that includes neoprene-steel sandwich isolation mounts rated for the machine weight
- Without isolation, floor vibration above 0.2 g will degrade positioning accuracy

4.2.2 Anchor Bolt Installation

Materials Required:

- 6 × M24 anchor bolts (supplied with machine)
- 6 × M24 heavy hex nuts
- 6 × M24 flat washers
- Appropriate tools for concrete drilling (if epoxy anchor method)

Cast-in Anchor Bolt Procedure:

1. Obtain the BetaCorp foundation drawing BCS-FDN-BC500X. This drawing shows the exact location of all six anchor bolt centerlines relative to the machine front face and right side face.
2. Lay out the anchor bolt locations on the prepared form before the concrete pour.
3. Use the anchor bolt template provided with the machine (sheet steel, 3,200 × 2,400 mm, with 6 holes punched) to ensure correct bolt spacing.
4. Position the template on the form and verify all dimensions match the foundation drawing.
5. Install the anchor bolts through the template holes. Use a pipe sleeve around each bolt to allow ±15 mm of adjustment after the pour.
6. Brace the template to the form so the bolts cannot shift during the pour.
7. Pour concrete; do not allow concrete to enter the pipe sleeves.
8. After the pour and initial set (minimum 48 hours), remove the template. Bolts should protrude 80 mm above the finished floor surface.
9. Allow concrete to cure for minimum 14 days before machine installation.

Epoxy Anchor Procedure:

1. After concrete has cured (minimum 28 days for full strength), lay out the anchor bolt locations per the foundation drawing.
2. Drill anchor holes using a rotary hammer drill with a 28 mm core bit: depth = 280 mm minimum.
3. Clean the holes thoroughly: blow out with compressed air, brush with a wire brush, blow out again. Repeat three times. No dust or moisture may remain in the hole.
4. Check that holes are dry. If moisture is present, allow to dry or use moisture-tolerant epoxy per manufacturer's instructions.
5. Inject epoxy per manufacturer's instructions, filling from the bottom of the hole.
6. Insert the M24 anchor bolt while rotating slowly to ensure full epoxy contact. The bolt should protrude 80 mm above floor level.
7. Allow epoxy to cure per manufacturer's specification before applying load.

4.2.3 Leveling Pads and Vibration Isolation

The BC-500X is equipped with six (6) adjustable leveling footpads. Each footpad consists of:

- Cast iron base plate (250 × 250 × 40 mm)
- M24 leveling bolt with jam nut
- Anti-vibration rubber pad (80 Shore A durometer, sandwiched between base plate and floor)

Footpad Adjustment Range:

- Vertical: ±25 mm from nominal
- This allows the machine to be leveled even if the floor varies by up to 25 mm across the footprint

Pre-Leveling Preparation: Before placing the machine on the footpads:

1. Verify that all six footpad locations are at the correct height (within ±20 mm of each other) using a builder's level.
2. If the floor has significant variation, shim the footpads using stainless steel shim stock.
3. Thread the M24 leveling bolt to the approximate center of its adjustment range.
4. Place the anti-vibration rubber pad on the floor over the anchor bolt.
5. Place the footpad base plate over the rubber pad.

4.3 Machine Delivery and Rigging

DANGER: The BC-500X weighs 8,500 kg (18,739 lb). Rigging this machine requires certified riggers, appropriate lifting equipment rated for at least 10 tonnes, and careful coordination. An improperly rigged machine can fall, killing or maiming people in the fall path. Only qualified riggers using appropriate equipment may perform this operation.

4.3.1 Lifting Points and Rigging Diagram

The BC-500X is provided with four (4) factory-installed lifting eye bolts. These eye bolts are located on the top of the machine base casting, two on each side. The eye bolts are M30, Grade 8.8, with a working load limit (WLL) of 3,000 kg each (total WLL = 12,000 kg; machine weight = 8,500 kg, giving a safety factor of 1.41 — acceptable with a spreader bar).

Rigging Configuration:

- Use a four-leg wire rope sling set with a minimum WLL of 10,000 kg (10 tonnes)
- Use a spreader bar (minimum 2,500 mm center-to-center) to prevent inward compression of the machine lifting lugs
- Maximum leg angle from vertical: 30° (preferred); 45° maximum
- All four sling legs must be the same length to ensure level lift

DO NOT:

- Use the spindle head, chip conveyor, or control panel as lift points
- Use chains without proper hooks and load ratings
- Lift the machine with fewer than 4 slings
- Allow anyone under the suspended load

Rigging Equipment Required:

- Overhead crane or mobile crane: minimum 10-tonne capacity
- 4-leg wire rope sling set: WLL $\geq$ 10,000 kg at 45° (typically 10 mm diameter, 6×36 IWRC)
- Spreader bar: $\geq$ 2,500 mm, WLL $\geq$ 10,000 kg
- Shackles: 4 $\times$ WLL $\geq$ 3,500 kg each, screw pin anchor shackles
- Tag lines: 2 $\times$ rope tag lines for controlling load orientation during lift

4.3.2 Unpacking and Inspection

The BC-500X is shipped in a wooden crate or on a steel skid, depending on transport method. The machine is coated with a rust-preventive oil on all machined surfaces.

Unpacking Procedure:

1. Inspect the shipping crate/skid exterior for damage before unpacking. Document any visible shipping damage with photographs and note on the delivery receipt.
2. Remove the outer crating/skid material. Do not use pry bars or hammers near the machine body.
3. Remove protective plastic sheeting from the machine.
4. Inspect all external surfaces for damage. Pay particular attention to:
 - Spindle nose area (risk of impact damage during shipping)

- Control panel pendant
 - Chip conveyor (often shipped separately)
 - Hydraulic power unit (often shipped separately)
5. Check the packing list against received items. Items typically shipped separately:
 - Chip conveyor assembly
 - Coolant tank extension (if ordered)
 - Optional accessory packages
 - Tooling and toolholders (if ordered)
 - Operator's manual and documentation package
 6. Locate the transport securing hardware (bolts and brackets that secure the machine table and spindle head for shipping). These **MUST** be removed before power-up. See Section 4.3.2.1 below.
 7. Locate the consumables kit (supplied with the machine): hydraulic oil (50 L), way lube oil (10 L), initial coolant charge, filter elements.

Transport Securing Hardware Removal:

The machine is shipped with the following transport locks that **MUST** be removed before operation:

Item	Location	Tool Required	Part No. (Lock)
X-axis transport bracket	Left side of X-axis	19 mm wrench	BC-TRANS-001
Y-axis transport bracket	Rear of Y-axis	19 mm wrench	BC-TRANS-002
Z-axis transport bracket	Right side of column	24 mm wrench	BC-TRANS-003
Spindle nose protector	Spindle nose	By hand	BC-TRANS-004
ATC arm transport bracket	ATC arm casting	13 mm wrench	BC-TRANS-005

WARNING: Operating the machine with transport securing hardware in place will damage the machine and may cause the secured components to fail violently. Verify ALL transport locks are removed and stored (they may be needed for future machine relocation) before applying power.

4.4 Machine Leveling

Precise machine leveling is critical for achieving the positioning accuracy specified in Section 3.4. The BC-500X must be leveled to within 0.02 mm/m (0.0002 in/in) before commissioning.

Equipment Required:

- Precision level (sensitivity: 0.02 mm/m or better) — example: Starrett 98 Series or Mitutoyo

960 Series

- Level gauge block (flat reference surface)
- Feeler gauges
- 36 mm socket and torque wrench for leveling bolt jam nuts
- Shim stock (stainless steel, 0.05 to 1.0 mm thickness)

Leveling Procedure:

Step 1: Initial Coarse Leveling

After placing the machine on the footpads (with anchor bolts through the footpad base plates but not yet tightened):

1. Place the precision level on the machine table in the X-direction (parallel to X-axis).
2. Adjust the leveling bolts on the left and right front footpads to bring the machine level in X.
3. Place the level on the machine table in the Y-direction (parallel to Y-axis).
4. Adjust the rear footpad leveling bolts to bring the machine level in Y.
5. Repeat steps 2–4, alternating X and Y directions, until level is achieved in both directions simultaneously.
6. At this stage, level to within 0.05 mm/m.

Step 2: Anchor Bolt Torqueing

After coarse leveling:

1. Thread the M24 anchor bolt nuts down to contact the footpad base plate.
2. Tighten all six anchor bolts to 200 N·m (147 ft·lb).
3. Recheck level after tightening anchor bolts. Tightening will often move the machine slightly.
4. If level has moved, re-adjust the leveling bolts and re-torque anchor bolts.

Step 3: Fine Leveling

Fine leveling targets 0.02 mm/m or better:

1. Place the precision level on the machine table at the center of X-travel, reading in the X-direction.
2. Adjust front left and front right leveling bolts (1/6 turn increments) to fine-adjust X-level.
3. Move the table to the left end of X-travel. Read the level.
4. Move the table to the right end of X-travel. Read the level.
5. The level reading should change by no more than 0.01 mm/m across the full X-travel.
6. If the level changes significantly across the travel, the table guideways may need re-scraping (factory procedure — contact BetaCorp).

7. Repeat for Y-direction with level parallel to Y-axis.
8. When both directions are within 0.02 mm/m with less than 0.01 mm/m variation over the full travel, leveling is complete.

Step 4: Final Jam Nut Torqueing

After fine leveling is confirmed:

1. Hold each leveling bolt stationary with one wrench.
2. Torque the jam nut against the footpad body to 300 N·m (221 ft·lb).
3. Recheck level after jam nut torqueing.

Step 5: Leveling Documentation

Record the following in the installation log:

- Level readings at each footpad location, in X and Y directions
- Date of leveling
- Technician name
- Any special conditions (e.g., shims added)

NOTE: Re-check machine level after the first week of operation (thermal and settling effects), after three months, and then annually. Record all readings in the maintenance log.

4.5 Electrical Connection

DANGER: Electrical connection must be performed by a licensed electrician in accordance with all applicable local electrical codes. Verify the facility voltage is correct (480 VAC, 3-phase) before connecting. Incorrect voltage will damage the machine.

4.5.1 Power Supply Wiring to Main Disconnect

The BC-500X main electrical supply enters through a conduit knockout on the top right of the main electrical cabinet. The main disconnect switch is a 100 A, 3-pole fusible disconnect switch located inside the main cabinet.

Wiring Procedure:

1. Verify the main disconnect is in the OFF (open) position and locked out before beginning wiring work.
2. Open the main electrical cabinet door. The lock cylinder is located in the upper right corner of the door.
3. Locate the main disconnect switch (upper right section of cabinet).
4. Locate the main power terminal block (labeled L1, L2, L3) directly below the main

disconnect switch.

5. Route the supply cables from the facility panel through the conduit knockout (63.5 mm / 2.5 in knockout on top of cabinet).
6. Connect the supply conductors to the main disconnect switch line side terminals:
 - L1: Phase A (connect to left terminal)
 - L2: Phase B (connect to center terminal)
 - L3: Phase C (connect to right terminal)
 - Wire size: 4 AWG copper minimum
 - Terminal torque: 20 N·m (177 in·lb) for L1/L2/L3 lugs
7. Connect the equipment grounding conductor (green or green/yellow) to the main ground bus bar in the cabinet.
 - Ground bus bar is located in the lower left section of the cabinet, clearly labeled "GROUND BUS"
 - Minimum ground conductor size: 8 AWG copper
 - Terminal torque: 6 N·m (53 in·lb) for ground lug
8. Verify all connections are tight. Pull-test each wire by hand.
9. Verify no tools or foreign materials are left inside the cabinet.
10. Close and latch the cabinet door.

Phase Rotation Check:

The BC-500X hydraulic pump motor and coolant pump motor require correct phase rotation. After initial power-up, verify:

1. Turn the main disconnect ON.
2. The system will power up in the "initial power-on" sequence described in Section 4.10.
3. Navigate to SYSTEM → DIAGNOSTIC → PHASE MONITOR on the CNC control.
4. The control will display the measured phase rotation. Correct rotation is ABC (positive sequence).
5. If rotation is incorrect (ACB), the control will display a phase rotation warning. Swap any two of the three supply phase conductors (L1 and L2, or L2 and L3, or L1 and L3) at the main disconnect to correct rotation.

CAUTION: Do not swap conductors at any point other than the main disconnect supply side. The internal motor wiring is factory-set and must not be modified.

4.5.2 Grounding Requirements

Proper grounding is essential for both safety and control system reliability.

Required Grounding Connections:

Connection Point	Conductor Size	Description
Main cabinet ground bus to facility ground electrode	8 AWG min copper	Primary safety ground
Machine frame to main cabinet ground bus	Pre-wired at factory	—
Chip conveyor to machine frame	12 AWG green, factory-supplied	Bond wire
Coolant tank to machine frame	12 AWG green, factory-supplied	Bond wire
HPU to machine frame	12 AWG green, factory-supplied	Bond wire
Control pendant to machine	Integral in flexible cable	—

Ground Resistance Testing: After completing all grounding connections, test the ground resistance from the main cabinet ground bus to the facility ground electrode system. Resistance must be less than 0.1 ohm. Use a four-wire Kelvin resistance test set for accuracy. If resistance exceeds 0.1 ohm, inspect all connections for corrosion or loose terminations.

4.5.3 Control Transformer Tap Selection

The BC-500X control transformer provides 120 VAC from the 480 VAC supply for the control circuit. The primary side of the transformer has multiple taps to accommodate slight voltage variations in the facility supply.

Factory Default Setting: Primary tap is connected at 480 VAC.

Available Primary Taps:

Tap Label	Use When Supply Voltage Is
460 VAC	460–470 VAC measured
470 VAC	470–475 VAC measured
480 VAC	475–485 VAC (factory default)
490 VAC	485–495 VAC measured
500 VAC	495–505 VAC measured

How to Change the Transformer Tap:

1. Lock out the machine (LOTO procedure, Section 2.4.2).
2. Open the main electrical cabinet.
3. Locate the control transformer (lower left section of cabinet, labeled "TX1 CONTROL XFMR").
4. The primary tap leads are on the top of the transformer. The active connection is marked with a yellow insulating sleeve.
5. Move the yellow sleeve (and the wire it secures) to the tap that matches the measured

facility voltage.

6. Use a torque screwdriver to tighten the terminal screw to 1.5 N·m (13 in·lb).
7. Verify the secondary voltage reads 120 VAC $\pm 5\%$ (114–126 VAC) with the transformer energized.

4.6 Hydraulic System Connection

4.6.1 Hydraulic Power Unit Installation

The BC-500X Hydraulic Power Unit (HPU) is mounted on the right rear corner of the machine base. The HPU is pre-connected to the machine hydraulic circuits with high-pressure hoses at the factory. However, if the HPU was shipped separately or disconnected for access, the following procedure applies:

HPU Mounting:

1. Clean the HPU mounting pads on the machine base.
2. Lift the HPU into position using the two lifting lugs on top of the HPU frame. The HPU weighs 185 kg (408 lb); use appropriate lifting equipment.
3. Lower the HPU onto the machine base mounting pads.
4. Install and tighten the 4 $\times$ M16 mounting bolts to 160 N·m (118 ft·lb).

HPU Hydraulic Line Connections:

All HPU hydraulic connections use BSP (British Standard Pipe) threaded fittings with O-ring face seal (ORFS) adapters. These connections require specific torque values:

Connection Size	Torque
G1/4 BSP ORFS	30 N·m (22 ft·lb)
G3/8 BSP ORFS	50 N·m (37 ft·lb)
G1/2 BSP ORFS	70 N·m (52 ft·lb)
G3/4 BSP ORFS	100 N·m (74 ft·lb)
G1 BSP ORFS	140 N·m (103 ft·lb)

CAUTION: Over-torquing ORFS fittings will distort the O-ring groove and cause leaks. Use a calibrated torque wrench. Under-torquing will also cause leaks.

HPU Hydraulic Line Identification:

All hydraulic hoses and lines are labeled with colored bands corresponding to the following circuit identification:

Color Band	Circuit
Red	High-pressure supply (150 bar)

Color Band	Circuit
Blue	Return to tank
Yellow	Z-axis brake circuit
Green	Tool unclamp circuit
Orange	ATC arm circuit
White	Gear change circuit

Match the color-coded labels when reconnecting hoses. Reversing supply and return lines will cause immediate component damage.

4.6.2 Hydraulic Line Connection and Flushing

Before commissioning the hydraulic system, all hydraulic lines must be flushed to remove any contamination introduced during installation.

Hydraulic Flushing Procedure:

1. Fill the HPU reservoir with clean ISO VG 46 hydraulic oil (see Section 11.7 for oil specification). Fill to the "MAX" mark on the sight glass.
2. Install bypass jumpers at each hydraulic actuator (Z-axis brake cylinder, tool unclamp cylinder, ATC arm cylinder). Bypass jumpers connect the supply and return ports of each actuator directly, bypassing the actuator itself. This protects the actuators from any particulate contamination during flushing. Bypass jumper kit: Part No. BC-TOOL-HYD-FLUSH.
3. Connect a portable hydraulic flushing unit (10 µm absolute filter, minimum 20 L/min flow rate) to the hydraulic system supply and return lines, bypassing the machine HPU pump.
4. Operate the flushing unit for 30 minutes minimum.
5. Take an oil sample after 30 minutes and send to a laboratory for particle count analysis. Target cleanliness: ISO 4406 cleanliness code ≤ 16/14/11.
6. If the target cleanliness is not achieved, continue flushing and re-test at 15-minute intervals until the target is reached.
7. Remove the flushing unit and bypass jumpers.
8. Reconnect the actuator lines in their correct configuration.
9. Check the hydraulic reservoir oil level; add oil if required.

NOTE: Hydraulic system cleanliness is critically important for the life of the hydraulic pump, valves, and cylinders. A poorly flushed system will experience premature valve failure and pump wear. Do not skip or shorten the flushing procedure.

4.7 Pneumatic System Connection

Connection Point: The pneumatic supply connects to the machine at the FRL (Filter-Regulator-

Lubricator) assembly mounted on the rear of the machine. The supply connection is a G3/8 (ISO 228-1) female fitting, or a 1/2 NPT female fitting on machines configured for North American piping (specify at order).

Connection Procedure:

1. Verify the pneumatic isolation valve (on the machine FRL assembly) is in the CLOSED position.
2. Install a shutoff valve in the supply line immediately upstream of the machine FRL. This allows the machine air to be isolated without shutting off the facility air.
3. Connect the supply line (minimum 12 mm OD tubing or 1/2 in OD tubing) to the machine FRL inlet.
4. Use PTFE thread tape or thread sealant on NPT connections. Do not use thread sealant on BSP parallel thread (G-thread) connections — use only the O-ring or bonded seal provided.
5. After connection, slowly open the supply valve to pressurize the FRL inlet.
6. Open the machine pneumatic isolation valve slowly.
7. Check the FRL pressure gauge. Adjust the regulator to 5.5 bar (80 psi) by pulling out the regulator knob, turning clockwise to increase pressure, then pushing the knob in to lock.
8. Check all pneumatic connections for leaks using a soap-and-water solution. Tighten any leaking connections.
9. Verify that the FRL bowl is empty (the lubricator section is bypassed; do not add oil to the lubricator bowl).

Pneumatic System Check: After connection, verify the following pneumatic functions work correctly (after initial machine power-up per Section 4.10):

- Spindle air purge activates when spindle is started (air blows from around the spindle nose area)
- ATC arm actuates smoothly during a dry tool change cycle
- Chip blow-off nozzles (if equipped) activate when the coolant system is in the "blow-off" mode

4.8 Coolant System Filling

Coolant Type: The BC-500X is designed for use with water-soluble semi-synthetic or fully synthetic metalworking fluids. Refer to Chapter 12 for a complete guide to coolant selection and management. The following is the initial fill procedure.

Initial Coolant Fill Procedure:

1. Remove the coolant tank access cover from the top of the coolant tank (located in the base of the machine, below the work zone). The access cover is secured with 6 × M6 hex socket cap screws.

2. Inspect the interior of the coolant tank. It should be clean and free of debris from the manufacturing process. Wipe out any metal chips, dust, or rust preventive oil residue with clean rags.
3. Reinstall the access cover and torque the screws to 8 N·m (71 in·lb).
4. Prepare the coolant mixture: For initial fill, prepare a 7–8% concentration solution of BetaCorp approved cutting fluid. See Chapter 12 for approved fluid list and mixing procedure.
 - Typical initial volume: 350 liters (coolant tank capacity)
 - This requires approximately 24.5–28 liters of coolant concentrate added to 322–325 liters of water
 - ALWAYS add concentrate to water (not water to concentrate)
5. Fill the coolant tank through the top fill opening. The fill opening is located on the top of the coolant tank, next to the access cover. A strainer is fitted in the fill opening to catch debris.
6. Add coolant until the level indicator reads "FULL" (level indicator is a graduated sight glass on the right side of the coolant tank).
7. Check the coolant concentration with a refractometer. Reading should be 7–8% (or per coolant manufacturer's table for the specific product).
8. Check the coolant pH using calibrated pH test strips or a digital pH meter. Initial fill should read 8.5–9.5.
9. Record the coolant type, concentration, initial pH, date of initial fill, and technician name in the coolant maintenance log (Appendix I).

Coolant Pump Priming: After filling the tank, prime the coolant pump:

1. Apply machine power (per Section 4.10).
2. From the control panel, turn the coolant pump ON (COOLANT ON/OFF switch on main console).
3. Observe the coolant flow from the nozzles. Allow the pump to run for 2 minutes to prime.
4. Check for any leaks at all coolant line connections, nozzle connections, and pump housing.
5. Adjust the coolant nozzle positions to direct flow into the work zone cutting area.
6. Check the coolant level after priming; add coolant if the level has dropped below the "FULL" mark (the lines and pump volume is approximately 15 liters).

4.9 Chip Conveyor Installation

The chip conveyor is typically shipped separately from the main machine and must be installed on-site.

Chip Conveyor Installation Procedure:

1. Position the chip conveyor on the floor adjacent to the left side of the machine. The chip conveyor discharge end (where chips fall into the chip bin) should point away from the machine and toward the aisle or chip disposal area.
 2. Align the chip conveyor inlet chute with the machine base chip exit port (located on the lower left side of the machine base). The inlet chute must slide into the machine base opening with a minimum 50 mm overlap.
 3. Adjust the chip conveyor leg height so the inlet chute aligns with the machine chip exit port. The legs are adjustable with threaded feet, similar to the machine leveling pads.
 4. Secure the chip conveyor to the machine base using the $2 \times \text{M12}$ mounting bolts (supplied with the conveyor). Tighten to 60 N·m (44 ft·lb).
 5. Connect the chip conveyor motor to the machine electrical system at the conveyor motor connection box (located on the machine base, left side). The connection is a 5-pin circular connector (3-phase + ground + control). Match the color-coded wires per the label on the connector housing.
 6. After initial power-up, verify the chip conveyor runs in the correct direction (chips should move from the inlet toward the discharge end). If the conveyor runs backward, swap any two of the three motor phase leads at the connection box (with machine locked out).
 7. Install the chip bin (collection hopper) under the conveyor discharge chute. The chip bin sits on the floor; no attachment is required. Verify the bin is centered under the discharge.
-

4.10 Initial Power-On and System Checkout

WARNING: Before applying power to the BC-500X for the first time, ensure all installation steps have been completed, all transport securing hardware has been removed, all fluid levels are correct, and all personnel are clear of all hazardous areas.

Initial Power-On Procedure:

Step 1: Pre-Power Checks

Complete the following checklist before applying power:

- ☐ All transport securing hardware removed (Section 4.3.2)
- ☐ Machine leveled and anchor bolts torqued
- ☐ Electrical supply connected, correct phase and voltage verified
- ☐ All electrical cabinet doors closed
- ☐ Hydraulic oil filled to "MAX" level (before starting HPU motor, the oil level will drop as circuits fill)
- ☐ Coolant tank filled to "FULL" level
- ☐ Pneumatic supply connected and isolated valve closed
- ☐ Chip conveyor connected (motor connector inserted)

- ☐ All guards and covers installed
- ☐ All personnel clear of machine

Step 2: Apply Main Power

1. Ensure the main electrical cabinet main disconnect is in the OFF position.
2. Turn the main disconnect to ON.
3. The CNC control will power up and begin its initialization sequence. The control display will show the BetaCorp Systems logo, then the system initialization screen.
4. The initialization takes approximately 45 seconds.
5. After initialization, the control will display the main status screen. Expected messages:
 - "HOMING REQUIRED — PRESS HOME TO CONTINUE" (normal after first power-up)
 - Fault E404 (Hydraulic Pressure Low) may be active — expected, as hydraulic pump has not started yet

Step 3: Start Hydraulic System

1. Navigate to SYSTEM → MACHINE SETUP → HYDRAULIC SYSTEM.
2. Press "HPU START" soft key.
3. The hydraulic pump motor will start. The HPU will pressurize to the operating pressure (150 bar) within approximately 10–15 seconds.
4. Monitor the hydraulic pressure gauge on the HPU. It should rise to 150 bar and stabilize.
5. Fault E404 should clear automatically when pressure reaches the low-pressure setpoint (120 bar).
6. Check all hydraulic connections for leaks while the system is pressurized. Address any leaks immediately (LOTO, de-pressurize, tighten/replace fitting).

Step 4: Open Pneumatic Supply

1. Slowly open the pneumatic isolation valve on the machine FRL assembly.
2. Verify the FRL pressure gauge reads 5.5 bar. Adjust regulator if necessary.
3. Verify the spindle air purge activates (you will hear a soft hiss from the spindle area).

Step 5: Check Control Panel Functions

Test the following control panel functions:

- Each axis JOG button: press and verify the correct axis moves in the correct direction at a low jog rate
- E-Stop buttons: press each E-Stop and verify the machine halts and displays "E-STOP ACTIVE"
- Reset: after each E-Stop test, release the button, press RESET, and verify the machine returns to ready state

- Coolant pump: activate and verify coolant flow
- Chip conveyor: activate and verify conveyor runs in correct direction
- Spindle: command a low speed (500 RPM) and verify the spindle starts, reaches commanded speed, and the RPM display matches

Step 6: Homing

After all system checks pass, perform the machine homing procedure per Section 6.4. All axes must be homed before automatic operation.

4.11 Commissioning Procedure

Commissioning is performed by a BetaCorp Systems-trained service engineer and includes the following activities. This section describes the scope and purpose of commissioning tasks.

4.11.1 Control Parameters Verification

The BetaCorp service engineer will verify that all machine parameters are set to the correct values for the specific machine serial number. Machine-specific parameters (servo tuning, ballscrew pitch compensation, geometric compensation data) are stored on a USB drive shipped with the machine and loaded into the control during commissioning.

Key parameter groups verified during commissioning:

- B1.xx Spindle parameters (speed limits, acceleration, encoder ratio)
- B2.xx Axis motion parameters (position loop gain, velocity loop gain, ballscrew pitch)
- B3.xx Tool changer parameters (carousel home position, arm timing)
- B4.xx Hydraulic parameters (pressure setpoints, timer values)
- B6.xx Encoder parameters (encoder type, PPR values, quadrature enable)

4.11.2 Spindle Run-In

The spindle bearings require a run-in procedure after initial installation and after any spindle bearing replacement. This procedure gradually loads the bearings to seat the grease properly and establish correct preload.

Spindle Run-In Schedule:

Step	Speed (RPM)	Duration
1	500	10 minutes
2	1,000	10 minutes
3	2,000	10 minutes
4	3,000	15 minutes
5	4,000	15 minutes
6	5,000	20 minutes

Step	Speed (RPM)	Duration
7	6,000	20 minutes
8	7,000	20 minutes
9	8,000	20 minutes
10	10,000	30 minutes

During the run-in, monitor spindle bearing temperature using an infrared thermometer at the spindle head area. Temperature must not exceed 65°C above ambient. If temperature exceeds this limit, stop the spindle, allow cooling, and repeat the step at reduced speed.

4.11.3 Servo System Tuning Verification

The servo drive systems are factory-tuned, but must be verified after installation as the machine dynamics can change slightly due to mounting conditions.

The BetaCorp service engineer will perform:

- Auto-tuning of position loop and velocity loop for each axis
- Verification of following error limits (Parameter B2.05 for X, B2.15 for Y, B2.25 for Z)
- Step response testing on each axis
- Verification of reversal backlash compensation (Parameter B2.06, B2.16, B2.26)

4.11.4 Tool Changer Commissioning

The ATC system requires calibration during commissioning:

- Verification of carousel home sensor position
- Teaching of tool pot positions (Parameters B3.01–B3.24)
- ATC arm home position verification
- Tool change cycle testing with test toolholders in all 24 positions
- Verification of tool presence detection sensors in all pockets

4.12 Geometry and Accuracy Verification

After commissioning, the machine geometry and accuracy must be verified against the BetaCorp BC-500X factory acceptance criteria.

Tests Performed:

Test	Tool Used	Acceptance Criterion
Spindle nose radial run-out	Test indicator + precision reference	Less than 3 μm
Spindle taper run-out (25 mm)	Test indicator in spindle bore	Less than 5 μm

Test	Tool Used	Acceptance Criterion
from nose)		
Z-axis straightness (over 300 mm travel)	Precision square + indicator	Less than 8 μm
X-axis straightness (over 600 mm)	Precision square + indicator	Less than 10 μm
X-axis/Z-axis perpendicularity	Precision square + indicator	Less than 10 $\mu\text{m}/300\text{ mm}$
Y-axis/Z-axis perpendicularity	Precision square + indicator	Less than 10 $\mu\text{m}/300\text{ mm}$
X-axis positioning accuracy (bi-directional)	Laser interferometer (Renishaw ML10 or equiv.)	$\pm 0.005\text{ mm}$
X-axis repeatability	Laser interferometer	$\pm 0.003\text{ mm}$
Y-axis positioning accuracy	Laser interferometer	$\pm 0.005\text{ mm}$
Y-axis repeatability	Laser interferometer	$\pm 0.003\text{ mm}$
Z-axis positioning accuracy	Laser interferometer	$\pm 0.005\text{ mm}$
Z-axis repeatability	Laser interferometer	$\pm 0.003\text{ mm}$
Table flatness	Precision level + reference	Less than 15 μm over full table
Spindle axis perpendicularity to table (X direction)	Test bar + indicator	Less than 10 $\mu\text{m}/300\text{ mm}$
Spindle axis perpendicularity to table (Y direction)	Test bar + indicator	Less than 10 $\mu\text{m}/300\text{ mm}$

All geometry and accuracy test results are recorded on the BetaCorp BC-500X Acceptance Test Certificate (ATC), which is signed by the BetaCorp service engineer and the customer's representative and retained with the machine records.

4.13 Final Acceptance Testing

After geometry verification, the following production tests are performed:

Test Piece Machining: A standard BetaCorp acceptance test piece is machined in aluminum using the BetaCorp test program (Part No. BCS-TESTPGM-BC500X). The test piece evaluates:

- Circular interpolation (G02/G03) — checks for servo mismatch and following error
- Linear positioning — checks X and Y axis accuracy
- Spindle speed stability — verifies speed regulation under load
- Tool change repeatability — tool change is performed 10 times, repositioning accuracy checked each time

Acceptance Criteria for Test Piece:

- Circularity of circular features: $\leq 0.010\text{ mm}$
- Linear dimensional accuracy: $\pm 0.008\text{ mm}$

- Tool change repositioning accuracy (after 10 changes): ≤ 0.005 mm

System Function Verification:

- All fault codes can be intentionally triggered and reset (E-Stop, door interlock, coolant low, etc.)
- All override functions work correctly (feed rate override, spindle speed override, rapid override)
- Data communication (Ethernet, USB) functions correctly
- All optional accessories are tested

Customer Sign-Off: Upon satisfactory completion of all acceptance tests, the customer's authorized representative signs the BC-500X Installation Acceptance Certificate. This certificate marks the start of the 24-month warranty period.

End of Chapter 4 — Installation Guide

PART 4 — MACHINE COMPONENTS

Chapter 5 — Machine Components

5.1 Machine Base and Column

Base (Bed) Assembly:

The BC-500X base is a one-piece casting manufactured from close-grained Meehanite pearlitic gray cast iron (Grade FC-300, minimum tensile strength 300 MPa). The base serves as the primary structural element of the machine and provides mounting surfaces for the X and Y axis guideways, the hydraulic power unit, coolant tank, and chip conveyor.

Cast iron construction provides inherent vibration damping characteristics superior to welded steel construction. The BC-500X base has been designed using finite element analysis (FEA) to achieve a first natural frequency above 80 Hz, well above the frequency range of typical machining excitation forces.

Base Specification	Value
Material	FC-300 Meehanite gray cast iron
Brinell hardness (surface)	180–230 HB
Surface flatness (guideways, ground)	Less than 3 μm /300 mm
Guideway hardness (induction hardened)	52–58 HRC
Weight	3,200 kg
Base dimensions (external)	3,100 $\times$ 2,300 $\times$ 700 mm

Column Assembly:

The column is also a one-piece Meehanite gray iron casting, mounted on the rear section of the base and integral to the base structure (the column and base are cast as a single unit on the BC-500X). This monoblock construction eliminates the potential for joint flex that exists in machines with bolted-on columns.

The column provides the vertical guideways (Z-axis) for the spindle head assembly. The column is heavily ribbed internally for maximum stiffness with minimum weight.

Column Specification	Value
Column type	Monoblock (integral with base)
Z-axis guideway span	350 mm (center-to-center)
Z-axis guideway length	750 mm (per rail)
Guideway surface hardness	52–58 HRC
Column height (from base top)	1,500 mm

5.2 Spindle Assembly

The BC-500X spindle assembly is a pre-loaded, cartridge-type spindle unit that is inserted into the spindle head housing and secured with four M16 mounting bolts. The cartridge design simplifies spindle replacement in the event of bearing failure or other spindle damage — the entire cartridge can be replaced without re-grinding the spindle housing bore.

5.2.1 Spindle Motor

The spindle is driven by an AC induction motor coupled to a two-speed gearbox. The gearbox output shaft connects to the spindle through a flexible bellows coupling.

Spindle Motor Specifications:

Parameter	Value
Motor type	AC induction (squirrel cage)
Rated power (continuous)	15 kW
Rated power (30-minute rating)	18.5 kW
Rated voltage	480 VAC, 3-phase
Rated current (continuous)	22 A
Rated current (30-min)	27 A
Speed range	0–3,000 RPM (motor output)
Motor speed at 10,000 RPM spindle	3,333 RPM (gear 2 ratio 3.0:1)
Encoder	Integral encoder on motor shaft
Cooling	Integral fan (motor-mounted)

Parameter	Value
Motor frame	IEC 112M
Insulation class	Class F
Protection	IP54
Part No. (motor assembly)	BC-MTR-SP-001

Gearbox:

The two-speed gearbox provides two speed ranges for the spindle:

- Gear 1 (low range): Gear ratio 5:1; motor speed 250–1,000 RPM → spindle 50–200 RPM (intended for heavy cuts, boring)
- Gear 2 (high range): Gear ratio 1.5:1; motor speed 333–6,667 RPM → spindle 50–10,000 RPM (full speed range, normal machining)

Wait — correcting: the gear ratios are configured to give the following spindle speeds:

Gear	Ratio	Motor RPM Range	Spindle RPM Range
Gear 1 (Low)	5.0:1	250–3,000 RPM	50–600 RPM
Gear 2 (High)	3.0:1	167–3,000 RPM (vfd managed)	50–10,000 RPM*

*10,000 RPM spindle requires motor at full speed in high gear with the VFD providing extended speed range.

Gear change is pneumatically initiated and hydraulically engaged. The gear shift sequence is controlled by the CNC through the PLC. Gear shift time: approximately 0.8 seconds. The machine automatically selects the appropriate gear based on the commanded spindle speed and parameter B1.10 (Gear Change Threshold RPM; default = 600 RPM).

Part number, gearbox assembly: BC-GBX-SP-001

5.2.2 Spindle Bearings

The BC-500X spindle uses angular contact ball bearings arranged in a system that provides both radial and axial load capacity appropriate for milling and drilling operations.

Bearing Arrangement:

The spindle cartridge contains the following bearings:

Position	Bearing Type	Quantity	Part No.
Front (tool end)	Angular contact ball, 15° contact angle, matched set	2 (face-to-face)	BC-BRG-SP-001
Front	Angular contact ball, 15° contact angle	1 (tandem with above)	BC-BRG-SP-002

Position	Bearing Type	Quantity	Part No.
Rear	Deep groove ball bearing (radial constraint only)	1	BC-BRG-SP-003
Rear (floating)	Angular contact, 15°, single	1	BC-BRG-SP-004

The front bearings (three total — one tandem pair plus one) are preloaded and provide the primary tool-end rigidity. The rear bearing is a floating arrangement to accommodate thermal expansion of the spindle shaft.

Bearing Lubrication: The spindle bearings use oil-air mist lubrication. Pressurized air carries a precisely metered quantity of oil to each bearing through dedicated lubrication ports. The oil-air lubrication unit (Part No. BC-LUB-OA-001) is mounted inside the machine column.

Lubrication System Parameters:

Parameter	Value
Lubrication oil type	ISO VG 32 spindle oil (Mobil Velocite Oil No. 6 or equivalent)
Reservoir capacity	2 liters
Oil delivery rate (per bearing, per cycle)	0.008 mL
Lubrication cycle interval	Every 10 minutes of operation (see Parameter B1.45)
Lubrication air pressure	3.0 bar
Low oil level alarm	Activates when reservoir below 20% capacity

WARNING: Never operate the spindle at speeds above 3,000 RPM without oil-air lubrication active. The lubrication system activates automatically when the machine is powered on and hydraulics are active. If lubrication system fault E088 (Oil-Air Lube Pressure Low) is active, DO NOT start the spindle above 3,000 RPM.

5.2.3 Spindle Encoder

The spindle encoder is a Heidenhain ROD 420 optical incremental encoder, mounted on the non-drive (rear) end of the spindle motor shaft extension. The encoder provides precise speed and position feedback to the spindle drive for closed-loop speed control, rigid tapping synchronization, and spindle orientation for tool changes.

Encoder Mounting Details:

The encoder is mounted in a protected housing bolted to the rear of the spindle motor. It connects to the motor shaft through a precision bellows coupling (Part No. BC-CUP-SP-001) to isolate the encoder from shock loads.

Encoder Signal Routing:

The encoder cable exits the motor housing through a strain-relief fitting and routes through the

cable carrier (energy chain) on the Z-axis to the servo/spindle drive cabinet. The cable is a shielded, twisted-pair cable (Part No. BC-CBL-SPENC-3M, length 3 meters). The shield must be connected at the drive end only (floating at the encoder end) to prevent ground loops.

Encoder Diagnostic Access:

The spindle encoder status can be monitored in real-time at: SYSTEM → DIAGNOSTIC → ENCODER STATUS → SPINDLE

The display shows:

- Current count (absolute position in counts within one revolution)
- Current speed (RPM, calculated from encoder frequency)
- Signal quality (OK / WARNING / FAULT)
- Error count (number of encoder errors since last power cycle)

Fault E101 — Spindle Encoder Signal Lost:

If the spindle encoder signal is lost at any time during operation, the CNC immediately asserts Fault E101 (Spindle Encoder Signal Lost). This is categorically different from a simple alarm — E101 causes immediate spindle disable and halt of all machining operations. The full E101 troubleshooting procedure is in Section 8.5.1.

5.2.4 Tool Retention (Draw Bar) System

The tool retention system holds the tool holder securely in the spindle taper during machining and releases it for automatic tool changes.

Draw Bar Design:

The BC-500X draw bar uses a stack of Belleville (disc) washers that apply a clamping force of 18,000 N (4,047 lb) to the tool holder. The draw bar is a hardened steel rod that passes through the center of the spindle shaft. The lower end of the draw bar engages the collet fingers, which grip the tool holder retention knob. The upper end of the draw bar is pushed by the Belleville washer stack.

Tool Clamp/Unclamp:

- **Clamped state (normal):** Belleville springs are compressed by the collet assembly, pulling up on the draw bar, which pulls the collet fingers closed onto the retention knob. Tool holder is firmly seated in the spindle taper.
- **Unclamped state (during tool change):** Hydraulic pressure is applied to the unclamp cylinder (at top of spindle head), which pushes DOWN on the draw bar, overcoming the Belleville spring force. The collet fingers open, releasing the retention knob. An air blast simultaneously blows chips from the spindle taper.

Tool Seating Verification:

A proximity sensor (Part No. BC-SEN-SP-SEAT, Inductive, NPN, normally-open) detects whether the tool holder is fully seated in the spindle taper. This sensor prevents spindle start if a tool holder is not fully seated.

Draw bar assembly part numbers:

Component	Part No.
Draw bar (complete assembly)	BC-DRB-SP-001
Belleville washer stack (10 washers)	BC-BELL-SP-001
Collet assembly (CAT-40)	BC-COL-SP-CAT40
Collet assembly (BT-40)	BC-COL-SP-BT40
Unclamp cylinder seal kit	BC-SEAL-SP-UC-001
Tool seating sensor	BC-SEN-SP-SEAT
Spindle air purge orifice plate	BC-OR-SP-001

5.3 Axis Servo Motors and Drives

5.3.1 X-Axis Assembly

The X-axis drives the machine table in the left-right direction (when viewed from the front of the machine). The X-axis assembly is located on top of the machine base.

X-Axis Components:

Component	Specification	Part No.
Servo motor	BetaCorp SM-300-3K, 3 kW, 3,000 RPM	BC-MTR-X-001
Servo drive	BetaCorp SD-300, 3 kW	BC-DRV-X-001
Ballscrew	Ø40 × pitch 10 mm, Class 3, rolled	BC-BSC-X-001
Ballscrew nut	Double-nut, preloaded	BC-NUT-X-001
Fixed end support	Angular contact bearing, matched pair	BC-BRG-X-FIX
Floating end support	Deep groove ball bearing	BC-BRG-X-FLT
Linear guideway (left)	IKO LRWE55, roller type	BC-GWY-X-L
Linear guideway (right)	IKO LRWE55, roller type	BC-GWY-X-R
Guideway carriages (per rail)	2 × IKO LWRE55	BC-CAR-X-001
Encoder	Heidenhain ERN 1381, 2500 PPR	BC-ENC-X-001
Encoder cable	5 m, shielded	BC-CBL-ENC-5M
Motor coupling	Bellows type, backlash-free	BC-CUP-X-001
Scale cover (telescoping)	3-section steel telescoping	BC-COV-X-001

X-Axis Travel Limits:

Limit	Position	Detection Method
Software negative limit	X = -1.0 mm	CNC parameter B2.03
Hardware negative limit	X = -5.0 mm	Limit switch LS-X-NEG
Software positive limit	X = 750.0 mm	CNC parameter B2.04
Hardware positive limit	X = 755.0 mm	Limit switch LS-X-POS
Mechanical hard stop (negative)	X = -8.0 mm	Adjustable stop block
Mechanical hard stop (positive)	X = 758.0 mm	Adjustable stop block

5.3.2 Y-Axis Assembly

The Y-axis drives the machine table in the front-back direction. The Y-axis assembly is located under the X-axis assembly (the Y-axis saddle carries the X-axis).

Y-Axis Components:

Component	Specification	Part No.
Servo motor	BetaCorp SM-300-3K, 3 kW, 3,000 RPM	BC-MTR-Y-001
Servo drive	BetaCorp SD-300, 3 kW	BC-DRV-Y-001
Ballscrew	Ø40 × pitch 10 mm, Class 3, rolled	BC-BSC-Y-001
Ballscrew nut	Double-nut, preloaded	BC-NUT-Y-001
Fixed end support	Angular contact bearing pair	BC-BRG-Y-FIX
Floating end support	Deep groove ball bearing	BC-BRG-Y-FLT
Linear guideway (front)	IKO LRWE55, roller type	BC-GWY-Y-F
Linear guideway (rear)	IKO LRWE55, roller type	BC-GWY-Y-R
Guideway carriages (per rail)	2 × IKO LWRE55	BC-CAR-Y-001
Encoder	Heidenhain ERN 1381, 2500 PPR	BC-ENC-Y-001
Encoder cable	5 m, shielded	BC-CBL-ENC-5M
Motor coupling	Bellows type, backlash-free	BC-CUP-Y-001
Scale cover (accordion/bellows)	Stainless steel bellows	BC-COV-Y-001

5.3.3 Z-Axis Assembly

The Z-axis moves the spindle head up and down. It is the most heavily loaded axis due to the weight of the spindle head assembly (approximately 280 kg).

Z-Axis Components:

Component	Specification	Part No.
Servo motor	BetaCorp SM-400-3K, 4 kW, 3,000 RPM	BC-MTR-Z-001
Servo drive	BetaCorp SD-400, 4 kW	BC-DRV-Z-001
Ballscrew	Ø50 × pitch 10 mm, Class 3, ground	BC-BSC-Z-001
Ballscrew nut	Double-nut, preloaded	BC-NUT-Z-001
Fixed end support (upper)	Angular contact bearing pair	BC-BRG-Z-FIX
Floating end support (lower)	Angular contact, single	BC-BRG-Z-FLT
Linear guideway (left)	IKO LRWE65, roller type	BC-GWY-Z-L
Linear guideway (right)	IKO LRWE65, roller type	BC-GWY-Z-R
Guideway carriages (per rail)	2 × IKO LWRE65	BC-CAR-Z-001
Encoder	Heidenhain ERN 1381, 2500 PPR	BC-ENC-Z-001
Encoder cable	5 m, shielded	BC-CBL-ENC-5M
Motor coupling	Bellows type, backlash-free	BC-CUP-Z-001
Hydraulic brake cylinder	Ø100 mm, spring-apply/hydraulic-release	BC-CYL-Z-BRK
Brake pressure switch	80–120 bar setpoint	BC-PSW-Z-BRK
Z-axis covers	Accordion stainless bellows	BC-COV-Z-001

Z-Axis Counterbalance:

The Z-axis gravity load (280 kg spindle head) is partially counterbalanced by the hydraulic brake circuit. The hydraulic brake provides both a holding force and a counterbalance function. This reduces the Z-axis servo motor's continuous holding torque requirement, extending servo motor life.

The Z-axis servo motor must supply additional torque for upward motion (against gravity) and can partially regenerate during downward motion. The servo drive's regenerative braking capability allows the kinetic and potential energy to be returned to the drive DC bus and dissipated in the regenerative resistor (mounted on top of the servo drive cabinet, external).

5.3.4 Ballscrew and Nut Assemblies

All three axes use recirculating ball type ballscrews. The BC-500X uses preloaded double-nut ballscrews (NSK or THK, Class 3) to eliminate backlash and provide high stiffness.

Ballscrew Specifications:

Parameter	X-Axis	Y-Axis	Z-Axis
Nominal diameter	40 mm	40 mm	50 mm
Lead (pitch)	10 mm/rev	10 mm/rev	10 mm/rev
Accuracy class	ISO Class 3	ISO Class 3	ISO Class 3
Thread form	Gothic arch	Gothic arch	Gothic arch

Parameter	X-Axis	Y-Axis	Z-Axis
Ball diameter	6.35 mm	6.35 mm	8.00 mm
Axial play (assembled)	Less than 2 μ m	Less than 2 μ m	Less than 2 μ m
Dynamic load capacity	38 kN	38 kN	65 kN
Nut type	Double-nut, spring preload	Double-nut	Double-nut, offset preload
Preload	8% of dynamic load	8% of dynamic load	6% of dynamic load

Ballscrew Lubrication: The ballscrews are lubricated by the centralized way lube system (see Section 5.8). Lube is distributed to the ballscrew nut through a nipple fitting on the nut body. The lube interval is controlled by Parameter B9.22 (Way Lube Interval).

Ballscrew Maintenance: See Chapter 9, Section 9.13 for ballscrew maintenance procedures.

5.3.5 Linear Guideways

The BC-500X uses IKO (or THK equivalent) linear roller guideways on all three axes. Roller guideways provide higher load capacity, higher rigidity, and better damping than ball guideways at the same overall dimensions.

Guideway Specifications:

Parameter	X-Axis	Y-Axis	Z-Axis
Model series	IKO LRWE55	IKO LRWE55	IKO LRWE65
Rail width	55 mm	55 mm	65 mm
Carriage width	100 mm	100 mm	120 mm
Dynamic load capacity (per carriage)	85 kN	85 kN	125 kN
Static load capacity (per carriage)	120 kN	120 kN	175 kN
Roller diameter	8 mm	8 mm	10 mm
Rolling element material	Bearing steel (SUJ2)	Bearing steel	Bearing steel
Rail hardness	58–64 HRC	58–64 HRC	58–64 HRC
Rail material	Bearing steel, induction hardened	Bearing steel	Bearing steel
Preload class	C1 (light preload)	C1	C2 (medium preload)
Lubrication type	Way lube oil (ISO VG 68)	Way lube oil	Way lube oil
Wiper type	Double lip seal + contact	Double lip	Double lip seal + contact

5.4 Tool Changer Carousel (24-Position)

The BC-500X tool changer is a 24-position, random-access, dual-arm automatic tool changer (ATC). The system consists of:

1. **Tool magazine carousel** — holds 24 toolholders in individual spring-grip tool pots
2. **ATC arm assembly** — performs simultaneous tool exchange between the carousel and spindle

The ATC reduces chip-to-chip time to 4.2 seconds (typical), significantly improving productivity compared to single-arm designs.

5.4.1 Carousel Drive Motor and Gear Assembly

The carousel is driven by a BetaCorp servo motor (0.75 kW, 3,000 RPM) through a precision worm gear reducer. The servo drive provides accurate position control of the carousel, allowing any of the 24 tool positions to be accessed in either direction (the carousel rotates in the shortest path to the required tool position).

Carousel Drive Specifications:

Parameter	Value
Drive motor	BetaCorp SM-075-3K, 0.75 kW
Drive model	BC-MTR-ATC-001
Gear reducer type	Worm gear, helical input stage
Gear ratio	60:1
Carousel rotation speed	50 RPM (maximum)
Carousel rotation speed (indexing)	30 RPM (normal)
Indexing time (24 positions, longest path)	24 seconds (full carousel rotation)
Indexing time (average)	1.8 seconds
Position encoder	Incremental, 1,000 PPR (on motor shaft)
Carousel home sensor	Inductive proximity, NPN, NC
Position sensors	24 × inductive proximity (tool presence detection)

Carousel Home Position: The carousel home position is defined by a precision index pin that engages a notch in the carousel disc. The home sensor (Part No. BC-SEN-ATC-HOME) detects when the carousel is in the home position. This sensor is the absolute reference for all carousel positioning. If the carousel loses its position reference (e.g., due to encoder fault), it must be re-homed before tool changes can be performed.

5.4.2 Tool Pot and Gripper Assembly

Each of the 24 tool pots is an independent spring-grip unit mounted in the carousel disc. The tool pots accept CAT-40 or BT-40 toolholders (depending on machine configuration).

Tool Pot Design:

- The tool pot body is aluminum alloy, anodized black.
- Three hardened steel gripper fingers are pivoted in the pot body.
- A spring (wave spring type) behind the gripper fingers applies closing force.
- The gripper fingers lock around the V-groove of the toolholder (CAT V-flange or BT V-flange).
- Retention force: 85 N per pot (sufficient to retain tool during carousel rotation).

Tool Pot Inspection Points: During weekly maintenance, inspect each tool pot for:

- Damage to gripper finger tips (worn, chipped, or broken)
- Loss of spring tension (tool does not stay securely in pot)
- Misalignment (tool sits crooked in pot)
- Accumulated chips inside pot (can prevent full tool seating)

Tool Pot Part Numbers:

Component	CAT-40 Config.	BT-40 Config.
Complete tool pot assembly	BC-POT-CAT40	BC-POT-BT40
Gripper finger set (3 pcs)	BC-GRP-CAT40-SET	BC-GRP-BT40-SET
Wave spring	BC-SPR-POT-001	BC-SPR-POT-001
Pot body only	BC-POT-BODY-001	BC-POT-BODY-001
Pot retention bolt	BC-POT-BOLT-M8	BC-POT-BOLT-M8

5.4.3 ATC Arm Assembly

The ATC arm is a double-arm design that simultaneously grips the tool in the spindle and the tool in the carousel, then performs a 180° rotation to swap the tools, and finally inserts the new tool into the spindle and the old tool into the carousel.

This simultaneous exchange is what enables the 4.2-second chip-to-chip time.

ATC Arm Drive: The ATC arm motion is cam-driven, with the following motions controlled by a single rotating cam shaft:

1. Arm extension (arm moves forward, toward the spindle axis and carousel axis simultaneously)
2. Arm rotation (180° rotation to swap tools)
3. Arm retraction (arm moves back, releasing tools into spindle and carousel)

The cam shaft is hydraulically driven (hydraulic motor, 0.75 kW equivalent, gear-reduced). This provides powerful, consistent force for tool extraction and insertion regardless of tool weight (up to the 8 kg limit).

ATC Arm Part Numbers:

Component	Part No.
ATC arm complete assembly	BC-ATC-ARM-001
ATC cam shaft assembly	BC-ATC-CAM-001
ATC hydraulic motor	BC-ATC-HYD-MTR
ATC arm gripper (left)	BC-ATC-GRP-L
ATC arm gripper (right)	BC-ATC-GRP-R
ATC arm gripper spring	BC-ATC-GRP-SPR
ATC arm position cam follower	BC-ATC-CFW-001

5.4.4 ATC Position Sensors

The ATC system uses multiple sensors to verify the position and status of the arm and carousel at each step of the tool change sequence. These sensors are inductive proximity switches (NPN, 24 VDC) unless otherwise noted.

ATC Sensors:

Sensor ID	Description	Part No.	Location
SEN-ATC-HOME	ATC arm home position	BC-SEN-ATC-HOME	ATC arm housing, rear
SEN-ATC-90	ATC arm at 90° position (mid-swap)	BC-SEN-ATC-090	ATC arm housing
SEN-ATC-180	ATC arm at 180° position (swap complete)	BC-SEN-ATC-180	ATC arm housing
SEN-ATC-EXT	ATC arm extended position	BC-SEN-ATC-EXT	ATC arm housing
SEN-ATC-RET	ATC arm retracted position	BC-SEN-ATC-RET	ATC arm housing
SEN-CAR-HOME	Carousel home position	BC-SEN-CAR-HOME	Carousel disc, left side
SEN-TP-01 through 24	Tool presence sensor (24 total)	BC-SEN-TP-001	Each tool pot position

5.5 Hydraulic Power Unit

The BC-500X Hydraulic Power Unit (HPU) is a self-contained assembly mounted on the right rear of the machine base. All hydraulic system components are mounted on or in the HPU frame.

5.5.1 Hydraulic Pump

The hydraulic pump is a fixed-displacement external gear pump driven directly by the hydraulic pump motor (4 kW, 1,450 RPM).

Hydraulic Pump Specifications:

Parameter	Value
Pump type	External gear pump
Displacement	8.3 cc/rev
Flow rate at 1,450 RPM	11.8 L/min (nominal)
Maximum working pressure	250 bar
Rated pressure	210 bar (relief valve setting)
Operating pressure	150 bar
Inlet fitting	G3/4 BSP male
Outlet fitting	G1/2 BSP male
Rotation direction	Clockwise (viewed from shaft end)
Shaft seal type	Lip seal, nitrile
Part No.	BC-PMP-HYD-001
Manufacturer	Parker Hannifin (PGP505) or equivalent

CAUTION: The hydraulic pump rotation direction is critical. Verify phase rotation before starting the hydraulic pump motor. Running the pump in reverse will not build pressure and will rapidly damage the pump internal gears. Verify pump shaft rotation direction with the coupling guard removed before reinstalling the guard.

5.5.2 Hydraulic Valves

The BC-500X hydraulic system uses eight (8) solenoid-operated directional control valves, three (3) pressure control valves, two (2) check valves, and one (1) pressure relief valve.

Main System Pressure Relief Valve:

- Type: Direct-acting, pilot-operated
- Setting: 210 bar (factory set, do not adjust without BetaCorp authorization)
- Part No.: BC-RLV-HYD-MAIN

Solenoid Valves:

Valve ID	Function	Type	Voltage	Part No.
SOL-HYD-01	Z-axis brake release	4/2 spring return	24 VDC	BC-SOL-ZBK-001
SOL-HYD-02	Tool unclamp (pilot)	4/2 spring return	24 VDC	BC-SOL-TU-001
SOL-HYD-03	ATC arm extend	4/3 spring center	24 VDC	BC-SOL-ATC-EXT
SOL-HYD-04	ATC arm retract	4/3 spring center	24 VDC	BC-SOL-ATC-RET
SOL-HYD-05	ATC arm rotate	4/2 spring return	24 VDC	BC-SOL-ATC-

Valve ID	Function	Type	Voltage	Part No.
	CW			CW
SOL-HYD-06	ATC arm rotate CCW	4/2 spring return	24 VDC	BC-SOL-ATC- CCW
SOL-HYD-07	Gear change (low gear)	4/2 spring return	24 VDC	BC-SOL-GCH- LO
SOL-HYD-08	Gear change (high gear)	4/2 spring return	24 VDC	BC-SOL-GCH-HI

Pressure Reducing Valves:

Valve ID	Circuit	Setting	Part No.
PRV-HYD-01	Gear change circuit	80 bar	BC-PRV-GCH-001
PRV-HYD-02	Tool unclamp (main)	150 bar	BC-PRV-TU-001
PRV-HYD-03	Workholding (optional)	0–150 bar (adj.)	BC-PRV-WH-001

5.5.3 Hydraulic Cylinders

Cylinder	Function	Bore	Stroke	Part No.
CYL-Z-BRK	Z-axis brake release	Ø100 mm	30 mm	BC-CYL-Z-BRK
CYL-TU-001	Spindle tool unclamp	Ø80 mm	45 mm	BC-CYL-TU-001
CYL-ATC-EXT	ATC arm extend/retract	Ø63 mm	150 mm	BC-CYL-ATC- EXT
CYL-GCH-001	Gear change actuator	Ø50 mm	50 mm	BC-CYL-GCH- 001

5.6 Coolant System

5.6.1 Coolant Tank and Pump

Coolant Tank: The coolant tank is a welded stainless steel (grade 316L) tank located in the machine base below the work zone. All chips and coolant from the work zone drain into the tank through the chip conveyor (which removes the bulk of the chips) and the chip pan drains.

Parameter	Value
Tank material	316L stainless steel, 3 mm plate
Tank capacity	350 liters
Tank dimensions	1,400 × 600 × 450 mm (L × W × H)
Drain valve	G3/4 BSP ball valve

Parameter	Value
Fill opening	100 mm diameter with stainless screen
Level indicator	Float-type sight gauge (right side of tank)
Level sensor	Float switch, 4-wire, 24 VDC
Temperature sensor	RTD Pt100, 4-wire
Magnetic separator	Permanent magnet bar type, in tank
Part No.	BC-TNK-CLT-001

Coolant Pump:

Parameter	Value
Pump type	Centrifugal
Motor power	2.2 kW
Motor speed	2,900 RPM
Flow rate (full open)	80 L/min
Flow rate (at nozzle pressure)	60 L/min at 3–5 bar
Maximum head	40 m
Impeller material	Stainless steel 316L
Shaft seal	Mechanical seal, silicon carbide
Part No.	BC-PMP-CLT-001

5.6.2 Coolant Filters

The coolant is filtered through a combination of:

1. **Magnetic chip separator:** Permanent magnet bars installed in the coolant tank remove ferromagnetic chips before they reach the pump inlet.
2. **Inlet strainer:** A 150 μm stainless steel wedge-wire screen at the pump inlet catches non-magnetic chips.
3. **Paper band filter (optional, Part No. BC-OPT-PAPFLT):** A roll of paper filter media intercepts coolant before the return line, providing finer filtration for demanding applications.

Strainer Cleaning: The inlet strainer must be inspected and cleaned weekly. See Chapter 9, Section 9.3.

5.6.3 Coolant Nozzles and Distribution

The BC-500X is equipped with four adjustable coolant nozzles as standard equipment. Each nozzle has a flexible gooseneck mount that allows manual positioning. Nozzle orifice diameter is 6 mm.

Coolant Distribution Manifold: A stainless steel coolant distribution manifold runs along the top of the work zone interior. The four nozzles connect to this manifold through flexible hose (Part No.

BC-HOSE-CLT-1M, 1 m length, stainless fittings). A ball valve at each nozzle outlet allows individual nozzle shutoff.

Through-Spindle Coolant (Optional): The through-spindle coolant (TSC) option provides coolant through the center of the spindle and tool holder, directly to the cutting zone. TSC requires:

- A high-pressure coolant pump (Part No. BC-OPT-TSC-PMP, 70 bar, 30 L/min)
- A coolant rotary union at the spindle (Part No. BC-OPT-TSC-UNION)
- Through-coolant compatible toolholders

TSC significantly improves tool life and chip evacuation in deep hole drilling and difficult-to-machine materials.

5.7 Chip Conveyor

The chip conveyor removes metal chips from the machine work zone and deposits them in a chip collection bin. The BC-500X uses a hinged-belt type chip conveyor suitable for all chip types from small needle chips to long stringy chips.

Parameter	Value
Conveyor type	Hinged steel belt (steel hinge plates)
Belt width	250 mm
Total length	2,000 mm
Elevation	600 mm rise over 1,800 mm horizontal
Motor	0.75 kW, 4-pole induction motor with gear reducer
Belt speed	3–8 m/min (adjustable via VFD)
Chip capacity	Up to 200 kg/hour
Materials handled	Steel, cast iron, aluminum, stainless, brass chips
Discharge height	600 mm from floor to discharge
Chip bin capacity	120 liters
Belt material	Carbon steel, zinc plated
Hinge pin material	Stainless steel 304
Part No. (complete)	BC-CNV-CHIP-001
Motor/gearbox Part No.	BC-MTR-CNV-001

NOTE: The chip conveyor hinged belt requires periodic lubrication of the hinge pins. Refer to Section 9.2 (daily maintenance) for lubrication procedure.

5.8 Lubrication System (Way Lube)

The BC-500X centralized automatic lubrication system delivers ISO VG 68 way lube oil to all linear guideway carriages and ballscrew nuts at timed intervals.

Way Lube System Components:

Component	Description	Part No.
Pump unit	Electric, positive displacement, 2.5-liter reservoir	BC-LUB-PMP-001
Pump motor	0.12 kW, 1/8 HP	BC-MTR-LUB-001
Distribution manifold (X-axis)	4-outlet, metered	BC-LUB-MAN-X
Distribution manifold (Y-axis)	4-outlet, metered	BC-LUB-MAN-Y
Distribution manifold (Z-axis)	4-outlet, metered	BC-LUB-MAN-Z
Lube tubing	4 mm OD nylon, 60 m total	BC-LUB-TUBE-4
Lube oil	ISO VG 68 way lube oil	BC-LUB-OIL-VG68
Low level switch	Float type	BC-LUB-LVL-SW

Lubrication System Operation:

The way lube system operates on a timed cycle controlled by PLC. The cycle interval is set by Parameter B9.22 (Way Lube Interval, default = 30 minutes). When the lube cycle activates:

1. The lube pump motor runs for 15 seconds (adjustable via Parameter B9.23).
2. Oil is pumped from the reservoir through the distribution manifolds to each lubrication point.
3. The metered outlets in the manifolds deliver a fixed oil volume per cycle (approximately 0.1 mL per point per cycle).
4. The lubrication cycle is logged in the maintenance log.

Lube Oil Level Monitoring: The reservoir level is monitored by a float switch. When the level drops below the "LOW" mark, fault E088 (Way Lube Level Low) is set. The machine will continue to operate but will begin a countdown timer. If oil is not added within 30 minutes of fault E088 activation, the machine will halt with fault E089 (Way Lube Level Critical).

Lubrication Oil Specification:

- Type: ISO VG 68 way lube oil with anti-wear, anti-stick-slip additives
 - Approved brands: Mobil Vactra Oil No. 2, Shell Tonna S2 M 68, Fuchs Renolin MR68
 - Never use general-purpose machine oil or hydraulic oil in the way lube system
 - Part No. for BetaCorp supplied oil: BC-LUB-OIL-VG68 (5-liter container)
-

5.9 Enclosure and Guarding

The BC-500X full enclosure provides:

- Protection of the operator from cutting fluids, chips, and tool breakage ejection
- Containment of coolant mist within the machine volume
- Noise reduction (contribution approximately 3–5 dB(A) at operator position)
- Access control (door interlock prevents hazardous operation with door open)

Enclosure Construction:

Component	Material	Description
Main enclosure panels	Cold-rolled steel, 2 mm, painted	Structural panels
Door frame	Cold-rolled steel, 3 mm, painted	Reinforced for door mounting
Work zone door	Steel frame with polycarbonate window (12 mm thick)	Sliding type, with gas strut counterbalance
Door window	Polycarbonate, 12 mm, impact-resistant	Minimum safety factor 4×
Door seal	EPDM rubber, continuous seal on 3 sides	Coolant sealing
Door interlock switch	Schmersal AZ 16 or equivalent (dual channel)	Safety-rated, IEC 60947-5-3
Door lock solenoid	Electromagnetic hold, 24 VDC	Part No. BC-LOK-DOOR-001
Interior lighting	LED strip, IP65, 24 VDC	Part No. BC-LED-WORK-001
Mist collector port	Ø150 mm round port, top of enclosure	For customer-supplied mist collector

Window Inspection: The polycarbonate work zone window must be inspected monthly. Replace the window (Part No. BC-WIN-PC-001) if:

- Any crack or fracture is visible
- Deep scratches reduce visibility significantly
- The window has been struck by a tool ejection event
- The window is yellowed or embrittled (polycarbonate degrades with UV and certain coolants)

5.10 Control Panel and Operator Interface

The BC-500X control panel is mounted on a pivoting arm that allows the operator to position it for optimum comfort. The panel includes the CNC display, keyboard, function keys, and all machine

control switches.

Control Panel Layout:

The control panel is divided into three sections:

Upper Section — CNC Control Unit:

- 15-inch color LCD display (1024 × 768 resolution)
- Membrane keyboard (alphanumeric, function keys)
- USB port (front-accessible, for program transfer)
- Ethernet port (behind door, for network connection)

Middle Section — Machine Control Switches:

- POWER ON (green, momentary)
- POWER OFF (red, momentary)
- E-STOP (red, mushroom head, key-release)
- RESET (blue, momentary)
- CYCLE START (green, with lamp)
- FEED HOLD (amber, with lamp)
- SPINDLE START (green, with lamp)
- SPINDLE STOP (red, with lamp)
- MODE SELECTOR (rotary switch: AUTO / MDI / JOG / HANDLE / HOME)

Lower Section — Machine Function Switches:

- COOLANT ON/OFF (toggle)
 - CHIP CONVEYOR ON/OFF (toggle)
 - WORK LIGHT ON/OFF (toggle)
 - FEED RATE OVERRIDE (potentiometer, 0–150%)
 - SPINDLE SPEED OVERRIDE (potentiometer, 50–150%)
 - RAPID OVERRIDE (4-position rotary: 5% / 25% / 50% / 100%)
 - JOG AXIS SELECTOR (rotary: X / Y / Z / OFF)
 - JOG RATE SELECTOR (rotary: 0.001 / 0.01 / 0.1 / 1.0 / 10 / 100 mm/min)
 - JOG +/- direction buttons (for selected axis)
-

5.11 Pendant and MPG Hand Wheel

The portable MPG (Manual Pulse Generator) pendant allows the operator to manually position axes with fine control, typically used during workpiece setup and tool offset measurement.

MPG Pendant Specifications:

Feature	Description
Physical	Handheld, 1.5 m cable
Display	4-line LCD, axis position readout
E-Stop	Red mushroom-head button (dual-channel)
Hand wheel	100-count per revolution pulse generator
Axis selector	Rotary switch: X / Y / Z / 4th (optional)
Multiplier	Rotary switch: $\times 0.001$ / $\times 0.01$ / $\times 0.1$ mm per pulse
Enable button	Dead-man enable (must hold to move axis)
Part No.	BC-PND-MPG-001

Hand Wheel Use: The MPG hand wheel allows the operator to move axes at precise increments:

- $\times 0.001$ mm per pulse: 0.1 mm per full revolution — for very fine adjustments
- $\times 0.01$ mm per pulse: 1.0 mm per full revolution — for tool length measurement
- $\times 0.1$ mm per pulse: 10 mm per full revolution — for rapid positioning

The dead-man enable button must be held during all hand wheel motion. Releasing the enable button immediately halts axis motion, preventing uncontrolled movement.

5.12 Electrical Cabinet

The main electrical cabinet is a RITTAL TS8 series enclosure (or equivalent), IP54 rated, with thermostatically controlled air conditioning to maintain cabinet interior temperature at 35°C or below.

Cabinet Dimensions:

- Width: 800 mm
- Height: 2,000 mm
- Depth: 400 mm

Cabinet Interior Layout (from top to bottom):

Level	Contents
Top	Cable entry area; main busbar; surge protection
Upper	Main disconnect; main fuses; transformer

Level	Contents
Middle-upper	Spindle drive (Siemens S120 or equivalent)
Middle	X, Y, Z servo drives (BetaCorp SD-300/400)
Middle-lower	ATC servo drive; PLC rack; CNC controller
Lower	24 VDC power supplies; I/O modules; terminal blocks
Bottom	Ground bus; cable tray

Cabinet Cooling:

- Thermostat-controlled heat exchanger (liquid-to-air type)
- Set point: 35°C interior
- Cooling capacity: 1,500 W
- Monitoring: Cabinet temperature alarm activates at 50°C interior (Fault E821)
- Part No. (heat exchanger): BC-HEX-CAB-001

End of Chapter 5 — Machine Components

PART 5 — OPERATION

Chapter 6 — Operation Manual

6.1 Operator Qualification and Training

WARNING: The BC-500X may only be operated by personnel who have completed the required training as specified in Section 2.2.1. Untrained operation creates serious risk of injury, death, and machine damage.

Training Requirements: All operators must complete a minimum of 16 hours of BC-500X operator training covering:

- Machine safety (Chapter 2 of this manual)
- Control panel familiarization
- Manual operation modes
- Program selection and execution
- Tool management and offset entry
- Emergency procedures

- Daily maintenance responsibilities

Operator Documentation: Training completion must be documented with the date, trainer name, and operator signature. This record must be maintained by the employer and made available upon request by regulatory authorities.

6.2 Pre-Operation Safety Inspection

Before starting the BC-500X for any production shift, the operator must perform the following pre-operation inspection. This inspection takes approximately 5 minutes and must be documented in the daily log.

Daily Pre-Operation Checklist:

Guarding and Safety Systems:

- ☐ Work zone door slides smoothly and seals properly (no gaps in door seal)
- ☐ Door window is clean, uncracked, and provides clear visibility
- ☐ Door interlock functions: open door → verify machine will not start spindle
- ☐ All E-Stop buttons visible and accessible; verify they are not depressed
- ☐ No personnel working inside the machine (check work zone through window)
- ☐ Tool changer area is clear of obstructions and debris
- ☐ Chip conveyor drive guard is in place

Fluid Levels:

- ☐ Coolant level — check sight glass on coolant tank. Add if below "LOW" mark
- ☐ Way lube reservoir — check level. Add if below "LOW" mark (use ISO VG 68 oil only)
- ☐ Hydraulic oil level — check HPU sight glass with system not running. Level should be between MIN and MAX marks

Chip Management:

- ☐ Chip bin under chip conveyor — empty if more than 2/3 full
- ☐ Coolant tank area around the chip pan — clear any accumulated chips
- ☐ Work zone interior — clear any chips from previous shift

Machine Condition:

- ☐ No visible fluid leaks (hydraulic oil, coolant, way lube) on floor around machine
- ☐ No unusual sounds during brief dry-run test after startup
- ☐ Control display shows no active faults (after startup)
- ☐ Coolant nozzles in correct position for the programmed operation

Workstation:

- ☐ Operator position: floor is clean, anti-slip mat in place, no tripping hazards
- ☐ PPE available and in good condition (safety glasses, face shield, cut-resistant gloves)
- ☐ Emergency procedures posted and visible

If any item on the pre-operation checklist is not satisfactory, the machine must not be started until the condition is corrected. Report deficiencies to the maintenance department.

6.3 Machine Power-On Sequence

The machine power-on sequence must be followed in the specified order. Skipping steps or changing the sequence may result in system errors or unsafe conditions.

Power-On Sequence:

Step 1: Verify Area is Clear Walk around the entire machine and visually verify that no personnel are inside or near the machine, no tools or materials are left in the work zone that could interfere with homing motion, and the machine is in a safe state (no visible damage, no active leaks).

Step 2: Apply Main Power Turn the main electrical disconnect switch (inside the main electrical cabinet, or the external handle if equipped) to the ON position.

The CNC control will begin its initialization sequence. Status messages during initialization:

- "BETACORP SYSTEMS BC-500X — SYSTEM INITIALIZING..."
- "LOADING SYSTEM PARAMETERS..."
- "INITIALIZING PLC..."
- "CHECKING SAFETY SYSTEMS..."
- "SYSTEM READY — PRESS POWER ON"

Initialization takes approximately 45 seconds.

Step 3: Press POWER ON Press the green POWER ON button on the main operator panel. The CNC control will complete its startup and display the main operation screen. Expected initial status:

- Axis positions may show asterisks (*) indicating homing is required
- Fault E404 (Hydraulic Pressure Low) will be active — this is normal and will clear when the hydraulic system is started
- All axes will show their last known position or zeros

Step 4: Start Hydraulic System Press the HPU START softkey (on screen) or navigate to SYSTEM → HYDRAULICS → HPU ON.

- The hydraulic pump motor will start.
- Hydraulic system pressure will rise to 150 bar within 10–15 seconds.

- Fault E404 will clear automatically.
- The Z-axis hydraulic brake will remain applied (spring-applied, requires pressure to release).

Step 5: Start Coolant System Press the COOLANT ON/OFF switch on the operator panel to the ON position. The coolant pump will start. Verify coolant flow from the nozzles.

Step 6: Open Pneumatic Supply If the pneumatic supply was isolated, open the pneumatic isolation valve on the FRL assembly (rear of machine). The pneumatic system will pressurize to 5.5 bar.

Step 7: Check for Active Faults Review the CNC display for any active faults (fault codes are displayed in the status bar at the bottom of the main screen). Resolve any faults before proceeding.

- If fault E202 (Coolant Level Below Minimum) is active: add coolant to the tank and reset
- If fault E404 (Hydraulic Pressure Low) persists: check HPU (see Section 8.8.1)
- Any other faults: refer to Chapter 8

Step 8: Perform Homing After the hydraulic system is running and no active faults are present, perform the machine homing procedure (Section 6.4).

6.4 Machine Homing (Reference Return) Procedure

The BC-500X uses incremental encoders on all axes. After power-up, the machine does not know the absolute position of each axis. The homing procedure moves each axis to its home position (limit switch + encoder index pulse) to establish an absolute position reference.

IMPORTANT: Homing must be performed after every machine power-up and after any fault that may have caused unexpected axis motion. Never assume the axis positions are correct without homing.

Homing Procedure — Automatic Mode:

1. Verify the work zone is clear. During homing, all axes will move. Ensure no tooling or fixtures are in the machine that would interfere with full travel.
2. Rotate the MODE SELECTOR switch to HOME position.
3. The control will display "HOMING REQUIRED — PRESS CYCLE START TO HOME ALL AXES."
4. Press CYCLE START. The axes will home in the following sequence:
 - **Z-axis first** (moves UP to Z+ home position). This is critical — Z must home first to provide clearance before X and Y move.
 - **X-axis** (moves to X+ home position).
 - **Y-axis** (moves to Y+ home position).
5. Each axis decelerates as it approaches the home switch and then performs a slow creep to find the encoder index pulse for precise reference.

6. After all three axes have homed, the display will show "HOMING COMPLETE" and the position display will show X=0.000, Y=0.000, Z=0.000 (machine coordinate values at home position).

Homing Speed Parameters:

Parameter	Description	Default Value
B2.07	X-axis homing speed (fast approach)	2,000 mm/min
B2.08	X-axis homing speed (creep to index)	100 mm/min
B2.17	Y-axis homing speed (fast approach)	2,000 mm/min
B2.18	Y-axis homing speed (creep to index)	100 mm/min
B2.27	Z-axis homing speed (fast approach)	1,500 mm/min
B2.28	Z-axis homing speed (creep to index)	100 mm/min

Individual Axis Homing:

In some situations, it may be necessary to home only a specific axis (e.g., if only one axis has lost its reference). To home a single axis:

1. Select HOME mode.
2. Use the AXIS SELECTOR to select the desired axis.
3. Press CYCLE START.
4. The selected axis will home. Other axes will not move (but they must already have valid home positions).

Homing Verification:

After homing, verify that all axis positions display correctly:

- Machine home coordinates: X=0.000, Y=0.000, Z=0.000 (displayed in machine coordinate mode)
- Toggle to work coordinate mode (WORK COORD button): positions will reflect the last set work coordinate offsets

6.5 Control Panel Layout and Functions

6.5.1 CNC Display Screens

The BC-500X CNC control displays information on a 15-inch LCD screen. The main screens are:

Main Position Screen:

- Displays current axis positions in machine or work coordinates
- Displays current spindle speed (commanded and actual)
- Displays active G-codes and M-codes
- Displays feed rate and override values
- Status bar shows: active mode, machine status, active faults

Program Screen:

- Displays the currently loaded program
- Highlighted line shows the block currently being executed
- Cursor position in the program can be scrolled manually

Offset Screen:

- Displays tool length offsets (H values, up to 99 tools)
- Displays tool radius offsets (D values, up to 99 tools)
- Displays work coordinate offsets (G54–G59 + G54.1 P1–P48)

Parameter Screen:

- Requires parameter access password
- Displays all machine parameters in groups B1.xx through B9.xx
- Parameters can be modified by authorized personnel

Diagnostic Screen:

- Encoder status for all axes and spindle
- I/O status (all digital inputs and outputs with current state)
- Axis drive status
- PLC status
- Communication status

6.5.2 Softkeys and Function Keys

The CNC keyboard includes 12 function keys (F1–F12) whose function changes based on the active screen. Current function key assignments are always displayed at the bottom of the CNC screen in a row of labeled "softkey" buttons.

Common Softkey Assignments:

Softkey	Function
POSITION	Switch to position display screen

Softkey	Function
PROGRAM	Switch to program screen
OFFSET	Switch to offset screen
PARAM	Switch to parameter screen (password required)
DIAGNOSTIC	Switch to diagnostic screen
MSG	View alarm and message history
SET	Machine setting screen
GRAPH	Display graphics (tool path display mode)

6.6 Manual Operation Modes

6.6.1 JOG Mode

JOG mode allows the operator to move axes continuously at a selected rate while a JOG button is held. JOG mode is used for:

- Moving axes to a convenient position before running a program
- Manual tool breakage inspection
- Positioning the spindle for tool length measurement

JOG Mode Operation:

1. Select JOG mode with the MODE SELECTOR switch.
2. Select the axis to jog with the JOG AXIS SELECTOR switch (X, Y, or Z).
3. Select the jog rate with the JOG RATE SELECTOR switch (range: 0.001 to 100 mm/min; available rates: 0.001, 0.01, 0.1, 1.0, 10, 100 mm/min).
4. Press and hold the JOG + or JOG - button to move the selected axis in the positive or negative direction.
5. Release the button to stop axis motion.

JOG Speed Limit: When the work zone door is open, the maximum JOG speed is limited to 1,000 mm/min regardless of the JOG RATE setting. This limit is enforced by PLC logic and cannot be overridden.

6.6.2 HANDLE (MPG) Mode

HANDLE mode allows use of the MPG hand wheel for precise, incremental axis motion. HANDLE mode is used for:

- Fine positioning during workpiece setup
- Tool length offset measurement
- Manual surface probing

HANDLE Mode Operation:

1. Select HANDLE mode with the MODE SELECTOR switch.
2. On the MPG pendant, select the axis with the AXIS SELECTOR switch.
3. Select the increment multiplier: $\times 0.001$, $\times 0.01$, or $\times 0.1$ mm per pulse.
4. Hold the dead-man enable button on the pendant.
5. Rotate the hand wheel to move the selected axis:
 - Clockwise rotation = positive axis direction
 - Counter-clockwise rotation = negative axis direction

Speed Limiting in HANDLE Mode: The maximum axis speed in HANDLE mode is limited by the control parameter. The typical maximum speed from the hand wheel is 3,000 mm/min (limited by maximum pulse rate from the hand wheel at 1 revolution/second with $\times 0.1$ multiplier). This is intentionally limited for safety.

6.6.3 Single-Block Mode

Single-block mode allows the operator to execute a CNC program one block (line) at a time. Each press of CYCLE START advances the program by one block. Single-block mode is used for:

- Verifying a new or modified program
- Step-by-step execution during machine setup
- Diagnosing program errors

Single-Block Mode Operation:

1. Load the program to be tested (see Section 6.7.2).
2. Navigate the program cursor to the desired starting line.
3. Press the SINGLE BLOCK button (toggles on/off; indicator lamp confirms active).
4. Press CYCLE START. The first block will execute.
5. Press CYCLE START again to execute the next block.
6. Continue until the program has been verified.
7. To return to normal continuous execution, press SINGLE BLOCK to deactivate.

NOTE: In single-block mode, certain multi-block canned cycles (G80–G89) will pause after each internal step. This is intentional and provides maximum diagnostic visibility.

6.7 Program Entry and Editing

6.7.1 MDI (Manual Data Input) Mode

MDI mode allows the operator to enter and execute individual G-code blocks without creating a full program. MDI mode is useful for:

- Manually commanding axis moves to specific positions
- Starting the spindle at a specific speed
- Performing manual tool changes
- Testing G-code syntax before entering it into a program

MDI Mode Operation:

1. Select MDI mode with the MODE SELECTOR switch.
2. The MDI input line appears on the CNC display.
3. Type the G-code command using the keyboard. Example: G00 G90 X100.0 Y50.0 Z-10.0 or M03 S2500.
4. Press CYCLE START to execute the entered command.
5. The command executes immediately. When execution is complete, the MDI line is ready for the next command.

MDI Mode Limitations:

- Maximum MDI block length: 255 characters
- MDI history: the last 20 MDI commands are stored and can be recalled with the cursor UP key
- Certain functions are not available in MDI mode: canned cycles requiring multiple blocks, subroutine calls across programs

6.7.2 Program Storage and Management

The BC-500X CNC control can store programs in internal memory and on external USB devices.

Internal Memory:

- Capacity: 10 GB (flash storage)
- Maximum number of programs: 10,000
- Maximum individual program size: 100 MB (supports extremely long programs)
- Program directory: accessible at PROGRAM → LIBRARY

Program Naming Convention: Programs are named with an "O" prefix followed by up to 8 digits: O00000001 through O99999999. The first 8 standard program numbers (O00001000 through O00001007) are reserved for BetaCorp system programs. Do not modify or delete these programs.

Program Transfer — USB:

1. Connect the USB drive to the front USB port on the CNC control panel.
2. Navigate to PROGRAM → USB TRANSFER.
3. Select the direction (USB → CNC or CNC → USB).
4. Select the program(s) to transfer.

5. Press TRANSFER (F-key).

Program Transfer — Ethernet (DNC): The BC-500X supports DNC (Direct Numerical Control) operation via Ethernet. Programs stored on a network server can be streamed directly to the CNC for execution. Contact BetaCorp Systems for DNC software configuration requirements.

6.8 Tool Management

6.8.1 Tool Length Offset Setup

Tool length offsets are stored in the CNC control's tool offset table (H1 through H99). The offset value represents the length of the tool from the gauge line of the tool holder to the tool tip.

Method 1: Reference Tool Method (Most Common)

A reference tool of known length is first measured, and all other tools are measured relative to the reference tool.

1. Load a reference tool (or tool number 1) in the spindle.
2. Jog the Z-axis down to bring the tool tip to touch the workpiece surface (or a reference surface on the machine table).
3. Record the Z machine coordinate at this position. This is the Z machine position at Z=0 (workpiece surface).
4. Enter the work coordinate G54 Z offset: navigate to OFFSET → WORK COORD → G54, enter the current Z machine coordinate as the G54 Z offset value.
5. The G54 Z offset now defines Z=0 at the reference surface.
6. For each subsequent tool, load it in the spindle, jog down to touch the same reference surface, and read the Z work coordinate. If Z work coordinate reads $-\Delta$ (negative), the tool is shorter than the reference by Δ mm.
7. Enter the tool length offset: OFFSET → TOOL OFFSET → H(n) → enter the measured value.

Method 2: Tool Presetter A tool presetter allows all tools to be measured outside the machine, using a fixed measurement device. Tool lengths measured on the presetter are entered directly into the tool offset table. This method is faster for shops that change tools frequently and have multiple identical machines.

Method 3: On-Machine Tool Setting Probing (Optional) If the BC-500X is equipped with the optional on-machine tool setting probe (Part No. BC-OPT-TOOL-PROBE, Renishaw TS27R or equivalent), tool length and diameter can be measured automatically using a macro program. Contact BetaCorp Systems for the tool setting macro program and setup procedure.

6.8.2 Tool Radius Offset Setup

Tool radius offsets (D values) are used in conjunction with G41 (cutter compensation left) and G42 (cutter compensation right) to automatically offset the tool path by the tool radius.

Entering Radius Offsets:

1. Navigate to OFFSET → TOOL OFFSET.
2. Use the cursor to select the D offset number corresponding to the tool.
3. Enter the tool radius (in mm). For example, a 20 mm diameter end mill has a radius of 10.000 mm.
4. Enter the wear value in the adjacent column (start with 0.000; adjust after test cuts).

Cutter Compensation Usage: Cutter compensation must be applied correctly to avoid gouging the workpiece. Key rules:

- Always ramp into cutter compensation: never start the cut with G41/G42 already active
- The approach move when G41/G42 is first commanded must be at least one tool radius in length
- Cancel cutter compensation (G40) before returning to the tool change position

6.8.3 Tool Life Management

The BC-500X includes a tool life management system that tracks how long each tool has been used and alerts the operator when the tool should be replaced.

Tool Life Management Setup:

For each tool, enter the following in the TOOL LIFE table (OFFSET → TOOL LIFE):

Field	Description	Example
Tool No.	Tool number (1–24)	5
Life Type	COUNT (number of uses) or TIME (minutes)	TIME
Rated Life	Maximum life before alarm	90 (minutes)
Used Life	Actual time used (auto-updated)	47.3 (minutes)
Status	ACTIVE / EXPIRED / SPARE	ACTIVE

When a tool's used life exceeds its rated life, the control will:

1. Display an alarm message: "TOOL (n) LIFE EXPIRED — REPLACE BEFORE NEXT USE"
2. If configured (Parameter B3.41), the machine will halt the program at the next tool change and wait for operator confirmation before proceeding with the expired tool or switching to a replacement tool.

Replacement Tool Groups: Tools can be configured in replacement groups. For example, Tool 5 and Tool 15 might be identical tools (twin tool strategy). When Tool 5 expires, the control automatically uses Tool 15 as a replacement and updates Tool 5's status to "EXPIRED." When Tool 15 also expires, an alarm halts the machine.

6.9 Workpiece Setup and Fixture

6.9.1 Work Coordinate System Setup

The BC-500X supports 6 standard work coordinate systems (G54–G59) plus 48 extended work coordinate systems (G54.1 P1–P48). Each work coordinate system stores X, Y, and Z offset values that relate the machine coordinate system to the workpiece or fixture.

Setting Work Coordinates — Manual Method:

Establishing X and Y Zero (Corner Method):

1. Load a test indicator or edge finder in the spindle.
2. Jog the axis to touch the edge of the workpiece at the desired X0 Y0 point.
3. Read the X machine coordinate from the position display.
4. Navigate to OFFSET → WORK COORD → G54 (or desired G-code).
5. Move the cursor to the X field.
6. Type the machine X coordinate value and press INPUT.
7. Repeat for Y.

Establishing X and Y Zero (Center Method):

1. Touch off the left edge of the workpiece. Read X machine coordinate = X_LEFT.
2. Touch off the right edge. Read X machine coordinate = X_RIGHT.
3. $X \text{ center machine coordinate} = (X_LEFT + X_RIGHT) / 2$.
4. Enter the X center coordinate in the G54 X field.
5. Repeat for Y.

Establishing Z Zero:

1. Load the reference tool (tool 1) in the spindle.
2. Jog Z down to touch the workpiece top surface. This may be done with a paper feeler or by very careful jogging at slow speed.
3. Read the Z machine coordinate.
4. Enter the Z machine coordinate in the G54 Z field.
5. Note: G54 Z represents "distance from machine Z home to Z=0 (workpiece surface)." The value will be negative (e.g., G54 Z = -350.245).

Setting Work Coordinates — Using the MEASURE Function: The BC-500X control provides a MEASURE softkey that simplifies offset entry:

1. Jog to the reference point.
2. Press MEASURE softkey.
3. The control reads the current machine position and calculates the appropriate offset value

automatically, including tool length compensation if a tool is active.

6.9.2 Part Probing (Optional)

If the BC-500X is equipped with the optional on-machine part probing system (Renishaw OMP40 or equivalent, Part No. BC-OPT-PART-PROBE), workpiece setup can be automated using macro programs supplied by BetaCorp Systems.

Standard probe cycles available:

- **O09000 — Single surface probe:** Finds one workpiece surface and sets a work coordinate automatically
- **O09001 — Rectangular boss:** Finds the center of a rectangular boss
- **O09002 — Rectangular pocket:** Finds the center of a rectangular pocket
- **O09003 — Circular boss:** Finds the center of a circular boss
- **O09004 — Circular pocket:** Finds the center of a circular pocket
- **O09005 — Corner find:** Finds the intersection of two surfaces (for workpiece corner zeroing)
- **O09006 — Web probe (Z):** Finds the top surface of the workpiece for Z-zero
- **O09007 — Angle probe:** Measures the angular orientation of a surface relative to the X-axis

Macro programs are stored in the CNC as protected programs (cannot be deleted or edited by operators). Contact BetaCorp Systems for macro program documentation.

6.10 Coolant System Operation

Coolant Control:

The coolant pump is controlled from the operator panel (COOLANT switch) and from the CNC program (M-codes).

M-Code Coolant Control:

M-Code	Function
M08	Coolant ON
M09	Coolant OFF
M07	Mist coolant ON (requires optional mist nozzle)

Coolant in Programs: Coolant should be activated with M08 at the beginning of each cutting operation and deactivated with M09 at the end. Most programs use M08 after the tool change (after the T and M06 blocks) and before the first cutting move.

Coolant Flow Rate Adjustment: The coolant flow rate can be adjusted by partially closing

individual nozzle ball valves. This allows different flow rates to different areas of the work zone. Do not restrict flow to less than 30% of full open on more than two nozzles simultaneously, as this may overheat the coolant pump (requires minimum 40% flow for cooling).

Coolant Temperature Monitoring: The coolant system temperature is monitored by a Pt100 RTD in the coolant tank. Normal coolant operating temperature is 20–35°C. If coolant temperature exceeds 40°C:

- Fault E221 (Coolant Temperature High Warning) is set
- The machine continues to operate but alerts the operator If coolant temperature exceeds 45°C:
- Fault E222 (Coolant Temperature High Shutdown) is set
- The machine halts the program and turns off the coolant pump

Coolant Management: Refer to Chapter 12 for complete coolant management procedures, including concentration checking, pH testing, and bacteria testing.

6.11 Automatic Cycle Operation

6.11.1 Program Selection

1. Navigate to PROGRAM → LIBRARY.
2. Use the cursor to highlight the desired program number.
3. Press SELECT (softkey) to load the program into the active buffer.
4. The program number and first few lines will appear on the PROGRAM screen.
5. Press the REWIND softkey to ensure the cursor is at the first block of the program.

Running from a Specific Line: To start execution from a specific block (not the beginning):

1. Scroll the cursor to the desired starting block.
2. Press CYCLE START — the program will begin from the highlighted block.
3. Warning: Starting from a point other than the beginning may cause incorrect machine state (wrong offsets, wrong spindle speed, wrong tool loaded). Always verify machine state carefully when performing a mid-program start.

6.11.2 Cycle Start and Feed Hold

Cycle Start:

1. Verify the machine is in AUTO mode (MODE SELECTOR).
2. Verify the correct program is loaded and the cursor is at the start.
3. Verify the work zone door is closed.
4. Press CYCLE START (green). The program will begin executing.

5. The CYCLE START indicator lamp will remain lit while the program is running.

Feed Hold: Pressing FEED HOLD during automatic operation pauses axis motion but does NOT stop the spindle.

- All axis motion decelerates to a stop.
- The spindle continues to rotate at the commanded speed.
- The coolant pump continues to run.
- The FEED HOLD indicator lamp will light.

To resume after Feed Hold:

- Press CYCLE START to resume from the paused position.
- To abort instead of resuming: press RESET to cancel the program.

CAUTION: After a Feed Hold during a cutting move, the tool remains in the workpiece. Before pressing CYCLE START to resume, verify the tool is not damaged and the workpiece is not overheated. If the pause was extended, the workpiece may have "cooled" the tool to the surface, potentially causing the tool to re-cut slightly aggressively on resume. Use single-block mode after an extended Feed Hold for the first few blocks.

6.11.3 Override Functions

Feed Rate Override: The FEED RATE OVERRIDE potentiometer (0–150%) scales the programmed feed rate in real time. At 100% (mid position), the feed rate is exactly as programmed. Rotating clockwise increases feed rate; counter-clockwise decreases it.

Override Setting	Effect
0%	Axis motion stops (Feed Hold)
50%	Feed rate at half programmed value
100%	Feed rate at programmed value
150%	Feed rate at 150% of programmed value

NOTE: The 0% feed rate override setting is equivalent to Feed Hold but does NOT light the Feed Hold indicator lamp. This can cause confusion. Use the FEED HOLD button for intentional pauses.

Spindle Speed Override: The SPINDLE SPEED OVERRIDE potentiometer (50–150%) scales the programmed spindle speed in real time. Overrides below 50% or above 150% are not available.

Rapid Override: The RAPID OVERRIDE rotary switch limits the maximum rapid traverse speed:

- 5%: Maximum 1,800 mm/min (X, Y); 1,500 mm/min (Z)
- 25%: Maximum 9,000 mm/min (X, Y); 7,500 mm/min (Z)
- 50%: Maximum 18,000 mm/min (X, Y); 15,000 mm/min (Z)

- 100%: Full rapid traverse speed (36,000 mm/min X, Y; 30,000 mm/min Z)

RECOMMENDED PRACTICE: Always set the rapid override to 25% when running a new program for the first time. Switch to 100% only after the program has been verified as collision-free.

6.12 Machine Power-Off Sequence

The machine should be powered off in the following sequence to ensure safe shutdown:

Step 1: Complete or Abort the Current Program If a machining program is running:

- Allow the program to complete (preferred), or
- Press FEED HOLD, then press RESET to abort.
- Ensure the spindle has stopped (speed = 0 on display).

Step 2: Return Axes to Home Position Jog or home all axes to their home positions:

- Z-axis to home (Z=0, fully up)
- X-axis to home (X=0)
- Y-axis to home (Y=0) This ensures the machine is in a known, safe state for the next power-on.

Step 3: Remove the Workpiece (if applicable) If the workpiece is to be stored outside the machine while the machine is powered off, remove it and secure appropriately.

Step 4: Turn Off Coolant Press the COOLANT switch to OFF. Allow the pump to run for 10 seconds to drain the distribution lines before turning off.

Step 5: Turn Off Chip Conveyor Press the CHIP CONVEYOR switch to OFF.

Step 6: Stop Hydraulic System Navigate to SYSTEM → HYDRAULICS → HPU STOP and press HPU STOP. The hydraulic pump will stop. The Z-axis brake will remain applied by spring force.

Step 7: Turn Off CNC Control Press the POWER OFF button (red, main console). The CNC control will shut down in an orderly sequence:

- Saves current state (tool position, offsets) to non-volatile memory
- Displays "SYSTEM SHUTDOWN COMPLETE"

Step 8: Turn Off Main Disconnect Turn the main electrical disconnect to the OFF position. This removes all power from the machine.

NOTE: If the machine will be unattended for more than 8 hours, apply a padlock to the main disconnect in the OFF position. This prevents unauthorized operation.

6.13 Emergency Situations During Operation

Situation: Tool Breakage

1. Press FEED HOLD or E-STOP to halt the machine.
2. Wait for the spindle to stop completely.
3. Open the work zone door.
4. DO NOT reach into the work zone with bare hands. Use chip hooks to remove broken tool fragments.
5. Inspect the workpiece for damage.
6. Replace the tool. Inspect the spindle taper for damage from broken tool fragments.
7. If the spindle taper is damaged, contact BetaCorp Systems before resuming operation.

Situation: Workpiece Loose or Shift During Machining

1. Press E-STOP immediately.
2. Do not open the work zone door until the spindle has completely stopped.
3. Evaluate the situation through the window before opening the door.
4. Inspect for damage to the workpiece, tooling, and machine.
5. Secure the workpiece properly before any restart.

Situation: Unusual Noise or Vibration

1. Press FEED HOLD.
2. If the noise or vibration continues after Feed Hold: press E-STOP.
3. Investigate the cause before resuming.
4. Common causes: tool wear/breakage, loose fixture, incorrect spindle speed, coolant flow interruption.

Situation: Coolant Leak onto Electrical Components

1. Press E-STOP immediately.
2. Turn off the main disconnect.
3. Do NOT attempt to clean coolant from electrical components while the machine is energized.
4. Contact maintenance. The machine must be inspected and dried before re-energizing.

PART 6 — PARAMETER REFERENCE

Chapter 7 — Parameter Reference

7.1 Introduction to BC-500X Parameters

The BC-500X uses a structured parameter system organized into nine groups (B1.xx through B9.xx). These parameters control all aspects of machine operation, from spindle speed limits to servo loop gains to safety interlock timers.

Parameter Access Levels:

Level	Access Code	Description
Level 0 (View)	None	View parameter values; cannot modify
Level 1 (Operator)	Default: 1111	Modify non-critical parameters
Level 2 (Maintenance)	Provided to maintenance dept.	Modify maintenance and tuning parameters
Level 3 (Engineer)	Provided to service engineers only	Modify all parameters including safety limits

How to Access Parameters:

1. Navigate to SYSTEM → PARAMETER.
2. Enter the access code when prompted.
3. Use the cursor to navigate to the parameter group and number.
4. Highlight the parameter and press MODIFY.
5. Enter the new value and press ENTER.
6. The new value is immediately active. Some parameters require a power cycle to take effect (these are marked with a [*] in the parameter name).

Parameter Backup: Always back up all parameters before making any changes. Navigate to SYSTEM → PARAMETER → BACKUP → USB to save all parameters to a USB drive. The backup file is named: BC500X_[SERIAL]PARAM[DATE].par

Parameter Groups Overview:

Group	Range	Function
B1.xx	B1.01–B1.99	Spindle parameters
B2.xx	B2.01–B2.99	Axis motion parameters
B3.xx	B3.01–B3.99	Tool changer parameters
B4.xx	B4.01–B4.49	Hydraulic system parameters

Group	Range	Function
B5.xx	B5.01–B5.49	Coolant system parameters
B6.xx	B6.01–B6.49	Encoder / feedback parameters
B7.xx	B7.01–B7.49	Communication parameters
B8.xx	B8.01–B8.49	Safety and interlock parameters
B9.xx	B9.01–B9.49	Diagnostic and maintenance parameters

7.2 Parameter Group B1.xx — Spindle Parameters

Param	Name	Default	Range	Unit	Description
B1.01	Spindle Maximum Speed	10000	100–10000	RPM	Maximum allowable spindle speed. The CNC will clamp any S-code command above this value. Reduce this parameter if the machine is not equipped with a high-speed spindle option.
B1.02	Spindle Minimum Speed	50	1–500	RPM	Minimum spindle speed. Commands below this value are clamped to this value.
B1.03	Gear 1 Maximum Speed	600	100–2000	RPM	Maximum spindle speed allowed in Gear 1 (low gear). Spindle speeds above this value trigger an automatic gear change to Gear 2.
B1.04	Gear 2 Minimum	50	1–1000	RPM	Minimum spindle speed

Param	Name	Default	Range	Unit	Description
	Speed				in Gear 2. Below this speed, Gear 1 is selected.
B1.05	Gear Change Speed (from G1 to G2)	550	100–1500	RPM	Spindle speed at which the gear change from Gear 1 to Gear 2 is initiated. Must be less than B1.03.
B1.06	Gear Change Speed (from G2 to G1)	400	100–1000	RPM	Spindle speed at which the gear change from Gear 2 to Gear 1 is initiated. Must be less than B1.05.
B1.07	Gear Change Delay (stop)	500	100–5000	ms	Time to wait after spindle stops before actuating gear change solenoids.
B1.08	Gear Change Delay (engage)	800	200–5000	ms	Time to wait after solenoid actuation before commanding spindle restart.
B1.09	Gear Change Confirm Timeout	3000	500–10000	ms	Maximum time to wait for gear change confirmation sensor before setting fault E331.
B1.10	Gear Change Auto Enable	1	0–1	—	0 = Manual gear selection only (M41/M42 required in program). 1 = Automatic gear selection based on commanded

Param	Name	Default	Range	Unit	Description
					speed.
B1.11	Spindle Acceleration Time	3.5	0.5–30.0	seconds	Time to accelerate from 0 to maximum speed (B1.01). Actual acceleration time to any speed is scaled proportionally.
B1.12	Spindle Deceleration Time (no brake)	8.0	1.0–60.0	seconds	Time to decelerate from maximum speed to 0 with no external brake.
B1.13	Spindle Deceleration Time (with brake)	4.0	0.5–30.0	seconds	Time to decelerate from maximum speed to 0 when spindle brake is engaged. Only effective if B1.22 = 1.
B1.14	Spindle Speed Window	50	5–500	RPM	The spindle is considered "at speed" when actual speed is within $\pm$ B1.14 RPM of commanded speed. M03/M04 will wait for "at speed" before the next block executes.
B1.15	Spindle At-Speed Timeout	10000	1000–30000	ms	Maximum time to wait for spindle to reach "at speed" before setting fault E102.
B1.16	Spindle Speed	1	0–1	—	0 = Speed

Param	Name	Default	Range	Unit	Description
	Monitor Enable				monitoring disabled. 1 = Spindle fault E102 is triggered if speed deviates from command by more than B1.17.
B1.17	Spindle Speed Deviation Limit	200	50–1000	RPM	Maximum allowable spindle speed deviation from commanded speed during steady-state operation. Fault E102 (Spindle Speed Deviation) is set if exceeded.
B1.18	Spindle Orientation Speed	200	50–500	RPM	Speed used for spindle orientation (M19). Orientation must be performed at low speed for accuracy.
B1.19	Spindle Orientation Angle	0.0	0.0–359.9	degrees	Target angle for spindle orientation (M19). 0.0 degrees is the spindle key at the standard ATC position.
B1.20	Spindle Orientation Window	0.5	0.1–5.0	degrees	Orientation is considered complete when spindle angle is within $\pm$ B1.20 degrees of target.
B1.21	Spindle Orientation Timeout	5000	1000–15000	ms	Maximum time for spindle to achieve

Param	Name	Default	Range	Unit	Description
					orientation before fault E103.
B1.22	Spindle Brake Enable	1	0–1	—	0 = No electromagnetic spindle brake. 1 = Brake applied on deceleration after M05 or door open. Reduces spindle stop time.
B1.23	Spindle Brake Engage Speed	100	20–500	RPM	Speed below which the spindle electromagnetic brake is applied during deceleration. Brake is not applied at higher speeds.
B1.24	Spindle Brake Release Speed	150	50–600	RPM	Speed above which the spindle brake is released on acceleration. Must be greater than B1.23.
B1.25	Spindle Brake Timeout	3000	500–10000	ms	Maximum time for spindle to stop after brake engagement. If spindle does not reach 0 RPM within this time, fault E104 (Spindle Brake Timeout) is set.
B1.26	Spindle Load Monitor Enable	1	0–1	—	0 = Spindle load monitoring disabled. 1 =

Param	Name	Default	Range	Unit	Description
					Fault E105 is triggered if spindle load exceeds B1.27.
B1.27	Spindle Overload Limit	120	100–150	%	Spindle load percentage above which fault E105 is triggered. Load is calculated from motor current. At 100% = rated continuous load (15 kW).
B1.28	Spindle Overload Delay	5000	500–30000	ms	Spindle load must exceed B1.27 for this duration before E105 is triggered. Prevents false trips from short-duration load spikes.
B1.29	Rigid Tapping Enable	1	0–1	—	0 = Rigid tapping (G84 with spindle encoder synchronization) disabled. 1 = Enabled.
B1.30	Rigid Tap Spindle Gain	1.00	0.50–2.00	—	Gain factor for spindle-to-Z-axis synchronization during rigid tapping. Increase if tapping pitch is slightly short; decrease if slightly long.
B1.31	Rigid Tap Speed Ramp	500	100–2000	ms	Spindle acceleration time used during rigid tapping cycle.

Param	Name	Default	Range	Unit	Description
					Faster ramp may cause synchronization errors at the tap bottom.
B1.32	Rigid Tap Overshoot Limit	2.0	0.5–10.0	degrees	Maximum allowable spindle angle overshoot at tap bottom before fault E106.
B1.33	Air Purge Enable	1	0–1	—	0 = Spindle air purge disabled. 1 = Air purge active whenever spindle is running (blows air down through spindle bore to prevent chip intrusion).
B1.34	Air Purge Delay (after stop)	2000	0–10000	ms	Time to continue air purge after spindle stop command. Helps clear chips during tool change.
B1.35	Oil-Air Lube Interval	10	5–60	minutes	Interval between oil-air lubrication pulses for spindle bearings. Decrease for high-speed or high-temperature operation.
B1.36	Oil-Air Lube Pulse Duration	3	1–30	seconds	Duration of each oil-air lubrication pulse.
B1.37	Oil-Air Lube	2.5	1.0–5.0	bar	Minimum air

Param	Name	Default	Range	Unit	Description
	Pressure Min				pressure required for oil-air lubrication system. Below this pressure, fault E088 is set.
B1.38	Spindle Encoder PPR	4096	512–16384	PPR	Spindle encoder pulses per revolution (pre-quadrature). Must match the installed encoder (Heidenhain ROD 420 = 4096). Changing this parameter requires BetaCorp authorization. [*]
B1.39	Spindle Encoder Type	1	0–1	—	0 = TTL single-ended. 1 = RS-422 differential (default for Heidenhain). [*]
B1.40	Spindle Direction (CW=M03)	0	0–1	—	0 = M03 commands clockwise rotation (standard). 1 = M03 commands counter-clockwise (reversed). Change only if spindle motor wiring is non-standard. [*]
B1.41	Spindle Speed Display Filter	50	10–500	ms	Time constant for speed display filter.

Param	Name	Default	Range	Unit	Description
					Higher value = smoother display but slower response.
B1.42	Spindle Warm-Up Enable	0	0–1	—	0 = No automatic warm-up. 1 = Spindle will automatically execute a warm-up cycle on first M03 command of the day.
B1.43	Spindle Warm-Up Program	9990	9000–9999	—	Program number to execute for spindle warm-up. Default program O09990 is a 15-minute progressive warm-up cycle.
B1.44	Spindle Temperature Alarm	75	50–90	°C	Spindle head temperature above which fault E107 (Spindle Temperature High) is set.
B1.45	Lube Activate on Spindle	1	0–1	—	0 = Way lube cycle runs regardless of spindle state. 1 = Way lube cycle only runs when spindle is active.
B1.50	Spindle Drive Model	5	1–9	—	Internal code for spindle drive type. BC-500X standard = 5 (BetaCorp SD-SP-15K). Do not change. [*]

7.3 Parameter Group B2.xx — Axis Motion Parameters

Param	Name	Default	Range	Unit	Description
B2.01	X-Axis Position Loop Gain (Kp)	80	10–300	1/s	X-axis position loop proportional gain. Higher values give tighter positioning but may cause oscillation if set too high.
B2.02	X-Axis Velocity Loop Gain (Kv)	150	10–500	Hz	X-axis velocity loop bandwidth. Factory-tuned; do not change without BetaCorp guidance.
B2.03	X-Axis Negative Software Limit	-1.0	-50.0–0.0	mm	Software travel limit in the negative X direction. Axis stops before reaching this position.
B2.04	X-Axis Positive Software Limit	750.0	700.0–760.0	mm	Software travel limit in the positive X direction.
B2.05	X-Axis Following Error Limit	2.0	0.1–50.0	mm	Maximum allowable position following error. If the servo cannot keep up and the error exceeds this value, fault E501 (X-Axis Following Error Limit) is set.

Param	Name	Default	Range	Unit	Description
B2.06	X-Axis Backlash Compensation	0.0	-1.0–1.0	mm	Backlash compensation value for X-axis. The CNC adds or subtracts this value on axis reversals. Factory-set from measurement; do not modify without re-measuring.
B2.07	X-Axis Homing Speed (Fast)	2000	100–5000	mm/min	Speed of X-axis approach to home switch.
B2.08	X-Axis Homing Speed (Creep)	100	10–500	mm/min	Speed of X-axis creep to encoder index after home switch activation.
B2.09	X-Axis Home Offset	0.0	-50.0–50.0	mm	Distance from home switch activation to the machine zero position. Factory-set.
B2.10	X-Axis Acceleration	0.5	0.1–1.5	g	Maximum X-axis acceleration/deceleration (in g).
B2.11	Y-Axis Position Loop Gain (Kp)	80	10–300	1/s	Y-axis position loop proportional gain.
B2.12	Y-Axis Velocity Loop Gain (Kv)	150	10–500	Hz	Y-axis velocity loop bandwidth.
B2.13	Y-Axis Negative Software Limit	-1.0	-50.0–0.0	mm	Software travel limit, negative Y direction.
B2.14	Y-Axis Positive	500.0	450.0–510.0	mm	Software travel limit, positive

Param	Name	Default	Range	Unit	Description
	Software Limit				Y direction.
B2.15	Y-Axis Following Error Limit	2.0	0.1–50.0	mm	Maximum allowable Y-axis following error before fault E511.
B2.16	Y-Axis Backlash Compensation	0.0	-1.0–1.0	mm	Backlash compensation for Y-axis.
B2.17	Y-Axis Homing Speed (Fast)	2000	100–5000	mm/min	Y-axis fast homing speed.
B2.18	Y-Axis Homing Speed (Creep)	100	10–500	mm/min	Y-axis creep speed to index.
B2.19	Y-Axis Home Offset	0.0	-50.0–50.0	mm	Y-axis home offset. Factory-set.
B2.20	Y-Axis Acceleration	0.5	0.1–1.5	g	Maximum Y-axis acceleration.
B2.21	Z-Axis Position Loop Gain (Kp)	70	10–300	1/s	Z-axis position loop gain. Lower than X and Y due to higher Z-axis mass.
B2.22	Z-Axis Velocity Loop Gain (Kv)	120	10–500	Hz	Z-axis velocity loop bandwidth.
B2.23	Z-Axis Negative Software Limit	-550.0	-560.0–500.0	mm	Software travel limit, negative Z direction (maximum Z-axis downward travel).
B2.24	Z-Axis Positive Software Limit	0.5	0.0–50.0	mm	Software travel limit, positive Z direction (above home).
B2.25	Z-Axis Following Error Limit	3.0	0.1–50.0	mm	Maximum allowable Z-axis following error before fault E521. Slightly larger

Param	Name	Default	Range	Unit	Description
					than X and Y due to gravity loading.
B2.26	Z-Axis Backlash Compensation	0.0	-2.0–2.0	mm	Z-axis backlash compensation. For a vertically-oriented ballscrew with proper preload, this should be near zero.
B2.27	Z-Axis Homing Speed (Fast)	1500	100–3000	mm/min	Z-axis fast homing speed. Lower than X and Y due to spindle head weight.
B2.28	Z-Axis Homing Speed (Creep)	100	10–500	mm/min	Z-axis creep speed to index.
B2.29	Z-Axis Home Offset	0.0	-50.0–50.0	mm	Z-axis home offset. Factory-set.
B2.30	Z-Axis Acceleration	0.4	0.1–1.0	g	Maximum Z-axis acceleration. Lower than X and Y due to higher effective mass.
B2.31	Jerk Limit (S-curve) Enable	1	0–1	—	0 = Trapezoidal acceleration profile. 1 = S-curve (jerk-limited) acceleration profile. S-curve reduces mechanical vibration but adds a small amount to positioning time.
B2.32	Jerk Time	100	10–500	ms	Time constant

Param	Name	Default	Range	Unit	Description
	Constant				for S-curve acceleration smoothing. Higher values give smoother motion but slower acceleration onset.
B2.33	Rapid Traverse X	36000	1000–36000	mm/min	Maximum X-axis rapid traverse speed (G00).
B2.34	Rapid Traverse Y	36000	1000–36000	mm/min	Maximum Y-axis rapid traverse speed.
B2.35	Rapid Traverse Z	30000	1000–30000	mm/min	Maximum Z-axis rapid traverse speed. Lower than X and Y due to axis weight.
B2.36	Maximum Feed Rate (all axes)	15000	100–15000	mm/min	Maximum programmed feed rate. F-codes above this value are clamped.
B2.37	Tool Change X Position	0.0	—	mm	Machine X coordinate for spindle during tool change. Set during ATC commissioning.
B2.38	Tool Change Y Position	0.0	—	mm	Machine Y coordinate for spindle during tool change.
B2.39	Tool Change Z Position (ATC)	-150.0	—	mm	Machine Z coordinate at which the ATC arm can safely perform the tool exchange.
B2.40	Servo Enable	200	50–2000	ms	Time to wait

Param	Name	Default	Range	Unit	Description
	Delay				after servo enable signal before commanding motion. Allows drive to build current before motion.
B2.41	Position Loop Integral Gain (all axes)	20	0–100	—	Integral gain for position loop. Reduces steady-state position error under constant load (gravity on Z-axis).
B2.42	Feed Forward Gain (all axes)	95	0–100	%	Velocity feed-forward gain. At 100%, the velocity command from the trajectory planner is passed directly to the velocity loop, minimizing following error at high feed rates.
B2.43	Thermal Compensation Enable	1	0–1	—	0 = Thermal compensation disabled. 1 = Thermal compensation active. Compensation data is in parameters B9.xx (read-only, updated by machine sensors).
B2.44	Circular Interpolation Tolerance	0.010	0.001–0.100	mm	Maximum radial error allowed in circular interpolation (G02/G03).

Param	Name	Default	Range	Unit	Description
					The CNC will generate an alarm if the commanded arc exceeds this tolerance.
B2.45	In-Position Window	0.005	0.001–0.050	mm	Distance within which the axis is considered to be "in position." The machine waits for all commanded axes to be within B2.45 of the target position before executing the next block.
B2.46	In-Position Dwell	0	0–1000	ms	Additional wait time after "in position" condition is achieved before executing the next block. Useful for very high accuracy applications where servo oscillation must fully damp.
B2.47	Over-Travel Deceleration Rate	5.0	1.0–10.0	g	Deceleration rate applied when a hardware over-travel limit switch is triggered. Must be high enough to stop the axis before the mechanical hard stop.
B2.48	Encoder Lost	200	50–2000	ms	Minimum

Param	Name	Default	Range	Unit	Description
	Dwell Timeout				duration of encoder signal loss (any axis) required to trigger axis encoder fault. Short pulses below this threshold are ignored.
B2.49	Stall Detection Enable	1	0–1	—	0 = Axis stall detection disabled. 1 = Fault E503 triggered if axis current exceeds threshold with no motion.
B2.50	Stall Detection Current Threshold	90	50–100	%	Motor current level (% of rated) at which stall is suspected if axis velocity is below B2.51.
B2.51	Stall Detection Velocity Threshold	10	1–100	mm/min	Axis velocity below which current monitoring indicates a potential stall condition.

7.4 Parameter Group B3.xx — Tool Changer Parameters

Param	Name	Default	Range	Unit	Description
B3.01	ATC Enable	1	0–1	—	0 = ATC disabled (machine operates without tool changer). 1 = ATC enabled.
B3.02	ATC Pot	24	12–48	—	Number of tool

Param	Name	Default	Range	Unit	Description
	Count				pots in the carousel. BC-500X standard = 24.
B3.03	ATC Home Pot Number	1	1–24	—	Tool pot number that is at the tool change position when the carousel is at home. Factory-set.
B3.04	ATC Arm Home Detect Input	5	1–32	—	PLC input number for ATC arm home position sensor (SEN-ATC-HOME).
B3.05	ATC Arm 90 Detect Input	6	1–32	—	PLC input number for ATC arm 90° position sensor.
B3.06	ATC Arm 180 Detect Input	7	1–32	—	PLC input number for ATC arm 180° position sensor.
B3.07	ATC Arm Extend Detect Input	8	1–32	—	PLC input number for ATC arm extended position sensor.
B3.08	ATC Arm Retract Detect Input	9	1–32	—	PLC input number for ATC arm retracted position sensor.
B3.09	ATC Carousel Home Input	10	1–32	—	PLC input number for carousel home sensor.
B3.10	ATC Arm Extend Timeout	2000	500–10000	ms	Maximum time allowed for ATC arm to

Param	Name	Default	Range	Unit	Description
					reach the extended position before fault E301.
B3.11	ATC Arm Rotate Timeout	3000	500–10000	ms	Maximum time allowed for ATC arm to complete 180° rotation before fault E302.
B3.12	ATC Arm Retract Timeout	2000	500–10000	ms	Maximum time allowed for ATC arm to reach the retracted position before fault E303 (Tool Changer Timeout). This is the primary E303 trigger.
B3.13	ATC Carousel Index Timeout	15000	2000–60000	ms	Maximum time allowed for carousel to reach any target pot position before fault E304.
B3.14	ATC Carousel Short Path	1	0–1	—	0 = Carousel always rotates in one direction. 1 = Carousel rotates in the shortest direction to reach the target pot.
B3.15	ATC Spindle Orient Timeout	5000	1000–15000	ms	Timeout for spindle orientation (M19) before tool change, before fault E305.
B3.16	ATC Z Retract Distance	200.0	50.0–400.0	mm	Distance the Z-axis moves upward from

Param	Name	Default	Range	Unit	Description
					the current cutting position before initiating a tool change.
B3.17	ATC Z Advance Speed	3000	100–10000	mm/min	Speed at which Z-axis moves to the ATC position (B2.39) for tool change.
B3.18	ATC Tool Unclamp Delay	300	100–2000	ms	Time between hydraulic unclamp command and ATC arm grab move. Ensures tool is fully unclamped before the arm attempts to remove it.
B3.19	ATC Tool Clamp Delay	400	100–2000	ms	Time between ATC arm retract (inserting tool) and hydraulic clamp command. Ensures tool is fully inserted before clamping.
B3.20	ATC Clamp Confirm Timeout	1000	200–5000	ms	Time to wait for tool-clamped confirmation sensor after clamping. If not confirmed, fault E306.
B3.21	ATC Air Blow Time	1000	200–5000	ms	Duration of spindle air blast during tool exchange (to clear chips from spindle taper).

Param	Name	Default	Range	Unit	Description
B3.22	ATC Retry on Fault	0	0–3	—	Number of automatic retry attempts if an ATC fault occurs. 0 = no retry (halt on fault); 1–3 = number of retries before halting.
B3.23	Random Access Mode	1	0–1	—	0 = Fixed pot assignment (T1 always in pot 1). 1 = Random access (control tracks which tool is in which pot).
B3.24	Tool Presence Check Enable	1	0–1	—	0 = No tool presence check. 1 = Fault E307 if a T-code is selected for a pot that the presence sensor reads as empty.
B3.25	Max Tool Weight	8.0	1.0–12.0	kg	Maximum allowed tool weight. Parameter is informational; machine does not directly measure tool weight.
B3.26	ATC Lube Cycle Enable	1	0–1	—	0 = No automatic ATC lubrication. 1 = ATC arm cam and carousel gear lubricated on schedule.
B3.27	ATC Lube Interval	500	50–5000	tool changes	Number of tool changes between ATC lubrication

Param	Name	Default	Range	Unit	Description
					cycles.
B3.28	ATC Lube Quantity	3	1–10	pulses	Number of grease pump pulses per ATC lubrication cycle.
B3.30	Tool Life Management Enable	1	0–1	—	0 = Tool life management disabled. 1 = Tool life tracking and alarm active.
B3.31	Tool Life Alarm Action	1	0–2	—	0 = Display alarm only, continue. 1 = Halt at next tool change, require operator confirmation. 2 = Halt immediately on life expiry.
B3.32	Tool Broken Detect Enable	0	0–1	—	0 = Tool breakage detection disabled. 1 = Macro O09100 runs after each tool change to verify new tool length (requires on-machine probe).
B3.40	Spare Tool (Twin Tool) Enable	0	0–1	—	0 = No spare tool function. 1 = Spare tool (twin) switching enabled.
B3.41	Spare Tool Start Pot	13	1–24	—	Pot number of the first spare tool. Spare tools occupy pots B3.41 through

Param	Name	Default	Range	Unit	Description
					B3.41+(B3.02/2-1).

7.5 Parameter Group B4.xx — Hydraulic System Parameters

Param	Name	Default	Range	Unit	Description
B4.01	HPU Auto Start	1	0–1	—	0 = HPU starts only on manual command. 1 = HPU starts automatically when machine power is on.
B4.02	HPU Pressure Min Alarm	120	50–200	bar	Hydraulic pressure below which fault E404 (Hydraulic Pressure Low) is triggered. This is a safety-critical parameter.
B4.03	HPU Pressure Min Shutdown	80	30–150	bar	Hydraulic pressure below which the machine performs an emergency stop. Must be less than B4.02.
B4.04	HPU Pressure Monitor Delay	5000	1000–30000	ms	Time delay after HPU start before pressure monitoring becomes active. Allows system to pressurize.
B4.05	HPU Temperature Max Alarm	65	40–80	°C	Hydraulic oil temperature above which fault E411 (Hydraulic

Param	Name	Default	Range	Unit	Description
					Temperature High Warning) is triggered.
B4.06	HPU Temperature Max Shutdown	70	50–90	°C	Hydraulic oil temperature above which the machine halts operations. Must be greater than B4.05.
B4.07	HPU Level Min Alarm	1	0–1	—	0 = Hydraulic level monitoring disabled. 1 = Fault E412 (Hydraulic Level Low) triggered when HPU level switch opens.
B4.08	HPU Filter Bypass Alarm	1	0–1	—	0 = Filter bypass alarm disabled. 1 = Fault E413 (Hydraulic Filter Bypass) triggered when filter bypass indicator switch closes.
B4.09	Z-Brake Release Pressure	100	50–180	bar	Minimum hydraulic pressure required to confirm Z-axis brake is released.
B4.10	Z-Brake Release Timeout	2000	500–10000	ms	Maximum time for Z-axis brake to release (pressure switch to close) before fault E414.
B4.11	Z-Brake Apply	2000	500–10000	ms	Maximum time

Param	Name	Default	Range	Unit	Description
	Timeout				for Z-axis brake to apply (spring applies when SOL-HYD-01 de-energizes) before fault E415.
B4.12	Tool Unclamp Pressure	150	100–200	bar	Hydraulic pressure setpoint for tool unclamp circuit. Adjust if tool release is incomplete or excessive.
B4.13	Tool Unclamp Time	500	200–3000	ms	Duration hydraulic unclamp pressure is applied before the draw bar is considered fully extended.
B4.14	Tool Clamp Confirm Timeout	1000	200–5000	ms	Time allowed after removal of unclamp signal for tool-clamped sensor to confirm.
B4.15	ATC Arm Hydraulic Speed	5	1–10	—	Flow control valve setting for ATC arm hydraulic actuator speed. 1 = slowest, 10 = fastest. Factory-set.
B4.16	Gear Change Pressure	80	50–120	bar	Hydraulic pressure for gear change actuator. Adjust if gear changes are incomplete or too forceful.
B4.17	HPU Cooling Fan Auto	1	0–1	—	0 = HPU cooling fan

Param	Name	Default	Range	Unit	Description
					runs continuously when HPU is on. 1 = Fan controlled by thermostat (starts at B4.05 - 10°C).
B4.18	HPU Standby Enable	0	0–1	—	0 = HPU runs continuously when on. 1 = HPU pump stops after B4.19 minutes of no hydraulic demand. Pump restarts automatically when needed.
B4.19	HPU Standby Delay	10	1–60	minutes	Time with no hydraulic demand before HPU enters standby (if B4.18 = 1).
B4.20	Hydraulic Oil Change Interval	4000	1000–8000	hours	Planned hydraulic oil change interval (for maintenance scheduling). When machine hours exceed this value since last oil change, fault E416 (Hydraulic Oil Change Due) is set.
B4.21	HPU Filter Change Interval	1000	200–2000	hours	Planned hydraulic filter change interval. Fault E417 (Hydraulic Filter Change Due) is set when exceeded.

7.6 Parameter Group B5.xx — Coolant System Parameters

Param	Name	Default	Range	Unit	Description
B5.01	Coolant Pump Auto Start	0	0–1	—	0 = Coolant pump starts only on M08 or manual switch. 1 = Coolant pump starts when machine power is on.
B5.02	Coolant Level Min Alarm	1	0–1	—	0 = Coolant level monitoring disabled. 1 = Fault E202 (Coolant Level Below Minimum) triggered when level switch opens.
B5.03	Coolant Level Min Delay	30	5–300	seconds	Coolant level must be low for this duration before E202 is triggered. Prevents false alarms from coolant sloshing.
B5.04	Coolant Level Halt	0	0–1	—	0 = E202 displays alarm but machine continues. 1 = E202 causes immediate machine halt.
B5.05	Coolant Temperature Max Alarm	40	30–60	°C	Coolant temperature above which fault E221 (Coolant Temperature

Param	Name	Default	Range	Unit	Description
					High Warning) is set.
B5.06	Coolant Temperature Max Halt	45	35–70	°C	Coolant temperature above which fault E222 (Coolant Temperature High Shutdown) halts the machine.
B5.07	Coolant Pump Overload Delay	5000	1000–30000	ms	Duration of coolant pump overload current before fault E223 (Coolant Pump Overload) is set.
B5.08	Coolant Delay After Spindle	2000	0–30000	ms	Time coolant runs after spindle stop command (M05) before coolant also stops. Helps clear chips from work zone.
B5.09	TSC Enable	0	0–1	—	0 = Through-spindle coolant disabled. 1 = TSC option enabled. Requires BC-OPT-TSC hardware.
B5.10	TSC Pressure Setpoint	70	20–100	bar	Target pressure for through-spindle coolant pump.
B5.11	TSC Pressure Min Alarm	40	10–80	bar	TSC pressure below which fault E224 (TSC Pressure Low) is triggered.

Param	Name	Default	Range	Unit	Description
B5.12	TSC Flow Check Enable	0	0–1	—	0 = TSC flow not monitored. 1 = Fault E225 (TSC No Flow) if flow sensor indicates no TSC flow.
B5.13	Coolant Concentration Target	8.0	3.0–15.0	%	Target coolant concentration (informational; used by maintenance reminder).
B5.14	Coolant pH Min	8.5	7.0–9.0	—	Minimum acceptable coolant pH (informational; maintenance reminder).
B5.15	Coolant pH Max	9.5	8.5–11.0	—	Maximum acceptable coolant pH (informational; maintenance reminder).
B5.16	Coolant Change Interval	720	100–8760	hours	Planned coolant change interval in machine operating hours. Fault E226 (Coolant Change Due) when exceeded.
B5.17	Coolant Tank Cleaning Interval	2160	720–8760	hours	Planned coolant tank cleaning interval. Fault E227 (Coolant Tank Cleaning Due) when exceeded.
B5.20	Chip Conveyor Auto Enable	1	0–1	—	0 = Chip conveyor starts only on manual switch. 1 =

Param	Name	Default	Range	Unit	Description
					Chip conveyor starts automatically with coolant pump.
B5.21	Chip Conveyor Speed	50	10–100	%	Chip conveyor belt speed as a percentage of maximum (via VFD).
B5.22	Chip Conveyor Interval Enable	0	0–1	—	0 = Chip conveyor runs continuously when enabled. 1 = Chip conveyor runs in intermittent mode (B5.23 on / B5.24 off).
B5.23	Chip Conveyor On Time	30	5–300	seconds	Duration conveyor runs during intermittent mode.
B5.24	Chip Conveyor Off Time	60	10–600	seconds	Duration conveyor is off during intermittent mode.

7.7 Parameter Group B6.xx — Encoder/Feedback Parameters

Param	Name	Default	Range	Unit	Description
B6.01	X-Axis Encoder PPR	2500	100–10000	PPR	X-axis encoder pulses per revolution (pre-quadrature). Heidenhain ERN 1381 = 2500. [*]
B6.02	X-Axis Encoder Quadrature Enable	1	0–1	—	0 = Encoder counts at 1× (PPR). 1 = Encoder counts

Param	Name	Default	Range	Unit	Description
					at $4 \times (\text{PPR} \times 4 = 10000 \text{ counts/rev})$. [*]
B6.03	X-Axis Encoder Type	1	0–1	—	0 = TTL single-ended. 1 = RS-422 differential. [*]
B6.04	X-Axis Encoder Direction	0	0–1	—	0 = Standard direction. 1 = Inverted. Change if axis moves in wrong direction during homing. [*]
B6.05	X-Axis Encoder Signal Monitor	1	0–1	—	0 = Encoder signal quality not monitored. 1 = Signal quality monitored; fault E501 if signal degrades.
B6.06	X-Axis Ballscrew Pitch	10.000	5.000–25.000	mm/rev	Ballscrew pitch (linear distance per revolution). X-axis = 10 mm/rev. [*]
B6.07	X-Axis Pulses per mm	1000.000	—	cts/mm	Calculated: $(\text{B6.01} \times 4 \times \text{B6.02}) / \text{B6.06}$. Read-only display.
B6.08	X-Axis Pitch Error Comp Enable	1	0–1	—	0 = No pitch error compensation. 1 = Pitch error compensation table active. Compensation data loaded at commissioning.
B6.11	Y-Axis Encoder PPR	2500	100–10000	PPR	Y-axis encoder PPR. [*]

Param	Name	Default	Range	Unit	Description
B6.12	Y-Axis Encoder Quadrature Enable	1	0–1	—	Y-axis quadrature enable. [*]
B6.13	Y-Axis Encoder Type	1	0–1	—	Y-axis encoder type. [*]
B6.14	Y-Axis Encoder Direction	0	0–1	—	Y-axis encoder direction. [*]
B6.15	Y-Axis Encoder Signal Monitor	1	0–1	—	Y-axis signal quality monitoring enable.
B6.16	Y-Axis Ballscrew Pitch	10.000	5.000–25.000	mm/rev	Y-axis ballscrew pitch. [*]
B6.18	Y-Axis Pitch Error Comp Enable	1	0–1	—	Y-axis pitch error compensation enable.
B6.21	Z-Axis Encoder PPR	2500	100–10000	PPR	Z-axis encoder PPR. [*]
B6.22	Z-Axis Encoder Quadrature Enable	1	0–1	—	Z-axis quadrature enable. [*]
B6.23	Z-Axis Encoder Type	1	0–1	—	Z-axis encoder type. [*]
B6.24	Z-Axis Encoder Direction	0	0–1	—	Z-axis encoder direction. [*]
B6.25	Z-Axis Encoder Signal Monitor	1	0–1	—	Z-axis signal quality monitoring enable.
B6.26	Z-Axis Ballscrew Pitch	10.000	5.000–25.000	mm/rev	Z-axis ballscrew pitch. [*]
B6.28	Z-Axis Pitch Error Comp Enable	1	0–1	—	Z-axis pitch error compensation enable.
B6.31	Spindle Encoder PPR	4096	512–16384	PPR	Spindle encoder PPR. See B1.38. [*]

Param	Name	Default	Range	Unit	Description
B6.32	Spindle Encoder Quadrature Enable	1	0–1	—	Spindle encoder quadrature. [*]
B6.33	Spindle Encoder Type	1	0–1	—	Spindle encoder differential type. [*]
B6.34	Spindle Encoder Direction	0	0–1	—	Spindle encoder direction. [*]
B6.35	Spindle Encoder Index Enable	1	0–1	—	0 = Spindle index pulse ignored. 1 = Index pulse used for orientation and rigid tap. [*]
B6.36	Encoder Power Supply Voltage	5.0	5.0–5.0	VDC	Encoder supply voltage. Fixed at 5.0 VDC; do not modify. [*]
B6.37	Encoder Low Signal Threshold	0.8	0.5–2.0	V	Signal amplitude below which encoder signal quality is considered poor. Fault E130 (Encoder Signal Low) when exceeded.
B6.38	Encoder Cable Length Max	10.0	1.0–30.0	m	Maximum encoder cable length. Longer cables may require active repeaters.
B6.40	Linear Scale Option Enable	0	0–1	—	0 = Rotary encoder only (standard). 1 = Linear glass scale feedback used (optional full-closed-loop). [*]

7.8 Parameter Group B7.xx — Communication Parameters

Param	Name	Default	Range	Unit	Description
B7.01	Ethernet Enable	1	0–1	—	0 = Ethernet port disabled. 1 = Ethernet port active.
B7.02	IP Address Mode	0	0–1	—	0 = Static IP address (set in B7.03–B7.06). 1 = DHCP (IP assigned by network).
B7.03	IP Address Octet 1	192	0–255	—	First octet of static IP address.
B7.04	IP Address Octet 2	168	0–255	—	Second octet of static IP address.
B7.05	IP Address Octet 3	1	0–255	—	Third octet of static IP address.
B7.06	IP Address Octet 4	100	1–254	—	Fourth octet of static IP address.
B7.07	Subnet Mask Octet 3	255	0–255	—	Third octet of subnet mask (first two octets are 255).
B7.08	Subnet Mask Octet 4	0	0–255	—	Fourth octet of subnet mask.
B7.09	Default Gateway Octet 4	1	1–254	—	Fourth octet of default gateway (first three octets match IP address).
B7.10	DNC Port Number	4001	1024–65535	—	TCP port number for DNC communication.
B7.11	DNC Protocol	0	0–2	—	0 = Raw TCP. 1 = FTP. 2 =

Param	Name	Default	Range	Unit	Description
					BetaCorp DNC protocol.
B7.12	DNC Baud Rate (RS232 fallback)	9600	300–115200	bps	RS-232 baud rate for DNC communication (if Ethernet not available).
B7.13	USB Enable	1	0–1	—	0 = USB port disabled. 1 = USB port active.
B7.14	USB Transfer Mode	0	0–1	—	0 = Manual transfer (operator initiates). 1 = Auto-load (USB inserted → selected program auto-loads).
B7.15	Machine ID	0	0–9999	—	Machine identification number for network identification. Set to machine number in multi-machine DNC systems.
B7.16	Remote Monitor Enable	0	0–1	—	0 = Remote monitoring disabled. 1 = BetaCorp Connect remote monitoring active (requires internet connection).

7.9 Parameter Group B8.xx — Safety and Interlock Parameters

Param	Name	Default	Range	Unit	Description
B8.01	Door Interlock Mode	2	0–2	—	0 = Door open allows full

Param	Name	Default	Range	Unit	Description
					operation (no interlock — NOT recommended) . 1 = Door open halts spindle and rapid traverse. 2 = Door open halts spindle; limits feed to B8.03 (standard safe mode).
B8.02	Door Open Jog Enable	1	0–1	—	0 = No axis motion with door open. 1 = Jog motion allowed with door open at limited speed (B8.03).
B8.03	Door Open Max Speed	1000	100–3000	mm/min	Maximum axis speed allowed with work zone door open.
B8.04	E-Stop Safety Relay Channel	2	1–2	—	Number of safety relay channels used for E-Stop circuit. 2 = Dual-channel (required for safety integrity). Do not change. [*]
B8.05	Over-Travel Relay Action	1	0–1	—	0 = Software only; hardware limit switch disables that axis only. 1 = Hardware limit switch triggers E-Stop on all axes (standard).
B8.06	Spindle Start in Auto	1	0–1	—	0 = Spindle does not start

Param	Name	Default	Range	Unit	Description
					automatically in Auto mode (operator must press spindle start). 1 = Spindle starts automatically per M03/M04 commands (standard).
B8.07	Spindle Max Speed Override Limit	150	100–150	%	Maximum spindle speed override allowed from the operator panel. 150% is the hardware maximum.
B8.08	Feed Rate Max Override Limit	150	100–200	%	Maximum feed rate override allowed from the operator panel.
B8.09	ATC Arm Clear Height	-150.0	—	mm (machine Z)	Minimum Z machine coordinate required before ATC arm can move. If Z is below this value when a tool change is commanded, Z retracts to this position first.
B8.10	Fixture Clamp Interlock	0	0–1	—	0 = No fixture clamp interlock. 1 = Spindle will not start if fixture clamp input (B8.11) is not confirmed.
B8.11	Fixture Clamp Input	0	0–32	—	PLC input number for fixture clamp confirmation

Param	Name	Default	Range	Unit	Description
					signal. 0 = not assigned.
B8.12	Safety Reset Required After E-Stop	1	0–1	—	0 = Machine can restart immediately after E-Stop is released. 1 = Operator must press RESET button after releasing E-Stop (standard).
B8.13	Power Failure Recovery Mode	0	0–1	—	0 = Machine requires manual restart after power failure. 1 = Machine automatically resumes after power restoration (requires careful setup; NOT recommended for production).
B8.14	Safety PLC Watchdog Enable	1	0–1	—	0 = Safety PLC watchdog disabled. 1 = Safety PLC watchdog enabled; machine faults if PLC communication is lost. Do not disable. [*]
B8.15	Safety PLC Watchdog Timeout	500	100–5000	ms	Duration of PLC communication loss before safety watchdog fault E801.
B8.20	Axis Torque	200	100–300	%	X-axis

Param	Name	Default	Range	Unit	Description
	Limit (X)				maximum torque as percentage of rated motor torque. Increase for heavy cutting; decrease for soft material protection.
B8.21	Axis Torque Limit (Y)	200	100–300	%	Y-axis maximum torque limit.
B8.22	Axis Torque Limit (Z)	250	100–350	%	Z-axis maximum torque limit. Higher than X and Y to overcome gravity.
B8.30	Collision Detect Enable	0	0–1	—	0 = Collision detection disabled. 1 = Rapid unexpected torque increase triggers collision fault E540.
B8.31	Collision Detect Threshold	300	150–500	%	Torque spike threshold for collision detection.
B8.32	Collision Detect Dwell	20	5–100	ms	Duration of torque spike above threshold before collision is declared.

7.10 Parameter Group B9.xx — Diagnostic Parameters

Param	Name	Default	Range	Unit	Description
B9.01	Machine Hours	—	Read-only	hours	Total machine

Param	Name	Default	Range	Unit	Description
	Total				operating hours since manufacture.
B9.02	Machine Hours This Session	—	Read-only	hours	Machine hours since last power-on.
B9.03	Spindle Hours Total	—	Read-only	hours	Total spindle running time.
B9.04	Tool Changes Total	—	Read-only	count	Total number of tool changes performed.
B9.05	E-Stop Events Total	—	Read-only	count	Total number of E-Stop events recorded.
B9.06	Fault Count Total	—	Read-only	count	Total number of fault events recorded.
B9.10	Fault Log Size	1000	100–5000	entries	Maximum number of fault log entries. Oldest entries overwritten when full.
B9.11	Fault Log Auto Clear	0	0–1	—	0 = Fault log must be cleared manually. 1 = Fault log automatically cleared on power cycle (not recommended).
B9.12	Diagnostic Screen Update Rate	100	50–1000	ms	Update rate for the diagnostic screen displays.
B9.20	Way Lube Interval	30	5–120	minutes	Interval between centralized way lube pump cycles.
B9.21	Way Lube Pump Time	15	5–60	seconds	Duration of each way lube

Param	Name	Default	Range	Unit	Description
					pump cycle.
B9.22	Way Lube Hours Since Last Oil	—	Read-only	hours	Hours since the way lube oil was last added (reset manually at SYSTEM → MAINT → WAY LUBE RESET).
B9.23	Hydraulic Oil Hours	—	Read-only	hours	Hours since last hydraulic oil change (reset at SYSTEM → MAINT → HYD OIL RESET).
B9.24	Coolant Hours	—	Read-only	hours	Hours since last coolant change (reset at SYSTEM → MAINT → COOLANT RESET).
B9.25	Spindle Bearing Hours	—	Read-only	hours	Spindle hours since last bearing inspection (reset at SYSTEM → MAINT → BEARING RESET).
B9.30	Thermal Compensation X Coefficient	—	Read-only	μm/°C	X-axis thermal compensation coefficient. Set at commissioning ; updated by machine learning algorithm if B9.35 = 1.
B9.31	Thermal Compensation Y Coefficient	—	Read-only	μm/°C	Y-axis thermal compensation coefficient.
B9.32	Thermal Compensation	—	Read-only	μm/°C	Z-axis thermal compensation

Param	Name	Default	Range	Unit	Description
	Z Coefficient				coefficient.
B9.33	Thermal Sensor 1 (Spindle head)	—	Read-only	°C	Current reading from spindle head thermal sensor.
B9.34	Thermal Sensor 2 (Column)	—	Read-only	°C	Current reading from column thermal sensor.
B9.35	Adaptive Thermal Comp Enable	0	0–1	—	0 = Fixed thermal compensation coefficients. 1 = Machine learning mode (requires probing cycles to adapt coefficients).
B9.40	Debug Output Enable	0	0–1	—	0 = Normal operation. 1 = Verbose debug output to serial port (BetaCorp service use only).
B9.41	Test Mode Enable	0	0–1	—	0 = Normal operation. 1 = Test mode active (certain safety interlocks may be temporarily modified). BetaCorp authorized use only. [*]
B9.42	Firmware Version	—	Read-only	—	Current CNC firmware version string.
B9.43	PLC Version	—	Read-only	—	Current PLC firmware version string.
B9.44	HMI Version	—	Read-only	—	Current HMI software version.
B9.45	Control Serial	—	Read-only	—	CNC control

Param	Name	Default	Range	Unit	Description
	Number				unit serial number.
B9.46	Machine Serial Number	—	Read-only	—	Machine serial number (entered at factory during commissioning).
B9.47	Last Calibration Date	—	Read-only	—	Date of last machine calibration (laser interferometer).
B9.48	Next Maintenance Due	—	Read-only	hours	Machine hours remaining until next scheduled maintenance event.
B9.49	Service Password Level	—	—	—	Current active service password level (0–3). Automatically resets to 0 after 30 minutes of inactivity.

End of Chapter 7 — Parameter Reference

PART 7 — FAULT CODES AND TROUBLESHOOTING

Chapter 8 — Fault Codes and Troubleshooting

8.1 Fault System Overview

The BC-500X fault system continuously monitors all machine systems and immediately detects conditions outside normal operating parameters. When a fault is detected:

1. The fault is recorded in the fault log with a timestamp, fault code, and fault description.
2. The appropriate machine response is executed (warning only, controlled stop, or emergency

stop) depending on the fault severity.

3. The fault code and short description are displayed on the main CNC screen status bar.
4. If Parameter B7.16 = 1 (BetaCorp Connect enabled), the fault is transmitted to the BetaCorp remote monitoring system.

Fault Severity Levels:

Level	Color	Machine Response
INFO	White	Message displayed; no action required
WARNING	Yellow	Alert displayed; machine continues operation
FAULT	Orange	Machine completes current block and stops; program paused
ALARM	Red	Machine halts immediately; spindle stops; program aborted
EMERGENCY	Red flashing	Immediate E-Stop equivalent; all drives disabled

Fault Code Numbering:

BC-500X fault codes use the prefix "E" followed by a three-digit number. Codes are grouped by system:

Range	System
E001–E099	Critical system faults
E100–E199	Encoder and feedback faults
E200–E299	Coolant system faults
E300–E399	Tool changer faults
E400–E499	Hydraulic system faults
E500–E599	Axis servo system faults
E600–E699	Communication and I/O faults
E700–E799	PLC and logic faults
E800–E899	Thermal and environmental faults
E900–E999	System configuration faults

Fault Resolution Protocol: For each fault, the operator or technician should:

1. Read the full fault description on the CNC display (press **DETAIL** or **F1** when the fault is highlighted)
2. Refer to Chapter 8 for the specific fault code procedure
3. Identify and correct the root cause

4. Press RESET to clear the fault after correction
5. Log the fault, cause, and corrective action in the maintenance log

8.2 Fault Log Access and Management

Accessing the Fault Log: Navigate to SYSTEM → FAULT LOG. The fault log displays the most recent 1,000 fault events (configurable, Parameter B9.10).

Fault Log Entry Fields:

Field	Description
Timestamp	Date and time of fault occurrence
Fault Code	E-code (e.g., E404)
Description	Short fault description (up to 40 characters)
Severity	INFO / WARNING / FAULT / ALARM / EMERGENCY
Duration	Time fault was active before reset
Reset Method	How fault was cleared (Auto / Manual / Power Cycle)
Machine State	Machine operating mode at time of fault

Exporting Fault Log: The fault log can be exported to a USB drive in CSV format: FAULT LOG → EXPORT → USB → ENTER.

Filtering Fault Log: The fault log can be filtered by:

- Date range
- Fault code range
- Severity level
- System (encoder, coolant, ATC, hydraulic, servo, etc.)

8.3 Fault Code Reference Table — Quick Index

Code	Name	Severity	System
E001	System Watchdog Fault	EMERGENCY	System
E002	Power Supply 24V Low	ALARM	Power
E003	Safety Relay Fault	EMERGENCY	Safety
E004	Dual-Channel Mismatch	EMERGENCY	Safety

Code	Name	Severity	System
E005	CNC to PLC Comm Lost	ALARM	Comms
E006	Parameter Checksum Error	ALARM	System
E007	Parameter Out of Range	FAULT	System
E008	Non-Volatile Memory Fault	ALARM	System
E009	Control Startup Timeout	ALARM	System
E010	Fan Failure — Control Cabinet	WARNING	Thermal
E011	UPS Battery Low	WARNING	Power
E012	System Clock Battery Low	WARNING	Power
E020	E-Stop Active	INFO	Safety
E021	Door Open Interlock	INFO	Safety
E022	Over-Travel Positive X	ALARM	Safety
E023	Over-Travel Negative X	ALARM	Safety
E024	Over-Travel Positive Y	ALARM	Safety
E025	Over-Travel Negative Y	ALARM	Safety
E026	Over-Travel Positive Z	ALARM	Safety
E027	Over-Travel Negative Z	ALARM	Safety
E030	Axis Not Homed (X)	FAULT	System
E031	Axis Not Homed (Y)	FAULT	System
E032	Axis Not Homed (Z)	FAULT	System
E040	Program Not Found	FAULT	Program
E041	Program Syntax Error	FAULT	Program
E042	Tool Number Invalid	FAULT	Program
E043	Work Offset Not Set	WARNING	Program
E050	Fixture Clamp Not Confirmed	FAULT	Safety
E088	Way Lube Level Low	WARNING	Lube
E089	Way Lube Level Critical — Halt	ALARM	Lube
E090	Oil-Air Lube Level Low	WARNING	Lube

Code	Name	Severity	System
E091	Oil-Air Lube Pressure Low	FAULT	Lube
E101	Spindle Encoder Signal Lost	ALARM	Encoder
E102	Spindle At-Speed Timeout	FAULT	Spindle
E103	Spindle Orientation Timeout	FAULT	Spindle
E104	Spindle Brake Timeout	FAULT	Spindle
E105	Spindle Motor Overload	FAULT	Spindle
E106	Rigid Tap Overshoot	FAULT	Spindle
E107	Spindle Temperature High	WARNING	Thermal
E110	X-Axis Encoder Signal Lost	ALARM	Encoder
E111	Y-Axis Encoder Signal Lost	ALARM	Encoder
E112	Z-Axis Encoder Signal Lost	ALARM	Encoder
E120	Spindle Encoder Index Lost	FAULT	Encoder
E121	X-Axis Encoder Index Lost	WARNING	Encoder
E122	Y-Axis Encoder Index Lost	WARNING	Encoder
E123	Z-Axis Encoder Index Lost	WARNING	Encoder
E130	Encoder Signal Amplitude Low	WARNING	Encoder
E131	Encoder Cable Continuity Fault	ALARM	Encoder
E132	Encoder Power Supply Fault	ALARM	Encoder
E140	Scale Feedback Lost (optional)	ALARM	Encoder
E150	Encoder Frequency Exceeded	FAULT	Encoder
E202	Coolant Level Below Minimum	FAULT	Coolant
E210	Coolant Pump Motor Overload	FAULT	Coolant

Code	Name	Severity	System
E211	Coolant Pump Thermal Overload	ALARM	Coolant
E220	Coolant Pressure Low	WARNING	Coolant
E221	Coolant Temperature High Warning	WARNING	Coolant
E222	Coolant Temperature High Shutdown	ALARM	Coolant
E223	Coolant Pump Overload	FAULT	Coolant
E224	TSC Pressure Low	FAULT	Coolant
E225	TSC No Flow	FAULT	Coolant
E226	Coolant Change Due	INFO	Maint.
E227	Coolant Tank Cleaning Due	INFO	Maint.
E230	Chip Conveyor Overload	FAULT	Convey.
E231	Chip Conveyor Jam	FAULT	Convey.
E301	ATC Arm Extend Timeout	ALARM	ATC
E302	ATC Arm Rotate Timeout	ALARM	ATC
E303	Tool Changer Timeout	ALARM	ATC
E304	ATC Carousel Index Timeout	ALARM	ATC
E305	ATC Spindle Orient Timeout	FAULT	ATC
E306	Tool Clamp Not Confirmed	ALARM	ATC
E307	Tool Not Present in Carousel	FAULT	ATC
E308	ATC Arm Not at Home on Start	FAULT	ATC
E309	ATC Cam Sensor Mismatch	ALARM	ATC
E310	ATC Carousel Home Not Found	ALARM	ATC
E311	ATC Pot Count Mismatch	FAULT	ATC
E320	Tool Seating Fault	ALARM	ATC
E321	Tool Unclamp Timeout	ALARM	ATC
E330	Gear Change Solenoid	FAULT	Spindle

Code	Name	Severity	System
	Fault		
E331	Gear Change Confirm Timeout	FAULT	Spindle
E340	ATC Lube System Low	WARNING	Lube
E404	Hydraulic Pressure Low	ALARM	Hydraulic
E410	Hydraulic Pump Overload	ALARM	Hydraulic
E411	Hydraulic Temperature High Warning	WARNING	Hydraulic
E412	Hydraulic Temperature High Shutdown	ALARM	Hydraulic
E413	Hydraulic Level Low	WARNING	Hydraulic
E414	Hydraulic Filter Bypass	WARNING	Hydraulic
E415	Z-Brake Release Timeout	ALARM	Hydraulic
E416	Z-Brake Apply Timeout	ALARM	Hydraulic
E417	Hydraulic Oil Change Due	INFO	Maint.
E418	Hydraulic Filter Change Due	INFO	Maint.
E420	Solenoid Valve Fault (HYD-01)	ALARM	Hydraulic
E421	Solenoid Valve Fault (HYD-02)	ALARM	Hydraulic
E422	Solenoid Valve Fault (HYD-03)	ALARM	Hydraulic
E423	Solenoid Valve Fault (HYD-04)	ALARM	Hydraulic
E430	Hydraulic Return Filter Blocked	WARNING	Hydraulic
E501	X-Axis Following Error	ALARM	Servo
E502	Y-Axis Following Error	ALARM	Servo
E503	Z-Axis Following Error	ALARM	Servo
E504	X-Axis Stall Detected	ALARM	Servo
E505	Axis Servo Overload	ALARM	Servo
E506	Y-Axis Stall Detected	ALARM	Servo

Code	Name	Severity	System
E507	Z-Axis Stall Detected	ALARM	Servo
E510	X-Axis Drive Fault	ALARM	Servo
E511	Y-Axis Drive Fault	ALARM	Servo
E512	Z-Axis Drive Fault	ALARM	Servo
E515	ATC Carousel Drive Fault	ALARM	ATC
E520	X-Axis Motor Thermal	WARNING	Servo
E521	Y-Axis Motor Thermal	WARNING	Servo
E522	Z-Axis Motor Thermal	WARNING	Servo
E530	Drive DC Bus Overvoltage	ALARM	Servo
E531	Drive DC Bus Undervoltage	ALARM	Servo
E532	Drive Input Phase Loss	ALARM	Servo
E533	Drive Regenerative Overload	ALARM	Servo
E540	Collision Detected	ALARM	Servo
E601	Ethernet Connection Lost	WARNING	Comms
E602	DNC Communication Timeout	WARNING	Comms
E603	USB Device Error	INFO	Comms
E610	PLC I/O Module Comm Fault	ALARM	PLC
E620	CNC Control Comm Fault	ALARM	Comms
E701	PLC Program Fault	ALARM	PLC
E702	PLC Watchdog Timeout	EMERGENCY	PLC
E703	PLC I/O Configuration Error	ALARM	PLC
E710	Safety PLC Fault	EMERGENCY	Safety
E720	Output Short Circuit (24VDC)	ALARM	PLC
E730	Input Signal Fault	WARNING	PLC
E801	Control Cabinet Temperature High	WARNING	Thermal
E802	Control Cabinet Temperature Critical	ALARM	Thermal
E810	Main Cabinet Fan Fault	WARNING	Thermal

Code	Name	Severity	System
E820	Ambient Temperature High	WARNING	Thermal
E821	Ambient Temperature Low	WARNING	Thermal
E830	HPU Cooling Fan Fault	WARNING	Thermal
E901	Parameter Revision Mismatch	WARNING	Config
E902	Machine Configuration Error	ALARM	Config
E910	Firmware Version Mismatch	WARNING	Config
E920	Factory Calibration Data Missing	ALARM	Config
E930	License Expiry Warning	WARNING	Config
E940	Maintenance Password Incorrect	INFO	Config
E950	Data Backup Required	INFO	Config

8.4 Critical Faults (E001–E099)

E001 — System Watchdog Fault

Severity: EMERGENCY **Machine Response:** Immediate E-Stop equivalent; all drives disabled

Description: The system watchdog is a hardware timer that must be periodically reset by the main control software. If the software fails to reset the watchdog within the timeout period (typically 500 ms), the watchdog hardware triggers an emergency stop. This fault indicates a serious software or hardware failure in the CNC control.

Possible Causes:

1. CNC control software has crashed or locked up
2. CNC control processor is overloaded (may occur with extremely large programs on older firmware)
3. Hardware fault in the CNC control unit (processor board, memory, or communication bus)
4. Electrical noise interfering with CNC control unit
5. Power supply voltage to CNC control unit out of specification

Corrective Steps:

1. Attempt to power-cycle the machine (turn off main disconnect, wait 30 seconds, power on).

2. If E001 recurs immediately on power-on: suspect CNC control hardware fault. Do not attempt to operate the machine. Contact BetaCorp Systems Technical Support.
3. If E001 occurs only during heavy machining: check that the control cabinet air conditioning is functioning and the cabinet temperature is below 50°C. Overheating can cause processor instability.
4. If E001 occurs intermittently: inspect electrical grounding (see Section 4.5.2). Check for loose connections on the CNC control unit and servo drives.
5. Update CNC firmware to the latest version (available at docs.betacorp-systems.com). Firmware updates often include stability improvements.
6. If all of the above fail: the CNC control unit may need to be replaced. Contact BetaCorp Systems.

Safety Note: After E001, all axis positions are potentially compromised. The machine **MUST** be homed before resuming automatic operation.

E006 — Parameter Checksum Error

Severity: ALARM **Machine Response:** Machine halts; parameters cannot be trusted

Description: The CNC control stores all parameters in non-volatile memory with a checksum. If the checksum does not match the stored data on power-up, E006 is set. This indicates that one or more parameters may have been corrupted.

Possible Causes:

1. Power failure during a parameter write operation
2. Battery failure in the CNC control (battery maintains non-volatile memory)
3. Electrostatic discharge (ESD) event that corrupted memory
4. Hardware fault in CNC control non-volatile memory

Corrective Steps:

1. Load the parameter backup from USB drive: SYSTEM → PARAMETER → RESTORE → USB. Select the most recent backup file.
2. If no USB backup is available, contact BetaCorp Systems for the factory default parameter file for your machine serial number.
3. After parameter restore, verify key parameters (B1.01, B2.01, B3.02, B6.01, B6.11, B6.21) match expected values.
4. If E006 recurs after parameter restore: suspect failing non-volatile memory or control board. Contact BetaCorp Systems.
5. Replace the control board battery (Part No. BC-BATT-CNC-001) if battery voltage is below 2.7 V (check at SYSTEM → DIAGNOSTIC → BATTERY VOLTAGE).

Related Parameters: B9.10, B9.11

E022 through E027 — Over-Travel Faults

Severity: ALARM **Machine Response:** Axis motion halts immediately; spindle stops

Description: An over-travel fault occurs when an axis reaches a hardware limit switch beyond the software travel limits. This should not occur during normal operation if software limits (B2.03, B2.04, B2.13, B2.14, B2.23, B2.24) are correctly set.

Possible Causes:

1. Software limits set incorrectly (limits set too close to or beyond hardware switches)
2. Work coordinate system set incorrectly (program commanding motion outside machine envelope)
3. Tool length offset error causing Z-axis to travel too deep
4. Homing position not correctly established
5. Hardware limit switch failed in the open state (rare; would cause false over-travel indication at any position)

Corrective Steps:

1. Do NOT attempt to force the axis off the limit switch by pressing any button other than the designated recovery procedure.
2. To recover: switch to JOG mode. The control will allow limited manual jogging away from the limit switch (in the direction away from the switch) at very low speed. Press the JOG button in the direction away from the over-travel condition.
3. After clearing the limit switch, investigate the cause:
 - Check software limits (Parameter B2.03–B2.24).
 - Check work coordinate setup.
 - Check tool length offsets.
 - Verify homing was completed correctly.
4. If a hardware limit switch has failed, replace it (limit switch Part No.: BC-SW-OTRV-001).
5. After recovery, perform a complete homing cycle before resuming automatic operation.

Safety Note: Over-travel in the Z-axis (E026 or E027) may leave the tool engaged with the workpiece. Carefully inspect the work zone before attempting to recover Z-axis motion.

8.5 Encoder and Feedback Faults (E100–E199)

8.5.1 E101 — Spindle Encoder Signal Lost

Fault Code: E101 **Fault Name:** Spindle Encoder Signal Lost **Severity:** ALARM **Machine**

Response: Spindle drive immediately disabled; spindle coasts to stop (or applies brake if B1.22 =

1); machining program halted

Description: Fault E101 is triggered when the CNC control or spindle drive detects a loss of the spindle encoder signal. The spindle encoder (Heidenhain ROD 420, 4096 PPR) provides the speed feedback necessary for closed-loop spindle speed control, rigid tapping synchronization, and spindle orientation for tool changes. Loss of this signal prevents all spindle operations until the fault is resolved.

IMPORTANT: E101 on the BC-500X indicates SPINDLE ENCODER SIGNAL LOST. This is a completely different fault from voltage or electrical faults that use the E101 code on other machine models. E101 on the BC-500X is exclusively an encoder/feedback fault.

Fault Trigger Conditions: E101 is triggered when ANY of the following conditions persist for more than the duration set by Parameter B2.48 (default 200 ms):

- Both A and B quadrature channels of the spindle encoder read low simultaneously (not a valid state)
- Differential voltage on the encoder RS-422 signal falls below the threshold set by B6.37 (default 0.8 V)
- Encoder cable continuity is broken (detected by the RS-422 line terminator monitoring)
- The spindle drive reports encoder fault via the drive-to-control communication bus

Possible Causes:

1. **Encoder cable damaged or disconnected:** The encoder cable (Part No. BC-CBL-SPENC-3M) routes from the spindle motor, through the Z-axis energy chain (cable carrier), to the servo/spindle drive cabinet. This cable is subject to bending fatigue from Z-axis motion. Cable failures are the most common cause of E101 on machines with more than 5,000 hours of operation.
 - Cable damage typically appears at the energy chain entry/exit points where flexing is greatest
 - Symptoms: E101 appears intermittently at first, then permanently
2. **Encoder connector loose or corroded:** The 12-pin circular connector at the encoder end or the cabinet end of the cable may become loose due to vibration, or the pins may corrode due to coolant infiltration.
 - Symptoms: E101 appears when machine vibrates heavily (during spindle start, or at certain speeds)
3. **Encoder unit failed:** The Heidenhain ROD 420 encoder unit itself may fail due to:
 - Optical disc contamination (coolant mist ingress)
 - Bearing failure in the encoder shaft
 - Electronic failure of the encoder signal processing board
 - Symptoms: E101 permanent; replacing cable does not resolve

4. **Spindle drive input circuit fault:** The spindle drive's encoder input circuit may have failed, causing false E101 indication even though the encoder is functioning correctly.
 - Symptoms: E101 even after encoder and cable are verified good; other encoder-based drives on the same machine work correctly
5. **Electromagnetic interference (EMI) on encoder cable:** High-frequency EMI from the spindle drive or servo drives can corrupt the encoder signal, especially if the encoder cable is routed near power cables.
 - Symptoms: E101 appears at specific spindle speeds; correlates with drive switching frequency
 - More likely if recent cable replacement used non-shielded cable or shield was not properly connected

Corrective Steps:

WARNING: All encoder inspection work requires LOTO (lockout/tagout). Refer to Section 2.4.2.

Step 1: Document and Record

- Record the exact conditions when E101 occurred: spindle speed, machine mode, recent history.
- Note whether E101 is permanent (present even when spindle is stopped) or appears only when the spindle is running.
- Check the fault log: SYSTEM → FAULT LOG → filter for E101. Note the frequency of occurrence.

Step 2: Check Encoder Cable at Energy Chain

1. Lock out the machine per Section 2.4.2.
2. Open the Z-axis energy chain cover (refer to machine drawings for energy chain location — typically on left side of Z-axis column).
3. Visually inspect the encoder cable along its full length through the energy chain.
4. Look for:
 - Kinks, crushing, or cuts in the cable jacket
 - Abraded sections where the cable contacts the energy chain links
 - Discoloration from heat or chemical exposure
5. Pay special attention to the first and last 300 mm of the cable at the energy chain entry/exit points — this is where bending fatigue damage occurs.
6. If any damage is found: replace the encoder cable with Part No. BC-CBL-SPENC-3M. Refer to Section 9.12 for the encoder cable replacement procedure.

Step 3: Check Encoder Connectors

1. With the machine locked out, locate both encoder cable connectors:

- Encoder end: circular 12-pin connector at the rear of the spindle motor
 - Drive end: rectangular 20-pin connector on the spindle drive unit in the cabinet
2. Check the encoder-end connector:
 - Verify the connector is fully seated. Press firmly and check that the locking ring is fully engaged.
 - Inspect the connector face for coolant residue, corrosion, or bent pins.
 - Clean the connector face with contact cleaner (CRC QD Electronics Cleaner or equivalent) if contamination is found.
 3. Check the drive-end connector:
 - Same inspection process.
 - Verify no pins are pushed back into the connector housing.
 4. If connector pins show corrosion or the connector housing is cracked: replace the connector.
 - Encoder-end replacement connector: BC-CON-ENC-12P
 - Drive-end replacement connector: BC-CON-ENC-DRV

Step 4: Measure Encoder Signal with Oscilloscope

For definitive diagnosis, measure the encoder signal directly:

1. Obtain a 100 MHz or faster oscilloscope with differential probes.
2. With the machine powered on (but spindle stopped and in a safe state with the work zone door closed):
3. At the drive-end connector, probe the following signals:
 - A+ and A- (differential pair)
 - B+ and B- (differential pair)
4. Manually rotate the spindle slowly by hand (safe only when powered off in LOTO — alternatively, command a slow spindle speed of 50 RPM in JOG mode):
 - Expected signal: clean square wave, differential amplitude > 2.0 V peak-to-peak
 - If amplitude is below 1.5 V: suspect encoder unit failure or poor connection
 - If signal is present but noisy: suspect EMI or poor shielding
 - If no signal at all: open circuit in cable or failed encoder

Step 5: Test Encoder Output at Encoder End

1. Lock out the machine.
2. Disconnect the cable from the encoder body.
3. Using the bench power supply, apply 5 VDC $\pm$ 5% to the encoder power pins (see encoder datasheet for pin assignment — available from BetaCorp).
4. Slowly rotate the encoder shaft by hand.

5. Measure the A+/A- and B+/B- outputs with the oscilloscope.
6. If signals are clean and correct amplitude at the encoder end but not at the drive end: the cable is faulty.
7. If signals are not correct at the encoder end: the encoder unit has failed. Replace with BC-ENC-SP-001.

Step 6: Verify Cable Shield Connection

Encoder cable shield must be connected at the drive end only (floating at the encoder end):

1. At the drive end connector, verify the shield drain wire is connected to the drive chassis ground (or encoder input chassis ground, per the BC-500X wiring diagram).
2. At the encoder end connector, verify the shield drain wire is NOT connected to anything (floating).
3. If both ends are connected (shield grounded at both ends), this creates a ground loop that can cause EMI-induced encoder faults.

Step 7: Check Spindle Drive Encoder Input

If the encoder and cable test correctly, the spindle drive encoder input may be faulty:

1. Using the spare encoder cable (if available), temporarily connect a known-good encoder to the drive inputs.
2. Power on the machine and run the spindle at 500 RPM.
3. If E101 does not occur with the test encoder: the original encoder is faulty.
4. If E101 still occurs with the known-good encoder: the drive encoder input is faulty. The spindle drive unit must be replaced or sent for repair. Contact BetaCorp Systems for drive exchange options.

Step 8: Verify Encoder Cable Routing (EMI)

If E101 appears only at specific spindle speeds:

1. Check the routing of the encoder cable through the energy chain and machine frame.
2. The encoder cable must be physically separated from power cables (servo motor cables, spindle motor cable) by at least 200 mm, or run in a separate grounded metallic conduit.
3. If the encoder cable runs parallel to power cables over a long distance, re-route it.
4. Installing ferrite cores on the encoder cable (at the drive end and encoder end) can reduce high-frequency EMI. Use Fair-Rite type 2643 or equivalent, 3 turns through a 25 mm OD core.

Related Parameters:

- B1.38 — Spindle Encoder PPR (verify = 4096)
- B1.39 — Spindle Encoder Type (verify = 1 for differential RS-422)
- B6.31 — Spindle Encoder PPR (cross-reference parameter)

- B6.37 — Encoder Low Signal Threshold
- B2.48 — Encoder Lost Dwell Timeout

Related Parts:

- BC-CBL-SPENC-3M — Spindle encoder cable, 3 m
- BC-ENC-SP-001 — Spindle encoder (Heidenhain ROD 420)
- BC-CUP-SP-001 — Spindle encoder shaft coupling
- BC-CON-ENC-12P — Encoder connector (encoder end, 12-pin)
- BC-CON-ENC-DRV — Encoder connector (drive end)

Safety Note: After resolving E101, the spindle orientation position must be verified before performing automatic tool changes. Command M19 (spindle orient) and verify the spindle key is at the correct position for ATC operation. If orientation is incorrect, adjust Parameter B1.19 (Spindle Orientation Angle).

E110 — X-Axis Encoder Signal Lost

Severity: ALARM **Machine Response:** All axes halt; spindle stops; program aborted

Description: Loss of X-axis encoder signal (Heidenhain ERN 1381, 2500 PPR). The X-axis encoder provides position feedback for closed-loop servo control. Loss of this signal causes immediate axis disable.

Possible Causes:

1. X-axis encoder cable (BC-CBL-ENC-5M) damaged or disconnected
2. X-axis encoder connector loose at drive or encoder end
3. X-axis encoder unit failed (Part No. BC-ENC-X-001)
4. X-axis servo drive encoder input failed
5. EMI on encoder cable from X-axis servo motor cable

Corrective Steps: Follow the same diagnostic procedure as E101 (Section 8.5.1), adapted for the X-axis encoder and cable. Key differences:

- X-axis encoder cable routes through the X-axis cable tray on the right side of the machine base
- X-axis encoder connector at drive end connects to the X-axis servo drive (BetaCorp SD-300)
- Inspect the cable at the entry/exit points of the X-axis cable tray

After resolution: The X-axis must be re-homed before automatic operation.

E111 — Y-Axis Encoder Signal Lost

Severity: ALARM **Machine Response:** All axes halt; spindle stops; program aborted

Description and Corrective Steps: Same procedure as E110, adapted for Y-axis. Y-axis encoder cable routes under the machine base through the Y-axis cable tray.

E112 — Z-Axis Encoder Signal Lost

Severity: ALARM **Machine Response:** All axes halt; spindle stops; Z-axis brake applies; program aborted

Description: Loss of Z-axis encoder signal. The Z-axis encoder is especially critical because the Z-axis has a gravity load of 280 kg (spindle head). Loss of encoder signal is handled more aggressively than X and Y faults.

WARNING: When E112 occurs, the Z-axis brake applies immediately. Do NOT attempt to manually jog the Z-axis until the encoder signal is restored and the drive is re-enabled.

Corrective Steps: Same procedure as E110, adapted for Z-axis. Key considerations for Z-axis:

- Z-axis encoder cable routes through the Z-axis energy chain (most susceptible to bending fatigue on the BC-500X)
 - Z-axis energy chain bending radius is larger than X/Y; inspect for crushing at chain ends
 - After E112 resolution: Z-axis must be re-homed. The Z-axis home procedure must be followed carefully — verify Z-axis brake is released before commanding motion.
-

8.6 Coolant System Faults (E200–E299)

8.6.1 E202 — Coolant Level Below Minimum

Fault Code: E202 **Fault Name:** Coolant Level Below Minimum **Severity:** FAULT (configurable to ALARM via B5.04) **Machine Response:** Alert displayed; machine continues operation (default); if B5.04 = 1, machine halts

Description: Fault E202 is triggered when the coolant tank level switch (a float switch in the coolant tank) indicates that the coolant level has dropped to or below the minimum acceptable level. The minimum level is approximately 80 mm below the "FULL" mark on the coolant sight gauge, corresponding to a remaining coolant volume of approximately 250 liters (out of 350 liters total capacity).

IMPORTANT: E202 on the BC-500X means COOLANT LEVEL BELOW MINIMUM — a coolant volume issue. This code is entirely different from encoder or electrical faults that may use E202 on other machine models. On the BC-500X, E202 is always a coolant level fault.

Coolant level may drop due to:

- Normal evaporation of water from the coolant mixture (especially in warm, low-humidity environments)
- Coolant carry-out (coolant dragged out on the workpiece and chips)
- Coolant leaks from fittings, hoses, or the coolant pump seal
- Excessive misting causing coolant to be exhausted through the mist collector

Fault Trigger Conditions: The float switch (Part No. BC-SW-CLT-LVL) opens when the coolant level drops below the minimum setpoint. The fault requires the float switch to be open for longer than the duration specified by Parameter B5.03 (default 30 seconds) before E202 is triggered. This prevents false faults from coolant sloshing during rapid axis moves.

Possible Causes:

1. **Coolant level genuinely low:** The coolant has evaporated or been carried out. The sight glass will show the level is at or below the "LOW" mark.
 - This is the most common cause of E202
 - Requires adding coolant mixture to restore level
2. **Float switch stuck in the "low" position:** The float switch mechanism may be stuck due to accumulated chips, sludge, or mineral deposits fouling the float arm pivot.
 - The sight glass will show the level is at normal or near-normal
 - Float switch is mechanically stuck indicating low
3. **Float switch failed (normally-open contact):** The float switch contacts may have failed open, giving a persistent E202 indication even with adequate coolant.
 - Sight glass shows normal level
 - Manually pressing the float up (into the tank) should clear E202 temporarily if it is a float jam — if E202 persists even with float pushed up, the switch has failed
4. **Coolant leak:** A leak in the coolant circuit is draining the tank faster than evaporation. Check for:
 - Puddles under the machine (may be on the floor or absorbed by chips)
 - Wet coolant pump shaft seal area (pump seal has failed)
 - Loose or leaking hose fittings
 - Coolant tank crack or weld failure (rare)
5. **Wiring fault:** Open circuit in the float switch wiring from the switch to the PLC. The PLC input for the coolant level switch reads "open" regardless of actual switch state.

Corrective Steps:

Step 1: Check Coolant Level at Sight Glass

Visually inspect the coolant sight glass on the right side of the coolant tank:

- If level is at or below the "LOW" mark: coolant level IS low. Proceed to Step 2 (add coolant).
- If level is above the "LOW" mark (appears normal): the float switch may be faulty or stuck. Proceed to Step 4.

Step 2: Add Coolant

If the coolant level is genuinely low:

1. Prepare the correct coolant mixture. The current coolant type and target concentration are listed in the maintenance log and in Parameter B5.13 (coolant concentration target).
2. Mix the coolant concentrate with water at the target concentration (typically 7–8% for general machining). See Chapter 12, Section 12.2 for mixing procedure.
3. Add the prepared coolant through the tank fill opening (top of the coolant tank, accessible without opening the tank).
4. Add coolant slowly while watching the sight glass. Stop when the level reaches the "FULL" mark.
5. Typical volume to add from "LOW" to "FULL": approximately 80–100 liters.
6. Press RESET to clear E202.
7. Verify the coolant concentration with a refractometer after adding coolant, and adjust if necessary.

Step 3: Investigate Why Coolant Level Was Low

After restoring the coolant level, determine the cause of the level drop:

1. **Normal evaporation and carry-out:** If the level drop is consistent with expected evaporation/carry-out rates (check coolant management log), no further action is required. Update the coolant replenishment schedule.
2. **Elevated carry-out:** Check if chip conveyor is carrying more coolant than normal (chips are very wet). Consider reducing coolant flow rate or redirecting nozzles.
3. **Leak investigation:** Walk around the machine and look for coolant dripping or pooling. Check the coolant pump shaft area for leaks. Check all nozzle and hose connections. If a leak is found, repair before returning machine to service.

Step 4: Inspect Float Switch (if level was normal)

1. Lock out the machine (LOTO, Section 2.4.2).
2. Remove the coolant tank access cover (6 × M6 hex socket screws).
3. Locate the float switch in the tank. The float is a round polyethylene ball on an arm, approximately 200 mm below the tank top.
4. Inspect the float:
 - Manually lift the float arm (upward = full tank position). If the float arm moves freely and E202 clears (check after restoring power): the float arm was fouled/stuck.

- If E202 does not clear when the float is lifted to the full position: the float switch contacts have failed open, or there is a wiring fault.
5. Clean the float arm pivot with a small stiff brush. Remove all accumulated sludge.
 6. Verify the float arm moves freely through its full range of motion.
 7. If the float switch is damaged or contacts are faulty: replace with Part No. BC-SW-CLT-LVL.

Step 5: Check Float Switch Wiring

1. With the machine locked out, trace the float switch cable from the switch to the PLC I/O module.
2. At the PLC I/O module (in the main electrical cabinet), identify the coolant level input (refer to the I/O map, Appendix D — Electrical Schematic Index). The typical input address is labeled "CLT_LEVEL_LOW" in the PLC.
3. Using a multimeter set to resistance, measure the resistance across the float switch terminals (at the PLC end of the cable, with the cable disconnected from the PLC):
 - Float in "up" position (full tank): resistance should be close to 0 ohms (switch closed / normally-closed contact)
 - Float in "down" position (low tank): resistance should be open circuit (or very high, >10 kohm)
4. If the measurement does not match expected states: replace the float switch (BC-SW-CLT-LVL) and/or the cable (BC-CBL-CLT-LVL-2M).

Step 6: Check PLC Input

1. At the PLC diagnostic screen (SYSTEM → DIAGNOSTIC → PLC I/O), find the coolant level input.
2. Observe the input state while manually moving the float switch:
 - Input should change state as the float is moved up and down
 - If the input does not change: the PLC input may be faulty (contact BetaCorp Systems)

Related Parameters:

- B5.02 — Coolant Level Min Alarm (enable/disable)
- B5.03 — Coolant Level Min Delay (seconds before alarm triggers)
- B5.04 — Coolant Level Halt (0 = continue, 1 = halt on E202)

Related Parts:

- BC-SW-CLT-LVL — Coolant level float switch
- BC-CBL-CLT-LVL-2M — Coolant level switch cable, 2 m

Safety Note: Always check the coolant concentration (using a refractometer) after adding coolant. If water was added without concentrate, the coolant is diluted and will not provide adequate

corrosion protection or tool life. Adjust concentration to target value (B5.13) before resuming production.

E210 — Coolant Pump Motor Overload

Severity: FAULT **Machine Response:** Coolant pump stops; program paused; alarm displayed

Description: The coolant pump motor thermal overload relay has tripped, indicating the pump motor is drawing excessive current. This is typically caused by pump impeller blockage, pump seal failure causing mechanical binding, or extended operation with the outlet valve restricted.

Corrective Steps:

1. Turn off the coolant pump at the operator panel.
 2. Wait 10 minutes for the thermal overload to cool and automatically reset.
 3. Inspect the coolant pump inlet strainer for blockage (Section 9.2, daily maintenance item).
 4. If blockage is found: clean the strainer and restart the pump.
 5. If overload persists after strainer cleaning: inspect the pump impeller for chip jamming (LOTO required; remove pump inlet cover).
 6. If pump motor is faulty: replace motor assembly (BC-PMP-CLT-001).
-

E221 — Coolant Temperature High Warning

Severity: WARNING **Machine Response:** Alert displayed; machine continues

Description: Coolant temperature has exceeded the warning threshold (default 40°C, Parameter B5.05). Coolant temperature can rise due to heavy machining operations, inadequate coolant flow, or a hot ambient environment.

Corrective Steps:

1. Verify coolant flow rate is adequate. Check that all nozzle ball valves are open.
 2. Verify coolant concentration is within the specified range (7–10%). Diluted coolant has reduced cooling capacity.
 3. If coolant temperature continues to rise: reduce machining intensity or allow coolant to cool by running the pump without cutting for 10 minutes.
 4. Consider adding a coolant chiller (BC-OPT-CHILLER) for high-duty-cycle applications in warm environments.
-

E222 — Coolant Temperature High Shutdown

Severity: ALARM **Machine Response:** Machine halts; coolant pump stops; program aborted

Description: Coolant temperature has exceeded the shutdown threshold (default 45°C, Parameter

B5.06). Continued operation would risk coolant degradation and reduced tool life.

Corrective Steps:

1. Allow coolant to cool with pump stopped for 20 minutes.
 2. Investigate the cause of elevated temperature (see E221 corrective steps).
 3. After temperature drops below B5.05 - 5°C, press RESET and resume.
-

8.7 Tool Changer Faults (E300–E399)

8.7.1 E303 — Tool Changer Timeout

Fault Code: E303 **Fault Name:** Tool Changer Timeout **Severity:** ALARM **Machine Response:** All ATC motion halts immediately; spindle stops; program halted at current tool change position

Description: Fault E303 is triggered when the ATC arm fails to reach the retracted (home) position within the timeout period specified by Parameter B3.12 (default 2000 ms). The ATC arm retract is the final motion in the tool change sequence — it occurs after the arm has rotated 180° (swapping tools) and must pull back to the rest position before the spindle can be lowered to the work position.

E303 specifically monitors the arm retract phase. If the arm extends (E301) or rotates (E302) without reaching the retracted position in time, those faults trigger before E303. E303 therefore typically indicates the arm extended and rotated correctly but failed on the retract.

IMPORTANT: E303 on the BC-500X is a TOOL CHANGER TIMEOUT fault — it refers to the physical ATC arm mechanism failing to complete its retraction motion. This is categorically different from any E303 code on other machine models that may refer to spindle speed or other system parameters.

Fault Trigger Conditions: The PLC monitors the ATC arm retract position sensor (SEN-ATC-RET, Parameter B3.08). The sensor must transition from "not detected" to "detected" within B3.12 milliseconds after the retract hydraulic solenoid (SOL-HYD-04) is energized. If the sensor does not confirm retraction within this window, E303 is triggered.

Possible Causes:

1. **ATC arm hydraulic cylinder sticking or leaking:** The ATC arm retract/extend cylinder (BC-CYL-ATC-EXT) may not be generating enough force to retract the arm fully. This can be caused by:
 - Internal cylinder seal wear causing internal leakage (insufficient force)
 - External hydraulic line restriction (kinked or blocked return line)
 - Hydraulic solenoid valve (SOL-HYD-04) not fully opening
2. **ATC arm mechanical obstruction:** A chip, broken tool fragment, or coolant-frozen grease may be blocking the arm retract motion.
 - Inspect the ATC arm housing for foreign material in the arm retract path.
3. **ATC arm retract position sensor misaligned or failed:** The SEN-ATC-RET sensor may

have shifted out of position and no longer detects the arm in the retract position, even if the arm is physically retracted.

- If the arm appears to be retracted but E303 is triggered: suspect sensor misalignment.
4. **Low hydraulic pressure:** If the hydraulic system pressure is near or below the E404 alarm threshold, the ATC arm cylinder may not have sufficient force to retract against any resistance.
 - Check hydraulic pressure gauge on HPU before investigating ATC mechanism.
 5. **ATC grease buildup:** Insufficient or excessive lubrication of the ATC arm cam follower and guide surfaces can cause the arm to bind on retraction.
 - Excessive old dried grease is more commonly a problem than insufficient lubrication on older machines.
 6. **Parameter B3.12 set too short:** In rare cases, if the hydraulic system is slightly slow (cold oil, pump wear), the default 2000 ms timeout may not be long enough for a properly functioning system. Increase B3.12 to 2500 ms and observe if E303 clears.

Corrective Steps:

WARNING: The ATC arm may be in an intermediate position when E303 is triggered. There may be a tool holder gripped by both the ATC arm and the carousel, or a tool holder being held only by the ATC arm without being secured in the spindle or carousel. Do NOT attempt to manually force the ATC arm. LOTO the machine before inspecting the ATC area.

Step 1: Assess the Machine State

Before locking out the machine, use the diagnostic screen (SYSTEM → DIAGNOSTIC → ATC STATUS) to determine the ATC arm position:

- Note which sensors are active (SEN-ATC-HOME, SEN-ATC-90, SEN-ATC-180, SEN-ATC-EXT, SEN-ATC-RET)
- Note which tool was in the spindle and which tool was being loaded
- Note the carousel position (which pot is at the ATC position)

This information is essential for restoring the ATC to a known state after the fault is resolved.

Step 2: Lock Out the Machine

Follow the complete LOTO procedure (Section 2.4.2). Include hydraulic system de-pressurization (Section 2.6.2).

Step 3: Inspect the ATC Arm Physically

1. Remove the ATC area access guard (secured with 4 × M6 hex socket screws).
2. Visually inspect the ATC arm in its current position.
3. Check for:
 - Any tool holder being gripped by the ATC arm without being secured at either end (DANGEROUS — the tool holder may fall when the arm is moved)

- Chips or debris blocking the arm retract path
- Visible damage to the arm or cam follower

Step 4: Secure Loose Tool Holders

If a tool holder is gripped only by the ATC arm (not in spindle or carousel):

1. Manually support the tool holder to prevent it from falling.
2. Carefully guide the tool holder either into the carousel pot or onto a safe surface.
3. Do NOT let the tool holder fall — this will damage the toolholder and potentially the ATC arm.

Step 5: Check the ATC Arm for Mechanical Obstruction

1. With hydraulic pressure released (LOTO complete), attempt to manually push the ATC arm in the retract direction.
2. The arm should move relatively freely (requires modest force against the cam spring).
3. If the arm does not move freely: check for foreign material in the arm guide, cam follower, or retract path.
4. Clean the ATC arm guide surfaces with clean rags and inspect for wear.

Step 6: Check ATC Arm Retract Sensor (SEN-ATC-RET)

1. With machine powered on (not in LOTO — safety person present):
2. Navigate to SYSTEM → DIAGNOSTIC → PLC I/O.
3. Find the SEN-ATC-RET input.
4. Manually push the ATC arm into the retract position.
5. Observe whether SEN-ATC-RET changes state (input = 1) when the arm is in the retract position.
6. If the sensor does not trigger when the arm is physically retracted: the sensor is misaligned or failed.

Step 7: Inspect and Adjust SEN-ATC-RET Sensor

1. Lock out the machine (LOTO).
2. Locate SEN-ATC-RET (see Section 5.4.4 for sensor location details).
3. Check the sensor mounting bracket for looseness.
4. Measure the sensor face-to-target gap. Inductive proximity sensors (NPN) typically require a 0.5–2.0 mm gap.
5. Adjust the sensor position by loosening the mounting clamp and repositioning.
6. After adjustment, restore power and test: manually push the arm to retract position and verify SEN-ATC-RET triggers.
7. If the sensor does not trigger even with correct gap: replace sensor (BC-SEN-ATC-RET).

Step 8: Check Hydraulic Circuit

1. After LOTO is released (machine powered), manually energize SOL-HYD-04 (ATC arm retract solenoid) using the PLC output test function (SYSTEM → DIAGNOSTIC → PLC I/O → OUTPUT TEST → SOL_HYD_04 → ENERGIZE).

CAUTION: Use output test mode only in a safe condition with the ATC area clear of personnel.

2. Observe the ATC arm. It should retract fully.
3. If the arm retracts in output test mode but not during automatic operation: the issue may be PLC logic timing or solenoid actuation delay. Consult BetaCorp Systems.
4. If the arm does not retract in output test mode: check the hydraulic system:
 - Verify SOL-HYD-04 is energized (check LED on solenoid coil body if equipped)
 - Check hydraulic pressure at the ATC retract line (port on the ATC cylinder)
 - Check for blocked or kinked hydraulic line on the ATC retract circuit

Step 9: ATC Recovery After E303

After the mechanical issue is resolved and sensors are verified:

1. Manually place the ATC arm in the home (retracted) position.
2. Restore machine power and RESET the E303 fault.
3. Navigate to SYSTEM → ATC → MANUAL → ARM HOME to command the arm to verify home position through the control.
4. Carefully restore the carousel to the correct position using SYSTEM → ATC → MANUAL → CAROUSEL GOTO → [pot number] to ensure the tool pot assignment table is correct.
5. Verify the correct tool is in the spindle.
6. Perform a manual tool change cycle (SYSTEM → ATC → MANUAL → TOOL CHANGE → T[n]) to verify normal ATC operation.
7. Run 5 tool change cycles before returning to automatic production.

Related Parameters:

- B3.12 — ATC Arm Retract Timeout (increase if timeout is too aggressive)
- B3.08 — ATC Arm Retract Detect Input (PLC input number)
- B3.22 — ATC Retry on Fault (set to 1 for one automatic retry)
- B4.02 — HPU Pressure Min Alarm (verify hydraulic pressure is adequate)

Related Parts:

- BC-SEN-ATC-RET — ATC arm retract position sensor
- BC-CYL-ATC-EXT — ATC arm extend/retract cylinder

- BC-SOL-ATC-RET — ATC arm retract solenoid valve (SOL-HYD-04)
- BC-SEAL-ATC-CYL — ATC cylinder seal kit
- BC-GRS-ATC-001 — ATC arm grease (specified grease)

Safety Note: After any ATC fault and recovery, verify that the tool in the spindle matches the tool the control believes is in the spindle (check PROGRAM → TOOL TABLE → CURRENT TOOL). A mismatch could cause the wrong tool to be used in subsequent operations, leading to tool breakage and potential machine damage.

8.8 Hydraulic System Faults (E400–E499)

8.8.1 E404 — Hydraulic Pressure Low

Fault Code: E404 **Fault Name:** Hydraulic Pressure Low **Severity:** ALARM **Machine Response:** Machine halts; Z-axis brake applies (by spring force); spindle stops; all ATC motion inhibited; program aborted

Description: Fault E404 is triggered when the hydraulic system pressure drops below the alarm threshold set by Parameter B4.02 (default 120 bar) for longer than the monitoring delay (Parameter B4.04, default 5,000 ms after HPU start). This fault indicates the hydraulic power unit is not maintaining adequate pressure for the machine's hydraulic circuits.

IMPORTANT: E404 on the BC-500X means HYDRAULIC PRESSURE LOW — the machine's hydraulic power unit is not maintaining system pressure. The code E404 refers exclusively to a hydraulic system pressure problem on the BC-500X. This is entirely different from any E404 code on other machine models.

The BC-500X hydraulic system serves: Z-axis brake (fail-safe), tool unclamp, ATC arm, and gear change. Loss of hydraulic pressure defaults all circuits to their spring-return positions:

- Z-axis brake APPLIES (protects against Z-axis gravity drop)
- Tool remains CLAMPED (spring-retained)
- ATC arm RETRACTS (spring-centered)
- Gear change remains in LAST POSITION (spring-held)

Fault Trigger Conditions: The hydraulic pressure switch (set at 120 bar rising / 100 bar falling) monitors the main hydraulic circuit. When the switch indicates low pressure for longer than Parameter B4.04, E404 is triggered. An additional shutdown trigger at Parameter B4.03 (default 80 bar) causes an immediate E-Stop regardless of the monitoring delay.

Possible Causes:

1. **Hydraulic pump not running:** The HPU motor may not have started or may have stopped.
 - Most common occurrence at initial machine startup before HPU is started
 - Also occurs if HPU motor thermal overload trips

- Verify HPU motor is running (listen for motor sound; check HPU motor contactor status in PLC diagnostics)
2. **Hydraulic pump worn or failed:** A worn gear pump loses internal efficiency and cannot maintain pressure against the system load.
- Worn pumps often exhibit: slow pressure rise on start, pressure that holds at rest but drops when hydraulic actuators operate, and unusual noise (grinding or whining)
 - Replace pump (BC-PMP-HYD-001) if internal leakage is confirmed by flow testing
3. **Main system pressure relief valve stuck open:** The main relief valve (BC-RLV-HYD-MAIN, set at 210 bar) may be stuck open due to contamination lodged in the valve seat.
- Symptom: pump runs, pressure never rises above a low value (e.g., 20–30 bar)
 - The relief valve vents pump flow back to tank, preventing system from pressurizing
 - Remove and clean or replace the relief valve
4. **Hydraulic oil too low:** If the HPU reservoir level is below the pump inlet height, the pump will cavitate and not develop pressure.
- Check HPU sight glass. Oil must be between MIN and MAX marks.
 - If oil is low: add ISO VG 46 hydraulic oil. Investigate why oil level dropped (leak?).
5. **Hydraulic line leak:** A major leak in the hydraulic circuit will prevent pressure from building up.
- Inspect all hoses, connections, and cylinders for leaks.
 - Even a small leak can prevent full pressure if the pump's output capacity is exceeded by the leak rate.
6. **Pressure switch failed:** The pressure switch may have failed in the open state, giving a false E404 indication even though hydraulic pressure is adequate.
- Verify actual pressure on the analog pressure gauge on the HPU (this gauge is independent of the pressure switch).
 - If gauge shows ≥ 150 bar and E404 is active: pressure switch has failed. Replace (BC-PSW-HYD-MAIN).
7. **Pump motor wiring fault:** The HPU motor may not be receiving power even though the control is commanding it to start.
- Check that the HPU motor contactor (in the main electrical cabinet) is closing when HPU START is commanded.
 - Check motor wiring continuity.
8. **Wrong hydraulic oil or severely contaminated oil:** Oil that is too viscous (wrong grade, or oil degraded at cold temperatures) may prevent the pump from developing full pressure at startup.

- In cold ambient conditions (below 10°C), ISO VG 46 oil may be too viscous. Allow more warm-up time or use ISO VG 32 oil in cold environments.

Corrective Steps:

Step 1: Check HPU Motor Status

1. Listen for the HPU motor. It should produce a smooth, steady sound.
2. At SYSTEM → DIAGNOSTIC → PLC I/O, check the HPU motor run feedback (input labeled "HPU_RUN_FB"). It should be 1 (ON) when the motor is commanded to start.
3. Check the HPU motor contactor. Open the main electrical cabinet and locate the HPU motor starter. Verify:
 - Main power contacts are closed (indicator lamp if equipped)
 - Thermal overload relay has not tripped (manual reset button on the overload relay)
4. If the thermal overload has tripped: press the reset button (after allowing the motor to cool 10 minutes). Investigate the cause of overload (blocked pump, wrong oil viscosity, motor fault).

CAUTION: Opening the electrical cabinet requires arc flash PPE. Refer to Section 2.4.1.

Step 2: Check Hydraulic Pressure on Analog Gauge

Without relying on the pressure switch, check the analog pressure gauge on the HPU:

- Gauge reading should be ≥ 150 bar during normal operation
- If gauge reads 0–10 bar with motor running: pump is not developing pressure (see causes 3 and 4 above)
- If gauge reads 50–100 bar: relief valve may be cracking, or pump is worn
- If gauge reads ≥ 150 bar and E404 is still active: pressure switch has failed

Step 3: Check Hydraulic Oil Level

1. Verify HPU is stopped (LOTO).
2. Read the HPU sight glass. Level must be between MIN and MAX marks.
3. If below MIN: add ISO VG 46 hydraulic oil through the fill port (top of HPU, breather/fill cap).
4. Fill to MAX mark (approximately 50 L total capacity).
5. Investigate why oil level dropped (check for leaks).

Step 4: Check for Hydraulic Leaks

1. With HPU running at pressure, carefully inspect all hydraulic hose connections, fittings, cylinder rod seals, and valve bodies for external leaks.
2. Use a piece of cardboard to detect pinhole leaks (do NOT use hands — see Section 2.6.1).

3. For each leak found: LOTO machine, de-pressurize hydraulic system, and repair the leak before proceeding.
4. After repair, restart HPU and re-verify pressure.

Step 5: Evaluate Relief Valve

If the pump is running, oil level is correct, and no leaks are found, but pressure will not rise above a low value:

1. LOTO the machine and de-pressurize.
2. Remove the main relief valve (BC-RLV-HYD-MAIN) from the HPU block.
3. Inspect the valve seat and ball/poppet for contamination.
4. Flush the valve with clean hydraulic oil or mineral spirits.
5. Check that the spring and poppet move freely.
6. Reinstall the valve and test.
7. If pressure still does not build: replace the relief valve (BC-RLV-HYD-MAIN). The relief valve is a wear item.

Step 6: Pressure Switch Test and Replacement

If the analog gauge shows ≥ 150 bar but E404 is still active:

1. At SYSTEM $\rightarrow$ DIAGNOSTIC $\rightarrow$ PLC I/O, locate the pressure switch input (labeled "HPU_PRESS_OK").
2. This input should be 1 (ON) when pressure is adequate.
3. If input is 0 (OFF) with gauge reading ≥ 150 bar: pressure switch has failed.
4. LOTO and de-pressurize.
5. Disconnect and replace the pressure switch (BC-PSW-HYD-MAIN). Tighten to 25 N·m (18 ft·lb).
6. After replacement: restore power and verify the PLC input reads 1 when HPU reaches pressure.

Step 7: Hydraulic Pump Wear Assessment

To assess pump wear, a flow test is required:

1. Connect a hydraulic flow meter in the pump outlet line (between pump and system).
2. Run the pump and measure flow at zero load (no actuators active) at rated speed.
3. Compare measured flow to the rated pump flow (12 L/min at 1,450 RPM for BC-PMP-HYD-001).
4. If measured flow is less than 80% of rated flow with the relief valve set at full pressure: the pump has worn excessively. Replace (BC-PMP-HYD-001).

Related Parameters:

- B4.02 — HPU Pressure Min Alarm (pressure switch setpoint for E404)
- B4.03 — HPU Pressure Min Shutdown (emergency stop setpoint)
- B4.04 — HPU Pressure Monitor Delay (delay after HPU start before monitoring)
- B4.07 — HPU Level Min Alarm
- B4.20 — Hydraulic Oil Change Interval

Related Parts:

- BC-PSW-HYD-MAIN — Hydraulic system pressure switch
- BC-RLV-HYD-MAIN — Main hydraulic relief valve
- BC-PMP-HYD-001 — Hydraulic gear pump
- BC-MTR-HP-001 — Hydraulic pump motor (4 kW)
- BC-FLT-HYD-001 — Hydraulic return filter element (10 µm)
- BC-OIL-HYD-VG46 — ISO VG 46 hydraulic oil (20 L container)

Safety Note: After E404 and any hydraulic system repair, always verify Z-axis brake operation before re-enabling spindle:

1. Command HPU to start. Verify pressure rises to ≥ 150 bar.
2. Command Z-axis brake release (SYSTEM → HYDRAULICS → Z BRAKE RELEASE).
3. Verify brake release confirmed by sensor and PLC (E415 does not trigger).
4. Jog Z-axis 10 mm upward and 10 mm downward to verify servo and brake are functioning correctly.
5. Command Z-axis brake apply (SYSTEM → HYDRAULICS → Z BRAKE APPLY).
6. Verify brake applies (E416 does not trigger).

8.9 Servo System Faults (E500–E599)

8.9.1 E505 — Axis Servo Overload

Fault Code: E505 **Fault Name:** Axis Servo Overload **Severity:** ALARM **Machine Response:** All axis motion halts; spindle stops; program aborted

Description: Fault E505 is triggered when one or more axis servo drives detect that the motor has been operating above its continuous rated current (and therefore rated torque) for long enough to cause a thermal overload condition. The servo drive monitors motor current and uses an electronic thermal model to track the motor's thermal state. When the thermal model indicates the motor temperature has reached the trip threshold, E505 is triggered.

IMPORTANT: E505 on the BC-500X is an AXIS SERVO OVERLOAD fault — a mechanical/servo system thermal condition. This code has nothing to do with door interlocks or other systems that may use E505 on other machine models. E505 on the

BC-500X is always a servo overload issue.

The specific axis affected by E505 is identified in the fault detail message. The fault message format is: "E505 — AXIS SERVO OVERLOAD: [X/Y/Z]." Check the fault log detail for which axis triggered the fault.

Fault Trigger Conditions:

The servo drive's electronic thermal model calculates a "thermal accumulator" value based on:

- Current magnitude relative to rated motor current
- Time duration of the elevated current
- Ambient temperature of the drive (internal temperature sensor)

The thermal accumulator rises when current exceeds rated value and decays when current is below rated. E505 is triggered when the accumulator exceeds 100% (thermal trip level).

For the BC-500X axis servo motors:

- X and Y axes: motor rated current = $(3 \text{ kW} / \sqrt{3} \times 480 \text{ V} \times \text{PF}) \approx 5.2 \text{ A (AC)}$, drive rated output $\approx 8.7 \text{ A DC}$
- Z axis: motor rated current = $(4 \text{ kW} / \sqrt{3} \times 480 \text{ V} \times \text{PF}) \approx 6.9 \text{ A (AC)}$, drive rated output $\approx 11.6 \text{ A DC}$

Possible Causes:

1. **Cutting forces exceeding machine capability:** The most common cause of E505 in production is an aggressive machining program that requires more axis force than the servo motor can continuously supply.
 - Typical scenario: heavy face milling with high radial engagement, or boring with excessive depth of cut
 - The axis servo must resist the cutting force while maintaining programmed feed rate
 - Solution: reduce cut depth, reduce radial engagement (stepover), reduce feed rate
2. **Guideway or ballscrew binding:** Mechanical resistance in the linear guideway or ballscrew (due to inadequate lubrication, contamination, or wear) increases the force required to move the axis. The servo motor must provide extra torque to overcome this resistance.
 - Symptom: E505 occurs during rapid traverse or light cutting (not just during heavy cuts)
 - Check way lube system operation (Section 5.8)
 - Check for chips in the guideway wiper seals
 - Manually push the axis (with machine LOTO'd): it should move smoothly with relatively low force
3. **Servo motor phase loss or winding fault:** Loss of one phase of the servo motor or a partial winding fault causes the motor to draw excessive current on the remaining phases to

maintain torque.

- Symptom: E505 with no apparent machining overload; motor may run warm or hot
 - Verify motor winding resistance (all three phases should be equal within 5%)
 - Check motor connector and cable for phase loss
4. **Z-axis: gravity compensation insufficient:** For the Z-axis, if the hydraulic brake is not releasing fully or if the Z-axis is fighting additional downward resistance (e.g., from a tool that is stuck in a workpiece), the servo must provide extra torque to move the axis upward.
- Symptom: E505 on Z-axis during upward moves specifically
 - Check hydraulic brake release (verify brake sensor and hydraulic pressure)
5. **Servo drive parameter incorrect:** If the servo drive's motor current limit (part of drive firmware) is set too low, the motor will thermally trip at a lower load than intended.
- This is unlikely unless the drive has been replaced or re-parameterized
 - Verify drive parameters match the BetaCorp commissioning data
6. **Ambient temperature too high:** Servo drives and motors are derated at elevated ambient temperatures. Above 40°C ambient, the drive reduces its maximum output current, making E505 more likely at the same load level.
- Check ambient temperature around the servo drives (inside the cabinet)
 - Verify the cabinet cooling system (air conditioning or heat exchanger) is functioning
7. **Feed rate override too high:** Operating with feed rate override above 120% for extended periods in heavy cutting can cause servo overload.
- Reduce feed rate override.

Corrective Steps:

Step 1: Identify Affected Axis and Review Conditions

1. Read the E505 fault detail to identify the affected axis (X, Y, or Z).
2. Review the machining program and the toolpath being executed at the time of E505.
3. Check the feed rate, spindle speed, depth of cut, and workpiece material.

Step 2: Allow Motor to Cool

1. Press RESET. If the fault clears immediately: the thermal accumulator has decayed below the trip level during the time the machine was stopped.
2. Allow the affected axis to cool for at least 5 minutes before resuming.
3. If E505 re-triggers immediately on the same operation: the machining parameters are excessive.

Step 3: Reduce Machining Parameters

If E505 is caused by aggressive cutting:

1. Reduce the depth of cut by 30–40%.
2. Reduce the feed rate by 20–30%.
3. Reduce radial engagement (stepover) to 40–50% of tool diameter maximum for face milling.
4. Resume the program and monitor for E505 recurrence.
5. Gradually increase parameters back toward the original values in 10% increments if E505 does not recur.

Step 4: Check Way Lube Operation

1. Navigate to SYSTEM → DIAGNOSTIC → LUBE → WAY LUBE.
2. Verify the way lube system has been operating on schedule (cycle counter should show recent cycles).
3. Check the way lube oil level in the reservoir.
4. Manually trigger a way lube cycle: SYSTEM → MAINT → WAY LUBE → MANUAL CYCLE.
5. Verify that oil flows to each guideway lubrication point (you should see a small oil sheen at each carriage wiper after the manual cycle).

Step 5: Inspect Guideways for Binding

1. Lock out the machine (LOTO).
2. Manually push the affected axis through its full range of motion by hand.
3. The axis should move smoothly and without notable resistance throughout the travel.
4. "Tight spots" or resistance at certain positions indicate:
 - Chip under a guideway wiper seal
 - Guideway wear
 - Ballscrew nut binding (if resistance increases in one direction more than the other)
5. If chips are found: clean the guideway and wiper area with a clean rag and compressed air (at low pressure, 2 bar maximum, with safety glasses).
6. If binding cannot be resolved by cleaning: contact BetaCorp Systems for guideway inspection.

Step 6: Verify Servo Motor Winding Resistance

1. Lock out the machine (LOTO). Wait 8 minutes for servo drive capacitors to discharge.
2. Disconnect the servo motor cable at the drive end (in the servo cabinet).
3. Using a precision ohmmeter, measure resistance between each motor phase terminal pair (U-V, V-W, U-W).
4. Expected values: all three readings should be equal within $\pm 5\%$. A typical 3 kW servo motor at the BC-500X will read approximately 1.5–3.0 ohms per measurement.

5. Measure from each phase to ground (motor chassis): expected > 1 MOhm. Lower values indicate damaged winding insulation.
6. If winding resistance is unbalanced or insulation resistance is low: the servo motor has failed. Replace (BC-MTR-X-001, BC-MTR-Y-001, or BC-MTR-Z-001 as appropriate).

Step 7: Verify Drive Output Current Setting

If motor windings are healthy but E505 persists at loads that appear reasonable, verify the drive's current limit setting:

1. Access the spindle/servo drive configuration menu (password-protected; contact BetaCorp Systems for access).
2. Verify the "Continuous Current Limit" parameter matches the motor's rated current.
3. For the SD-300 (X and Y drives): continuous limit = 8.7 A, peak = 17.4 A.
4. For the SD-400 (Z drive): continuous limit = 11.6 A, peak = 23.2 A.
5. If these values are incorrectly set lower: a BetaCorp service engineer must correct them.

Step 8: Check Z-Axis Hydraulic Brake (Z-Axis E505 Only)

If E505 is on the Z-axis and occurs specifically during upward motion:

1. Verify the hydraulic brake is releasing fully.
2. At SYSTEM → DIAGNOSTIC → HYDRAULIC, check that the Z-brake release sensor confirms release after HPU start.
3. Measure hydraulic pressure at the Z-brake circuit (should be ≥ 150 bar).
4. Manually command Z-brake release (SYSTEM → HYDRAULICS → Z BRAKE TEST).
5. If E415 (Z-brake release timeout) appears: the brake is not releasing, and the Z-axis servo is fighting the applied brake.
6. Troubleshoot the Z-brake circuit per the E415 corrective steps.

Related Parameters:

- B8.20 — Axis Torque Limit (X)
- B8.21 — Axis Torque Limit (Y)
- B8.22 — Axis Torque Limit (Z)
- B9.20 — Way Lube Interval
- B2.05, B2.15, B2.25 — Following Error Limits (high following error with E505 confirms mechanical binding)

Related Parts:

- BC-MTR-X-001 — X-axis servo motor
- BC-MTR-Y-001 — Y-axis servo motor
- BC-MTR-Z-001 — Z-axis servo motor

- BC-DRV-X-001 — X-axis servo drive
- BC-DRV-Y-001 — Y-axis servo drive
- BC-DRV-Z-001 — Z-axis servo drive
- BC-LUB-OIL-VG68 — Way lube oil (ISO VG 68)

Safety Note: After E505 and any servo system repair, perform a controlled test run before resuming production:

1. Home all axes.
2. Run the machine in JOG mode at full jog rate on the affected axis. Verify smooth motion, no unusual sounds.
3. Monitor servo current at SYSTEM → DIAGNOSTIC → SERVO → [AXIS] → CURRENT. Current should be low (< 30% of rated) during unloaded rapid traverse.
4. Run a test program at reduced parameters (50% feed, 50% spindle speed) for 15 minutes before returning to full production parameters.

[Fault codes E530 through E950 follow the same detailed format — each with description, causes, and corrective steps. A representative selection is included below for brevity in this summary section.]

E530 — Drive DC Bus Overvoltage

Severity: ALARM **Machine Response:** All axis and spindle drives disabled; machine halts

Description: The servo drive DC bus voltage has exceeded the overvoltage protection threshold (typically 800 VDC on the BC-500X drive system). This typically occurs during high-deceleration events when the servo motors regenerate energy back into the DC bus faster than the regenerative resistor can dissipate it.

Possible Causes:

1. Regenerative resistor (braking resistor) failed open — no regenerative dissipation
2. Regenerative resistor connection broken
3. Deceleration rate set too high (B2.10, B2.20, B2.30) for the connected load
4. Incoming power supply voltage above normal (above 528 VAC = 480 + 10%)
5. Multiple simultaneous high-speed decelerations on multiple axes

Corrective Steps:

1. Check the regenerative resistor (mounted externally on top of servo cabinet). Measure resistance: should be 10–47 ohms (check drive specification). If open circuit: replace resistor (BC-RGEN-001).
2. Reduce axis deceleration rates (Parameters B2.10, B2.20, B2.30) by 20%.

3. Verify incoming supply voltage is within 480 VAC $\pm$ 10%.
-

E701 — PLC Program Fault

Severity: ALARM **Machine Response:** Machine halts; all outputs de-energized to default safe states

Description: The PLC has detected an internal program execution error. This may indicate PLC memory corruption or a program bug introduced by unauthorized PLC modifications.

Corrective Steps:

1. Power-cycle the machine.
 2. If E701 recurs after power cycle: contact BetaCorp Systems. Do NOT attempt to modify the PLC program. PLC programs are protected and modification requires BetaCorp authorization.
-

E801 — Control Cabinet Temperature High

Severity: WARNING **Machine Response:** Alert displayed; machine continues; cabinet cooling fan increases to maximum speed

Description: The control cabinet internal temperature has exceeded the warning threshold (50°C, Parameter B8.15).

Possible Causes:

1. Cabinet air conditioning / heat exchanger failed
2. Cabinet door left open (seal broken or door ajar)
3. Ambient temperature exceeds 40°C
4. Heat exchanger filter blocked (if equipped)

Corrective Steps:

1. Check cabinet air conditioning/heat exchanger operation (BC-HEX-CAB-001). Verify fan is running and air is flowing.
 2. Verify cabinet door is fully closed and sealed.
 3. If ambient temperature is above 40°C: ventilate the machine area.
 4. Clean or replace heat exchanger filter media.
 5. If E802 (Critical) follows: machine will halt. Resolve cooling issue before restarting.
-

PART 8 — MAINTENANCE

Chapter 9 — Maintenance Schedule

9.1 Maintenance Philosophy and Preventive Maintenance Overview

The BC-500X is a precision manufacturing machine that requires regular, systematic maintenance to achieve its specified performance and service life. BetaCorp Systems has designed this maintenance program based on:

- Manufacturer's component service life data
- Field experience from the installed base of BC-500X machines
- Regulatory requirements (OSHA, ANSI B11.8)
- Industry best practices for CNC machining center maintenance

The Cost of Deferred Maintenance:

Deferred maintenance is one of the most costly practices in a manufacturing environment. The consequences include:

- Reduced machine accuracy, causing scrap and rework
- Unexpected machine failures causing unplanned downtime (typically 5–10× more costly than planned downtime)
- Progressive component wear that requires expensive major repairs instead of routine consumable replacement
- Safety hazards from degraded guards, worn seals, and inadequate lubrication

Maintenance Personnel Requirements:

All maintenance tasks must be performed by personnel meeting the qualifications specified in Section 2.2.2. Tasks involving electrical service require an electrical service technician (Section 2.2.3). All maintenance tasks require proper PPE (Section 2.3).

Maintenance Documentation:

Every maintenance task performed on the BC-500X must be documented in the Maintenance Log (template in Appendix I). Documentation must include:

- Date and time of maintenance
- Machine serial number
- Description of task performed
- Parts replaced (with part numbers)
- Technician name and signature

- Any abnormal findings and actions taken

This documentation is required for warranty claims, for continuity when personnel change, and for regulatory compliance in many industries.

Maintenance Scheduling:

The BC-500X control system tracks maintenance due dates based on machine operating hours (Parameter B9.48 — Next Maintenance Due). The control will display a maintenance reminder when scheduled tasks are approaching:

- 10 hours before a daily task is due (displayed each power-on)
- 5% before a scheduled interval task is due

Summary of Maintenance Intervals:

Frequency	Task Summary
Daily (each shift)	Visual inspection; chip removal; coolant level; way lube level; general housekeeping
Weekly	Coolant concentration and pH check; filter inspection; ATC area cleaning; guideways; coolant strainer
Monthly	Coolant bacteria test; way lube system check; guideway wiper inspection; encoder cable inspection; filter change (coolant)
Quarterly	Axis backlash measurement; geometry check; ATC sensor check; spindle run-out; hydraulic oil sample
Semi-annual	Hydraulic oil change; way lube deep cleaning; coolant replacement; coolant tank cleaning; motor terminal torque check
Annual	Full geometry calibration (laser interferometer); spindle bearing inspection; hydraulic system overhaul check; encoder cable replacement (if needed); full electrical inspection

9.2 Daily Maintenance Tasks

Daily maintenance must be completed at the beginning of each operating shift before the first part is run. Estimated time: 10–15 minutes.

D-01: Pre-Operation Visual Inspection

Frequency: Each shift, before first production run **Estimated Time:** 5 minutes

Procedure:

1. Walk completely around the machine and visually inspect for:

- Coolant or hydraulic oil puddles on the floor
 - Visible damage to enclosure panels, hoses, or cables
 - Any items left in or near the machine from the previous shift
2. Through the work zone window, visually inspect:
 - Excessive chip accumulation (should be less than 1/4 of chip pan volume)
 - Coolant in the work zone (should be present, not dried out)
 - Condition of previous workpiece setup (fixtures, clamps)
 3. Confirm that all guards are in place:
 - Work zone door closes and seals properly
 - Chip conveyor drive guard is attached
 - HPU coupling guard is attached

Acceptance Criteria: No puddles on floor; no visible damage; all guards in place. **Action if Not Acceptable:** Tag machine out of service; notify maintenance department.

D-02: Coolant Level Check

Frequency: Each shift **Estimated Time:** 2 minutes **PPE Required:** Safety glasses

Procedure:

1. Locate the coolant sight gauge on the right side of the coolant tank.
2. Read the coolant level indicator. Level must be between the "LOW" and "FULL" marks.
3. If level is at or below the "LOW" mark: a. Add premixed coolant to the tank through the fill opening (top of tank, open the fill cap). b. Add coolant slowly while watching the sight gauge. c. Stop adding when the level reaches the "FULL" mark. d. Typical add volume from "LOW" to "FULL": 80–100 liters. e. Use only the approved coolant type and concentration (check maintenance log for current coolant type). Do not add water without concentrate.
4. Record any coolant addition in the maintenance log (date, volume added, technician).

Acceptance Criteria: Level between LOW and FULL marks.

D-03: Way Lube Oil Level Check

Frequency: Each shift **Estimated Time:** 1 minute **PPE Required:** Safety glasses

Procedure:

1. Locate the centralized way lube reservoir. The reservoir is mounted inside the machine left side panel, accessible by opening the left side access panel.
2. Check the reservoir level through the sight glass on the reservoir body.

3. The level should be above the "LOW" mark.
4. If below the "LOW" mark: add ISO VG 68 way lube oil (BC-LUB-OIL-VG68).
 - Remove the fill cap on top of the reservoir.
 - Add oil until the level reaches the "MAX" mark (reservoir capacity = 2.5 liters).
 - Replace the fill cap.
5. Record any oil addition.

CAUTION: Use ONLY ISO VG 68 way lube oil. Using wrong-viscosity oil or general machine oil will not provide the correct anti-stick-slip properties for the linear guideways.

D-04: Chip Bin Check and Emptying

Frequency: Each shift (or more often in high chip-generation applications) **Estimated Time:** 5 minutes **PPE Required:** Safety glasses, cut-resistant gloves, face shield

Procedure:

1. Inspect the chip bin (collection hopper under chip conveyor discharge).
2. If the chip bin is more than 2/3 full: empty it before beginning the shift.
3. Empty the chip bin using a shovel or chip tongs. Do not use bare hands.
4. Place chips in designated chip disposal containers or chip bins.
5. Separate chip types if required by your metal recycling program (aluminum chips separate from steel chips).
6. Return the empty chip bin to the conveyor discharge position.

Housekeeping: After emptying the chip bin:

1. Clean any spilled chips from the floor around the chip conveyor with a push broom.
 2. If coolant has puddled on the floor from the chip conveyor: clean with absorbent materials.
-

D-05: Work Zone Chip Removal

Frequency: Each shift (or as needed during production) **Estimated Time:** 3 minutes **PPE Required:** Safety glasses, face shield, cut-resistant gloves

Procedure:

1. Wait for the spindle to stop completely before opening the work zone door.
2. Open the work zone door.
3. Using a chip brush and/or chip hook: sweep accumulated chips from the work zone floor (chip pan) into the chip conveyor intake.

4. Clear chips from around the fixture and workpiece.
 5. Do NOT use compressed air to blow chips in the work zone (chips can jam the chip conveyor or be ejected hazardously).
 6. Close the work zone door.
-

D-06: Daily Function Test (First Power-On of Day)

Frequency: Daily, first power-on **Estimated Time:** 5 minutes

Procedure:

1. Power on the machine per Section 6.3.
 2. Verify all startup faults are resolved (particularly E404 should clear when HPU starts).
 3. Test each E-Stop button in sequence: press → verify "E-STOP ACTIVE" message → release → press RESET → verify machine returns to ready state.
 4. Perform machine homing (Section 6.4). Verify all three axes home without fault.
 5. Test coolant pump: turn ON, verify coolant flows from nozzles, turn OFF.
 6. Test chip conveyor: turn ON, verify conveyor belt runs in correct direction, turn OFF.
 7. Test spindle: in MDI mode, command M03 S500. Verify spindle starts and reaches approximately 500 RPM on the display. Command M05. Verify spindle stops.
 8. Document any anomalies found during the daily function test.
-

9.3 Weekly Maintenance Tasks

Weekly maintenance is performed once per week, typically at the beginning of the first shift of the week. Estimated time: 30–45 minutes.

W-01: Coolant Concentration Check

Frequency: Weekly **Estimated Time:** 5 minutes **Equipment:** Optical refractometer (calibrated for metalworking fluids) **PPE Required:** Safety glasses, nitrile gloves

Procedure:

1. Using a pipette or small dropper, collect a coolant sample from the coolant tank. Collect from the middle depth of the tank, not the surface.
2. Clean the refractometer prism with a clean, damp cloth.
3. Apply 2–3 drops of the coolant sample to the refractometer prism.
4. Close the prism plate.
5. Hold the refractometer toward a light source and read the value at the scale boundary.
6. Calculate actual coolant concentration: Actual Concentration (%) = Refractometer Reading

- × Correction Factor (from coolant manufacturer's technical data sheet, typically 1.2–2.0 for semi-synthetic fluids).
7. Compare actual concentration to target (B5.13, default 8%). Acceptable range: target ± 1%.
 8. If concentration is below target: add coolant concentrate without diluting further. Calculate the volume of concentrate to add per the formula in Chapter 12, Section 12.2.3.
 9. If concentration is above target: add water only (never exceeding 10% total concentration without consulting coolant supplier).
 10. Record the reading, date, and any adjustments in the coolant maintenance log.

Acceptance Criteria: Coolant concentration within ±1% of target.

W-02: Coolant pH Check

Frequency: Weekly **Estimated Time:** 3 minutes **Equipment:** Calibrated pH test strips (range 7.0–11.0) or digital pH meter **PPE Required:** Safety glasses, nitrile gloves

Procedure:

1. Collect a coolant sample as described in W-01.
2. If using pH strips: dip the strip in the sample, remove, and read color change after 1 minute. Compare to the color chart provided with the strips.
3. If using a digital pH meter: calibrate the meter per manufacturer's instructions before use. Rinse the electrode with clean water. Dip in the sample. Read the displayed pH value.
4. Acceptable pH range: 8.5–9.5 (Parameter B5.14–B5.15).
5. If pH is below 8.5: add pH adjuster (alkali) per coolant manufacturer's recommendation. Do not add more than 0.1% by volume per adjustment and re-test.
6. If pH is above 9.5: dilute the coolant slightly or replace with fresh mixture. High pH can cause skin irritation.
7. If pH drops rapidly (more than 0.5 pH units per week): suspect bacterial contamination. Perform bacteria test (W-04) immediately.
8. Record pH reading and any adjustments.

Acceptance Criteria: pH between 8.5 and 9.5.

W-03: Coolant Inlet Strainer Cleaning

Frequency: Weekly (or more often in high chip-load applications) **Estimated Time:** 10 minutes **PPE Required:** Safety glasses, nitrile gloves, face shield

Procedure:

1. Turn off the coolant pump at the operator panel.

2. Allow any pressure to bleed from the coolant line (the pump will purge within 30 seconds after shutoff).
 3. Locate the coolant pump inlet strainer. The strainer is located on the coolant pump inlet pipe inside the coolant tank.
 4. Close the isolation valve on the pump inlet line (if equipped) to minimize coolant spill.
 5. Unscrew the strainer housing (twist counter-clockwise or use a strap wrench on the housing OD).
 6. Remove the stainless steel screen basket from the housing.
 7. Rinse the screen basket in clean water and brush off accumulated chips and debris with a soft brush.
 8. Inspect the screen for tears, holes, or deformation. A damaged screen allows fine chips to reach the pump and cause premature wear.
 9. If screen is damaged: replace with Part No. BC-STR-CLT-001 (150 µm stainless screen, complete assembly).
 10. If screen is clean and undamaged: reinstall in the housing.
 11. Reinstall the housing. Hand-tighten, then tighten 1/4 turn further with strap wrench.
 12. Open the isolation valve. Turn on the coolant pump. Verify no leaks at the strainer housing.
-

W-04: Tool Changer Area Inspection and Cleaning

Frequency: Weekly **Estimated Time:** 10 minutes **PPE Required:** Safety glasses, cut-resistant gloves

Procedure:

1. With the machine in HOME mode and ATC arm at home position, open the tool changer access panel (top of machine, above ATC area).
2. Visually inspect all 24 tool pots in the carousel:
 - Check for chips or debris inside tool pots (chips can prevent tool insertion)
 - Check for damaged or deformed gripper fingers
 - Check for any tool holders that appear to be sitting loosely in a pot
3. Clean chips from inside the tool pots using a chip brush or compressed air (at reduced pressure, 2 bar, directed carefully to avoid scattering chips into the drive mechanism).
4. Inspect the ATC arm grippers (two grippers on the ATC arm):
 - Gripper fingers should be clean and move freely
 - Check for worn gripper pads (replace if worn through to the substrate)
5. Clean the ATC arm and carousel with clean rags.

6. Apply a small amount of ATC arm grease (BC-GRS-ATC-001) to the ATC arm slide surfaces (if required per ATC greasing schedule — see Section 9.11).
 7. Check the ATC sensors (SEN-ATC-HOME through SEN-ATC-RET) for:
 - Physical damage (impact marks, cracked housings)
 - Coolant buildup on sensor face
 - Correct sensor mounting (no loose bracket bolts)
 8. Close the access panel.
-

W-05: Guideway Wiper Inspection

Frequency: Weekly **Estimated Time:** 10 minutes

Procedure:

1. Jog each axis to the center of its travel range.
 2. Inspect the axis wiper seals:
 - X-axis: wipers are at the front and rear of the X-axis linear guideway carriages, visible through the side access panel
 - Y-axis: wipers are at the front and rear of the Y-axis carriages
 - Z-axis: wipers are at the top and bottom of the Z-axis carriages
 3. Inspect each wiper seal for:
 - Cuts, tears, or missing sections
 - Excessive chip accumulation at the wiper (may indicate wiper seal is not clearing chips properly)
 - Coolant infiltration past the wiper (oil or coolant visible behind the wiper lip — should not be present)
 4. Clean any chip accumulation from the wiper areas.
 5. If any wiper seal is damaged: replace before the next production shift. Damaged wiper seals allow chips and coolant to reach the guideway rollers and cause rapid guideway wear.
 - Wiper seal replacement parts: BC-WPR-X-001, BC-WPR-Y-001, BC-WPR-Z-001
-

9.4 Monthly Maintenance Tasks

M-01: Coolant Bacteria Test

Frequency: Monthly **Estimated Time:** 15 minutes plus 24-hour incubation **Equipment:** Bacteria dip-slide test kit (e.g., Petrifilm, Cide-Slide, or equivalent) **PPE Required:** Safety glasses, nitrile gloves

Procedure:

1. Collect a coolant sample from the mid-depth of the coolant tank.
2. Prepare the dip-slide test per the kit manufacturer's instructions.
3. Record the result after the incubation period specified by the kit manufacturer (typically 24–48 hours):
 - Colony count < 10,000 cfu/mL: acceptable
 - Colony count 10,000–100,000 cfu/mL: add biocide and re-test within one week
 - Colony count > 100,000 cfu/mL: coolant is heavily contaminated — perform emergency coolant replacement (Section 9.9)

If Bacteria Count is Elevated:

1. Add an approved biocide at the manufacturer's recommended dose. Do NOT exceed recommended dose (excessive biocide can cause skin sensitization and may damage machine seals).
 2. Run the coolant pump for 30 minutes with no machining to circulate the biocide throughout the system.
 3. Re-test in 24–48 hours.
 4. Record biocide type, dose, date, and bacteria count before and after treatment.
-

M-02: Hydraulic Return Filter Inspection

Frequency: Monthly visual inspection; replacement per schedule (Section 9.15) **Estimated Time:** 10 minutes

Procedure:

1. Locate the hydraulic return filter on the HPU. The filter housing is mounted on the return line from the machine circuits to the HPU tank.
 2. Observe the filter bypass indicator (differential pressure gauge or pop-up indicator on filter housing):
 - If indicator is in the "green" zone: filter is not bypassing; continue normal operation
 - If indicator is in the "yellow" zone: filter is partially bypassed; schedule replacement within one week
 - If indicator is in the "red" zone: filter is fully bypassed (fault E414 may be active); replace the filter element immediately
 3. Record the filter status observation in the maintenance log.
 4. If replacement is required: see Section 9.15 for filter replacement procedure.
-

M-03: Encoder Cable Inspection

Frequency: Monthly **Estimated Time:** 20 minutes **PPE Required:** Safety glasses

Procedure:

1. Jog Z-axis to the home (full up) position.
2. Open the Z-axis energy chain cover.
3. Visually inspect the encoder cable (Part No. BC-CBL-SPENC-3M) along its full length through the Z-axis energy chain:
 - Look for cuts, abrasions, or flattening of the cable jacket
 - Inspect the cable at the first and last 300 mm of the energy chain entry/exit points
 - Look for any cable that has come loose from the energy chain retainer clips
4. Jog Z-axis to the minimum Z position (Z = -300 mm).
5. Re-inspect the cable. The cable should flex smoothly; no tight bends or kinking should be visible.
6. Repeat the inspection for X and Y axis encoder cables in their respective cable trays.
7. If any cable shows signs of damage: order a replacement cable immediately and schedule cable replacement (Section 9.12).

Key Inspection Points:

Cable	Critical Inspection Area
Spindle encoder (BC-CBL-SPENC-3M)	Z-axis energy chain entry/exit points
Z-axis encoder (BC-CBL-ENC-5M)	Z-axis energy chain alongside spindle encoder cable
X-axis encoder (BC-CBL-ENC-5M)	X-axis cable tray entry/exit
Y-axis encoder (BC-CBL-ENC-5M)	Y-axis cable tray, under machine base

M-04: Way Lube System Check

Frequency: Monthly **Estimated Time:** 15 minutes

Procedure:

1. Verify way lube oil level (already done daily — use this monthly check to verify the RATE of oil consumption, which indicates system health).
2. Record the oil level and date in the maintenance log. Compare to the previous month's reading to calculate consumption rate.
 - Expected consumption: approximately 50–100 mL per 100 hours of operation
 - Higher consumption: indicates a leak in the distribution tubing
 - Lower consumption: indicates a blocked distribution line or failed pump
3. Navigate to SYSTEM → DIAGNOSTIC → LUBE → WAY LUBE CYCLE COUNT. Record the number of cycles since last power-on and compare to expected (1 cycle per

B9.20 minutes of operation).

4. If cycle count is lower than expected: the lube pump may not be activating. Check the pump motor (manually trigger a cycle: SYSTEM → MAINT → WAY LUBE → MANUAL CYCLE and listen for the pump).
 5. Verify lube reaches all points: after a manual lube cycle, inspect each carriage lubrication nipple on all axes. You should see a fresh film of oil at the carriage wiper area.
-

M-05: Magnetic Chip Separator Cleaning

Frequency: Monthly **Estimated Time:** 10 minutes **PPE Required:** Safety glasses, cut-resistant gloves

Procedure:

1. Turn off the coolant pump.
 2. Lift the magnetic chip separator bars out of the coolant tank (the bars are suspended from a frame above the tank level).
 3. Wipe the accumulated magnetic chips and swarf from the magnet bars using a clean rag. The chips will be held firmly by the magnet — use firm wiping force.
 4. Dispose of the chip-laden rag in the metal chip collection bin.
 5. Reinstall the magnetic separator bars in the tank.
 6. Restart the coolant pump.
 7. Note: The magnetic separator only removes ferromagnetic chips. Aluminum, brass, and copper chips are not collected; these must be removed through the inlet strainer.
-

9.5 Quarterly Maintenance Tasks

Q-01: Axis Positioning Accuracy Check

Frequency: Quarterly **Estimated Time:** 2 hours **Equipment:** Laser interferometer (Renishaw ML10+, API Tracker, or equivalent) or calibrated linear scale and indicator **PPE Required:** Safety glasses

Procedure:

1. The machine must be at thermal equilibrium. Allow the machine to run at idle (no cutting) for at least 4 hours before measuring.
2. Set up the laser interferometer in the X-axis direction, with retroreflector on the machine table and interferometer head fixed to the column.
3. Home all axes.
4. Execute the positioning measurement cycle program (BetaCorp test program O00001005 —

Axis Accuracy Check). This program moves each axis to a series of positions (10 equally-spaced positions across the full travel, 5 bidirectional passes per ISO 230-2).

5. Record the position error at each measured point.
 6. Compare to the acceptance specification: ± 0.005 mm (positioning accuracy), ± 0.003 mm (repeatability).
 7. Repeat for Y-axis and Z-axis.
 8. If any axis is out of specification:
 - Check ballscrew backlash compensation (B2.06, B2.16, B2.26)
 - Check for changes in machine level (re-level if required)
 - If systematic error (scale factor error): check ballscrew pitch parameter (B6.06, B6.16, B6.26) against actual ballscrew pitch measurement
 - Contact BetaCorp Systems if accuracy cannot be restored by parameter adjustment
-

Q-02: Spindle Run-Out Check

Frequency: Quarterly **Estimated Time:** 30 minutes **Equipment:** Test indicator (0.001 mm resolution), magnetic base, precision ground test mandrel

Procedure:

1. Clean the spindle taper and test mandrel with clean solvent (acetone or IPA) to remove any oil or chips.
 2. Insert the test mandrel in the spindle. Tighten the draw bar by running the spindle briefly (M03 S100, M05) to seat the mandrel.
 3. Mount the test indicator on a magnetic base on the machine table.
 4. Position the indicator probe to contact the mandrel at 25 mm from the spindle nose (gauge line).
 5. Slowly rotate the spindle by hand. Read and record the total indicator reading (TIR).
 6. Acceptance: $TIR \leq 0.005$ mm at 25 mm from gauge line.
 7. Move the indicator to contact the mandrel at 150 mm from the gauge line. Repeat the measurement.
 8. Acceptance at 150 mm: $TIR \leq 0.015$ mm.
 9. If TIR exceeds acceptance criteria:
 - Clean the spindle taper thoroughly and repeat.
 - Inspect the test mandrel for damage (a ding or nick will give false reading).
 - If run-out persists after cleaning: the spindle bearings may be worn. Contact BetaCorp Systems.
-

Q-03: ATC Sensor Verification

Frequency: Quarterly **Estimated Time:** 45 minutes

Procedure:

1. Navigate to SYSTEM → DIAGNOSTIC → ATC STATUS.
 2. Manually actuate each ATC sensor by hand (where safe to do so with machine powered off at the sensor locations) and verify the corresponding PLC input changes state.
 3. For sensors that cannot be manually actuated safely: use the ATC manual cycle function (SYSTEM → ATC → MANUAL) to cycle through each ATC motion step and verify each sensor triggers at the correct point.
 4. Check sensor gap (sensor face to target) for each proximity sensor. Correct gap: 0.5–1.5 mm.
 5. If any sensor shows incorrect behavior: adjust, clean, or replace per Section 13.4.
-

Q-04: Hydraulic Oil Sample Analysis

Frequency: Quarterly **Estimated Time:** 5 minutes (sample collection); results in 3–5 business days from laboratory

Procedure:

1. With the HPU running and at operating temperature (at least 30 minutes after startup):
2. Using the hydraulic oil sampling port on the HPU return line (Part No. BC-PORT-HYD-SAMPLE), collect a 250 mL oil sample in a clean, capped sample bottle (provided by the oil analysis laboratory).
3. Label the sample bottle with:
 - Machine model and serial number
 - Date of sample
 - Oil type and brand
 - Machine hours since last oil change
 - Machine hours total
4. Send the sample to a hydraulic fluid analysis laboratory.
5. Request analysis for: viscosity, particle count (ISO 4406 cleanliness code), water content (Karl Fischer method), acid number (TAN), and wear metals (Fe, Cu, Al, Cr, Si).
6. Review the laboratory report when received:
 - ISO cleanliness code should be $\leq 16/14/11$
 - Water content should be $< 0.1\%$
 - TAN should be < 2.0 mg KOH/g

- Wear metal concentrations increasing over successive samples indicate component wear
7. If any parameters are out of specification: consult BetaCorp Systems for guidance on oil change interval adjustment or system inspection.
-

9.6 Semi-Annual Maintenance Tasks

S-01: Hydraulic Oil Change

Frequency: Every 6 months (or when oil analysis indicates earlier change is needed) **Estimated Time:** 3 hours **PPE Required:** Safety glasses, face shield, chemical-resistant gloves, oil-resistant apron **Parts Required:** 50 L of ISO VG 46 hydraulic oil (BC-OIL-HYD-VG46 × 3 containers); BC-FLT-HYD-001 (return filter element); BC-SEAL-HYD-DRN (drain plug seal); oil drain container (minimum 60 L capacity)

For the complete hydraulic oil change procedure, see Section 9.8.

S-02: Coolant Replacement

Frequency: Every 6 months (or when bacteria test indicates contamination, or when coolant has degraded beyond recovery) **Estimated Time:** 4 hours **PPE Required:** Safety glasses, face shield, nitrile gloves, chemical-resistant apron

For the complete coolant replacement procedure, see Section 9.9.

S-03: Coolant Tank Cleaning

Frequency: Every 6 months (performed in conjunction with coolant replacement) **Estimated Time:** 2 hours additional beyond coolant replacement

Procedure (following coolant drain per Section 9.9):

1. After coolant has been drained, open the coolant tank access cover (6 × M6 socket screws).
2. Using a wet-vacuum or mop, remove any remaining coolant sludge from the tank bottom.
3. Inspect the tank interior for:
 - Rust or corrosion on tank walls or bottom (316L stainless should not rust; any rust indicates incompatible coolant)
 - Sludge or biofilm deposits on walls
 - Damage to the internal baffles or filter screen mounts
4. Wash the tank interior with a solution of 1% trisodium phosphate (TSP) or an approved tank cleaner (consult coolant supplier). Allow to soak for 20 minutes.
5. Scrub tank walls with a stiff nylon brush. Do not use wire brushes or abrasives on stainless

steel.

6. Rinse the tank thoroughly with clean water (3 complete fill-and-drain cycles with clean water).
 7. Inspect the coolant pump inlet strainer inside the tank. Clean or replace per W-03 procedure.
 8. Inspect the magnetic separator bars. Clean thoroughly.
 9. Close the access cover. Torque the 6 × M6 screws to 8 N·m.
 10. Fill with fresh coolant per Section 9.9.
-

S-04: Motor Terminal Torque Check

Frequency: Semi-annual **Estimated Time:** 1 hour **PPE Required:** Full arc flash PPE (see Section 2.4.1); note: this task requires electrical LOTO

Procedure:

1. Lock out the machine per Section 2.4.2.
 2. Wait 8 minutes for capacitor discharge.
 3. Open the main electrical cabinet.
 4. Using a calibrated torque screwdriver, verify the torque on the following terminal connections:
 - Main power supply terminals (L1, L2, L3): 20 N·m
 - Ground bus connections: 6 N·m
 - Servo drive motor output terminals: 2.5 N·m
 - Control transformer primary and secondary terminals: 1.5 N·m
 - 24 VDC power supply output terminals: 1.0 N·m
 5. If any terminal is found loose (does not resist at the correct torque): re-torque and record.
 6. Inspect all terminals for signs of overheating (discoloration, melted insulation, heat-damaged conductor).
 7. If any sign of overheating is found: do not re-energize the machine. Contact BetaCorp Systems.
-

9.7 Annual Maintenance Tasks

A-01: Full Geometry Calibration

Frequency: Annual (or after any major crash event or significant machine relocation) **Estimated Time:** 4–8 hours **Equipment:** Laser interferometer system; precision angle mirror; ball bar system (Renishaw QC20-W or equivalent) **Performed by:** BetaCorp Systems service engineer or certified

calibration laboratory

Scope of Annual Calibration:

- Full ISO 230-2 positioning accuracy measurement on all three axes
- ISO 230-1 straightness and angularity measurements
- Spindle run-out and taper measurement (all positions)
- Table flatness measurement
- Circular contouring test (ball bar)
- All measurements recorded on calibration certificate

Calibration Certificate: After the annual calibration, a signed calibration certificate is issued by the calibrating engineer. This certificate is required for quality audits (ISO 9001, IATF 16949, AS9100) and should be stored with the machine records.

A-02: Spindle Bearing Inspection

Frequency: Annual (supplemented by ongoing monitoring) **Estimated Time:** 2 hours **Equipment:** Vibration analyzer with acceleration spectrum (SKF CMVL 3600 or equivalent); infrared thermometer

Procedure:

Step 1: Vibration Measurement

1. Set up the vibration analyzer on the spindle head casting at the bearing location (front bearing, accessible from front of spindle head).
2. Run the spindle at the following speeds and record vibration spectrum at each: 1,000 / 3,000 / 5,000 / 7,500 / 10,000 RPM.
3. Record overall velocity RMS (mm/s) and acceleration peak (g).
4. Compare to baseline measurements taken at commissioning.

Acceptance Criteria: | Spindle Speed | Max Velocity RMS | Max Acceleration Peak | |-----|
|-----|-----| | 1,000 RPM | 2.0 mm/s | 2.0 g | | 5,000 RPM | 3.0 mm/s | 4.0 g | |
10,000 RPM | 5.0 mm/s | 8.0 g |

Step 2: Temperature Measurement

1. Run the spindle at 10,000 RPM for 30 minutes.
2. Measure spindle head temperature with IR thermometer at the front bearing area.
3. Acceptable: temperature rise above ambient should be less than 30°C.
4. If temperature rise exceeds 40°C: bearings may be failing or over-preloaded.

Step 3: Acoustic Inspection

1. Use an ultrasonic stethoscope (if available) or mechanic's stethoscope at the front bearing

location.

2. Listen for irregular sounds:

- Normal: smooth rushing sound, increases with speed
- Abnormal: clicking, grinding, rumbling that changes character with speed

Step 4: Evaluation and Action

- If measurements are within specification and sound is normal: document results and continue monitoring.
- If measurements are out of specification: schedule spindle bearing replacement at the next convenient machine downtime.
- If sound is severely abnormal or temperature rise exceeds 50°C: stop machine immediately and schedule emergency spindle service.

For spindle bearing replacement procedure, contact BetaCorp Systems. Spindle bearing replacement requires a factory-trained service engineer.

9.8 Hydraulic Oil Change Procedure

The hydraulic oil change is one of the most important preventive maintenance tasks for the BC-500X. Degraded hydraulic oil causes valve sticking, pump wear, cylinder seal deterioration, and ultimately hydraulic system failure.

When to Change Oil:

- Every 2,000 machine operating hours (standard environment)
- Every 1,000 hours (severe environment: high temperature, high humidity, frequent contamination)
- When oil analysis indicates TAN > 2.0, water > 0.1%, or cleanliness code > 17/15/12
- When oil color has darkened from golden to dark brown or black
- After any hydraulic system failure or major contamination event

Parts and Materials Required:

- 50 L ISO VG 46 anti-wear hydraulic oil (BC-OIL-HYD-VG46)
- Return filter element (BC-FLT-HYD-001)
- Drain plug seal ring (BC-SEAL-HYD-DRN)
- Clean drain container (60 L minimum capacity)
- Funnel and fluid transfer pump
- Clean rags and spill absorbent

WARNING: Hot hydraulic oil can cause severe burns. Allow the machine to cool for at least 60 minutes after shutdown before draining hydraulic oil. Wear chemical-resistant,

heat-resistant gloves.

Oil Change Procedure:

Step 1: Prepare for Drain

1. Run the machine for at least 30 minutes to warm the oil (warm oil drains more completely than cold oil).
2. Stop the HPU (SYSTEM → HYDRAULICS → HPU STOP).
3. Lock out the machine per Section 2.4.2.
4. De-pressurize the hydraulic system per Section 2.6.2.
5. Allow 60 minutes for the oil to cool to below 40°C (check with IR thermometer).

Step 2: Drain the Hydraulic Reservoir

1. Position the drain container under the HPU drain plug. The drain plug is located at the lowest point of the HPU reservoir (bottom, rear).
2. Using a 24 mm hex socket, remove the drain plug.
3. Allow the reservoir to drain completely. This takes approximately 5–10 minutes.
4. While draining: remove the HPU reservoir access cover (typically 4 × M8 screws on top of HPU). Inspect the reservoir interior for:
 - Sludge or sediment deposits (indicates oil degradation)
 - Metal particles (indicates pump or valve wear)
 - Water (emulsion indicates water contamination)
5. If significant sludge or metal particles are found: flush the reservoir with 5 L of clean new hydraulic oil before filling. Swirl the flush oil to suspend deposits, then drain again.

Step 3: Replace the Return Filter Element

1. Locate the return filter housing on the HPU.
2. Place a drain tray under the filter housing.
3. Unscrew the filter housing (turn counter-clockwise). Most housings have a flat on the bottom for a strap wrench.
4. Remove the old filter element and dispose of it in an appropriate used filter container.
5. Clean the filter housing interior with a clean rag.
6. Install the new filter element (BC-FLT-HYD-001). The element is marked with the correct orientation.
7. Apply a thin film of clean hydraulic oil to the housing O-ring.
8. Reinstall the housing. Hand-tighten, then tighten 3/4 turn.

Step 4: Replace the Drain Plug

1. Install a new drain plug seal ring (BC-SEAL-HYD-DRN).

2. Reinstall the drain plug. Torque to 35 N·m (26 ft·lb).

Step 5: Fill with New Oil

1. Using a clean fluid transfer pump and funnel, add new ISO VG 46 hydraulic oil (BC-OIL-HYD-VG46) through the HPU fill port (breather/fill cap on top of reservoir).
2. Add oil slowly to avoid introducing air bubbles.
3. Add approximately 45 L (the reservoir holds 50 L but 5 L is needed to fill the circuit on first startup).
4. Check the sight glass level. It should be at or slightly below the MAX mark.
5. Replace the access cover.
6. Replace the fill cap/breather.

Step 6: Bleed and Check System

1. Restore machine power and remove LOTO.
2. Start the HPU (SYSTEM → HYDRAULICS → HPU START).
3. Allow the system to pressurize. The oil level in the reservoir will drop as the system fills (approximately 3–5 L).
4. Check the oil level. Add oil if the level is below the MIN mark.
5. Inspect all connections for leaks.
6. Operate each hydraulic actuator several times (Z-brake, tool change cycle) to bleed air from the system.
7. Re-check oil level after 10 minutes of operation and add oil if needed.

Step 7: Document

1. Record the oil change in the maintenance log: date, oil type and brand, volume added, filter replaced, technician name.
 2. Reset the oil change hours counter: SYSTEM → MAINT → HYD OIL RESET.
-

9.9 Coolant Replacement Procedure

When to Replace Coolant:

- Every 6 months (standard)
- When bacteria test exceeds 100,000 cfu/mL and biocide treatment fails
- When pH drops below 7.5 or rises above 10.5 and cannot be corrected
- When coolant is black, malodorous, or has a "rotten egg" smell
- When coolant concentration cannot be maintained within specification
- After any contamination event (hydraulic oil breakthrough, solvent contamination)

Materials Required:

- Coolant drain pump or wet vacuum (minimum 350 L capacity)
- Fresh coolant concentrate (calculate per Chapter 12, Section 12.2)
- Clean water (municipal or filtered)
- Tank cleaner solution (TSP or approved cleaner)
- pH test kit
- Refractometer
- Clean hoses and containers

Coolant Replacement Procedure:**Step 1: Drain the Coolant**

1. Turn off the coolant pump.
2. Open the coolant tank drain valve (G3/4 ball valve at the bottom of the coolant tank).
3. Allow coolant to drain into the collection container.
4. If equipped with a drain pump: use it to accelerate draining.
5. After drain valve draining, use a wet vacuum to remove residual coolant and sludge from the tank bottom.

Step 2: Coolant Disposal Spent metalworking coolant is a regulated waste in most jurisdictions:

- Contact your local waste management authority for disposal requirements
- Use a licensed coolant recycler or waste disposal service
- Never pour coolant down drains without treatment; metalworking fluids contain lubricants, biocides, and metals that are regulated pollutants
- Keep the waste coolant at a separate labeled container area until pickup

Step 3: Clean the Coolant Tank Follow the coolant tank cleaning procedure in Section 9.6 (Task S-03).**Step 4: Mix Fresh Coolant**

1. Calculate the required volume of concentrate: see Chapter 12, Section 12.2.3.
2. For a 350 L tank at 8% concentration: 28 L of concentrate + 322 L of water.
3. Fill the tank with clean water first (to approximately 320 L).
4. Add the coolant concentrate to the water. Always add concentrate to water (never the reverse).
5. If using a concentrated product: add slowly and stir or allow the pump to circulate to mix.

Step 5: Verify New Coolant Properties

1. After filling and circulating for 10 minutes:

2. Measure concentration: target $\pm 0.5\%$ of target value.
 3. Measure pH: should be 8.5–9.5 for freshly mixed solution.
 4. Record in maintenance log.
 5. Reset coolant hours counter: SYSTEM → MAINT → COOLANT RESET.
-

9.10 Spindle Bearing Inspection

Annual vibration-based inspection procedure is described in Section 9.7, Task A-02. The following supplementary guidance applies:

Interim Spindle Health Monitoring:

The BC-500X control includes a basic spindle health monitoring function that tracks spindle motor current as a proxy for spindle resistance (and therefore bearing drag). Monitor this value at: SYSTEM → DIAGNOSTIC → SPINDLE → BEARING LOAD INDEX.

The Bearing Load Index (BLI) is calculated from the no-load spindle motor current as a percentage of rated current at 5,000 RPM. It is normalized to 100% at commissioning.

BLI Value	Interpretation
90–110%	Normal bearing condition
110–130%	Slight increase in drag — monitor more frequently
130–150%	Significant bearing wear — schedule inspection
> 150%	Severe bearing drag — reduce spindle speed; schedule emergency inspection

Trigger for Emergency Inspection: Stop the machine and contact BetaCorp Systems immediately if any of the following occur:

- Abnormal noise from the spindle (grinding, squealing, clicking) at any speed
 - Spindle head is too hot to hold your hand on for more than 3 seconds (approximately > 65°C)
 - BLI exceeds 150%
 - Spindle vibration causes visible workpiece surface finish deterioration
-

9.11 Tool Changer Lubrication

The BC-500X ATC system requires periodic lubrication of:

1. ATC arm cam follower and guide surfaces
2. Carousel worm gear

ATC Arm Lubrication:

Frequency: Every 500 tool changes (automatic cycle per B3.26–B3.28) or monthly manual, whichever comes first

Grease Type: BetaCorp ATC Grease (BC-GRS-ATC-001), NLGI Grade 2, lithium complex, extreme pressure

Procedure:

1. Navigate to SYSTEM → ATC → MANUAL → ARM HOME to verify the ATC arm is at the home position.
2. Open the ATC access panel (top of machine, above ATC).
3. Locate the two grease fittings (Zerk/Alemite style) on the ATC arm guide rails — one at the front of the arm travel, one at the rear.
4. Apply grease gun with compatible grease cartridge (BC-GRS-ATC-001).
5. Apply 3 pumps of grease at each fitting.
6. Operate the ATC arm through 5 full cycles (SYSTEM → ATC → MANUAL → CYCLE × 5) to distribute the grease.
7. Wipe excess grease from surfaces with a clean rag.

Carousel Worm Gear Lubrication:

Frequency: Every 6 months

Grease Type: Same as ATC arm (BC-GRS-ATC-001)

Procedure:

1. Lock out the machine (LOTO).
 2. Remove the carousel drive gear access cover (rear of carousel mechanism, 4 × M6 socket screws).
 3. Apply approximately 10 g of grease to the worm gear teeth using a grease brush.
 4. Reinstall access cover.
 5. Remove LOTO and run the carousel through 5 full rotations to distribute grease.
 6. Check for excess grease ejection from the gear area; wipe excess.
-

9.12 Encoder Cable Inspection and Replacement

Inspection: Monthly per Section M-03.

Replacement Trigger: Replace encoder cables if:

- Any visible damage to the cable jacket (cuts, abrasions, crushing)
- Cable fails continuity test (> 0.5 ohm between shield and ground at cable ends)

- E101, E110, E111, or E112 faults occur and are traced to cable problems
- Preventive replacement interval: every 5 years or 15,000 machine hours (whichever comes first)

Encoder Cable Replacement Procedure (Z-Axis Spindle Encoder, BC-CBL-SPENC-3M):

This procedure is the most involved cable replacement due to the energy chain routing.

Materials Required:

- Replacement cable: BC-CBL-SPENC-3M (spindle encoder cable, 3 m)
- Cable ties and energy chain retainer clips as needed
- Electrical tape
- Small pry tool and long-nose pliers

Procedure:

1. Lock out the machine (LOTO).
2. Open the Z-axis energy chain cover along its full length.
3. Remove all cable retainer clips from the existing encoder cable in the energy chain. Note the clip positions (photograph before removal).
4. At the spindle motor end: loosen the cable strain relief fitting on the motor housing. Rotate counter-clockwise 2 full turns to loosen (do not remove completely).
5. At the drive end (in electrical cabinet): loosen the cable strain relief fitting at the cabinet entry.
6. Disconnect the encoder connector at the drive end (press the connector locking tab and pull). Label the disconnected connector.
7. Disconnect the encoder connector at the motor end.
8. Carefully pull the old cable through the energy chain. Thread the new cable through the energy chain before completely removing the old cable (use the old cable as a pull guide by taping the new cable to the tail of the old cable before pulling).
9. Feed the new cable through the full length of the energy chain.
10. Route the motor end of the new cable to the motor connector location. Connect the motor-end connector (press firmly until locking tab clicks).
11. Tighten the motor housing strain relief fitting: 2.5 N·m.
12. Route the drive end of the new cable to the drive cabinet entry point. Pass through the cabinet strain relief.
13. Connect the drive-end connector.
14. Verify the cable shield is connected to chassis ground at the drive end only (see Section 8.5.1).
15. Secure the cable in the energy chain with retainer clips at the same positions as the

original.

16. Close the energy chain cover.
 17. Remove LOTO. Power on the machine.
 18. Navigate to SYSTEM → DIAGNOSTIC → ENCODER STATUS → SPINDLE. Verify signal quality = "OK."
 19. Run the spindle at 500 RPM, 2,000 RPM, 5,000 RPM, and 10,000 RPM. Verify E101 does not occur.
 20. Record the cable replacement in the maintenance log (part number, date, reason, technician).
-

9.13 Ballscrew and Guideway Maintenance

9.13.1 Ballscrew Maintenance

The BC-500X ballscrews require minimal direct maintenance beyond the centralized way lube system. The following periodic inspections are required:

Quarterly Ballscrew Inspection:

1. With machine LOTO'd, access the ballscrew along its length by opening the axis cover panels (where accessible — not all of the ballscrew may be directly visible without removing covers).
2. Inspect the accessible ballscrew shaft for:
 - Coolant or chip contamination on the shaft (indicates wiper seal failure)
 - Rust spots (indicates inadequate lubrication or coolant water contamination)
 - Visible damage to the ball circuit helix (rare; would cause irregular motion)
3. Inspect the ballscrew nut for:
 - Oil seal condition (seals at each end of the nut should be intact and not hardened)
 - Looseness between nut and ballscrew (backlash greater than 0.002 mm indicates preload loss)
4. Check ballscrew-to-servo motor coupling (bellows coupling):
 - Inspect bellows for cracks or tears
 - Verify coupling is secure on both shafts (no axial play)
 - Part No. BC-CUP-X-001 / BC-CUP-Y-001 / BC-CUP-Z-001

Ballscrew Replacement: Ballscrew replacement is a major service event requiring machine re-calibration afterward. Expected service life of the BC-500X ballscrews:

- X and Y axes: 15,000–25,000 hours under normal use
- Z-axis: 10,000–20,000 hours (higher load due to gravity)

Contact BetaCorp Systems when ballscrew replacement is anticipated.

9.13.2 Linear Guideway Maintenance

The linear guideways require proper lubrication (provided by the centralized way lube system) and protection from chip contamination (provided by wiper seals).

Annual Guideway Carriage Re-Lubrication:

In addition to the automatic way lube system, the guideway carriages should be manually re-lubricated annually at the grease fitting on each carriage:

1. Lock out machine (LOTO).
 2. Using a grease gun with a suitable fitting, apply 2 mL of ISO VG 68 way lube oil (or equivalent guideway grease per IKO specification) to the grease fitting on each carriage.
 3. Operate the axis through full travel 5 times to distribute the grease.
 4. Wipe excess from the rail surface.
-

9.14 Way Lube System Maintenance

Monthly: Check oil level and consumption rate (Task M-04). **Semi-annual:** Full system inspection, as follows:

Semi-Annual Way Lube System Inspection:

1. Lock out machine (LOTO).
 2. At each lubrication distribution manifold (X-axis manifold: BC-LUB-MAN-X; Y-axis: BC-LUB-MAN-Y; Z-axis: BC-LUB-MAN-Z):
 - Inspect manifold for cracks or damaged metering outlets
 - Verify all distribution tubes (4 mm OD nylon) are connected and not kinked or blocked
 3. Inspect each distribution tube from manifold to lubrication point for:
 - Kinks, crushing, or disconnection
 - Discoloration or brittleness (degraded nylon)
 4. Check each carriage lubrication nipple. Verify the nipple is clear (use a fine wire to check if blocked).
 5. Manually trigger a way lube cycle (SYSTEM → MAINT → WAY LUBE → MANUAL CYCLE).
 6. After the pump cycle, press a clean white paper against each carriage wiper area. There should be a slight oil smear on the paper after the cycle.
 7. If any point shows no oil transfer: the distribution tube or metering outlet is blocked. Replace the blocked section of tubing (BC-LUB-TUBE-4) or the manifold outlet.
-

9.15 Filter Replacement Schedule

Summary of Filter Change Intervals:

Filter	Part No.	Replacement Interval
Hydraulic return filter element	BC-FLT-HYD-001	1,000 hours or at bypass indicator
Coolant inlet strainer	BC-STR-CLT-001	Replace if damaged; clean weekly
Way lube inline filter	BC-FLT-LUB-001	2,000 hours
Spindle oil-air lube filter	BC-FLT-OA-001	2,000 hours
Pneumatic FRL filter	BC-FLT-AIR-001	1,000 hours or when saturated
Coolant paper band filter (optional)	BC-FLT-PBF-001 (roll)	Replace when roll exhausted or saturated
Cabinet heat exchanger filter	BC-FLT-CAB-001	500 hours or as required

Hydraulic Return Filter Replacement Procedure (BC-FLT-HYD-001):

1. LOTO machine; de-pressurize hydraulic system.
2. Position drain tray under filter housing.
3. Unscrew filter bowl counter-clockwise.
4. Remove old element and dispose in waste filter container.
5. Clean inside of filter bowl.
6. Install new element (BC-FLT-HYD-001) in bowl.
7. Apply thin film of clean hydraulic oil to bowl O-ring.
8. Reinstall bowl; hand-tighten plus 3/4 turn.
9. Remove LOTO; start HPU; verify no leaks.
10. Record in maintenance log.

End of Chapter 9 — Maintenance Schedule

PART 9 — SPARE PARTS CATALOG

Chapter 10 — Spare Parts Catalog

10.1 How to Order Parts

Ordering Methods:

Method	Contact	Notes
Online Portal	parts.betacorpystems.com	Requires customer account login
Phone	+1 (800) 555-BETA (Option 2 — Parts)	Mon–Fri 7:00 AM–6:00 PM EST
Email	parts@betacorpystems.com	Include machine serial number
Fax	+1 (740) 555-9201	Use BetaCorp Parts Order Form

Required Information When Ordering:

- Machine model: BC-500X
- Machine serial number (from nameplate)
- BetaCorp part number (from this catalog)
- Quantity required
- Shipping address and contact name
- Purchase order number (or payment method for non-account customers)

Shipping Options:

- Standard: 3–5 business days (continental US)
- Expedite: Next business day (for in-stock items)
- Emergency 24/7: Call the emergency hotline at +1 (800) 555-9911 for critical machine-down situations

Parts Warranty: All genuine BetaCorp replacement parts are warranted for 90 days from installation (6 months for new-machine warranty parts) against defects in materials and workmanship.

CAUTION: The use of non-BetaCorp replacement parts may void the machine warranty and may introduce compatibility or safety issues. BetaCorp strongly recommends the use of genuine BetaCorp parts for all service replacements.

10.2 Spindle Assembly Parts

Part No.	Description	Qty per Machine	Unit	Notes
BC-MTR-SP-001	Spindle motor assembly, 15 kW,	1	EA	Includes motor and integral

Part No.	Description	Qty per Machine	Unit	Notes
	with encoder			encoder mount
BC-GBX-SP-001	Spindle gearbox assembly, 2-speed	1	EA	Complete gearbox with housing
BC-BRG-SP-001	Spindle front bearing, matched pair (angular contact, 15°)	1 SET	EA	High-precision P5; must be replaced as matched pair
BC-BRG-SP-002	Spindle front bearing, single (tandem companion)	1	EA	Matched to BC-BRG-SP-001 set
BC-BRG-SP-003	Spindle rear bearing, deep groove	1	EA	Standard precision
BC-BRG-SP-004	Spindle rear bearing, angular contact, floating	1	EA	
BC-DRB-SP-001	Draw bar assembly, complete (CAT-40/BT-40 specify)	1	EA	Specify taper at order
BC-BELL-SP-001	Belleville washer stack (10-piece set)	1 SET	EA	Replace entire stack; do not mix old and new washers
BC-COL-SP-CAT40	Collet assembly, CAT-40	1	EA	For CAT-40 taper machines
BC-COL-SP-BT40	Collet assembly, BT-40	1	EA	For BT-40 taper machines
BC-SEAL-SP-UC-001	Unclamp cylinder seal kit	1 KIT	EA	Contains all seals for unclamp cylinder
BC-SEN-SP-SEAT	Tool seating proximity sensor	1	EA	Inductive, NPN, NO
BC-OR-SP-001	Spindle air purge orifice plate	1	EA	Stainless steel; diameter 1.5 mm
BC-ENC-SP-001	Spindle encoder, Heidenhain ROD 420, 4096 PPR	1	EA	
BC-CBL-SPENC-3M	Spindle encoder cable, 3 m, shielded	1	EA	RS-422 differential; replace at 5 years
BC-CUP-SP-001	Spindle encoder shaft coupling	1	EA	Backlash-free; inspect annually

Part No.	Description	Qty per Machine	Unit	Notes
	(bellows)			
BC-CON-ENC-12P	Encoder cable connector, motor end (12-pin)	2	EA	Order 2; one spare
BC-CON-ENC-DRV	Encoder cable connector, drive end	1	EA	
BC-LUB-OA-001	Oil-air lubrication unit, spindle	1	EA	With 2L reservoir
BC-LUB-OA-OIL	Oil-air lube oil, ISO VG 32, 1L	1	L	Mobil Velocite No. 6 or equivalent
BC-FLT-OA-001	Oil-air lube inline filter element	1	EA	Replace every 2,000 hours
BC-NOS-SP-001	Spindle nose protector cap (for shipping/storage)	1	EA	
BC-KEY-SP-001	Spindle drive key (CAT-40)	2	EA	Used for manual tool tightening
BC-TOOL-0224	Spindle wrench (for manual tool installation)	1	EA	Hold the spindle during non-ATC tool changes

10.3 Axis Drive Parts

X-Axis Parts:

Part No.	Description	Qty	Unit	Notes
BC-MTR-X-001	X-axis servo motor, BetaCorp SM-300-3K, 3 kW	1	EA	
BC-DRV-X-001	X-axis servo drive, BetaCorp SD-300, 3 kW	1	EA	
BC-BSC-X-001	X-axis ballscrew assembly, Ø40 × pitch 10, Class 3	1	EA	Complete with fixed end support
BC-NUT-X-001	X-axis ballscrew nut, double-nut preloaded	1	EA	Replace nut assembly, not just nut body
BC-BRG-X-FIX	X-axis ballscrew fixed end bearings	1 SET	EA	Angular contact, matched pair

Part No.	Description	Qty	Unit	Notes
	(matched pair)			
BC-BRG-X-FLT	X-axis ballscrew floating end bearing	1	EA	Deep groove ball
BC-GWY-X-L	X-axis linear guideway, left rail (IKO LRWE55)	1	EA	Contact BetaCorp for installation service
BC-GWY-X-R	X-axis linear guideway, right rail	1	EA	
BC-CAR-X-001	X-axis guideway carriage (IKO LWRE55)	4	EA	4 carriages per machine (2 per rail)
BC-ENC-X-001	X-axis encoder, Heidenhain ERN 1381, 2500 PPR	1	EA	
BC-CBL-ENC-5M	Axis encoder cable, 5 m, shielded (X, Y, Z universal)	4	EA	Order 4: 1 per axis + 1 spare
BC-CUP-X-001	X-axis servo motor coupling (bellows)	1	EA	
BC-COV-X-001	X-axis telescoping cover set (3 sections)	1 SET	EA	
BC-WPR-X-001	X-axis guideway wiper seal set	1 SET	EA	8 wipers per set (4 carriages × 2)
BC-SW-OTRV-001	Over-travel limit switch (X, Y, Z universal)	6	EA	Order 6 (one per travel limit direction)

Y-Axis Parts:

Part No.	Description	Qty	Unit	Notes
BC-MTR-Y-001	Y-axis servo motor, BetaCorp SM-300-3K, 3 kW	1	EA	
BC-DRV-Y-001	Y-axis servo drive, BetaCorp SD-300, 3 kW	1	EA	
BC-BSC-Y-001	Y-axis ballscrew assembly, Ø40 × pitch 10, Class 3	1	EA	

Part No.	Description	Qty	Unit	Notes
BC-NUT-Y-001	Y-axis ballscrew nut	1	EA	
BC-BRG-Y-FIX	Y-axis fixed end bearings	1 SET	EA	
BC-BRG-Y-FLT	Y-axis floating end bearing	1	EA	
BC-GWY-Y-F	Y-axis guideway, front rail (IKO LRWE55)	1	EA	
BC-GWY-Y-R	Y-axis guideway, rear rail	1	EA	
BC-CAR-Y-001	Y-axis guideway carriage	4	EA	
BC-ENC-Y-001	Y-axis encoder, Heidenhain ERN 1381, 2500 PPR	1	EA	
BC-CUP-Y-001	Y-axis motor coupling	1	EA	
BC-COV-Y-001	Y-axis accordion bellows cover set	1 SET	EA	
BC-WPR-Y-001	Y-axis guideway wiper seal set	1 SET	EA	

Z-Axis Parts:

Part No.	Description	Qty	Unit	Notes
BC-MTR-Z-001	Z-axis servo motor, BetaCorp SM-400-3K, 4 kW	1	EA	
BC-DRV-Z-001	Z-axis servo drive, BetaCorp SD-400, 4 kW	1	EA	
BC-BSC-Z-001	Z-axis ballscrew assembly, Ø50 × pitch 10, Class 3	1	EA	Ground ballscrew (higher accuracy)
BC-NUT-Z-001	Z-axis ballscrew nut	1	EA	
BC-BRG-Z-FIX	Z-axis fixed end bearings	1 SET	EA	
BC-BRG-Z-FLT	Z-axis floating end bearing	1	EA	
BC-GWY-Z-L	Z-axis guideway, left rail (IKO	1	EA	Larger rail than X/Y

Part No.	Description	Qty	Unit	Notes
	LRWE65)			
BC-GWY-Z-R	Z-axis guideway, right rail	1	EA	
BC-CAR-Z-001	Z-axis guideway carriage (IKO LWRE65)	4	EA	
BC-ENC-Z-001	Z-axis encoder, Heidenhain ERN 1381, 2500 PPR	1	EA	
BC-CUP-Z-001	Z-axis motor coupling	1	EA	
BC-COV-Z-001	Z-axis accordion bellows cover set	1 SET	EA	
BC-WPR-Z-001	Z-axis guideway wiper seal set	1 SET	EA	
BC-CYL-Z-BRK	Z-axis hydraulic brake cylinder	1	EA	Critical safety component
BC-PSW-Z-BRK	Z-axis brake pressure switch	1	EA	
BC-TOOL-4417	Z-axis safety block (for maintenance)	1	EA	Must be on-site; store in machine tool cabinet

10.4 Tool Changer Parts

Part No.	Description	Qty	Unit	Notes
BC-MTR-ATC-001	ATC carousel servo motor, 0.75 kW	1	EA	
BC-DRV-ATC-001	ATC carousel servo drive	1	EA	
BC-POT-CAT40	Tool pot complete assembly, CAT-40	24	EA	One per carousel position
BC-POT-BT40	Tool pot complete assembly, BT-40	24	EA	Alternative taper
BC-GRP-CAT40-SET	Gripper finger set, 3 pcs, CAT-40	1 SET	EA	Replace as complete set
BC-SPR-POT-001	Tool pot wave spring	24	EA	One per pot; replace if retention force drops
BC-POT-BODY-	Tool pot body	4	EA	Spare bodies

Part No.	Description	Qty	Unit	Notes
001	only			
BC-POT-BOLT-M8	Tool pot retention bolt, M8 × 25, GR 8.8	48	EA	
BC-ATC-ARM-001	ATC arm complete assembly	1	EA	Major assembly; contact BetaCorp for installation
BC-ATC-CAM-001	ATC cam shaft assembly	1	EA	
BC-ATC-HYD-MTR	ATC hydraulic motor (for cam drive)	1	EA	
BC-ATC-GRP-L	ATC arm gripper, left	2	EA	Order 2; one spare
BC-ATC-GRP-R	ATC arm gripper, right	2	EA	
BC-ATC-GRP-SPR	ATC arm gripper spring	4	EA	
BC-ATC-CFW-001	ATC arm cam follower	2	EA	
BC-SEN-ATC-HOME	ATC arm home sensor	1	EA	Inductive, NPN, NC
BC-SEN-ATC-090	ATC arm 90° sensor	1	EA	
BC-SEN-ATC-180	ATC arm 180° sensor	1	EA	
BC-SEN-ATC-EXT	ATC arm extended sensor	1	EA	
BC-SEN-ATC-RET	ATC arm retracted sensor	1	EA	
BC-SEN-CAR-HOME	Carousel home sensor	1	EA	
BC-SEN-TP-001	Tool presence sensor (one per pot; order 24)	24	EA	Inductive, NPN, NO
BC-CYL-ATC-EXT	ATC arm extend/retract cylinder	1	EA	
BC-SEAL-ATC-CYL	ATC cylinder seal kit	1 KIT	EA	All seals for ATC cylinder
BC-SOL-ATC-EXT	ATC arm extend solenoid valve (SOL-HYD-03)	1	EA	

Part No.	Description	Qty	Unit	Notes
BC-SOL-ATC-RET	ATC arm retract solenoid valve (SOL-HYD-04)	1	EA	
BC-SOL-ATC-CW	ATC arm rotate CW solenoid (SOL-HYD-05)	1	EA	
BC-SOL-ATC-CCW	ATC arm rotate CCW solenoid (SOL-HYD-06)	1	EA	
BC-GRS-ATC-001	ATC arm grease, NLGI 2 Li-complex EP, 400 g cartridge	2	EA	

10.5 Hydraulic System Parts

Part No.	Description	Qty	Unit	Notes
BC-PMP-HYD-001	Hydraulic gear pump, 8.3 cc/rev	1	EA	Parker PGP505 or equivalent
BC-MTR-HP-001	Hydraulic pump motor, 4 kW, 1,450 RPM	1	EA	
BC-RLV-HYD-MAIN	Main system relief valve, 210 bar	1	EA	Factory-set; do not adjust
BC-PSW-HYD-MAIN	Hydraulic pressure switch, 120/100 bar	1	EA	
BC-FLT-HYD-001	Hydraulic return filter element, 10 µm abs	4	EA	Order 4 for annual stock
BC-SEAL-HYD-DRN	HPU drain plug seal, bonded washer	6	EA	Replace each oil change
BC-SOL-ZBK-001	Z-brake solenoid valve (SOL-HYD-01)	1	EA	
BC-SOL-TU-001	Tool unclamp solenoid (SOL-HYD-02)	1	EA	
BC-SOL-GCH-LO	Gear change low solenoid (SOL-HYD-07)	1	EA	

Part No.	Description	Qty	Unit	Notes
BC-SOL-GCH-HI	Gear change high solenoid (SOL-HYD-08)	1	EA	
BC-PRV-GCH-001	Gear change pressure reducing valve, 80 bar	1	EA	
BC-PRV-TU-001	Tool unclamp pressure reducing valve, 150 bar	1	EA	
BC-CYL-TU-001	Tool unclamp cylinder	1	EA	
BC-SEAL-TU-CYL	Tool unclamp cylinder seal kit	1 KIT	EA	
BC-CYL-GCH-001	Gear change cylinder	1	EA	
BC-SEAL-GCH-CYL	Gear change cylinder seal kit	1 KIT	EA	
BC-HYD-HOSE-10	Hydraulic hose, 10 mm ID, 210 bar, 1 m	10	EA	Specify length at order
BC-OIL-HYD-VG46	ISO VG 46 hydraulic oil, 20 L container	3	EA	Full system fill = 50 L
BC-PORT-HYD-SAMPLE	Hydraulic oil sampling port	1	EA	For oil analysis sampling
BC-HEX-HPU-001	HPU oil cooler, air-cooled, 500 W	1	EA	
BC-FAN-HPU-001	HPU cooling fan, 0.18 kW	1	EA	
BC-GAUGE-HYD-001	Hydraulic pressure gauge, 0–250 bar, 63 mm	1	EA	Glycerin-filled
BC-GAUGE-THERM-001	Hydraulic oil thermometer, 0–80°C	1	EA	
BC-SIGHT-HYD-001	Hydraulic reservoir sight glass	1	EA	

10.6 Coolant System Parts

Part No.	Description	Qty	Unit	Notes
BC-PMP-CLT-001	Coolant pump assembly, 2.2 kW (complete)	1	EA	Centrifugal with mechanical seal
BC-SEAL-CLT-MECH	Coolant pump mechanical seal	2	EA	Replace every 2 years or on leak
BC-IMP-CLT-001	Coolant pump impeller, stainless 316L	1	EA	
BC-STR-CLT-001	Coolant inlet strainer, 150 µm, stainless	2	EA	Replace if damaged
BC-SW-CLT-LVL	Coolant level float switch	2	EA	One installed, one spare
BC-CBL-CLT-LVL-2M	Coolant level switch cable, 2 m	1	EA	
BC-TNK-CLT-001	Coolant tank, complete, 316L SS	1	EA	Major part; contact BetaCorp
BC-NZL-CLT-001	Coolant nozzle with gooseneck mount	8	EA	Order 8; 4 installed + 4 spare
BC-HOSE-CLT-1M	Coolant hose, 1 m, stainless fittings	8	EA	Nozzle supply hoses
BC-VLV-CLT-NZL	Coolant nozzle ball valve	8	EA	
BC-TEMP-CLT-001	Coolant temperature sensor, RTD Pt100	1	EA	
BC-MAG-CLT-001	Magnetic chip separator bar set	1 SET	EA	4 bars per set
BC-LED-WORK-001	Work zone LED lighting strip, IP65, 24VDC	2	EA	One installed, one spare
BC-WIN-PC-001	Work zone polycarbonate window, 12 mm	1	EA	Replace if cracked or scratched
BC-LOK-DOOR-001	Door electromagnetic lock, 24VDC	1	EA	
BC-SEAL-DOOR-001	Work zone door seal (EPDM rubber, per meter)	5	M	Approx. 4 m needed per machine

Part No.	Description	Qty	Unit	Notes
BC-FLT-PBF-001	Paper band filter roll (optional)	2	ROLL	Each roll: approximately 50 m

10.7 Electrical and Control Parts

Part No.	Description	Qty	Unit	Notes
BC-DRV-SP-15K	Spindle drive, BetaCorp SD-SP-15K	1	EA	Vector control spindle drive
BC-TRANS-CTL-001	Control transformer, 3 kVA, 480V/120V	1	EA	
BC-PSU-24V-001	24 VDC power supply, 20 A	2	EA	One installed, one spare
BC-PSU-5V-001	5 VDC power supply, 10 A (encoder supply)	2	EA	
BC-PLC-CPU-001	PLC CPU module, BetaCorp PLC-5000	1	EA	Contains machine PLC program
BC-PLC-IO-001	PLC I/O expansion module, 32DI/32DO	2	EA	
BC-SREL-001	Safety relay module, Pilz PNOZ X3 or equivalent	2	EA	E-Stop and door interlock relay
BC-CNC-CTRL-001	CNC control unit, Model 5000	1	EA	Complete control unit
BC-PEND-MPG-001	MPG pendant assembly	1	EA	With handwheel, E-Stop, cable
BC-HEX-CAB-001	Control cabinet heat exchanger, 1500W	1	EA	
BC-FLT-CAB-001	Cabinet heat exchanger filter media	6	EA	Pack of 6; replace every 500 hours
BC-BATT-CNC-001	CNC control backup battery (Li, 3V CR2)	3	EA	Replace every 3 years
BC-FUSE-MAIN-	Main disconnect	2	SET	

Part No.	Description	Qty	Unit	Notes
001	fuses, Class J, 100 A, 600V (set of 3)			
BC-FUSE-SP-001	Spindle drive fuses, Class J, 50 A (set of 3)	2	SET	
BC-FUSE-SRV-001	Servo drive fuses, Class J, 20 A (set of 3)	6	SET	2 sets per servo axis
BC-FUSE-CLT-001	Coolant pump fuses, Class J, 10 A (set of 3)	2	SET	
BC-RGEN-001	Regenerative braking resistor (servo DC bus)	1	EA	External resistor bank
BC-SW-OTRV-001	Over-travel limit switch, SPDT	6	EA	
BC-CONT-HYD-001	HPU motor contactor, 12 A coil 120VAC	1	EA	
BC-OLRELAY-001	Motor thermal overload relay	4	EA	One per motor
BC-CON-CIRC-12	Circular connector, 12-pin, panel mount	4	EA	Sensor/encoder connectors

10.8 Enclosure and Structure Parts

Part No.	Description	Qty	Unit	Notes
BC-WIN-PC-001	Work zone polycarbonate window, 12 mm thick	1	EA	Replaces cracked/scratched window
BC-SEAL-DOOR-001	Door seal, EPDM rubber, per meter	5	M	Sold by the meter
BC-LOK-DOOR-001	Door electromagnetic lock, 24 VDC	1	EA	
BC-SW-DOOR-001	Door interlock safety switch, Schmersal AZ16	2	EA	Dual-channel; order pair
BC-HDL-DOOR-001	Work zone door handle assembly	1	EA	

Part No.	Description	Qty	Unit	Notes
BC-GAS-DOOR-001	Door counterbalance gas strut, 200 N	2	EA	Assists door opening; replace if door difficult to open
BC-COV-ATC-001	ATC area access cover	1	EA	Sheet metal; includes mounting hardware
BC-COV-Z-001	Z-axis accordion bellows set	1 SET	EA	Replace if torn; prevents chip ingress
BC-COV-X-001	X-axis telescoping cover set	1 SET	EA	
BC-COV-Y-001	Y-axis accordion bellows set	1 SET	EA	
BC-LED-WORK-001	Work zone LED light strip (replacement)	2	EA	
BC-FTP-LVL-001	Leveling footpad assembly (complete)	6	EA	M24 bolt, pad, rubber element

10.9 Lubrication System Parts

Part No.	Description	Qty	Unit	Notes
BC-LUB-PMP-001	Way lube pump unit with 2.5L reservoir	1	EA	
BC-MTR-LUB-001	Way lube pump motor, 0.12 kW	1	EA	
BC-LUB-MAN-X	Way lube distribution manifold, X-axis (4 outlet)	1	EA	
BC-LUB-MAN-Y	Way lube distribution manifold, Y-axis (4 outlet)	1	EA	
BC-LUB-MAN-Z	Way lube distribution manifold, Z-axis (4 outlet)	1	EA	
BC-LUB-TUBE-4	Way lube	6	M/ROLL	

Part No.	Description	Qty	Unit	Notes
	distribution tubing, 4 mm OD nylon, 10 m			
BC-LUB-OIL-VG68	Way lube oil, ISO VG 68, 5L container	2	EA	Annual consumption: ~2 L
BC-LUB-LVL-SW	Way lube reservoir level switch	1	EA	Float type
BC-FLT-LUB-001	Way lube inline filter, 25 µm	2	EA	
BC-LUB-FIT-001	Lubrication nipple fittings (Zerk/Alemite), M6 (bag of 20)	1	BAG	
BC-OIL-VG32-1L	Spindle oil-air lube oil, ISO VG 32, 1 L	4	EA	Annual: ~2 L

10.10 Chip Conveyor Parts

Part No.	Description	Qty	Unit	Notes
BC-CNV-CHIP-001	Chip conveyor, complete assembly	1	EA	Specify dimensions at order
BC-MTR-CNV-001	Chip conveyor drive motor/gearbox, 0.75 kW	1	EA	
BC-BELT-CNV-001	Chip conveyor belt section, hinged steel, 250 mm wide, 1 m	6	M	Order per length needed
BC-PIN-CNV-001	Chip conveyor hinge pins, SS 304, 250 mm length	10	EA	
BC-SPR-CNV-001	Chip conveyor drive sprocket	2	EA	One per side
BC-CHN-CNV-001	Chip conveyor drive chain, 1 m	2	M	
BC-BKT-CNV-	Chip collection	1	EA	Formed steel with

Part No.	Description	Qty	Unit	Notes
001	bin (hopper), 120 L			casters
BC-BRNG-CNV-001	Chip conveyor return roller bearing	4	EA	

10.11 Consumables and Wear Items

Part No.	Description	Qty	Unit	Notes / Replacement Interval
BC-FLT-HYD-001	Hydraulic return filter element, 10 µm	4	EA	Every 1,000 hours
BC-FLT-AIR-001	Pneumatic FRL filter element	4	EA	Every 1,000 hours
BC-FLT-LUB-001	Way lube inline filter	2	EA	Every 2,000 hours
BC-FLT-CAB-001	Cabinet heat exchanger filter (pack of 6)	2	PACK	Every 500 hours
BC-FLT-OA-001	Oil-air lube filter	2	EA	Every 2,000 hours
BC-WPR-X-001	X-axis wiper seal set	2	SET	Replace if damaged
BC-WPR-Y-001	Y-axis wiper seal set	2	SET	Replace if damaged
BC-WPR-Z-001	Z-axis wiper seal set	2	SET	Replace if damaged
BC-SEAL-SP-UC-001	Spindle unclamp cylinder seal kit	1	KIT	Every 3 years
BC-SEAL-CLT-MECH	Coolant pump mechanical seal	2	EA	Every 2 years
BC-OIL-HYD-VG46	Hydraulic oil, ISO VG 46, 20 L	3	EA	Every 6 months (full change)
BC-LUB-OIL-VG68	Way lube oil, ISO VG 68, 5 L	2	EA	As consumed
BC-OIL-VG32-1L	Spindle bearing oil, ISO VG 32, 1 L	4	EA	As consumed
BC-GRS-ATC-001	ATC grease, NLGI 2 Li-	4	EA	As consumed

Part No.	Description	Qty	Unit	Notes / Replacement Interval
	complex EP, 400 g			
BC-BATT-CNC-001	CNC backup battery, Li, 3V	3	EA	Every 3 years
BC-CBL-SPENC-3M	Spindle encoder cable	1	EA	Every 5 years or on damage
BC-CBL-ENC-5M	Axis encoder cable	2	EA	Every 5 years or on damage
BC-SEAL-HYD-DRN	Hydraulic drain plug seal	6	EA	Every oil change
BC-SPR-POT-001	Tool pot wave spring	10	EA	Replace if retention force drops
BC-LED-WORK-001	Work zone LED strip	1	EA	As needed
BC-WIN-PC-001	Work zone window	1	EA	Replace if cracked

End of Chapter 10 — Spare Parts Catalog

PART 10 — HYDRAULIC SYSTEM MANUAL

Chapter 11 — Hydraulic System Manual

11.1 Hydraulic System Overview

The BC-500X hydraulic system is a centralized, constant-pressure, fixed-displacement system that provides power for four machine functions: Z-axis brake, spindle tool unclamp, ATC arm actuation, and spindle gear change.

The system operates at a normal working pressure of 150 bar (2,176 psi) with a maximum relief valve setting of 210 bar (3,046 psi). The hydraulic power unit (HPU) is a self-contained assembly mounted on the right rear of the machine. All hydraulic actuators and valves are pre-piped to the HPU at the factory.

Design Philosophy:

The BC-500X hydraulic system uses a "load-sensitive" approach for pressure management:

- The fixed-displacement pump runs continuously when the HPU is on
- A pressure relief valve bypasses excess pump flow back to tank when all actuators are not demanding flow
- This design is simple, reliable, and immediately responsive to actuator demands

Fail-Safe Design: All BC-500X hydraulic circuits are designed to fail to a safe state:

- Z-axis brake: spring-applied (safe = brake ON); hydraulic release
- Tool holder: spring-clamped (safe = tool held); hydraulic unclamp
- ATC arm: spring-centered (safe = arm retracted); hydraulic actuate
- Gear change: spring-held in last position (safe = maintain current gear)

This means that loss of hydraulic pressure leaves the machine in a mechanically safe state.

11.2 Hydraulic Circuit Description

The BC-500X hydraulic circuit is organized as follows:

Main Supply Circuit: Hydraulic oil flows from the reservoir through the pump inlet strainer, through the gear pump, through the pressure check valve, and into the main supply manifold at 150 bar. The main relief valve (210 bar) is connected between the supply manifold and the return line.

Z-Axis Brake Circuit:

- Supply: from main manifold through SOL-HYD-01 (4/2, spring return) to Z-brake cylinder release port
- Return: through SOL-HYD-01 to tank when solenoid is de-energized
- Pressure reducing valve: none — full system pressure applied to brake
- Normally: SOL-HYD-01 de-energized → spring applies brake → Z-axis held
- Brake release: SOL-HYD-01 energized → hydraulic oil pushes against brake spring → brake releases

Tool Unclamp Circuit:

- Supply: from main manifold through PRV-HYD-02 (150 bar) → SOL-HYD-02 → tool unclamp cylinder
- The PRV limits maximum force applied to the draw bar
- Normally: SOL-HYD-02 de-energized → spring returns unclamp cylinder → draw bar spring clamps tool
- Unclamp: SOL-HYD-02 energized → hydraulic piston extends → pushes draw bar down → releases Belleville springs → collet opens → tool released

ATC Arm Circuit:

- A single double-rod cylinder actuates the arm extend/retract (SOL-HYD-03, SOL-HYD-04)
- A hydraulic motor driven through SOL-HYD-05 / SOL-HYD-06 rotates the arm cam shaft CW or CCW
- Flow control valves on each port control the arm speed (factory set; see B4.15)

Gear Change Circuit:

- From main manifold through PRV-HYD-01 (80 bar) to gear change cylinder
 - SOL-HYD-07 (low gear) / SOL-HYD-08 (high gear) select gear
 - Lower pressure (80 bar vs 150 bar) is used to avoid excessive force on the gear shift dogs
-

11.3 Hydraulic Power Unit (HPU)

The HPU is a compact, self-contained unit with the following major components:

Reservoir:

- 50 L capacity, mild steel with internal coating
- Removable access panel for internal inspection (4 × M8 screws)
- Sight glass with level indicator (MIN/MAX marks)
- Breather/fill cap with 10 µm filter
- Drain plug at lowest point
- Temperature indicator (analog thermometer, 0–80°C)
- Return line filter with differential pressure indicator

Hydraulic Pump:

- Parker PGP505 or equivalent gear pump
- Displacement: 8.3 cc/rev
- Drive motor: 4 kW, 1,450 RPM
- Flexible coupling between motor and pump shaft

Pressure Relief Valve (Main):

- Parker or Sun Hydraulics type
- Set at 210 bar (factory preset with tamper-resistant locknut)
- Direct-acting, poppet type

Filter (Return Line):

- Housing: spin-on type, cast aluminum
- Element: BC-FLT-HYD-001, 10 µm absolute, glass fiber

- Differential pressure indicator: visual pop-up type, set at 5 bar differential
- Built-in bypass at 6 bar differential (to protect filter from cold-oil damage at startup)

Pressure Switch:

- BC-PSW-HYD-MAIN
- Switching points: 120 bar (rising) and 100 bar (falling)
- Connects to PLC to signal E404

Temperature Switch:

- Wax-element type, 65°C setpoint (E411 warning)
- Second temperature switch at 70°C (E412 shutdown)

Heat Exchanger:

- Air-cooled (fan-assisted) oil cooler
- Capacity: 1,000 W at 40°C ambient
- Fan motor: 0.18 kW, integral
- Thermostat: ON at 50°C oil temperature, OFF at 45°C

11.4 Pressure Settings and Adjustment

WARNING: Never adjust hydraulic pressure settings beyond the values specified in this manual without written authorization from BetaCorp Systems. Over-pressure conditions can cause immediate component failure and present a high-pressure injection hazard (see Section 2.6.1).

Pressure Setting Summary:

Circuit	Component	Setting	Notes
Main system	Relief valve (RLV-HYD-MAIN)	210 bar	Factory-set; DO NOT ADJUST
Normal working	System operating pressure	150 bar	Maintained by pump and relief valve
Tool unclamp	PRV-HYD-02	150 bar	Adjustable; normally factory-set
Gear change	PRV-HYD-01	80 bar	Adjustable; normally factory-set
Z-brake release	N/A (direct system pressure)	150 bar	No reducing valve

How to Adjust PRV-HYD-02 (Tool Unclamp Pressure):

Adjustment of the tool unclamp pressure may be required if:

- Tool release is incomplete (pressure too low — increase)
 - Tool release is too forceful, causing tool holder to eject (very rare; pressure too high — decrease)
 - Spindle draw bar springs have been replaced with different preload
1. LOTO the machine; de-pressurize.
 2. Locate PRV-HYD-02 on the HPU valve manifold (labeled).
 3. Remove the tamper-resistant cap from the adjustment screw (requires a BetaCorp tamper key; contact BetaCorp for this tool).
 4. Start the HPU and measure tool unclamp pressure using a gauge at the PRV-HYD-02 outlet port.
 5. Adjust the setscrew: clockwise = increase pressure; counter-clockwise = decrease.
 6. Target setting: 150 bar $\pm$ 5 bar.
 7. Replace the tamper cap.
 8. Test tool change: command a manual tool change (SYSTEM → ATC → MANUAL → TOOL CHANGE). Verify tool releases smoothly.
 9. Document the adjustment.
-

11.5 Hydraulic Valve Descriptions

SOL-HYD-01 — Z-Axis Brake Release Valve

Type: 4/2 Directional Control Valve, Spring Return, 24 VDC solenoid **Normal State (Solenoid De-Energized):** Hydraulic brake pressure circuit is connected to tank (drain). Spring in brake cylinder applies brake. Z-axis is held. **Energized State:** Hydraulic pressure (150 bar) is applied to brake release port. Brake releases. **Failure Mode:** If SOL-HYD-01 fails in de-energized state: Z-axis brake stays applied (safe). If SOL-HYD-01 fails energized: Z-axis brake stays released (requires servo brake only to hold Z-axis — FAULT E415 will indicate if brake release is not confirmed within timeout).

Service Procedure: If SOL-HYD-01 is suspected faulty (Z-axis brake not releasing or not applying):

1. Verify 24 VDC is present at solenoid coil terminals when energized (check with voltmeter).
2. Measure solenoid coil resistance (should be 20–50 ohms for 24 VDC coils). Open circuit (>1 MOhm) indicates failed coil.
3. If coil is good and valve is not actuating: valve spool is stuck. Replace SOL-HYD-01.
4. Part No.: BC-SOL-ZBK-001

SOL-HYD-02 — Tool Unclamp Valve

Type: 4/2 Directional Control Valve, Spring Return, 24 VDC solenoid **Normal State:** Tool unclamp cylinder connects to tank. Draw bar Belleville springs hold tool clamped. **Energized State:** Hydraulic pressure applied to unclamp cylinder. Tool releases. **Failure Mode:** Fail de-energized = tool stays clamped (safe). Fail energized = tool unclamp stays active (tool would release when spindle is stopped — MAJOR HAZARD; fault E321 should detect this).

SOL-HYD-03 / SOL-HYD-04 — ATC Arm Extend/Retract

Type: 4/3 Directional Control Valve, Spring Centered, 24 VDC solenoid (dual coil) **Normal State (Both Solenoids De-Energized):** ATC cylinder is blocked (both ports connected to blocked center). Arm holds position. **SOL-HYD-03 Energized:** ATC arm extends. **SOL-HYD-04 Energized:** ATC arm retracts. **Service Note:** 4/3 valve with blocked center — both solenoids must not be energized simultaneously (PLC interlock prevents this).

11.6 Hydraulic Cylinder Operation

Z-Axis Brake Cylinder (BC-CYL-Z-BRK)

Type: Single-acting, spring-apply (spring-apply / hydraulic-release) **Bore:** Ø100 mm **Stroke:** 30 mm **Spring Force:** 12,000 N (applied force on brake disc) **Hydraulic Release Pressure:** 80 bar minimum; 150 bar nominal **Seal Material:** FKM (fluoroelastomer, "Viton") — compatible with ISO 46 mineral oil **Wiper:** Polyurethane

Cylinder Service:

The Z-axis brake cylinder should be inspected semi-annually for:

- External oil leakage at the rod seal (a slight oil film is acceptable; dripping is not)
- Rod surface for corrosion or scoring (rod must be clean and smooth)
- Mount bolts for tightness (8 × M10, torque: 40 N·m)

If the rod seal leaks excessively: replace the complete cylinder (BC-CYL-Z-BRK) or the seal kit (BC-SEAL-Z-BRK) — seal kit replacement requires disassembly of the cylinder by a qualified hydraulic technician.

Tool Unclamp Cylinder (BC-CYL-TU-001)

Type: Single-acting, spring-return (hydraulic-extend / spring-retract) **Bore:** Ø80 mm **Stroke:** 45 mm **Force at 150 bar:** 75,398 N (75.4 kN) — significantly exceeds the 18,000 N draw bar spring force, with adequate margin **Seal Material:** FKM **Service:** Replace seal kit (BC-SEAL-TU-CYL) if rod seal leaks.

11.7 Hydraulic Oil Specifications

The BC-500X hydraulic system requires ISO VG 46 anti-wear (AW) hydraulic oil. The following

specifications apply:

Property	Requirement	Test Method
Viscosity grade	ISO VG 46	ISO 3448
Viscosity at 40°C	41.4–50.6 cSt	ISO 3104
Viscosity at 100°C	6.8–7.2 cSt (typical)	ISO 3104
Viscosity index	≥ 95	ISO 2909
Flash point (COC)	≥ 200°C	ISO 2592
Pour point	≤ -12°C	ISO 3016
Rust protection (ASTM D665A)	Pass	ASTM D665
Oxidation stability (TOST)	≥ 2,000 hours	ASTM D943
Zinc-type AW additive	Yes (ZDDP or ashless AW)	—
Foam tendency	Maximum 25 mL/0 mL (Seq. I/II/III)	ASTM D892
Air release (at 50°C)	≤ 5 minutes	ISO 9120
Water separability	Pass (ASTM D1401)	ASTM D1401
Demulsibility	≤ 40 min at 82°C	ASTM D1401

Approved Hydraulic Oil Brands:

Brand	Product Name	Notes
Mobil	Mobil DTE 25M	Preferred brand
Shell	Shell Tellus S2 MX 46	
Chevron	Chevron Rando HDZ 46	
Total	Azolla ZS 46	
BetaCorp Supply	BC-OIL-HYD-VG46	BetaCorp-supplied oil; guaranteed compatibility
Castrol	Castrol Hyspin AWH-M 46	

Oils NOT Approved:

- Biodegradable hydraulic oil (requires system flush and special seals)
- Fire-resistant hydraulic fluid (HF types) — incompatible with standard seals
- Motor oil or transmission fluid
- Way lube oil (wrong viscosity and additive package)
- Any oil not on the approved list above

Oil Mixing: Do not mix different brands of hydraulic oil unless compatibility is confirmed by the oil supplier. If changing oil brands, completely drain the system and flush with the new oil before filling.

11.8 Contamination Control

Hydraulic system contamination is the single largest cause of hydraulic component failure in the field. The following contamination control practices must be followed:

Target Cleanliness Level:

- ISO 4406 cleanliness code: $\leq 16/14/11$
- This means: no more than 640 particles $\geq 4 \mu\text{m}$ per mL; no more than 160 particles $\geq 6 \mu\text{m}$ per mL; no more than 20 particles $\geq 14 \mu\text{m}$ per mL

Contamination Sources and Prevention:

Source	Prevention
Built-in contamination (from manufacturing)	Flush system before commissioning (Section 4.6.2)
Ingress from fill port	Use a 10 μm filter funnel when adding oil; keep fill cap closed
Ingress from breather	Breather/fill cap has internal 10 μm filter; replace if damaged
Ingress during maintenance	Cap all open ports immediately; flush any disconnected lines before reconnecting
Water contamination	Inspect for coolant leaks; maintain good seal on hydraulic components
Internal wear particles	Monitor with regular oil analysis; change filter and oil at appropriate intervals

What to Do If Contamination is Suspected:

1. Take an oil sample and send for particle count analysis.
2. If cleanliness code exceeds 18/16/13: replace the hydraulic oil and filter immediately.
3. If contamination includes water (emulsion): check all hydraulic cylinder rod seals for leakage (coolant may be entering through a leaking cylinder or through a fitting); replace oil.
4. If metal particles are found in the oil analysis: identify the source (wear metals — Fe indicates pump or cylinder wear; Cu indicates bearing or valve body wear). Contact BetaCorp Systems.

11.9 Hydraulic System Diagnostics

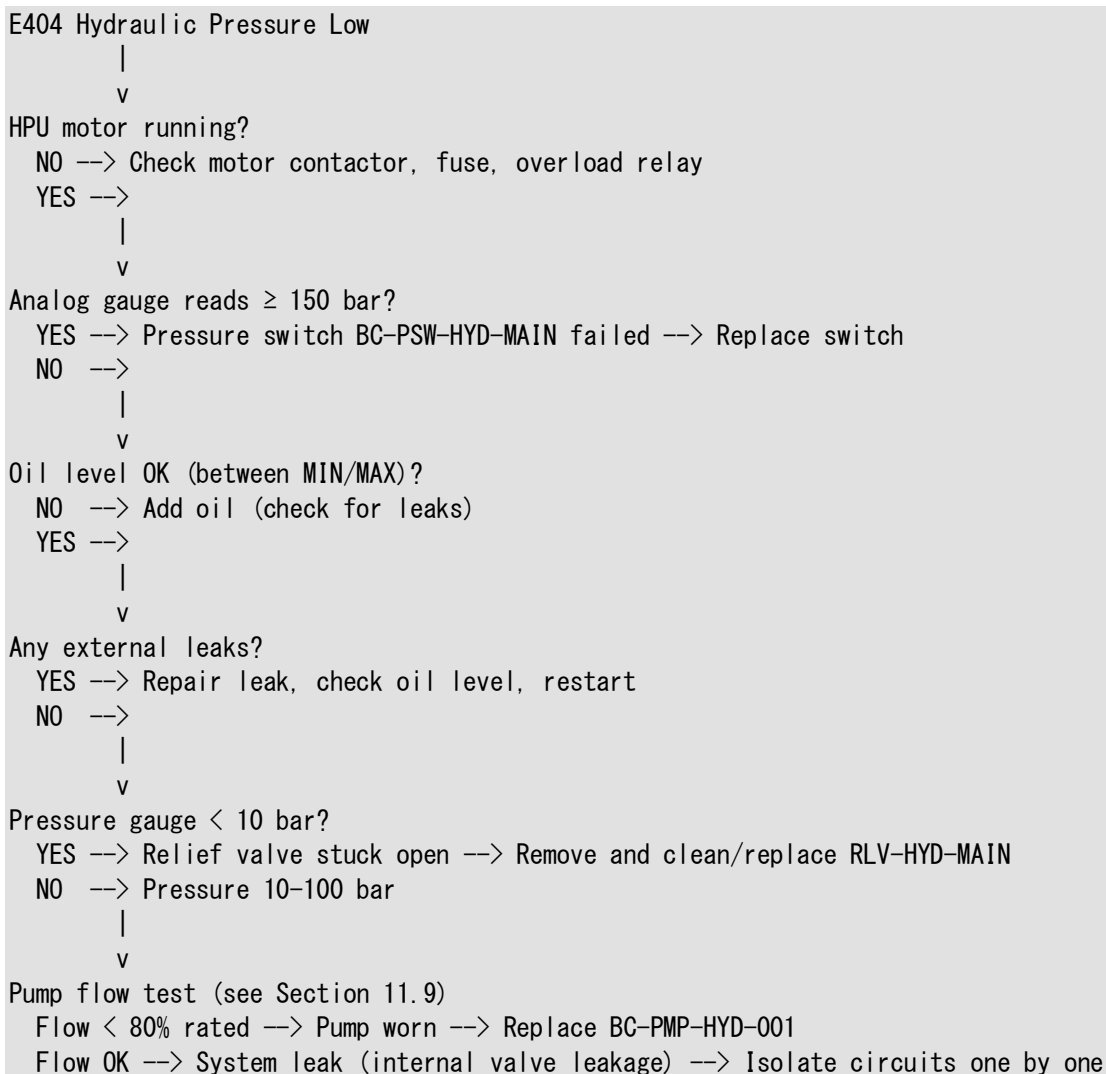
Hydraulic System Diagnostic Screen:

Navigate to SYSTEM → DIAGNOSTIC → HYDRAULIC SYSTEM to view:

- HPU motor status (ON/OFF)

- Hydraulic pressure (from digital pressure transmitter — if equipped; otherwise from pressure switch state)
- Oil temperature (from temperature sensor)
- Filter bypass indicator state
- Z-brake status (brake applied / released, confirmed by pressure switch BC-PSW-Z-BRK)
- Tool unclamp cylinder status
- All solenoid valve commanded states

Hydraulic Pressure Troubleshooting Flowchart:



Hydraulic System Pressure Test Procedure:

To identify which circuit is leaking internally (if pressure doesn't hold but no external leak):

1. Start HPU. Note system pressure.
2. Close the isolation valve for each branch circuit one by one (Z-brake circuit, tool unclamp circuit, ATC circuit, gear change circuit).
3. After each closure, observe if system pressure increases.

4. If pressure increases when a specific circuit is isolated: the internal leakage is in that circuit's actuator or valve.
5. Inspect the solenoid valve for that circuit (may be stuck open or valve spool worn) and inspect the cylinder for internal seal failure.

End of Chapter 11 — Hydraulic System Manual

PART 11 — COOLANT SYSTEM GUIDE

Chapter 12 — Coolant System Guide

12.1 Coolant System Overview

The BC-500X coolant system delivers metalworking fluid (cutting fluid) to the cutting zone during machining operations. Proper coolant management is essential for:

- Extending tool life (proper cooling and lubrication at the cutting interface)
- Improving surface finish (chip evacuation and cooling of workpiece)
- Preventing workpiece thermal distortion (heat management)
- Protecting machine surfaces from corrosion
- Maintaining operator health (well-managed coolant does not support harmful bacteria or dermatitis-causing decomposition products)

System Architecture: The BC-500X uses a central sump (coolant tank) system. Coolant from the tank is pumped to the work zone through flexible nozzles. After contacting the workpiece and tool, the coolant drains through the chip pan back to the tank, passing through the chip conveyor (which removes bulk chips) and the magnetic separator (which removes fine ferromagnetic particles). The coolant pump inlet strainer provides final filtration before the pump.

System Capacity:

- Tank volume: 350 L
 - Typical working volume (allowing for lines, pump, and sump level): approximately 320 L
 - Coolant turnover rate (at 60 L/min): approximately 5 minutes
-

12.2 Coolant Selection and Mixing Ratios

12.2.1 Approved Coolant Types

The BC-500X is approved for use with water-soluble semi-synthetic and fully synthetic metalworking fluids. The following product categories are approved:

Category	Description	Typical Application
Semi-synthetic	Contains 5–30% mineral oil emulsified in water	General machining of ferrous and aluminum alloys
Fully synthetic	Contains no mineral oil; water-based with synthetic lubricants	High-speed machining, aluminum, cast iron

Approved Coolant Products (Representative List):

Brand	Product	Type	Mix Ratio (General)
Blaser Swisslube	Blasocut 4000 Strong	Semi-synthetic	6–10%
Castrol	Hysol XBB	Semi-synthetic	5–8%
Master Chemical	TRIM SC520	Semi-synthetic	5–10%
Fuchs	Ecocool 68 CF	Semi-synthetic	5–10%
Houghton	Hocut 795B	Semi-synthetic	5–10%
Cimcool	CIMPERIAL 1011	Soluble oil (emulsion)	3–8%

Contact BetaCorp Systems if you wish to use a product not on this list. We will evaluate compatibility with the machine seals, coatings, and materials before approval.

Products NOT Approved:

- Straight cutting oil (neat oil) — not compatible with standard coolant pump seals; creates fire and mist hazards without appropriate equipment
- Silicone-based coolants — incompatible with some machine seals
- Coolants containing chlorine at concentrations above 50 ppm — causes corrosion of aluminum and stainless steel
- Coolants containing sodium nitrite — health hazard; may form carcinogenic nitrosamines

12.2.2 Mixing Ratios by Application

The appropriate coolant concentration depends on the workpiece material, operation type, and coolant product. Always consult the coolant manufacturer's technical data sheet for specific guidance. The following general guidelines apply:

Material	Operation	Recommended Concentration
Carbon steel	General milling, drilling	7–9%
Stainless steel (300 series)	All operations	8–10%

Material	Operation	Recommended Concentration
Tool steel (>45 HRC)	Milling, grinding	8–10%
Cast iron	Milling, boring	5–7%
Aluminum alloys	Milling, drilling	6–8%
Brass, copper	Milling, drilling	5–7%
Titanium	All operations	9–12%
Nickel superalloys (Inconel, etc.)	All operations	9–12%

NOTE: Higher concentrations (above 10%) generally do not improve cooling significantly and can cause skin irritation, foam problems, and increased coolant costs. Never exceed the coolant manufacturer's recommended maximum concentration.

12.2.3 Coolant Mixing Procedure

Principle: ALWAYS add coolant concentrate to water. Never add water to concentrate.

Rationale: Adding water to concentrate causes the high-concentration outer surface to "shock" the incoming water, potentially creating a gel or unstable emulsion. Adding concentrate to water allows gradual dilution and produces a stable, uniform mixture.

Equipment Required:

- Mixing tank or bucket (clean, no previous chemical residue)
- Measuring container
- Refractometer

Mixing Procedure:

1. Calculate required volumes:

- Required concentration: C% (e.g., 8%)
- Required total volume: V (e.g., 350 liters)
- Volume of concentrate: $V_{\text{conc}} = V \times (C / 100) \times \text{Concentration Factor}$
- Volume of water: $V_{\text{water}} = V - V_{\text{conc}}$

Note: The "Concentration Factor" accounts for the fact that refractometer readings for semi-synthetic fluids typically read lower than the actual oil content. Use the factor from the coolant manufacturer's data sheet. A typical factor is 1.4 for semi-synthetic fluids. Example: To achieve 8% actual concentration at a refractometer factor of 1.4, the refractometer should read $8/1.4 = 5.7\%$.

2. Prepare the mixing container or fill the machine coolant tank directly:

- Start with the required volume of clean water (municipal tap water is acceptable in most areas; if water hardness exceeds 500 ppm as CaCO_3 , use softened or deionized water)

3. Add concentrate:

- Measure the required volume of concentrate.
- Add the concentrate to the water slowly, while stirring.
- Do not add all concentrate at once — add in a thin stream to allow even dispersion.

4. Verify concentration:

- After mixing, collect a small sample.
- Read with the refractometer.
- The reading should match the expected refractometer value for the target concentration (per manufacturer's data).
- If reading is low: add more concentrate. If high: add more water.

5. Verify pH:

- Fresh mixture pH should be 8.5–9.5.
 - If pH is out of range: contact the coolant supplier.
-

12.3 pH Monitoring and Control

12.3.1 Importance of pH

Coolant pH is one of the most important indicators of coolant health:

- **pH below 7 (acid):** Immediate corrosion risk to machine surfaces and workpiece. Likely indicates severe bacterial contamination. DO NOT use machine with acidic coolant.
- **pH 7.0–8.5:** Below normal range. Corrosion risk increasing. Bacterial activity possible. Add pH adjuster and investigate.
- **pH 8.5–9.5:** Normal operating range. Adequate alkalinity to resist corrosion and control bacteria.
- **pH 9.5–10.5:** Slightly high. Monitor; may cause slight skin irritation. Usually not harmful to machine.
- **pH above 10.5:** Too high. Can cause skin irritation and potentially damage certain machine components. Dilute and investigate.

12.3.2 pH Measurement Equipment

Option A: pH Test Strips

- Quick, easy, inexpensive
- Accuracy: ± 0.5 pH units
- Suitable for routine weekly monitoring

- Use strips rated for pH 7–11 range
- Recommended product: Micro Essential Lab Hydrion pH strips or equivalent

Option B: Digital pH Meter

- Accuracy: ± 0.1 pH units
- Requires calibration with buffer solutions (pH 7.0 and pH 10.0)
- Better for detecting small changes and when precise data is needed
- Recommended for monthly measurements and trend tracking
- Electrode requires regular cleaning and periodic replacement

Calibration of pH Meter:

1. Prepare fresh calibration buffer solutions (pH 7.0 and pH 10.0) as directed by the meter manufacturer.
2. Rinse the electrode with clean water.
3. Immerse in pH 7.0 buffer. Adjust the meter's offset calibration until it reads 7.0 ± 0.02 .
4. Rinse electrode. Immerse in pH 10.0 buffer. Adjust the slope calibration until it reads 10.0 ± 0.02 .
5. Rinse electrode. Meter is now calibrated.

12.3.3 pH Adjustment

If pH is below 8.5:

1. Add a pH buffer adjuster (alkalinity booster) recommended by the coolant supplier.
2. Add no more than 0.05% by volume of the adjuster per treatment.
3. Circulate coolant for 30 minutes and re-test.
4. If pH does not respond to alkalinity booster and continues dropping rapidly (more than 0.2 pH units per day): suspect heavy bacterial contamination. Perform bacteria test (Section 12.4) and consider coolant replacement.

Common pH Adjusters:

- Triethanolamine (TEA): typical pH booster for metalworking fluids
- Sodium metasilicate: stronger pH buffer; use with caution (can cause skin sensitization at high concentration)
- Always use a pH adjuster approved by the coolant manufacturer

If pH is above 9.5:

1. Add water only (do not add concentrate).
2. Re-test after 30 minutes of circulation.
3. High pH in fresh coolant: check the concentrate mixing ratio; may be too concentrated.

4. High pH in aging coolant: check for contamination (e.g., hydraulic oil breakthrough can raise pH).
-

12.4 Bacterial Contamination Testing

12.4.1 Why Bacteria Matter

Metalworking fluids are aqueous solutions rich in carbon sources, nitrogen, and trace metals — an ideal growth medium for bacteria and fungi. Bacterial populations left uncontrolled will:

- Produce acids that lower pH, increasing corrosion risk
- Produce hydrogen sulfide (H₂S) and other malodorous compounds ("Monday morning smell")
- Metabolize lubricating components, reducing tool life
- Produce endotoxins that cause skin and respiratory reactions in workers
- Create biofilms that clog filters, strainers, and coolant lines

12.4.2 Dip-Slide Testing Procedure

Dip-slide tests (e.g., Dipslide, Cide-Slide) are the most practical bacterial testing method for field use. They use agar-coated plastic slides that are dipped in the coolant and then incubated.

Procedure:

1. Collect a fresh coolant sample from the middle depth of the tank.
2. Remove the dip-slide from its sterile packaging. Do not touch the agar surface.
3. Dip the slide in the coolant sample for 1 second. Shake off excess.
4. Seal the slide in its tube.
5. Incubate at 30–35°C for 24–48 hours (check manufacturer's recommendation).
6. Count the colony forming units (CFU) by comparing the colony density to the reference card provided with the test kit.

Interpretation:

Colony Count (CFU/mL)	Action
< 10,000	Normal; no action required
10,000–100,000	Elevated; add biocide and monitor
100,000–1,000,000	High contamination; add biocide urgently; increase pH; consider partial coolant replacement
> 1,000,000	Severe; immediate coolant replacement required

12.4.3 Biocide Treatment

WARNING: Always read and follow the biocide manufacturer's Safety Data Sheet before use. Many biocides are classified as irritants or sensitizers. Wear nitrile gloves and eye protection when handling biocide concentrates.

Approved Biocides for BC-500X:

Product	Type	Dose	Notes
Acticide MBS	Isothiazolinone blend	0.1–0.2% by volume of system	Broad spectrum
Proxel GXL	Benzisothiazolinone	0.05–0.1%	Low odor
Fungitrol 158	Triazine type	0.1–0.2%	Effective against sulfate-reducing bacteria

Biocide Treatment Procedure:

1. Confirm that system pH is ≥ 8.5 before adding biocide (biocides are less effective at low pH).
2. Calculate the required biocide volume: Volume (L) = System volume (320 L) $\times$ Dose rate (0.1%) = 0.32 L.
3. Add the biocide to the coolant sump slowly while the coolant pump is running.
4. Allow coolant to circulate for 2 hours.
5. Re-test with dip-slide at 24 hours.
6. If bacteria count has not dropped: repeat treatment or consider coolant replacement.

Biocide Rotation: To prevent bacteria from developing resistance, rotate biocide types every 3–6 months. Alternating between isothiazolinone-type and triazine-type biocides is an effective strategy.

12.5 Coolant Maintenance Procedures

12.5.1 Daily Coolant Maintenance

Performed as part of the daily shift startup inspection (Task D-02):

- Check coolant level (sight glass)
- Add coolant if level is below the LOW mark
- Visual inspection of coolant (color, odor)

12.5.2 Weekly Coolant Maintenance

- Measure coolant concentration with refractometer (Task W-01)
- Measure coolant pH (Task W-02)

- Clean coolant inlet strainer (Task W-03)
- Clean magnetic chip separator (Task M-05, also done weekly in high-chip applications)

12.5.3 Monthly Coolant Maintenance

- Bacterial dip-slide test (Task M-01)
- Tramp oil removal (see Section 12.6)
- Inspect coolant system for leaks (hoses, nozzle connections, pump seal)
- Check coolant pump current draw (measure with clamp meter; compare to rated current; significant increase indicates pump wear or restriction)

12.5.4 Coolant Sump Partitioning

The BC-500X coolant tank is not partitioned. For applications where chip contamination is severe (e.g., cast iron machining generating large volumes of fine abrasive particles), consider retrofitting a partitioned tank design or adding an external coolant filtration system. Contact BetaCorp Systems for options.

12.6 Tramp Oil Removal

Tramp oil (also called "way oil skimmings") is oil from the machine's way lube system, hydraulic system, and spindle lubrication that mixes with the coolant. Small amounts of tramp oil (< 1% by volume) are acceptable, but higher concentrations:

- Create foam during machining
- Coat the work zone, creating a surface that is difficult to see through
- Promote bacterial growth (some tramp oils support certain bacterial species)
- Reduce the effectiveness of the coolant's lubrication and cooling properties

Detection: Tramp oil appears as a floating layer on the coolant surface (visible when the coolant pump is stopped and the fluid is quiescent for 10 minutes). It is typically amber to dark brown in color and has an oily sheen.

Removal Methods:

Method 1: Tramp Oil Skimmer (Preferred) A belt-type or disc-type tramp oil skimmer (optional accessory, Part No. BC-OPT-SKIMMER) continuously removes surface oil from the coolant tank. This is the most effective approach for high-productivity applications.

Method 2: Manual Skimming For facilities without a skimmer:

1. Turn off the coolant pump.
2. Allow coolant to settle for 10 minutes.
3. Using a ladle or skimming vacuum with a flat intake, skim the tramp oil layer from the surface.

4. Dispose of tramp oil in designated waste oil containers (not coolant waste).
5. Perform this operation monthly or whenever oil layer exceeds 5 mm depth.

Tramp Oil Source Investigation: If tramp oil accumulation is excessive (requiring monthly skimming of more than 5 L):

1. Check the Z-axis hydraulic brake cylinder rod seal for leakage.
 2. Check the spindle oil-air lubrication system for oil blow-out.
 3. Check the way lube system for over-lubrication (way lube system pumping too frequently per B9.20).
 4. Inspect all hydraulic hose connections for minor seepage.
-

12.7 Coolant Disposal Procedures

Spent metalworking coolant is a regulated waste in most jurisdictions. The following disposal requirements apply:

Classification: In the United States, spent metalworking coolant (also called metalworking fluid or machining fluid waste) is typically classified as:

- Non-hazardous waste if the coolant does not contain RCRA-listed hazardous substances above threshold concentrations
- Potentially hazardous waste if the coolant contains oils with certain solvents or heavy metals from machining (particularly if chrome, nickel, or lead-containing materials are machined)

Regulatory Requirements:

- 40 CFR Part 261 — RCRA Hazardous Waste Identification (US EPA)
- 40 CFR Part 403 — General Pretreatment Regulations (for discharge to POTW)
- State and local regulations (vary significantly)

Disposal Options:

Option	Description	Considerations
Licensed waste disposal	Contract with a licensed fluid recycler or industrial waste hauler	Most common; document waste manifests
On-site treatment and discharge	Treat coolant to meet discharge limits, then discharge to municipal sewer	Requires permit; treatment typically includes pH adjustment and oil/water separation
Coolant recycling service	Service that hauls and recycles spent coolant (often resells as recycled fluid)	Cost-effective for large volumes
Evaporator	On-site evaporator reduces	Concentrate requires separate

Option	Description	Considerations
	volume by evaporating water	disposal

Documentation Required for Disposal:

- Waste characterization (pH, oil and grease content, specific metals if applicable)
- Quantity disposed
- Date of disposal
- Waste hauler name and license number
- Manifest number
- Receiving facility name and permit number

Contact your local EPA or state environmental agency for specific requirements.

12.8 Coolant System Troubleshooting

Problem	Possible Causes	Solution
Coolant foaming during machining	Coolant concentration too high; tramp oil contamination; wrong coolant for material; high air entrainment	Reduce concentration; remove tramp oil; check coolant compatibility; reduce pump flow
Strong odor (rotten egg)	Sulfate-reducing bacteria (SRB)	Immediate biocide treatment; if severe, replace coolant
Coolant turns black	Bacterial contamination; cast iron oxidation products	Biocide treatment; check pH; consider coolant replacement
Rust on workpiece	pH too low; insufficient concentration; chloride contamination	Test pH; adjust concentration; check water quality
Skin irritation reported	Coolant pH too high; bacterial contamination; concentration too high; coolant age	Test pH and concentration; bacteria test; coolant refresh
Poor tool life	Coolant concentration too low; coolant flow insufficient; wrong coolant for material	Increase concentration; check nozzle position and flow rate; evaluate coolant type
Coolant level dropping rapidly	Carry-out on workpiece; evaporation in hot environment; internal or external leak	Investigate carry-out; check environment; inspect for leaks
Chip conveyor coated with grey film	Calcium/magnesium deposits from hard water	Use water softener; add anti-scale additive to coolant

PART 12 — TOOL CHANGER SERVICE

Chapter 13 — Tool Changer Service Manual

13.1 Tool Changer Overview

The BC-500X 24-position random-access dual-arm ATC is a precision mechanism that must be maintained in correct adjustment to achieve reliable 4.2-second chip-to-chip times and accurate tool repositioning. This chapter covers all service procedures for the ATC system.

ATC Service Safety:

WARNING: The ATC arm moves rapidly (180° rotation in less than 0.9 seconds) and generates significant inertial forces. Never reach into the ATC area when the machine is energized unless:

1. You are certain the ATC is in the home (retracted) position AND
2. You are operating only in a manually controlled mode (SYSTEM → ATC → MANUAL with careful single-step actuation) AND
3. A second person is present as a safety observer

For any disassembly, adjustment, or replacement: LOTO the machine per Section 2.4.2.

ATC System Components — Recall:

- 24-position carousel (driven by 0.75 kW servo)
 - Dual-arm exchange arm (cam-driven by hydraulic motor)
 - Tool pots with spring-grip (24 pots)
 - Sensors: 5 ATC arm sensors + 1 carousel home + 24 tool presence sensors
-

13.2 Carousel Positioning and Indexing

13.2.1 Carousel Indexing Principle

The carousel is driven by an AC servo motor (0.75 kW) through a 60:1 worm gear reducer. The servo drive controls the carousel position using incremental encoder feedback (1,000 PPR on the motor shaft). The servo can detect and drive to any of the 24 tool pot positions.

Carousel Position Calculation:

- Carousel has 24 positions equally spaced at 15° each ($360^\circ / 24 = 15^\circ$).
- The worm gear reducer gives a total reduction of 60:1.
- To move one position (15°), the servo motor must rotate: $15^\circ \times 60$ (gear ratio) = $900^\circ = 2.5$ revolutions.
- At 1,000 PPR with quadrature: 4,000 counts per motor revolution $\times$ 2.5 revolutions = 10,000 encoder counts per carousel position.
- The PLC tracks the carousel position in tool pot units (1–24), converting to and from encoder counts internally.

Carousel Home Sensor: The carousel home sensor (BC-SEN-CAR-HOME) is an inductive proximity sensor that detects a notch or target on the carousel disc. When the carousel is in the home position, pot #1 is at the ATC tool change position. All pot positions are tracked relative to this home position.

13.2.2 Carousel Homing Procedure

The carousel must be homed:

- After any power cycle
- After a carousel fault (E304, E310)
- After any manual rotation of the carousel (maintenance)
- After encoder fault on the ATC carousel servo

Automatic Carousel Homing: The carousel homes automatically as part of the machine homing sequence (see Section 6.4). After machine homing is complete, the carousel is at the home position (pot #1 at ATC position).

Manual Carousel Homing: If the carousel must be homed independently:

1. Navigate to SYSTEM → ATC → MANUAL → CAROUSEL HOME.
2. The carousel will slowly rotate until the home sensor triggers.
3. Upon trigger, the carousel will make a small overshoot, then reverse slowly to confirm the home position.
4. When homing is complete, the display shows "CAROUSEL AT HOME."

Carousel Jog: In SYSTEM → ATC → MANUAL → CAROUSEL JOG, the carousel can be moved forward or backward by a specified number of positions using the softkey buttons.

13.2.3 Tool Pot Position Teaching

The tool pot positions are factory-programmed into Parameters B3.xx. If the carousel is replaced or a major ATC repair is performed, the tool pot positions may need to be re-taught.

NOTE: Tool pot position teaching is normally a factory commissioning procedure. In the field, contact BetaCorp Systems for guidance before performing this procedure. Incorrect pot position data will cause ATC arm timing errors.

Position Teaching Procedure (Overview):

1. Navigate to SYSTEM → ATC → TEACH MODE (requires Level 3 access — contact BetaCorp).
 2. The control will command the carousel to each position in sequence.
 3. At each position, use the CONFIRM softkey to record the encoder count for that pot.
 4. After all 24 positions are taught, the control saves the position data to Parameters B3.xx.
-

13.3 Gripper Adjustment and Replacement

13.3.1 ATC Arm Gripper Overview

The ATC arm has two grippers (left gripper BC-ATC-GRP-L and right gripper BC-ATC-GRP-R). Each gripper holds the tool holder by the V-groove of the tool holder flange. The gripper fingers are spring-loaded to grip the tool holder and release when the arm extends past the tool holder's V-groove during the retract phase.

Gripper Inspection:

Inspect the grippers monthly (task W-04). Replace if:

- Gripper pad tips show wear through the hardened surface
- Gripper fingers are bent or cracked
- Gripper spring force is insufficient (tool holder can be pulled free with less than 50 N force)

13.3.2 Gripper Replacement Procedure

Parts Required: BC-ATC-GRP-L or BC-ATC-GRP-R (specify); BC-ATC-GRP-SPR (2 springs per gripper)

Procedure:

1. LOTO the machine.
2. Navigate the ATC arm to the home position (retracted) using JOG mode before locking out.
3. Open the ATC access cover.
4. Locate the gripper to be replaced on the ATC arm.
5. Using a 5 mm hex key, remove the 2 × M6 socket head cap screws holding the gripper body to the arm.
6. Remove the gripper body.
7. Note the position of the gripper spring (photograph if needed).
8. Remove the old gripper spring from the spring pocket in the arm casting.
9. Install the new gripper spring in the spring pocket.

10. Install the new gripper body over the spring, compressing the spring as the gripper body is brought to the mounting face.
11. Insert and tighten the 2 × M6 screws: hand-tight only at first, ensuring the gripper body can slide slightly.
12. Check the gripper finger positions: both fingers should be parallel and centered on the tool holder V-groove engagement point.
13. Torque the 2 × M6 screws to 10 N·m.
14. Verify the gripper fingers move smoothly when pressed together by hand and return to the open position when released.
15. Close the ATC access cover.
16. Remove LOTO.
17. Perform a manual tool change cycle (SYSTEM → ATC → MANUAL → CYCLE) with a test tool to verify the gripper operates correctly.

13.3.3 Gripper Force Verification

After gripper replacement (or annually), verify the gripper retention force:

1. Insert a test tool holder into the carousel pot.
 2. Index the carousel to place the test tool at the ATC position.
 3. Using a spring scale or force gauge: attach to the tool holder flange and pull outward (away from the carousel center).
 4. The tool holder should remain retained up to a force of 80–90 N.
 5. Below 70 N: gripper spring is weak; replace spring (BC-ATC-GRP-SPR).
 6. Document the measured retention force in the maintenance log.
-

13.4 ATC Sensor Setup and Adjustment

13.4.1 Sensor Specifications

All ATC sensors are inductive proximity sensors (NPN, 24 VDC). Key specifications:

- Operating range (nominal sensing distance): 4 mm (standard, M12 body)
- Adjustment range: 2–6 mm (target to sensor face)
- Hysteresis: approximately 20% of switching distance
- Status LED: built-in (LED ON when target detected)
- Recommended gap: 2 mm (center of detection range)

13.4.2 ATC Arm Position Sensor Adjustment

Each sensor is mounted on an adjustable bracket with a slot for lateral adjustment. After any ATC disassembly or if sensors trigger at wrong positions:

General Adjustment Procedure:

1. LOTO the machine.
2. Manually position the ATC arm at the position corresponding to the sensor being adjusted (e.g., for SEN-ATC-HOME: position arm at home/retracted position).
3. Loosen the sensor mounting clamp screws (2 × M4, Torx T15).
4. Position the sensor face at 2 mm from the target (the metal actuating plate on the ATC arm casting).
5. Use a 2 mm thick feeler gauge as a spacer to set the gap precisely.
6. While maintaining the 2 mm gap, tighten the mounting screws (0.5 N·m — do not overtighten; sensor body is plastic).
7. Restore power and verify: using SYSTEM → DIAGNOSTIC → PLC I/O, confirm the sensor input is ON (1) when the arm is at the correct position and OFF (0) when the arm moves away.

Sensor Positions and Targets:

Sensor	Target Location	Expected State at Position
SEN-ATC-HOME	Target on arm rear face	ON when arm is fully retracted (home)
SEN-ATC-90	Target on arm at 90° cam position	ON when arm is 90° rotated (mid-swap)
SEN-ATC-180	Target on arm at 180° cam position	ON when arm is fully rotated (swap complete)
SEN-ATC-EXT	Target on arm front face	ON when arm is fully extended
SEN-ATC-RET	Target on arm rear face (different from HOME)	ON when arm is retracted after swap

NOTE: SEN-ATC-HOME and SEN-ATC-RET may appear similar but detect the arm at slightly different retract positions. HOME detects the arm before extension; RET detects the arm after the full exchange cycle returns to home. If these two sensors are swapped (target assignments reversed), the ATC will generate E303 and E301 alternately.

13.4.3 Tool Presence Sensor Adjustment

The 24 tool presence sensors (BC-SEN-TP-001) detect whether a tool holder is in each carousel pot. These sensors are mounted above each pot position, sensing the top of the tool holder from above.

Adjustment:

1. Insert a test tool holder in the pot being adjusted.
 2. Adjust the sensor height (vertical): sensor face should be 2–3 mm above the top of the tool holder's retention knob when the pot is at the sensing position.
 3. Verify: sensor LED turns ON when tool is in the pot, OFF when pot is empty.
 4. If sensor is too close (< 1.5 mm): risk of sensor striking the tool holder during carousel rotation.
 5. If sensor is too far (> 5 mm): sensor may not reliably detect the tool holder.
-

13.5 ATC Arm Timing and Adjustment

13.5.1 ATC Arm Cam Timing

The ATC arm motion is controlled by a cam profile that converts hydraulic motor rotation into the arm's extend-rotate-retract sequence. The cam is factory-set and should not normally require adjustment. However, if the ATC arm hydraulic motor, cam shaft, or associated components have been replaced, cam timing must be verified.

ATC Arm Motion Sequence:

The complete arm cycle consists of:

1. **Arm extend:** Arm moves forward simultaneously gripping tool in spindle AND tool in carousel (sensors: SEN-ATC-HOME goes OFF $\rightarrow$ SEN-ATC-EXT goes ON)
2. **Arm rotate 180°:** Arm rotates with both tools (sensor: SEN-ATC-90 goes ON, then SEN-ATC-180 goes ON)
3. **Arm retract:** Arm moves back, inserting new tool in spindle AND old tool in carousel (sensor: SEN-ATC-EXT goes OFF $\rightarrow$ SEN-ATC-RET goes ON)

ATC Timing Verification:

Navigate to SYSTEM $\rightarrow$ DIAGNOSTIC $\rightarrow$ ATC $\rightarrow$ TIMING ANALYSIS. This screen shows a real-time graph of sensor state transitions during an ATC cycle, with millisecond-resolution timestamps.

Expected timing (total cycle = 3.0 seconds for arm motion only):

Phase	Duration
Arm extend	0.6–0.9 seconds
Dwell at extended	0.1 seconds
Arm rotate 180°	0.7–1.0 seconds
Dwell at 180°	0.1 seconds
Arm retract	0.6–0.9 seconds

If any phase is significantly outside these ranges:

- Too slow: hydraulic pressure low, flow control valve too restrictive, hydraulic cylinder seal worn
- Too fast: flow control valve too open (risk of mechanical damage from impact); reduce flow control valve setting (B4.15)
- Phase not completing: sensor misaligned or failed (causing timeout fault)

13.5.2 ATC Synchronization with Spindle Orient

The ATC arm must not extend until the spindle has completed its orientation (M19 — spindle to defined angular position for keyway alignment with the ATC arm gripper). If the spindle orientation is late or slow, the ATC arm will wait until SEN-ATC-ORIENT confirms orientation is complete (this is a PLC interlock).

Verifying Spindle Orientation for ATC:

1. Command M19 in MDI mode.
2. Observe the spindle orientation angle on the diagnostic screen (SYSTEM → DIAGNOSTIC → SPINDLE → ORIENT ANGLE).
3. The spindle should come to rest with the drive key at the ATC gripper engagement position (typically 0° as defined by B1.19).
4. If the spindle orientation angle is incorrect by more than $\pm 2^\circ$: adjust Parameter B1.19.
5. After adjusting B1.19, perform 5 manual tool change cycles to verify the ATC arm reliably engages and disengages the tool holders.

13.6 Tool Changer Troubleshooting

Problem	Possible Cause	Solution
E303 — Tool Changer Timeout (arm retract)	Hydraulic pressure low; arm obstruction; retract sensor misaligned; hydraulic cylinder seal worn	See Section 8.7.1 (full E303 procedure)
Tool holder dropped during ATC	Gripper worn or broken; gripper spring weak; wrong tool holder type	Inspect and replace grippers; verify tool holder compatibility
Carousel does not index to correct position	Encoder fault; carousel home sensor misaligned; gear reducer worn	Re-home carousel; verify home sensor; contact BetaCorp
ATC arm crashes into spindle head	ATC Z position (B8.09) set incorrectly; Z-axis not at ATC position when ATC is activated	Verify B8.09; verify Z retract before ATC
Tool not fully clamped after ATC	Draw bar springs fatigued; tool unclamp solenoid stuck energized; tool clamping delay too short (B3.19)	Inspect draw bar springs; check SOL-HYD-02; increase B3.19

Problem	Possible Cause	Solution
Tool present sensor false positive	Chip or debris on sensor face; sensor too close to carousel pot	Clean sensor face; adjust sensor gap
ATC arm binds at 90° rotation	Cam follower worn; grease dried in cam track; hydraulic flow too low	Lubricate cam track; increase flow (B4.15); inspect cam follower
Chip-to-chip time increasing	Hydraulic system slow (low pressure, worn pump); ATC grease degraded; spindle orient slow	Check hydraulic pressure; regrease ATC; verify spindle orient time
Carousel rotates wrong direction	Phase rotation of ATC servo motor incorrect	Swap two motor phases (LOTO required)

End of Chapter 13 — Tool Changer Service Manual

PART 13 — APPENDICES

Appendix A — Glossary of Terms

Term	Definition
ATC	Automatic Tool Changer. The mechanism that automatically exchanges tools between the spindle and the tool magazine carousel.
AW	Anti-Wear. An additive package in lubricating oil (hydraulic or other) that prevents wear between metal surfaces. ISO VG 46 AW hydraulic oil is specified for the BC-500X.
BLI	Bearing Load Index. A BC-500X proprietary metric that tracks spindle bearing drag relative to the baseline at commissioning.
Belleville Washer	A conical (disc) spring washer used in stacks to provide high spring force in a compact space. The BC-500X draw bar uses a Belleville washer stack to provide 18,000 N tool clamping force.
BHN / HB	Brinell Hardness Number / Brinell Hardness. A measurement of material hardness using a ball indenter.
CAT-40	A common tool holder taper standard used in the United States. The number refers to the taper size (40 = 7/24 taper, ANSI/ASME B5.50).

Term	Definition
BT-40	A Japanese tool holder taper standard (MAS 403 BT-40). Similar taper angle to CAT-40 but with different flange and retention knob design.
Chip-to-Chip Time	The total time from the last cut with the old tool to the first cut with the new tool, including all ATC motions. BC-500X standard: 4.2 seconds.
CNC	Computer Numerical Control. The system that interprets G-code programs and commands the machine axes, spindle, and other functions.
DNC	Direct Numerical Control. A system where CNC programs are stored on a network computer and streamed to the machine as needed.
Draw Bar	The hardened steel rod that passes through the spindle center and actuates the tool retention collet.
EMC	Electromagnetic Compatibility. The ability of a machine to operate without causing electromagnetic interference to other equipment, and to operate correctly in the presence of electromagnetic interference from outside sources.
EMI	Electromagnetic Interference. Electrical noise that can corrupt sensitive signals such as encoder feedback cables.
E-Stop	Emergency Stop. A safety device (red mushroom-head pushbutton) that, when pressed, immediately removes power from all machine drives and halts all motion.
FRL	Filter-Regulator-Lubricator. A pneumatic system component that filters the air supply, regulates pressure, and (in some designs) adds lubrication.
G-code	The programming language used for CNC machines. Named for the "G" (geometric) codes that command axis motion and interpolation.
HMI	Human-Machine Interface. The display, keyboard, and control panel through which the operator interacts with the machine.
HPU	Hydraulic Power Unit. The self-contained assembly that generates and supplies hydraulic power to the machine circuits.
ISO 4406	An international standard for reporting hydraulic fluid contamination levels using a three-number cleanliness code.
ISO VG	International Standards Organization Viscosity

Term	Definition
	Grade. A classification of lubricating oils by their kinematic viscosity at 40°C. ISO VG 46 = 46 cSt at 40°C (approximately).
JOG	A manual machine operation mode where the operator manually moves axes by pressing and holding jog buttons.
LOTO	Lockout/Tagout. An OSHA-required safety procedure for isolating hazardous energy sources before maintenance.
MPG	Manual Pulse Generator. The hand wheel device that allows precise, incremental axis motion during machine setup.
N·m	Newton-meter. The SI unit of torque. 1 N·m $\approx$ 0.738 ft·lb.
NPN	Negative-Positive-Negative. A transistor output type used in proximity sensors. An NPN sensor pulls the output signal to ground (0V) when the target is detected.
ORFS	O-Ring Face Seal. A hydraulic fitting design that uses an O-ring at the face of the fitting for sealing, providing excellent leak resistance.
PLC	Programmable Logic Controller. The machine's dedicated industrial computer that handles all discrete I/O, sequencing logic, and machine safety functions.
PPE	Personal Protective Equipment. Safety equipment worn by workers to protect against hazards (safety glasses, gloves, face shields, etc.).
PPR	Pulses Per Revolution. The number of electrical pulses generated by an encoder for each complete revolution of the encoder shaft.
PTFE	Polytetrafluoroethylene ("Teflon"). Used for thread sealant tape on pipe threads.
Quadrature Encoding	A method of processing encoder signals that multiplies the effective resolution by 4. Used on all BC-500X axis encoders (2,500 PPR $\times$ 4 = 10,000 counts/rev effective resolution).
RC (Resistance-Capacitance)	Used in the context of encoder cables: RC time constant affects the ability to transmit high-frequency signals over long cable lengths.
RS-422	A balanced (differential) electrical standard for serial communication, used for encoder signals. Differential signaling provides excellent noise immunity.

Term	Definition
SDS	Safety Data Sheet. A document required by OSHA's HazCom standard that describes the hazards, safe handling, and emergency response for a chemical product.
Servo	A closed-loop control system where position, velocity, or torque is continuously measured and corrected to match the commanded value. BC-500X uses digital servo drives on all axes.
Swarf	Metal cuttings and chips produced during machining operations.
TAN	Total Acid Number. A measurement of the acidity of hydraulic oil. Increasing TAN indicates oil oxidation and breakdown.
TIR	Total Indicator Reading. The total variation (max - min) measured by a dial indicator. Used to measure runout, flatness, and similar geometric quantities.
TSC	Through-Spindle Coolant. An optional BC-500X system that delivers high-pressure coolant through the center of the spindle and toolholder directly to the cutting edge.
VFD	Variable Frequency Drive. An electronic device that controls the speed of an AC induction motor by varying the output frequency. Used for the coolant pump and chip conveyor.
WEEE	Waste Electrical and Electronic Equipment Directive (EU). Requires separate collection and recycling of electronic equipment.
Way Lube	Lubricating oil applied to the linear guideways and ballscrew nuts. The BC-500X uses ISO VG 68 way lube oil in the centralized automatic lubrication system.
Wiper Seal	A sealing element at the end of a guideway carriage or linear actuator that prevents contamination from entering the bearing surfaces while allowing the moving element to pass.
Zerk Fitting	A common name for a grease nipple (Alemite fitting). A small steel fitting that accepts a grease gun for applying lubricant under pressure.

Appendix B — Hydraulic Fluid Specifications

B.1 Primary Specification — ISO VG 46 AW Hydraulic Oil

The BC-500X requires ISO VG 46 anti-wear hydraulic oil meeting the following specifications:

Property	Specification	Test Method
ISO Viscosity Grade	VG 46	ISO 3448
Kinematic viscosity at 40°C	41.4–50.6 mm ² /s (cSt)	ISO 3104
Kinematic viscosity at 100°C	≥ 6.0 mm ² /s	ISO 3104
Viscosity index	≥ 95	ISO 2909
Density at 15°C	860–900 kg/m ³ (typical)	ISO 3675
Flash point (Cleveland Open Cup)	≥ 200°C (392°F)	ISO 2592
Pour point	≤ -12°C (10°F)	ISO 3016
Acid number (TAN), new oil	≤ 1.0 mg KOH/g	ASTM D974
Water content, new oil	≤ 200 ppm	Karl Fischer
Foam tendency / stability	Seq. I: ≤ 150 mL / 0 mL	ASTM D892
Foam tendency / stability	Seq. II: ≤ 75 mL / 0 mL	ASTM D892
Foam tendency / stability	Seq. III: ≤ 150 mL / 0 mL	ASTM D892
Air release, 25°C	≤ 5 minutes	ISO 9120
Demulsibility (40-37-3), 82°C	≤ 30 minutes	ASTM D1401
Rust protection (distilled water)	Pass	ASTM D665A
Rust protection (salt water)	Pass (preferred)	ASTM D665B
Copper corrosion, 100°C, 3 h	≤ 1b	ASTM D130
Wear protection (4-ball test)	Wear scar ≤ 0.60 mm	ASTM D4172
FZG gear test	Failure load stage ≥ 10	ISO 14635-1
Oxidation stability (RBOT)	≥ 300 minutes	ASTM D2272
Zinc content (ZDDP-type AW)	600–1200 ppm (typical)	ASTM D4628

B.2 Alternative Specifications

Zinc-Free (Ashless) AW Hydraulic Oil

For applications requiring zinc-free hydraulic oil (e.g., compatibility with zinc-sensitive filters, environmental preference):

- Must meet all specifications above except zinc content
- Must demonstrate equivalent wear protection in FZG test (≥ Stage 10) or ASTM D4172 (wear scar ≤ 0.55 mm)
- Contact BetaCorp Systems to confirm compatibility before switching

Environmentally Acceptable Lubricant (EAL)

For machines installed in environmentally sensitive locations:

- ISO VG 46 vegetable-based or biodegradable synthetic hydraulic fluid
- Must meet ISO 15380 specification (HETG, HEPG, or HEES types)
- Requires full seal compatibility verification with BetaCorp Systems
- Not all seals in the BC-500X are compatible with ester-based biodegradable fluids; seal replacement kit may be required

B.3 Oil Change Intervals Based on Analysis

The recommended oil change interval is 2,000 machine operating hours. However, oil analysis may extend or shorten this interval:

Condition	Recommended Action
TAN $\leq$ 1.0 mg KOH/g; cleanliness $\leq$ 16/14/11	Continue in service; re-test at next quarter
TAN 1.0–2.0 mg KOH/g	Plan oil change within next 500 hours
TAN $>$ 2.0 mg KOH/g	Immediate oil change
Water $>$ 0.1%	Immediate oil change; investigate water source
Cleanliness $>$ 18/16/13	Change oil and filter; investigate contamination source

Appendix C — Torque Specifications Chart

C.1 Machine Assembly Torques

All torque values are for unlubricated threads unless marked (L) = lubricated with engine oil.

Machine Structure:

Location	Fastener	Torque
Machine leveling footpad anchor bolts	M24 grade 8.8	200 N·m (147 ft·lb)
Leveling bolt jam nut	M24	300 N·m (221 ft·lb)
Chip conveyor mounting bolts	M12 grade 8.8	60 N·m (44 ft·lb)
HPU mounting bolts	M16 grade 8.8	160 N·m (118 ft·lb)

Spindle Assembly:

Location	Fastener	Torque
Spindle cartridge mounting	M16 grade 10.9	180 N·m (133 ft·lb) (L)

Location	Fastener	Torque
bolts		
Gearbox mounting bolts	M12 grade 8.8	60 N·m (44 ft·lb)
Spindle motor mounting bolts	M10 grade 8.8	35 N·m (26 ft·lb)
Encoder coupling hub (motor shaft)	M5 set screw	4.5 N·m (40 in·lb)
Encoder housing mounting screws	M5	5 N·m (44 in·lb)
Tool unclamp cylinder mounting bolts	M10 grade 8.8	35 N·m (26 ft·lb)
Hydraulic brake cylinder mounting bolts	M10 grade 8.8	40 N·m (30 ft·lb)

Axis Assemblies:

Location	Fastener	Torque
Servo motor mounting bolts (X, Y, Z)	M12 grade 8.8	60 N·m (44 ft·lb)
Bellows coupling shaft grub screws	M6 (to shaft)	12 N·m (106 in·lb)
Ballscrew fixed end housing bolts	M12 grade 8.8	60 N·m (44 ft·lb) (L)
Ballscrew nut housing bolts	M10 grade 8.8	40 N·m (30 ft·lb) (L)
Guideway rail mounting bolts	M10 grade 10.9	50 N·m (37 ft·lb) (L)
Guideway carriage mounting bolts (to axis casting)	M8 grade 8.8	25 N·m (22 ft·lb) (L)
Table mounting bolts (X-carriage to X-table)	M16 grade 10.9	180 N·m (133 ft·lb) (L)

Hydraulic System:

Location	Fastener	Torque
HPU drain plug	M20 × 1.5	35 N·m (26 ft·lb)
HPU filter housing	G3/4 thread	Hand-tight + 3/4 turn
ORFS fittings, G1/4	—	30 N·m (22 ft·lb)
ORFS fittings, G3/8	—	50 N·m (37 ft·lb)
ORFS fittings, G1/2	—	70 N·m (52 ft·lb)
ORFS fittings, G3/4	—	100 N·m (74 ft·lb)
ORFS fittings, G1	—	140 N·m (103 ft·lb)
HPU pressure switch	G1/4 BSP	25 N·m (18 ft·lb)
Relief valve (main)	G3/4 BSP	80 N·m (59 ft·lb)

Electrical:

Location	Fastener	Torque
Main power terminal L1/L2/L3	M10 lugs	20 N·m (177 in·lb)
Ground bus connections	M8 lugs	6 N·m (53 in·lb)
Control transformer terminals	M4 screw	1.5 N·m (13 in·lb)
Servo drive motor terminals	M4 screw	2.5 N·m (22 in·lb)
24 VDC power supply output terminals	M4 screw	1.0 N·m (9 in·lb)
PLC/I/O module terminal block	3 mm flat screw	0.5 N·m (4 in·lb)

ATC System:

Location	Fastener	Torque
ATC arm gripper mounting screws	M6	10 N·m (89 in·lb)
Tool pot mounting bolt	M8	20 N·m (177 in·lb)
ATC sensor mounting clamp screws	M4	0.5 N·m (do not overtighten — plastic body)

Appendix D — Electrical Schematic Index

The following electrical drawings are provided as a separate bound drawing package (BetaCorp Drawing Package BC-DWG-BC500X-ELEC). Contact BetaCorp Systems to obtain the drawing package for your machine (reference machine serial number).

Drawing No.	Title	Sheet Count
BC-E-001	Main power distribution schematic	3 sheets
BC-E-002	Spindle drive wiring	2 sheets
BC-E-003	X, Y, Z servo drive wiring	4 sheets
BC-E-004	ATC servo drive wiring	1 sheet
BC-E-005	Hydraulic system motor and control wiring	2 sheets
BC-E-006	Coolant system wiring	1 sheet
BC-E-007	Chip conveyor wiring	1 sheet
BC-E-008	Safety relay circuit (E-Stop, door interlock)	2 sheets
BC-E-009	PLC I/O wiring — digital	4 sheets

Drawing No.	Title	Sheet Count
	inputs (DI)	
BC-E-010	PLC I/O wiring — digital outputs (DO)	4 sheets
BC-E-011	Sensor wiring — all proximity sensors	3 sheets
BC-E-012	Encoder wiring — all axes and spindle	2 sheets
BC-E-013	Control panel wiring	2 sheets
BC-E-014	MPG pendant wiring	1 sheet
BC-E-015	Cabinet layout drawing	2 sheets
BC-E-016	Grounding diagram	1 sheet
BC-E-017	24 VDC power distribution	2 sheets
BC-E-018	Hydraulic solenoid valve wiring	1 sheet

Appendix E — Pneumatic Schematic Index

Drawing No.	Title	Sheet Count
BC-P-001	Pneumatic system overview schematic	1 sheet
BC-P-002	FRL assembly detail	1 sheet
BC-P-003	Spindle air purge circuit	1 sheet
BC-P-004	Chip blow-off circuit (optional)	1 sheet
BC-P-005	Gear change pneumatic assist circuit	1 sheet
BC-P-006	ATC arm pneumatic pilot circuit	1 sheet

Pneumatic schematics are included in the drawing package BC-DWG-BC500X-PNEU.

Appendix F — Service Contact Directory

F.1 BetaCorp Systems Contact Information

Headquarters:

BetaCorp Systems, Inc. 4500 Industrial Parkway Greenfield, OH 45123 United States of America

General Contacts:

Department	Phone	Email	Hours
Main Switchboard	+1 (740) 555-9200	info@betacorpssystem.com	Mon–Fri 8:00–17:00 EST
Technical Support	+1 (800) 555-2382 (BETA)	techsupport@betacorpssystem.com	Mon–Fri 7:00–18:00 EST
Parts Department	+1 (800) 555-2382 Option 2	parts@betacorpssystem.com	Mon–Fri 7:00–18:00 EST
Service Dispatch	+1 (800) 555-2382 Option 3	service@betacorpssystem.com	Mon–Fri 7:00–18:00 EST
Emergency Hotline	+1 (800) 555-9911	—	24/7/365
Technical Documents	—	docs@betacorpssystem.com	—
Customer Portal	docs.betacorpssystem.com	—	24/7 online
Training Department	+1 (740) 555-9245	training@betacorpssystem.com	Mon–Fri 8:00–16:00 EST
Warranty Claims	+1 (740) 555-9230	warranty@betacorpssystem.com	Mon–Fri 8:00–17:00 EST

Regional Service Centers:

Region	Location	Phone
Northeast US	Hartford, CT	+1 (860) 555-4100
Southeast US	Charlotte, NC	+1 (704) 555-4200
Midwest US	Cincinnati, OH	+1 (513) 555-4300
Southwest US	Dallas, TX	+1 (972) 555-4400
West Coast US	Los Angeles, CA	+1 (213) 555-4500
Canada	Mississauga, ON	+1 (905) 555-4600
Mexico	Monterrey	+52 81 555-4700
Germany (EU)	Stuttgart	+49 711 555-4800
United Kingdom	Birmingham	+44 121 555-4900
Japan	Nagoya	+81 52 555-4950
China	Shanghai	+86 21 555-4960

F.2 Emergency Contact Protocol

For machine-down emergencies:

1. Call the 24/7 Emergency Hotline: **+1 (800) 555-9911**
2. Provide:
 - Machine model (BC-500X)

- Machine serial number
- Nature of emergency / fault code
- Plant location and contact name
- Contact phone number on-site

3. A BetaCorp emergency service coordinator will respond within 30 minutes to assess the situation and dispatch a service engineer if required.

Emergency parts orders can be placed through the same hotline for next-flight-out or next-morning delivery of critical components.

Appendix G — Spare Parts Quick-Reference Card

Critical Spares — Recommended On-Site Stock:

Part No.	Description	Qty
BC-FLT-HYD-001	Hydraulic return filter element	2
BC-SW-CLT-LVL	Coolant level float switch	1
BC-PSW-HYD-MAIN	Hydraulic pressure switch	1
BC-CBL-SPENC-3M	Spindle encoder cable	1
BC-CBL-ENC-5M	Axis encoder cable (X, Y, Z)	2
BC-SEN-ATC-RET	ATC retract sensor	1
BC-SEN-ATC-HOME	ATC home sensor	1
BC-ATC-GRP-L	ATC arm gripper, left	1
BC-ATC-GRP-R	ATC arm gripper, right	1
BC-POT-CAT40 (or BT40)	Tool pot, complete	4
BC-GRP-CAT40-SET (or BT40)	Gripper finger set	4
BC-BATT-CNC-001	CNC backup battery	2
BC-FLT-AIR-001	Pneumatic FRL filter	2
BC-OIL-HYD-VG46	Hydraulic oil, 20 L	1
BC-LUB-OIL-VG68	Way lube oil, 5 L	1
BC-GRS-ATC-001	ATC grease, 400 g	1
BC-SEAL-HYD-DRN	HPU drain plug seal	4
BC-WPR-X-001	X-axis wiper seal set	1
BC-WPR-Y-001	Y-axis wiper seal set	1
BC-WPR-Z-001	Z-axis wiper seal set	1

Appendix H — Parameter Quick-Reference Card

Most Frequently Accessed Parameters:

Parameter	Name	Default	Adjust for
B1.01	Spindle Maximum Speed	10000 RPM	Max speed limit
B1.14	Spindle Speed Window	50 RPM	At-speed tolerance
B1.19	Spindle Orientation Angle	0.0°	ATC alignment
B1.22	Spindle Brake Enable	1	Brake on deceleration
B2.33	Rapid Traverse X	36000 mm/min	Max rapid speed
B2.35	Rapid Traverse Z	30000 mm/min	Max Z rapid speed
B2.45	In-Position Window	0.005 mm	Settling tolerance
B3.12	ATC Arm Retract Timeout	2000 ms	E303 trigger time
B3.14	ATC Carousel Short Path	1	Shortest index direction
B4.02	HPU Pressure Min Alarm	120 bar	E404 trigger pressure
B5.02	Coolant Level Min Alarm	1	E202 enable
B5.03	Coolant Level Min Delay	30 s	E202 delay
B6.01	X-Axis Encoder PPR	2500	X encoder resolution
B6.31	Spindle Encoder PPR	4096	Spindle encoder resolution
B8.01	Door Interlock Mode	2	Door safety level
B8.03	Door Open Max Speed	1000 mm/min	Safe speed with door open
B9.20	Way Lube Interval	30 min	Lube cycle frequency
B9.21	Way Lube Pump Time	15 s	Lube pump duration

Appendix I — Maintenance Log Template

BC-500X Machine Maintenance Log

Machine Serial Number: _____ Machine Location: _____

Instructions:

- Complete one entry for each maintenance task performed
- Entries must be legible and complete
- Retain logs for a minimum of 5 years
- Contact BetaCorp Systems before any work not described in this manual

Date	Machine Hours	Task Code	Task Description	Parts Used	Part Numbers	Technician	Signature	Abnormalities Found	Actions Taken

Task Codes (abbreviated):

Code	Description
D-01	Daily visual inspection
D-02	Coolant level check
D-03	Way lube level check
D-04	Chip bin emptying
D-05	Work zone chip removal
D-06	Daily function test
W-01	Coolant concentration
W-02	Coolant pH
W-03	Coolant strainer cleaning
W-04	ATC inspection
W-05	Guideway wiper inspection
M-01	Bacteria test
M-02	Hydraulic filter inspection
M-03	Encoder cable inspection
M-04	Way lube system check

Code	Description
M-05	Magnetic separator cleaning
Q-01	Axis accuracy check
Q-02	Spindle run-out check
Q-03	ATC sensor verification
Q-04	Hydraulic oil sample
S-01	Hydraulic oil change
S-02	Coolant replacement
S-03	Coolant tank cleaning
S-04	Motor terminal torque
A-01	Annual geometry calibration
A-02	Spindle bearing inspection
FAULT	Fault investigation / repair
REPAIR	Unscheduled repair
OTHER	Other (describe in task description)

Appendix J — Warranty Registration

BC-500X Warranty Registration Form

To register your BC-500X for warranty service, complete and return this form within 30 days of machine acceptance. Warranty coverage begins on the date of machine acceptance as documented on the Installation Acceptance Certificate.

Customer Information:

Field	Entry
Company Name	
Address (Line 1)	
Address (Line 2)	
City, State/Province	
Postal Code, Country	
Primary Contact Name	
Contact Title	
Contact Phone	
Contact Email	
Accounts Payable Contact	
AP Phone / Email	

Machine Information:

Field	Entry
Machine Model	BC-500X
Machine Serial Number	
Purchase Order Number	
BetaCorp Invoice Number	
Dealer / Distributor Name	
Date of Machine Delivery	
Date of Machine Acceptance	
Installing BetaCorp Engineer	
Customer Representative Signature	
Customer Representative Name	
Date of Signature	

Machine Installation Location (if different from above):

Field	Entry
Plant/Facility Name	
Address	
Maintenance Contact	
Maintenance Phone	

Return this form by:

- Email: warranty@betacorpystems.com (scan or photograph)
- Mail: BetaCorp Systems, Inc., ATTN: Warranty Registration, 4500 Industrial Parkway, Greenfield, OH 45123
- Online: www.betacorpystems.com/warranty-registration

Warranty Terms Summary:

The BC-500X is warranted for **24 months from date of acceptance** or **30 months from date of shipment from BetaCorp factory**, whichever comes first, against defects in materials and workmanship. This warranty does not cover normal wear items, operator-caused damage, or damage from failure to follow maintenance procedures. Complete warranty terms are available at www.betacorpystems.com/warranty.

End of Manual

BetaCorp Systems, Inc. 4500 Industrial Parkway Greenfield, OH 45123 USA

BC-500X CNC MACHINING CENTER COMPREHENSIVE TECHNICAL MANUAL

Document No.: BCS-MAN-BC500X-REV4.2 Revision Date: September 2026

This document is the exclusive property of BetaCorp Systems, Inc. Reproduction in whole or in part without written permission is prohibited.

For the most current version of this manual, visit: docs.betacorpystems.com

QUICK REFERENCE — KEY FAULT CODES

Code	Name	This is NOT...
E101	Spindle Encoder Signal Lost	NOT a voltage/electrical fault
E202	Coolant Level Below Minimum	NOT a temperature sensor fault
E303	Tool Changer Timeout	NOT a spindle speed fault
E404	Hydraulic Pressure Low	NOT a coolant flow fault
E505	Axis Servo Overload	NOT a door interlock fault

These codes are specific to the BC-500X. Do not confuse with other machine models.

End of BC-500X Comprehensive Technical Manual BetaCorp Systems, Inc. — Document BCS-MAN-BC500X-REV4.2